TopNET

User’s Manual

TOPCON
POSITIONING SYSTEMS

TopNET User's Manual

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Preface

Thank you for purchasing your Topcon receiver, survey product or accessory (the "Product"). The materials available in this manual (the "Manual") have been prepared by Topcon Positioning Systems, Inc. ("TPS") for owners of Topcon products. This Manual is designed to assist owners with the use of software (the "Software") to be used with the Product and its use is subject to these terms and conditions (the "Terms and Conditions").

NOTICE *NOTICE*

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Manual Conventions

This manual uses the following conventions:

Example	Explanation
File ▸ Exit	Click the File menu and click Exit.
Enter	Press the button labeled Enter.
Topo	Indicates the name of a dialog box, or screen.
Notes	Indicates a field on a dialog box or screen, or a tab within a dialog box or screen.

TIP

Supplementary information that can help you configure, maintain, set up, or use a system.

NOTICE

Supplementary information that can have an affect on system operation, system performance, measurements, personal safety.

CAUTION

Notification that an action has the potential to adversely affect system operation, system performance, data integrity, or personal health.

Notes:

TOPCON

POSITIONING SYSTEMS

Chapter 1. TopNET-S

User's Manual

List of contents:

Introduction

Welcome to TopNET-S™, the server portion of Topcon's unique and powerful software package designed to offer the easiest and most reliable method of managing any number of Continuously Operating Reference Stations (CORS).

TopNET-S Server connects TopNET applications with TPS receivers. Figure 1-1 illustrates the layout of a TopNET CORS™ network, which consists of multiple reference stations connected to a central server station. At the same time multiple computers (clients) running TopNET applications connect to the server station to receive and monitor CORS data.

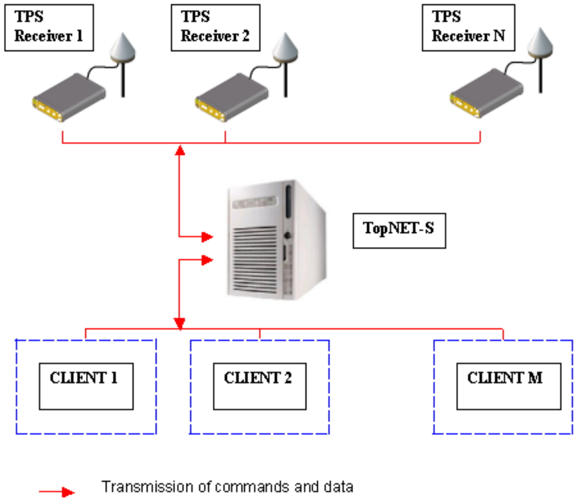


Figure 1-1. TopNET CORS Network Configuration

NOTICE

Refer to the TopNET Reference Manual for installing TopNET-S on the desired computer.

TPS Receiver Requirements

To connect with a TPS receiver used in TopNET CORS, the receiver must have firmware of version 2.4, or higher.

Connecting Receiver and TopNET-S server

The TopNET-S server requires internet access and high-speed data transfer ports to monitor and manage the CORS network. The following sections describe the hardware and software requirements for connecting the receiver and TopNET-S.

Ethernet Connection Requirements

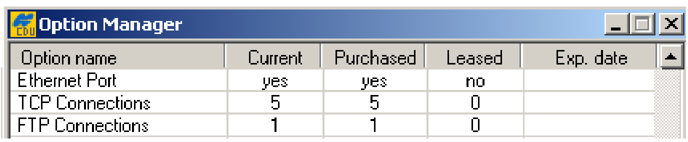
Ethernet supports a fast data transfer rate between the computer with the TopNET-S server and TPS receiver. When connecting via an Ethernet port, the receiver must have the following options:

- A physically mounted ethernet port on the bottom right of the receiver panel (Figure 1-2).

Figure 1-2. TPS Receiver Ethernet Port

- The Option Authorization File (OAF) must have the required options enabled, either Purchased or Leased.
  - – Ethernet Port – yes
  - – TCP Connections – from 1 to 5
  - – FTP Connections – either 0 or 1

Figure 1-3 shows the maximum possible configuration of options that allows the TopNET-S server to communicate with TPS receivers connected via the Internet and through the Ethernet port.



Option name	Current	Purchased	Leased	Exp. date
Ethernet Port	yes	yes	no	
TCP Connections	5	5	0	
FTP Connections	1	1	0	

Figure 1-3. Maximum OAF Requirements for Ethernet

Use one of the following two methods for connecting a TPS receiver to the TopNET-S server via the Internet using the Ethernet option.

Method 1: TCP Only Ethernet Connection

One Ethernet connection method uses the TCP protocol to transfer control commands to the receiver and download TPS receiver files from the receiver to the client. This method requires the following minimum configuration options:

- Ethernet Port – yes
- TCP Connections – 2
- FTP Connections – 0

Method 2: TCP and FTP Ethernet Connection

Another Ethernet connection method uses the TCP protocol to transfer control commands to the receiver and the FTP protocol to download TPS receiver files from the receiver to the client. This method requires the following minimum configuration options:

- Ethernet Port – yes
- TCP Connections – 1
- FTP Connections – 1

Starting TopNET-S

NOTICE

See the TopNET Reference Manual for installing TopNET-S on the desired computer.

Once installed, the TopNET-S server is started as an Automatic Startup Windows NT service. That means, it operates continuously until uninstalled, or stopped manually. Current status of the TopNET-S server service can be seen in the list of available services of the computer.

TopNET-S requires a Hardware Protection Key to be inserted in the appropriate port of the computer. The initial TopNET package contains one key specifically for TopNET-S. Refer to the *TopNET Reference Manual* for details on the Hardware Protection Key. If needed, contact your local Topcon representative to purchase more keys.

To view and control operation of the server, the TopNET-S application is used as an interface. Double-click a TopNET-S icon on the desktop, or select it from the Start menu.

When running for the first time, the TopNET-S server is available for users who try to connect through port 8000 to all network IP addresses of the computer, that is displayed on the title bar of the TopNET-S interface (Figure 1-4).

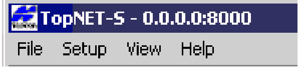


Figure 1-4. Configuration After First Start of TopNET-S

Getting Acquainted

The TopNET-S interface main window has the following components (Figure 1-5):

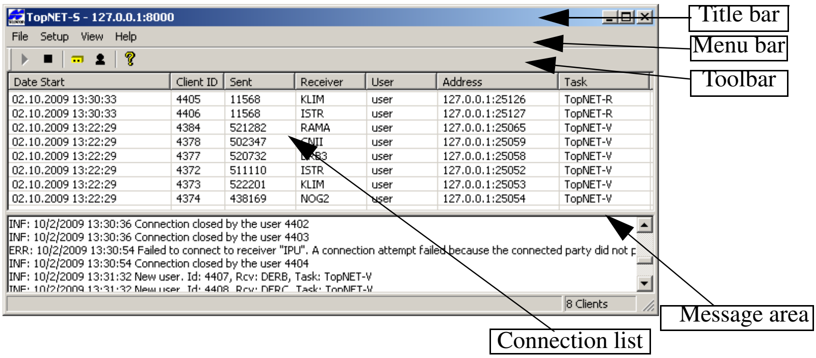


Figure 1-5. Main Window

Here:

- Title bar – displays the name of the application, server's IP address and port number.
- Menu bar – contains menus used to run TopNET-S server functions.
- Toolbar – contains shortcut buttons to commands.
- Connection list – refreshes every second, displays the following information:
  – *Date Start*: time when a client had been connected to the TopNET-S server.
  – *Client ID* - a serial identification number of a client assigned by the server.
  – *Sent* - the number of bytes sent by the receiver to a user. This field displays "–" if a user logged on to the system but did not connect to any receiver.
  – *Receiver* - the name of the receiver. This field displays "–" if a user logged on to the system but did not connect to any receiver.

– *User* - the name of the authorized user requesting the receiver.

– *Address* - the IP address and connection port of the user's computer where the application runs.

– *Task* - the name of the client program connected to the TopNET-S server.

Below the Connection list, a Message area is located. It displays service information about communication between the receivers, TopNET-S server, and clients.

Menu bar

The menu bar provides access to options available in the TopNET-S server.

File Menu

The **File** menu (Figure 1-6) is used to start or stop operation of the server, and close the TopNET-S interface application.

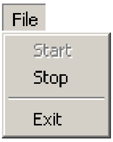


Figure 1-6. File Menu

Setup Menu

The **Setup** menu (Figure 1-7 on page 1-9) provides the following functions:

- selects an IP address and sets the port number (via Configuration dialog box. For details see "Choosing a Configuration" on page 1-11).
- edits users' list. See "Configuring Users" on page 1-21.

- edits receivers' list. See "Connecting TPS Receivers" on page 1-13.

Figure 1-7. Setup Menu

View Menu

The **View** menu (Figure 1-8) provides an opportunity to hide Toolbar and Status Bar.

Figure 1-8. View Menu

To hide a bar, remove the check mark from the corresponding menu entry by clicking it.

Help Menu

The Help menu (Figure 1-9) provides access to TopNET-S version and build date information.

Figure 1-9. Help Menu

Toolbar

The toolbar for the TopNET-S interface (Figure 1-10) contains buttons for frequently used functions.

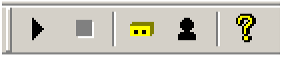


Figure 1-10. Toolbar

Table 1-1 lists and describes the various Toolbar buttons.

Table 1-1. Toolbar Buttons

Button	Description
	Starts operation of the server. Duplicates **File** ▸ **Start** selection.
	Stops operation of the server. Duplicates **File** ▸ **Stop** selection.
	Receivers – Opens the **Setup Receiver** dialog box. Duplicates **Setup** ▸ **Receivers...** selection.
	Users – Opens the **Setup Users** dialog box.Duplicates **Setup** ▸ **Users...** selection.
	About TopNET-S – Displays the **About TopNET-S** dialog box. Duplicates **Help** ▸ **About** selection.

Choosing a Configuration

Configuration Notes

- If a user has a Network Interface Card (NIC) connected to a Local Area Network, then the connected computer is identified by an IP address, say, 111.111.11.187. This address is assigned either by the network administrator, or by an automatic service running in the LAN.
- If a user has more than one NIC connected, each card's IP addresses will be listed in the TopNET-S interface (Figure 1-13).
- If the computer has a modem, a wireless (WiFi, 802.11) card, or a Bluetooth device, they can be assigned IP addresses when they are connected to a TCP/IP network service (dial-up, WAN, etc.), and these IP addresses will be listed in the TopNET-S interface as well. Two special addresses are always available (even when the computer is not connected to any network). These are 127.0.0.1, also known as the "loopback" address, and 0.0.0.0, also known as "any".
  - If a loopback address (127.0.0.1) is selected, the TopNET-S server will respond only to connection requests received from TopNET clients running on the same computer.
  - If the address 0.0.0.0 is selected, then all clients regardless of their IP addresses will be allowed to connect.
- If a user selects a specific address, only clients accessing this address have permission to connect to the server.

Choosing an IP Address and Port

As mentioned above, when running for the first time, the TopNET-S server is available for users who try to connect through the 8000 port to any network IP address of the computer, that is displayed on the title bar of the TopNET-S interface (Figure 1-11).

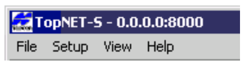


Figure 1-11. Configuration After First Start of TopNET-S

1. To select an IP address from the list on the computer, click **Setup ▸ Configuration** (Figure 1-12).

Figure 1-12. Open Boot Configuration

In the *Configuration* screen, select the desired IP address and port of connection (Figure 1-13).

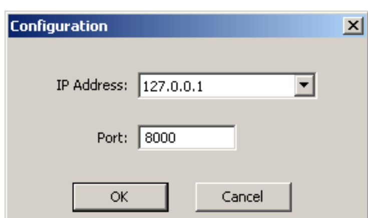


Figure 1-13. Select an IP Address

Press **OK** to save the settings.

2. Press **OK** at the information message and restart the TopNET-S server using **Start** and **Stop** buttons, or menu items.

Connecting TPS Receivers

Creating a Receiver List

When running TopNET-S for the first time, no receivers are connected to the TopNET-S server, and therefore the list of receivers in the TopNET-S interface is empty. Use the following procedure to setup the receivers used in the CORS network.

1. Click **Setup ▸ Receivers...** or click **Receivers** icon on the toolbar (Figure 1-1).

Figure 1-1. Setup Receivers

2. On the **Setup Receiver** dialog box a list of receivers is shown. (Figure 1-2 on page 1-13). At the first start, the list is empty. To add a receiver, press **Add**.

Figure 1-2. Setup Receiver

In the **Add Receiver** screen enter a name for the receiver.

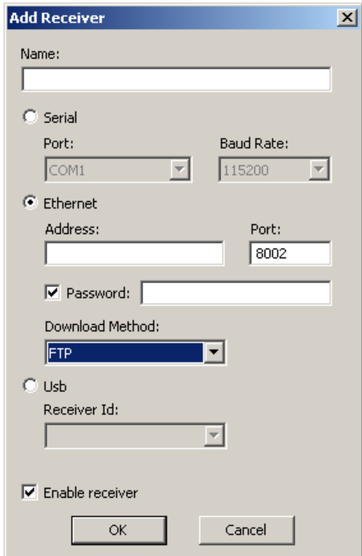


Figure 1-3. Add Receiver

NOTICE

Use dashes or underscores instead of space in the receiver name. For example, use "Receiver_1" instead of "Receiver 1".

Also select the type of receiver connection, Serial, Ethernet or USB, and enter the corresponding parameters.

For Serial type, these are Port and Baud Rate.

For Ethernet, enter the Address (IP address or host name), Port number, also Password and select the Download method: *FTP* (recommended), *DTP*[1] *new connection*, or *DTP same connection*.

1. DTP - Data Transfer Protocol. See GRIL for details.

Here:

FTP - is the default method for downloading files. It is the fastest one, and it does not interfere with other processes taking place in the receiver.

If it is impossible to use FTP, one of the following methods can be selected:

DTP new connection - is used, for example, when the receiver's security settings forbid FTP connection. In this case a new connection for data download is opened through the port noted in the **Add Receiver** screen (i.e., 8002 port). The disadvantages of this method are:

- reduced download speed, and
- the necessity to use additional connection.

NOTICE

NOTICE

By default, only five connections to a receiver are allowed.

DTP same connection - less preferred method, the only one possible for the CDMA modems. If this method is used, the receiver is considered to be disabled, no commands can be sent to the receiver, no messages are received until the download process is finished.

In case of Airlink CDMA modem being used for Ethernet connection, the *Password* checkmark should be removed.

Selecting USB connection, ensure that your receiver is connected to the USB port of the computer. In this case, the *Receiver Id* field shows the ID number of the receiver, and an information string displaying the model of your receiver, appears below. If several receivers are connected to the computer via USB, the information string shows the model of the receiver which ID is selected in the menu.

For details, see "Establishing a Connection to a TPS Receiver" on page 1-16.

To enable the receiver being added, place a corresponding checkmark.

Establishing a Connection to a TPS Receiver

To establish a connection between a computer and a TPS receiver (Figure 1-4), the TopNET-S server uses either Ethernet network, or serial port,

Figure 1-4 provides a connection diagram for these two ways.

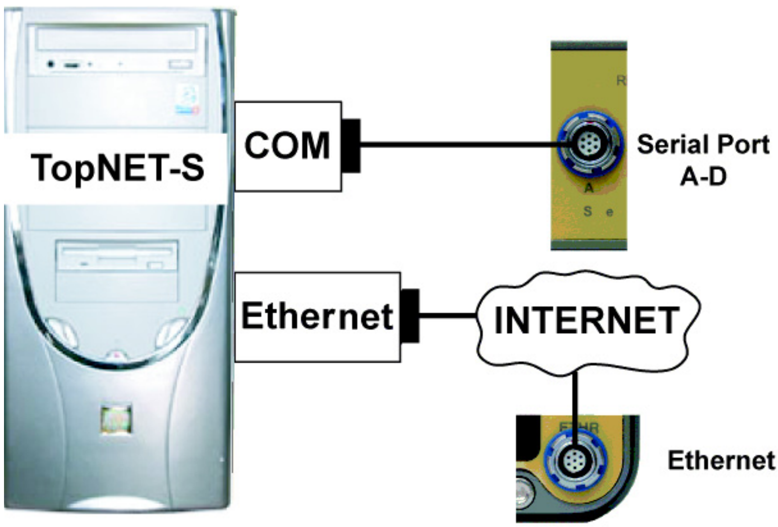


Figure 1-4. Connecting TopNET-S server to TPS Receiver

1. To specify the connection method to the TopNET-S server, select a receiver in the **Setup Receiver** dialog box and press **Edit**.
2. On the **Edit Receiver - [Receiver Name]** dialog box similar to that on Figure 1-3, change necessary parameters. See the following section for setting connection parameters for the different connection methods.

Connecting to a TPS receiver via an Ethernet Port

To connect a computer running the TopNET-S server to a TPS receiver via an Ethernet network, enable or select the following parameters when adding or modifying a receiver (Figure 1-3 on page 1-14):

- enable Ethernet radio button;
- enter the receiver's IP address,
- enter the receiver's Port,
- if the receiver requires a password, enter the password,
- if necessary, change the default FTP/Telnet/DTP download method by selecting the desired item from the drop-down menu.

Press **OK** to save these settings.

TIP

To review the settings for the Ethernet port of the TPS receiver use the PC-CDU software (version 2.1.11 or higher).

To review the Ethernet port settings of the receiver, do the following:

1. Connect the TPS receiver to the computer via either a COM port or USB.
2. Turn on the TPS receiver and run the PC-CDU program.
3. On the **Connection Parameters** dialog box, select the mode of connection (Figure 1-5).
4. Select the computer port used for connection, either COM or USB (Figure 1-5). Press **Connect**.

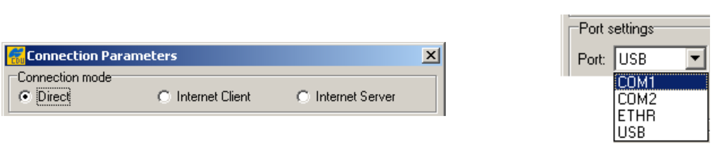


Figure 1-5. Connection Mode and Computer Port in PC-CDU

5. Once a connection gets established, click **Configuration ▸ Receiver** in PC-CDU.
6. In the **Receiver Configuration** dialog box, select the *Ports* tab then the *Ethernet* tab. The Ethernet connection settings entered in the TopNET-S interface should be reflected on this tab (Figure 1-6).

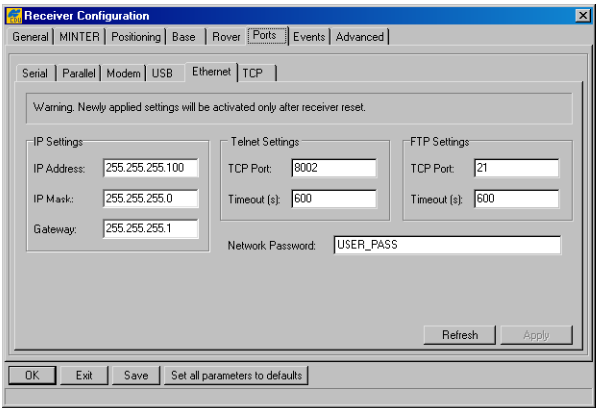


Figure 1-6. Check Ethernet Parameters in PC-CDU

Use these settings for the Ethernet port of the TPS receiver when configuring the receiver for communication with the TopNET-S server.

See “Configuring the Ethernet Port of a TPS Receiver” on page I-1 for details on setting up the receiver’s ethernet port using PC-CDU.

Connecting to a TPS receiver via a Com Port

To connect a computer running the TopNET-S server to a TPS receiver via a COM port, enable or select the following parameters when adding or modifying a receiver (Figure 1-3 on page 1-14):

- enable the Serial radio button;
- select the computer serial port to which the TPS receiver is connected;
- select a Baud rate.

Press **OK** to save these settings.

Notes:

Configuring Users

The following sections describe setting up users for the TopNET-S server.

Creating the User List

When running the TopNET-S server for the first time, the user list is empty. Use the following procedure to add users who have TopNET-S connection privileges.

1. Click **Setup** ▸ **Users...** or click **Users** icon on the toolbar (Figure 1-14).

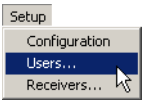


Figure 1-14. Open Operators List

2. On the **Setup Users** dialog box a list of users existing in the system is displayed (Figure 1-15).

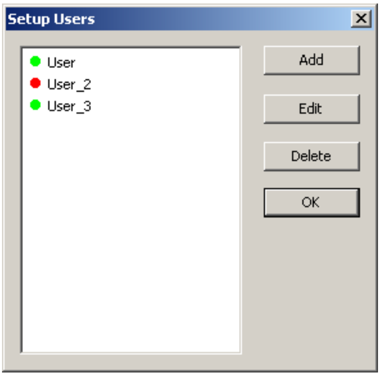


Figure 1-15. User List

At the first start, the screen is empty.

3. Press **Add** and enter a name for the new user, password and check if the user is enabled.

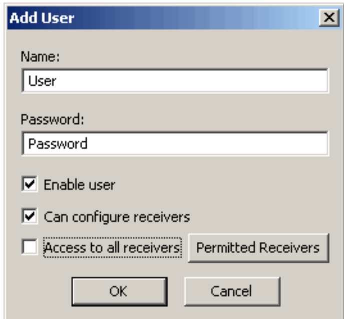


Figure 1-16. Add User

NOTICE

Use dashes or underscores instead of spaces in the user name and password fields. For example, use "User_1" instead of "User 1".

If a user is not allowed to configure receivers, remove the corresponding checkmark.

If it is necessary to limit the number of the receivers being used by this user, remove a check mark from the *Access to all receivers* field and press the **Permitted Receivers** button. The **Permitted Receivers** screen (Figure 1-17) opens:

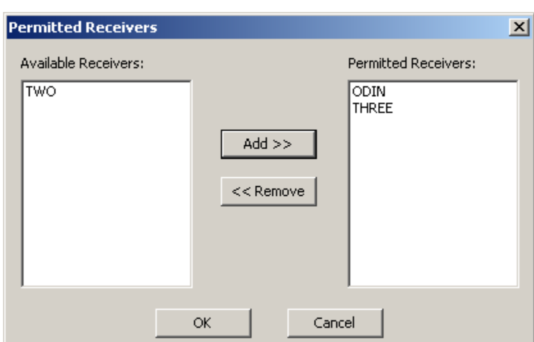


Figure 1-17. Permitted Receivers

The **Permitted Receivers** screen displays two lists of receivers. The *Permitted Receivers* list shows all receivers assigned for the user. The *Available Receivers* panel shows the entire list of receivers.

To assign a receiver to a user, highlight it in the left panel and press the **Add** button. Receiver name will appear in the right panel.

To remove a receiver from the list of receivers assigned to a user, highlight its name in the right panel and press **Remove**.

Press **OK** to save changes.

NOTICE *NOTICE*

When changing authorization status for a user, some tasks running in a TopNET client will be terminated. The other run until finished. With a TopNET-R enabled hardware key, an enabled user can run any number of client programs.

TopNET-S

Notes:

TOPCON

POSITIONING SYSTEMS

Chapter 2. TopNET-R

User's Manual

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Introduction

Welcome to TopNET-R™, the client portion of Topcon's unique and powerful software package designed to offer the easiest and most reliable method of managing any number of Continuously Operating Reference Stations (CORS).

TopNET-R is a shell program providing the primary interface for several application programs (clients) that manage and control TPS receivers connected through the server application TopNET-S™. Figure 2-1 illustrates the layout of a TopNET CORS™ network, which consists of multiple reference stations connected to a central server station. Multiple computers (clients) can also connect with the server station to receive and monitor CORS data.

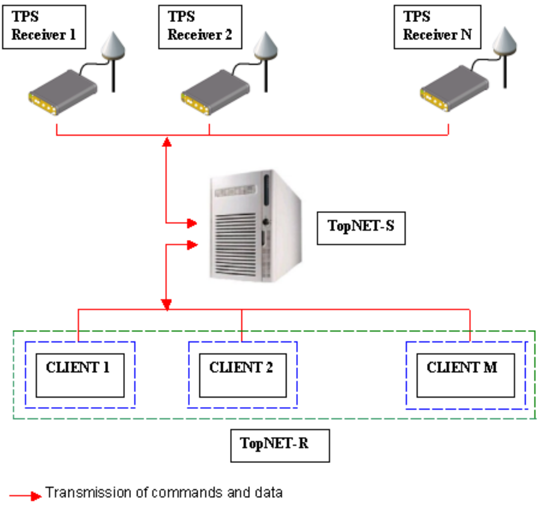


Figure 2-1. TopNET CORS Network Configuration

NOTICE

Refer to the TopNET Reference Manual for installing TopNET-R on the desired computer(s).

Starting TopNET-R

Each TopNET-R client requires a Hardware Protection Key inserted in the appropriate port of the computer. The initial TopNET package contains one key for the TopNET-R client. Refer to the *TopNET Reference Manual* for details on the Hardware Protection Key. If the network uses multiple clients, contact your local Topcon representative to purchase more keys.

Once the key is inserted into the appropriate port, double-click the TopNET-R icon on the computer's Desktop. TopNET-R can be used either with or without a connection to the TopNET-S server.

- If the client is registered with the TopNET-S server, see "Connecting to TopNET-S" on page 2-5.
- To begin using TopNET-R and the associated client applications, see "Getting Acquainted" on page 2-9.

Connecting to TopNET-S

First, before connecting to a TopNET-S server, either contact the TopNET-S system administrator to create a new account and provide you with login and password, or refer to “Chapter 1. TopNET-S” for details.

The **Select Server and Receivers** dialog box displays the server to which the receivers are connected (Figure 2-2).

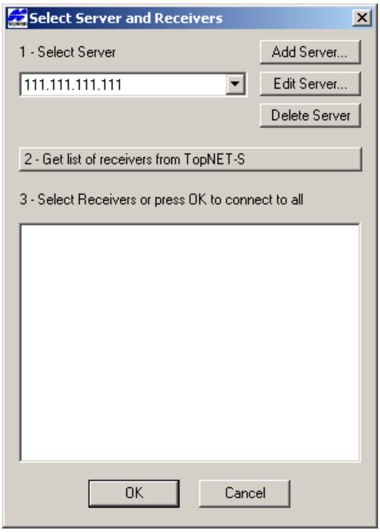


Figure 2-2. Select Server

When running for the first time, the list of servers is empty. In order to start working, one should first define at least one server.

To define a server:

1. Press **Add Server**.
2. Enter the server address to which the TPS receivers are connected, the user name, and the password for access to the TopNET-S server (Figure 2-3). To set a port for connection, define its number in the address field using colon, i.e. 123.456.456.321:8010. By default, the port number 8000 is used for connection.

Use dash lines or underscores instead of spaces.

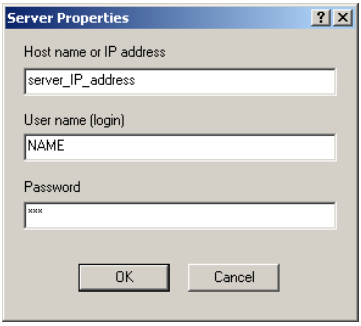


Figure 2-3. Server Properties

3. Press **OK** to add the server to the server list.

An unlimited number of servers can be added to the list.

To connect to a server:

1. Select a server from the *Select Server* drop-down list.

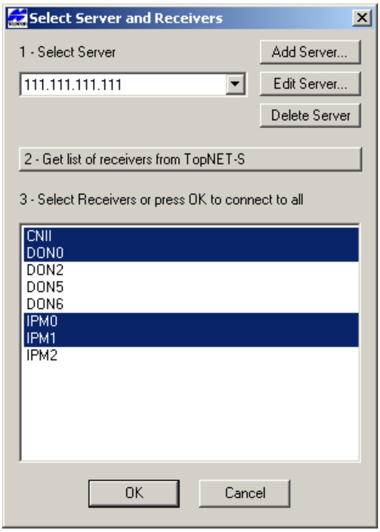


Figure 2-4. Select Receivers

2. Press **Get list of receivers from TopNET-S**.

3. In the receiver listing panel, select individual receivers to connect to and press **OK**, or press **OK** without selecting a particular one to start polling receivers (Figure 2-4 on page 2-6).

To edit a server:

1. Press **Edit Server**.
2. Edit the desired server properties (Figure 2-3 on page 2-6).
3. Press **OK** to save changes and exit.

To delete a server:

1. Press **Delete Server**. The currently selected server will be deleted.

NOTICE *NOTICE*

If any connection parameters are incorrect or communication problems exist between the client and server when connecting to the TopNET-S server, an error message will be displayed.

Notes:

Getting Acquainted

TopNET-R forms a framework for a suite of client programs. These programs can be opened from the TopNET-R main screen using Toolbar or Tools menu (Figure 2-5).

TopNET-R also establishes a connection to the TopNET-S server. When connected to the server, TopNET-R has the following features:

- displays all receivers connected to the server and select desired receivers;
- displays the current navigational coordinates for TPS receivers, the number of tracked satellites, and current receiver time;
- displays TPS receivers located on a map;
- synchronizes the PC clock with UTC (USNO) time.

The TopNET-R main screen (Figure 2-5 on page 2-10) consists of the following components:

- Title bar – displays the name of the active program;
- Menu bar – contains menus for the various TopNET-R applications and other functions;
- Toolbar – contains shortcut buttons to frequently used options
- Receiver table – lists each receiver and information for each receiver connected to TopNET-R;
- Event log – contains information about connections/ reconnections with the receivers;
- Status bar – displays a description of the selected toolbar button (screen tip) and UTC and local time according the computer’s clock.

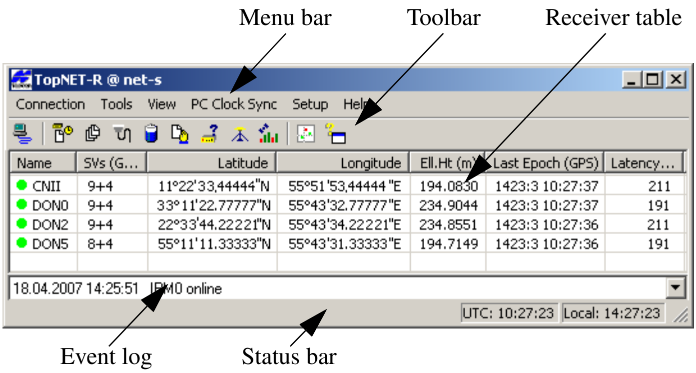


Figure 2-5 TopNET-R Main Window

Receiver table shows the following:

- The *Name* of the reference station;
- The *SVs* (satellite vehicles) column lists information about the number of satellites received (both GPS and GLONASS).
- *Latitude, Longitude, Height* - coordinates of the reference station;
- The *Last Epoch* column shows the parameters of the last received epoch; the GPS week number, GPS week day (Sunday = 0) and GPS time
- *Latency* (*L*) value, calculated as:

$$L = Tr - Tt + Ls$$

where:

Tr - the UTC time when data from the receiver arrived to TopNET-R - according to the PC clock of the computer running TopNET-R,

Tt - time tag in the data message (GPS time according to receiver clock), and

Ls - Leap Seconds correction.

To have PC clock synchronized, you can use features of TopNET-R (see below “PC Clock Synchronization”), or of Microsoft Windows operating system (consult with your IT specialist).

NOTICE

Latency depends on PC clock. If PC clock is not synchronized to UTC, Latency may show large values (if clock is running ahead) or may be negative (if PC clock is behind).

From the TopNET-R main screen a set of independent clients can be accessed through the **Tools** menu. These are used for monitoring data and controlling various receivers in the CORS network. The availability of clients depend on the desired task. Each client performs a specific function when working with TPS receivers. If the remote client is registered with TopNET-S, launching a client will also connect the computer to the TopNET-S server.

Menu Bar

The TopNET-R menu bar provides access to most functions. If client programs exist in the same computer directory as TopNET-R, these programs are accessible from an additional Tools menu. Each of these clients will open in a separate window.

Connection

The Connection menu (Figure 2-6 on page 2-11) is used to:

- select the Server, connect to the Server, and select receivers from the list of receivers
- reconnect to the Server
- close the program

Figure 2-6 Connection Menu

Tools

If the computer directory where TopNET-R resides also includes executable client programs, the Tools menu provides access to these client programs.

Figure 2-7 shows an example with all the available client programs.

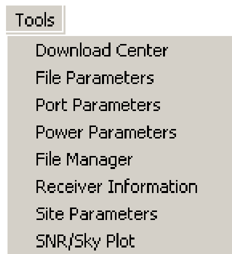


Figure 2-7 Tools Menu – Example

View

The View menu (Figure 2-8) displays one Map item. It shows the receivers located on a graphical map in the WGS84 system.

Figure 2-8 View Menu

PC Clock Sync

The PC Clock Sync menu (Figure 2-9) synchronizes the computer’s internal clock with the UTC (USNO) time scale of receiver.

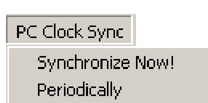


Figure 2-9 PC Clock Sync Menu

Setup

The Setup menu (Figure 2-10) sets the following parameters:

- time setting for receiver data sending period, PC clock synchronization period, and timeouts when no data are sent from the receiver
- E-mail address and schedules for sending alarm messages.

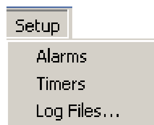


Figure 2-10 Setup Menu

Help

The Help menu (Figure 2-11) provides information about the TopNET-R version and build date and the Activated Options of the Hardware Key

Figure 2-11 Help Menu

Any computer running a TopNET client requires a Hardware Protection Key. A Hardware Protection Key enabled for the program must be inserted into the appropriate USB port.

The Activated Options menu item opens the information **Authorization Options** screen.

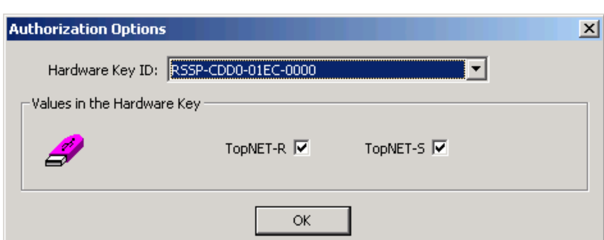
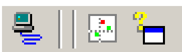


Figure 2-12 Authorization Options

Up to 4 Hardware keys can be used simultaneously on one computer. The list of their IDs is shown in the **Hardware Key ID** drop-down list. The **Values in Hardware Key** field correspond to the currently selected key.

Toolbar

The standard TopNET-R toolbar contains buttons for frequently used functions. If client programs are available, the toolbar will also include quick access buttons to those programs (Figure 2-13).



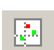
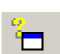
Standard Toolbar

Example Toolbar

Figure 2-13 Toolbar – Standard and Example (with Client Programs)

Table 2-1 describes the standard toolbar buttons.

Table 2-1. Standard Toolbar Buttons

Button	Description
	Connect – opens the **Select Server and Receivers** dialog box to connect to the server and receiver(s)
	Map view – opens the map view
	About – displays TopNET-R copyright and version information

Receiver Table

The receiver table displays the following information for each connected receiver:

- the name of the receiver
- the number of satellites currently being tracked for both GPS and GLONASS

- the current navigational position (latitude, longitude and ellipsoidal height) of the receiver
- GPS time reported by the receiver

The icon next to the receiver's name indicates the status of communication between TopNET-R and the receiver (Table 2-2).

Table 2-2. Receiver Communication Status Icons

Icon	Description	Icon	Description
	initial stage of communication with a receiver		yellow icon – TopNET-R is experiencing a delay in communication
	green icon – the receiver is "on-line" and TopNET-R has received data during the last few seconds		red icon – the receiver is "off-line" and TopNET-R has not received data during the last few seconds

To set time delays for receiver connections mentioned above, click **Setup ▸ Timers** (see "Setup Period and Timers" on page 2-23).

Event Log

The event log contains information about connections/reconnections with the receivers. Table 2-3 lists and comments on the various messages in the event log.

Table 2-3. Event Log Messages

Message	Comments
"PC time synchronized with receiver <RCV>" "PC clock was <N> ms ahead." or "PC clock was <N> ms behind."	Synchronization of the PC clock with the receiver time scale completed successfully. It is shown on how much the PC clock was ahead / behind relatively to the UTC/USNO receiver time scale before synchronization.

Table 2-3. Event Log Messages (Continued)

Message	Comments
“Synchronization with <RCV> failed:” “Unable to get current UTC time from receiver” or “Receiver response time greater than 20 seconds” or “Receiver time and PC time differ by more than 60 hours” or “Cannot set PC time”[a]	Synchronization of the PC clock with the receiver time scale failed. The reason is given.
“Opening <RCV>”	TopNET-R is trying to connect to the receiver.
“<RCV> online”	TopNET-R is connected to the receiver, and the receiver is transferring data at the specified interval.
“<RCV> failed to open: <ERR message from Server or DLL>”	TopNET-R failed to connect to the receiver.
“Closing <RCV> to reconnect”	The receiver did not received any data for several minutes[b]. The receiver is disconnected from TopNET-R, and TopNET-R is trying to reconnect to the receiver.
“Failed to send Alarm E-mail. Error code: <N>”	TopNET-R cannot send an error message email for problems that occurred while polling the receivers. The reason is given.
“Alarm E-mail sent”	TopNET-R has sent an E-mail with error message for problems that occurred while polling the receivers.
“Alarm posted: “<sending text>”	Adding a new alarm message to the alarms file.

a. Check for appropriate access rights on your computer (problem fixing may require Administrator privileges).
b. Use the *Set Periods and Timeouts* dialog box to set the timeout when TopNET-R automatically tries to reconnect to a receiver(s).

PC Clock Synchronization

The clock synchronization function in TopNET-R synchronizes the computer's internal clock with the UTC (USNO) time scale.

Single Synchronization

To synchronize immediately, select a receiver and click **PC Clock Sync ▸ Synchronize Now!** (Figure 2-14).

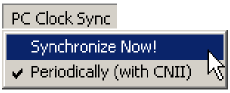


Figure 2-14 Synchronize Now!

When synchronization of the computer's internal clock with the receiver's time scale is completed, the *Event Log* displays a message (Figure 2-17 on page 2-18) with the receiver and the PC synchronization function that was performed, as well as the time in milliseconds that the PC clock was early or late.

NOTICE

NOTICE

PC Clock Synch requires Administrator privileges. If it fails, the following message will be displayed in TopNET-R Event log:

Synchronization with <rcv name> failed: Cannot set PC time

Periodic Synchronization

1. To set a period of synchronization, click **Setup ▸ Timers**.
2. In the **Set Period and Timeouts** dialog box and in the *Sync period* field, enter the period for repeating PC clock synchronization and press **OK** (Figure 2-15).

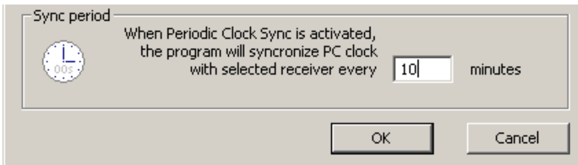


Figure 2-15 Clock Synchronization Period

3. Select a receiver and click **PC Clock Sync ▸ Periodically** (Figure 2-16).

Figure 2-16 Start Periodical Synchronization

When synchronization of the computer's internal clock with the receiver's time scale is completed, the *Event Log* displays a message (Figure 2-17) with the receiver and the PC synchronization function that was performed, as well as the time in milliseconds that the PC clock was early or late.

Figure 2-17 Synchronization Complete Message

After a receiver has been selected for periodic synchronization, the receiver's name displays next to the **PC Clock Sync ▸ Periodically** menu item (Figure 2-18 on page 2-18). This means that the mode for periodic synchronization is activated and the PC clock will be synchronized using the specified receiver ("DON5" in this example).

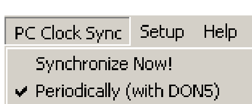


Figure 2-18 Periodical Synchronization with an Example Receiver

To perform synchronization with another receiver, select the desired receiver in the list and click **PC Clock Sync ▸ Periodically** (Figure 2-19).

Synchronization will commence with the selected receiver and the receiver's name will display next to the **PC Clock Sync ▸ Periodically** menu option (Figure 2-19).

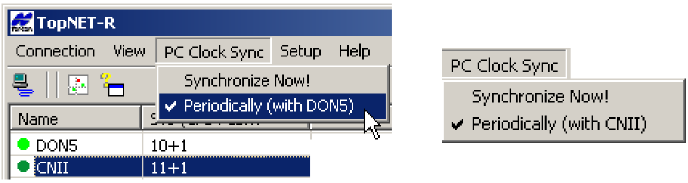


Figure 2-19 Swap Receiver for Synchronization PC Clock

To deactivate the synchronization mode, click on an empty cell, select **PC Clock Sync ▸ Periodically** and press **Yes** at the confirmation dialog box (Figure 2-20).

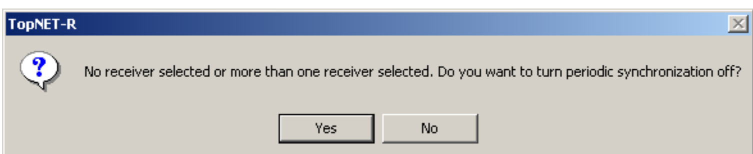


Figure 2-20 Deactivate Synchronization Mode

Periodic synchronization will be canceled, and the menu **PC Clock Sync ▸ Periodically** no longer displays a receiver's name.

Setup Alarm Mail

An alarm is used for notification about some event that may request reaction from the customer. The notification is sent by e-mail.

Click **Setup ▸ Alarms** to display the **Alarm Mail Setup** dialog box (Figure 2-21).

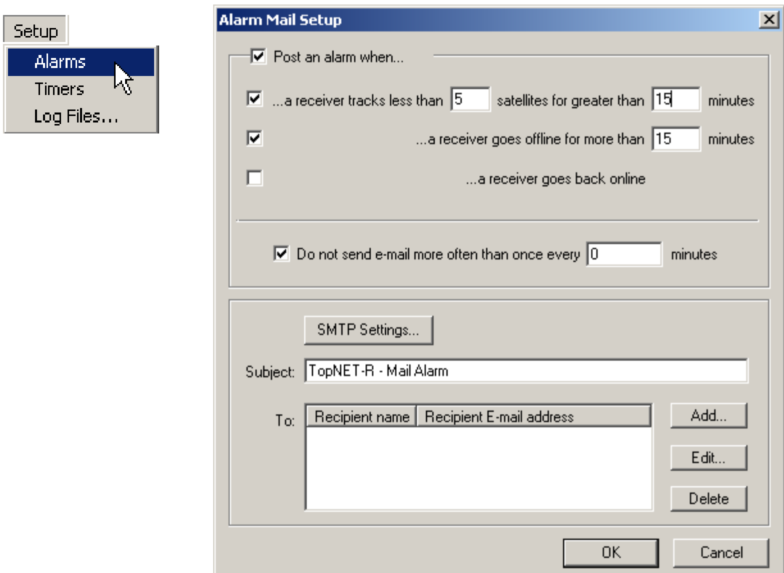


Figure 2-21 Alarms Mail Setup

The **Alarm Mail Setup** screen consists of two parts.

The first part of the screen (Figure 2-21) helps to select the contents of the alarm mail. To make application send notification e-mails, select the **Post an alarm when...** check box.

This tool can be used for notification about:

- lack of the satellites being tracked, for a considerable period of time. Place a checkmark in the corresponding field and set the minimum number of satellites and a maximum admissible tracking period of a less number of satellites.

- a receiver is offline for an extended period of time. Place a checkmark in the corresponding field and set the idle period for a receiver being offline.
- a receiver goes back online. If the previous check box is selected, this field becomes active. It allows you to get notifications about receivers coming back from the offline state.

To make the application generate error reports once at a time, i.e. 45 minutes, place a checkmark in the **Do not send e-mail more often than once every ... minutes** field and set the desired period. If there is no checkmark, the notification alarms are sent as soon as event occurs.

The second part of the **Alarm Mail Setup** screen (Figure 2-22) includes common parameters of the sent e-mails:

- an **SMTP Settings** button - opens a **SMTP Settings** screen, where the parameters of the SMTP server and sender's details can be configured:

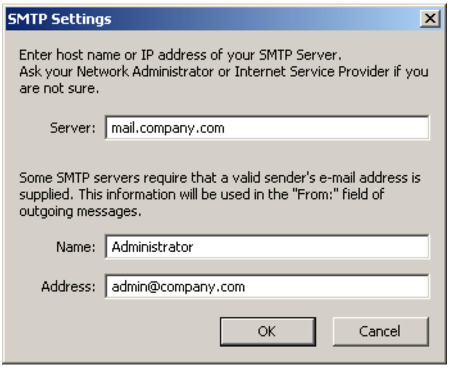


Figure 2-22 SMTP Settings

This module allows a user to send alarm messages, even if no mail client is installed on the PC.
Press **OK** to save the changes and return to the **Alarm Mail Setup** screen.

- *Subject* - the automatic subject of all the e-mails being sent. You can change it, if desired;

- *To* - the mailing list. You can add an address to the list with the help of the **Add** button. In the Type in a user name and address manually, as shown in Figure 2-23.

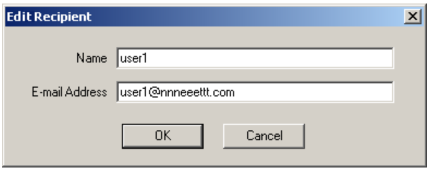


Figure 2-23 Edit Recipient

To make changes in the address entry, select an existing address and press **Edit** button (Figure 2-21) and call the **Edit Recipient** screen (Figure 2-23) again. You can change only one address at a time.

To delete one or several addresses, select the address (if you need to delete several addresses, select them using Shift), and press **Delete** (Figure 2-21). Confirm your decision.

All the generated error reports are displayed in Event log of the main screen and saved in the Log file.

Setup Period and Timers

The period and timers function sets the period of receiver data, the conditions of the receiver icon coloring, and the clock synchronization period.

1. Click **Setup ▸ Timers** (Figure 2-24).

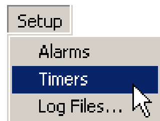


Figure 2-24 Setup Timers

2. Enter the desired information for receiver data and timeouts (Figure 2-25). Note the following requirements for entering values:

    - The minimum period for updating data from network receivers is 1 second.
    - Zero entries in the *Timeouts* and *Sync Period* fields are not allowed. For *Timeouts* the entered values should be set in ascending order. If incorrect values are entered, an error message will be displayed.

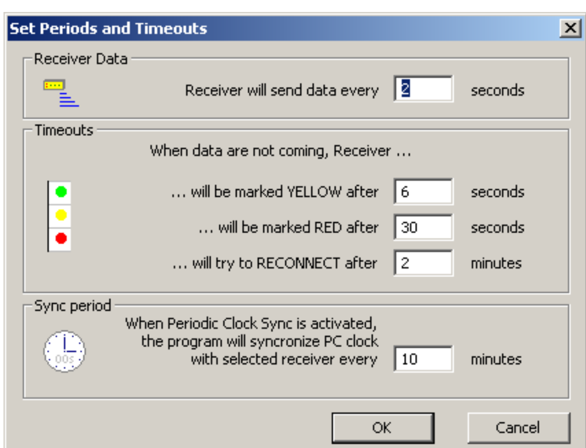


Figure 2-25 Set Periods and Timeouts

Setup Log Files

While operation, the TopNET-R application produces a set of log files containing information about performed actions, received responses and occurring situations. A new set of log files is created every midnight. By default, the screen is empty and all the log files are stored in the same folder as *.exe file.

To change the default location for the log files being created, click **Setup ▸ Log Files** menu (Figure 2-26)

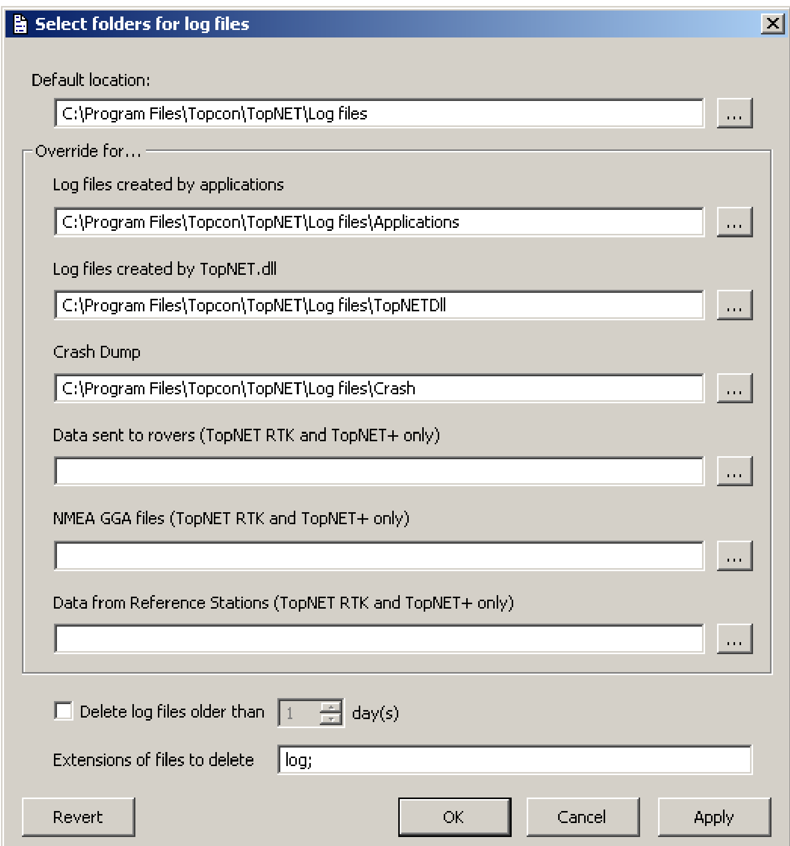


Figure 2-26 Select Folders for Log Files

In the **Default Location** field of the **Select folders for log files** screen, press ... to select a folder on your PC. To select specific folders for log files, enter the overriding locations for the desired log file type:

- Log files created by applications;
- Log files created by TopNET.dll;
- Crash Dump.

The other settings are available only for TopNET RTK and TopNET+ packages.

Any field of the **Select folders for log files** screen may contain the path to existing folder, to non-existing folder that will be created at the moment of the first corresponding log file being written, or contain a macros, for example:

C:\tmp\%DATE%\.

For detailed description of the available macros, see page 2-35.

In order not to store large amount of log files, check the **Delete log files older than** field and select the number of days to store the files. Specify the extensions of the files to be deleted by typing them in the *Extension of files to delete* field, divided by semicolons.

To save the data, press **Apply**.

To save the data and exit, press **OK**.

To revert to the last saved data, press **Revert**.

To exit without saving data, press **Cancel**.

Notes:

Download Center

Download Center is a TopNET-R client that establishes a connection to a TopNET-S server, providing communication with TPS receivers. Download Center provides the following functions:

- downloading files from TPS receivers to the computer where Download Center is installed, including creation of Daily files;
- uploading files from the computer to defined FTP servers;
- converting files to RINEX while uploading them from the computer to an FTP server;

Working with Download Center includes the following two stages:

- setting necessary parameters
- starting *Service* to download, upload, and translate files into RINEX

Figure 2-27 shows an example of how Download Center works with a TopNET-S server, TPS receivers, and an FTP server.

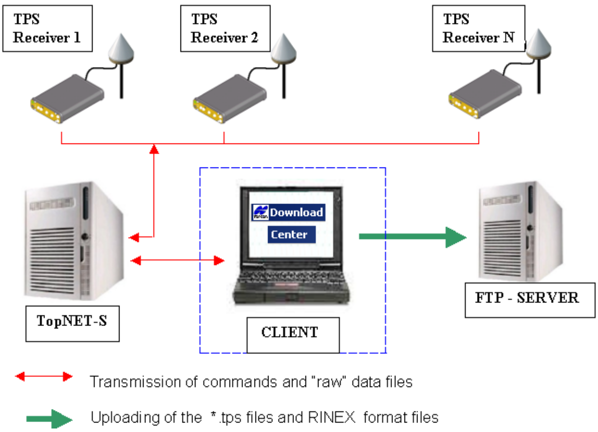


Figure 2-27. Managing Files Using Download Center

Starting Download Center

With TopNET-R running, click **Tools ▸ Download Center** or click the **Download Center** icon on the toolbar (Figure 2-28).

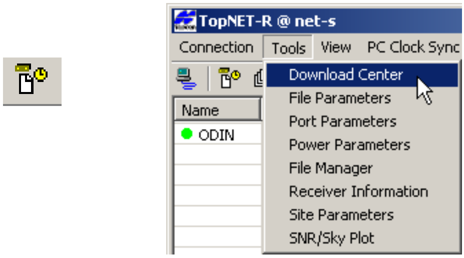


Figure 2-28. Start Download Center

If an older version of Download Center had been installed on the PC, the following message appears (Figure 2-29) and the **FTP Upload Setup** screen will be shown (see "Tasks for Storing Files" on page 2-43).

Figure 2-29. Upload Parameters Conversion Message

Otherwise the **Download Center** main screen (Figure 2-30)opens. It has the following components:

- Title bar – displays the program name and the Server address.
- Menu bar – contains three menus for Download Center functions.
- Toolbar – contains three shortcut buttons to execute the Download Center functions.
- Receiver status area – displays a list of receivers whose raw data files can be downloaded, and the local time of the downloading process beginning.
- FTP status area – displays a list of the files located on a local disk to be uploaded to the FTP server, the FTP server name and a directory where the uploaded files will be stored.

- System log area – displays information about operations executed in the TopNET-S server and receivers.
- Status bar – displays the computer's local time.

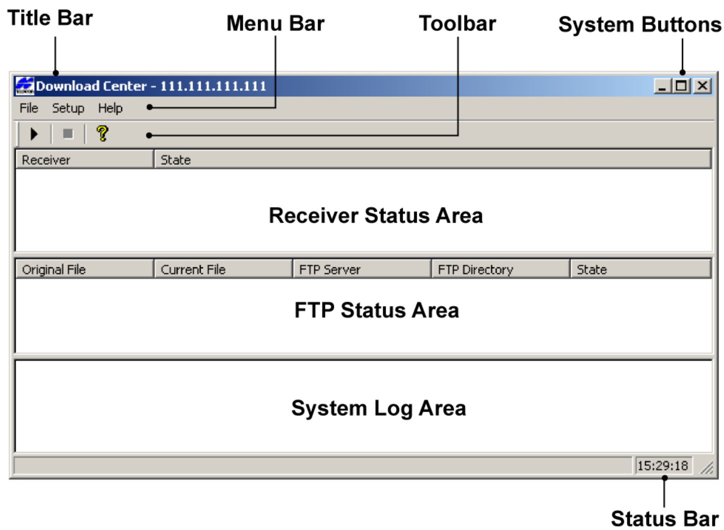


Figure 2-30. Main Screen

The following sections describe the parts of the Download Center main screen.

Menu Bar

The menu bar has three menus used to manage files, set up the Server, and open the help file (Figure 2-31).

Figure 2-31. Menu Bar

File Menu

- The File menu (Figure 2-32 on page 2-30) includes the following menu items:

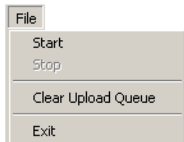


Figure 2-32. File Menu

- Start – begins Service
- Stop – terminates Service
- Clear Upload Queue - clears the queue of files to be uploaded. This option is used if the FTP server is not responding for some reason and a large upload queue appears. The queue is displayed in the FTP Status Area.
- Exit – closes the Download Center

Setup Menu

- The Setup menu (Figure 2-33) includes the following items:

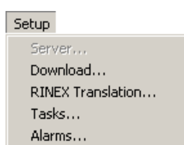


Figure 2-33. Setup Menu

- Server – sets the Server's parameters to establish a connection to the Server. The Server item is disabled if Download Center starts from the TopNET-R main screen as a Client. Download Center will be connected to the Server defined in TopNET-R.
- Download – downloads the selected receivers' files, sets the start time and period of downloading for each receiver, and specifies a directory to store the file.
- RINEX Translation – sets up conversion of the selected receivers' and Daily files in RINEX format while uploading them to an FTP server, and edits information in the header of the RINEX file.

- Tasks – uploads selected receivers' files and Daily Merged Files to an FTP server, sets appropriate parameters to gain access to the FTP server, sets the utilities for the receivers' and Daily files compression, sets up the merging of the downloaded files into one Daily file, and specifies a directory on the FTP server to store *.tps files and corresponding RINEX files.

Help Menu

The Help menu (Figure 2-34) provides information about the version and build date for Download Center.

Figure 2-34. Help Menu

Toolbar

The toolbar contains three buttons to execute the Download Center main functions (Figure 2-35).

Figure 2-35. Toolbar

Table 2-4 describes the toolbar buttons.

Table 2-4. Toolbar Buttons

Button	Description
	Start - Turns on the service responsible for downloading, uploading and translating of receiver files to RINEX
	Stop - Stops the service
	Displays information about the Download Center version and build date

Server Setup

This section describes establishing a connection between Download Center and a TopNET-S server with TPS receivers if the Download Center is launched outside the TopNET-R application.

1. When starting the Download Center from TopNET-R, the Server item is disabled, and the Download center operates with the server that has been set in the TopNET-R application.
2. If you start the Download Server independently, you can use another server. Click **Setup** ▸ **Server** to enter settings for the Server. The **Select Server** dialog box displays (Figure 2-36).

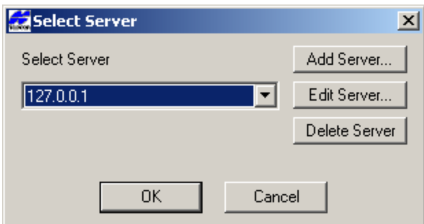


Figure 2-36. Select Server

3. Press **Add Server** to open the **Server Properties** dialog box (Figure 2-37). Enter the server address where TPS receivers are connected, the user name and the password for access to the TopNET-S server.

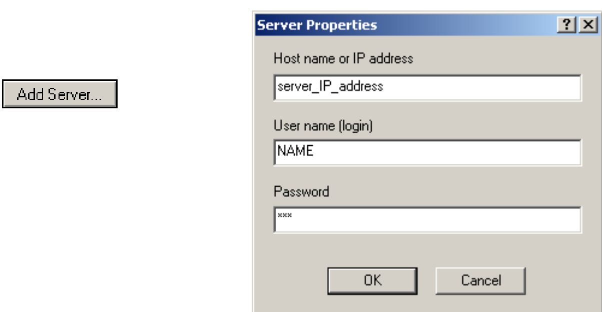


Figure 2-37. Server Properties

NOTICE

Do not use white space symbols for User name and Password: “User 1” is wrong, “User_1” is OK.

4. Press **OK** to save the server in the *Select Server* list and use the server when restarting Download Center.

Downloading Raw Data to a PC

Download Center can download files from TPS receivers connected to the TopNET-S server. Consider the following when downloading files to the computer:

- files are downloaded from a receiver’s internal memory;
- files are not saved on the Server when downloading to the computer;
- when downloaded, files are not removed from a receiver’s internal memory;
- files are allowed to be downloaded only if they have been closed by the time of polling of the receiver;

1. To select receivers for downloading files, click **Setup ▸ Download** (Figure 2-38).

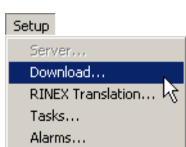


Figure 2-38. Start Download

TopNET-R

The *File Download Setup* dialog box reflects all TPS receivers available for the TopNET-S server (Figure 2-39).

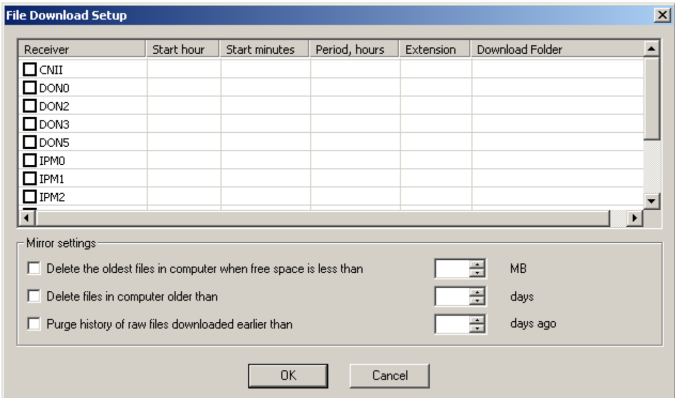


Figure 2-39. File Download Setup

Select the check boxes of the desired receivers to enable them for downloading (Figure 2-40):

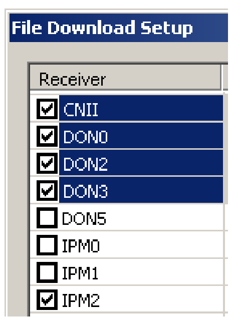


Figure 2-40. Enable Receivers for Downloading

2. After selecting the receiver(s), enter the following parameters into the corresponding columns in the *File Download Setup* screen for each receiver (open the cell for editing by clicking on it):

the beginning hour of the download (in the local computer time)

the minutes of the download start time

the interval between download sessions; possible selections: 15 min, 30 min, 1 h, 2 h, 3 h, 4 h, 5 h, 6 h, 8 h, 12 h, 24 h.

the extension of the downloaded files, typically *tps*

the full path to the directory where the downloaded files will be stored

To include the date of the file creation and the receiver name into the destination directory name, use the following macros (Table 2-5):

Table 2-5. Available Macros

Macros	Definition
%YYYY%	long format of the year, e.g. “2005”
%YY%	short format of the year, e.g. “05”
%MM%	short numerical format of the month, e.g. “09” (September)
%MMM%	short literal format of the month English name, e.g. “SEP”
%mmmm%	full month name according to regional settings of the computer, e.g. “September”
%mmm%	abbreviated month name according to regional settings of the computer, e.g. “Sep”
%DD%	day of month, e.g. “25”
%DDD%	day of the year, e.g. “268” is equal to September 25, 2005
%WEEK%	number of GPS week, e.g. “1342”
%WD%	number of the week day (“0” for Sunday)
%DATE%	date in “YY-MM-DD” format, identical to %YY%-%MM%-%DD%
%RCV%	name of the receiver providing the files

For example: Entering a path shown on Figure 2-41 a) will download files to the indicated directory (Figure 2-41 b).

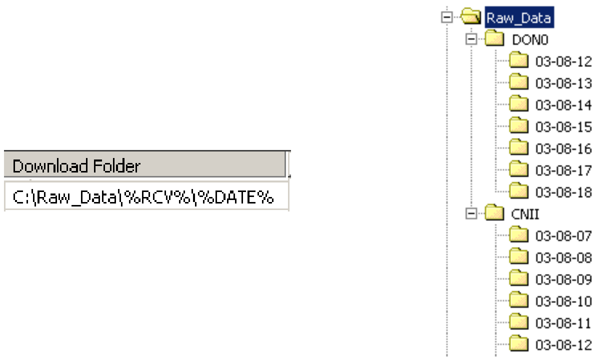


Figure 2-41. Entering a Path to Download Folder

If the amount of free space on the local disk is insufficient to save a file, the file will not be downloaded and an standard error message will be displayed. However, the downloading process will continue and the program will retry downloading the file at three minute intervals. The corresponding message is displayed in the *Receiver Status Area* (Figure 2-42).

Figure 2-42. “Retry” Message

The downloaded files can be automatically removed. The least recent downloaded files can be removed automatically by two rules:

- If available disk space is less than a threshold specified in the *File Download Setup* dialog box (Figure 2-43).

Figure 2-43. Set a Free Space Threshold

- Files older than N days will be removed from the local disk. The number of days (N) is entered in the *File Download Setup* dialog box (Figure 2-44):

Figure 2-44. Set a File Storage Interval (days)

NOTICE

If the "Delete files in computer older than.....days" check box is selected, the files created more than N days before starting Download Center will not be downloaded from TPS receivers.

The history list of the downloaded files is stored on your computer and each time the download starts, the application checks if the name of the file to be downloaded exists in the history list. If the name is not found, the download starts. To reduce the time on checkup, check the *Purge history of raw files downloaded earlier than* field and set the number of days to store the file name in the history:

Figure 2-45. Set a Raw File History Storage Interval (days)

NOTICE

The number of days to store the file name in the history must be larger than the number of days to keep files in computer. Otherwise older files will not be deleted.

Example of incorrect setting:

Delete files in computer older than 90 days.
Purge history of raw files downloaded earlier than 30 days ago.

To delete files older than 90 days, the program goes through history and deletes files listed there (with file date, download date and full path to the file), but after 30 days, file name is purged from history, so the program will not know what files must be deleted.

3. After the settings in the *File Download Setup* screen have been entered, press **OK** to apply. Figure 2-46 shows an example of settings in the *File Download Setup* screen for a selected receiver.

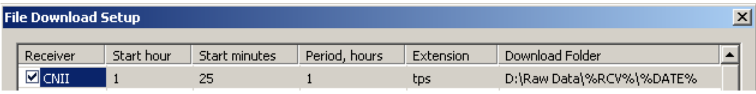


Figure 2-46. Example – Settings in the File Download Setup

The names of the downloaded files and raw data files created by the receivers are identical. A value entered in the *Extension* column sets the extension of the downloaded file; for example, *CNII-0818a.tps*.

If a file has the same name as a downloaded file, “(0)” will be added to the end of the downloaded file name. For example, *CNII-0818c(0).tps*.

NOTICE *NOTICE*

Check the remaining space on the local disk before starting Service.

4. Press **Start** on the toolbar to begin downloading data according to the download setup.

   Depending on the values entered in the *Period, hours* column and the moment the *Service* is activated, the downloading process can be different.

   - For intervals (periods) of 1 hour, the file downloading begins at the moment shown by a value of minutes in the *Start minutes* field in spite of the user setting in the *Start hour* field and will be performed every hour after starting the program.

     For example: For the start time of 22 minutes and an interval of 1 hour, downloading will begin every 22 minutes past every hour 24 hours a day.

   - For intervals (periods) of more then one hour (2, 3, 4, etc.), file downloading begins at the time defined by both hours and minutes in the fields, and will be repeated at the specified interval.

For example: For the start time of 14 hour and 30 minutes with an interval of 3 hours, downloading will begin at 14:30 hours. The download will repeat at 17:30, 20:30, etc.

NOTICE

NOTICE

When editing parameters in the File Download Setup screen, be sure that Service is inactive (the Start button on the toolbar is disabled).

If *Service* is active, the modified parameters will become operative only after the *Service* restarts. In this case, a confirmation message is displayed.

5. Press **Start** on the toolbar to begin the downloading process. The following messages will display as applicable (Table 2-6).

   Once files are downloaded they will not be downloaded again.

Table 2-6. Service and Download Messages

Service Message Examples	Description
2/2/2004 15:09:48 Service started	The date and time when *Service* was activated.
Receiver / State IPMCE Wait until 21:00 CNII Wait until 21:22 8NPC Wait until 21:22	The earliest time downloading files will start.
2/2/2004 16:16:00 Time has come for DONO 2/2/2004 16:16:00 Time has come for CNII	The time came to download files.
8/20/2003 18:37:34 Time has come for "IPU" 8/20/2003 18:38:01 Failed to open receiver "IPU"	If the receiver does not respond, the corresponding disconnect message is displayed. The Download Center will try to connect every three minutes until the receiver is powered on or *Service* stops.

Table 2-6. Service and Download Messages (Continued)

)4 16:22:28 New file DON0-0130v found on DON0)4 16:22:29 New file DON2-0130u found on DON2)4 16:22:31 File DON0-0130v has been successfully downloaded frc)4 16:22:32 File DON2-0130u has been successfully downloaded frc	A new file is found in the receiver. File(s) successfully downloaded.
11:25:06 New file "MAI-0818b" found on "MAI" 11:25:27 Failed to download file "MAI-0818b" from "MAI". Error me	A new file cannot be downloaded and the reason is listed. Download Center will retry downloading every three minutes.
Setup Help eiver State Downloaded 81920 bytes file "CNII_0613s"	When downloading files, the *Receiver Status Area* reflects the amount of file data downloaded.
/21/2003 11:36:00 Time has come for "DON5" /21/2003 11:36:00 Time has come for "CNII" /21/2003 11:36:05 No more file for download on receiver: DON5 /21/2003 11:36:54 No more file for download on receiver: CNII	No new download file is found.

During the first downloading session ALL closed files existing in the selected receivers will be downloaded to the local disk.

6. Press **Stop** on the toolbar to have *Download Center* gradually terminate the downloading process and communication with the receivers (Figure 2-47).

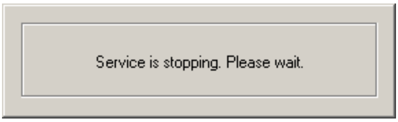


Figure 2-47. Service is stopping

The **Start** button is enabled and the *System Log Area* displays a “service stopped” message (Figure 2-48).

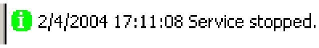


Figure 2-48. Service Stopped

Converting Files to RINEX

The *Download Center* application translates the *.tps files to the RINEX format during the upload process to an FTP server.

1. Configure the **Tasks** screen as described in the “Tasks for Storing Files” on page 2-43.
2. Click **Setup ▸ RINEX Translation** (Figure 2-49)

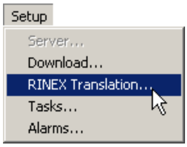


Figure 2-49. Setup Menu, RINEX Translation Selection.

The **RINEX Translation Setup** screen opens:

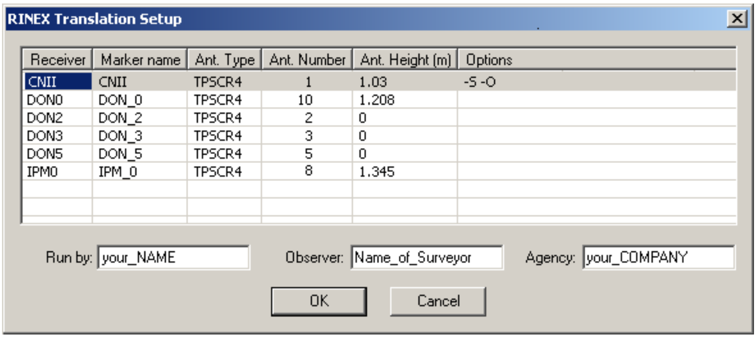


Figure 2-50. RINEX Translation Setup

3. On the *RINEX Translation Setup* screen (Figure 2-50 on page 2-41), enter or select the following information:

- Marker name – name of antenna marker,
- Antenna type – standard name of antenna type according to the NGS[1] list;
- Antenna Number - antenna serial number;

1. See, for example, Complete Antenna Calibration Results at http://www.ngs.noaa.gov/ANTCAL.

- Antenna height – antenna vertical height (from Marker up to antenna reference point (ARP). See Figure 2-51);

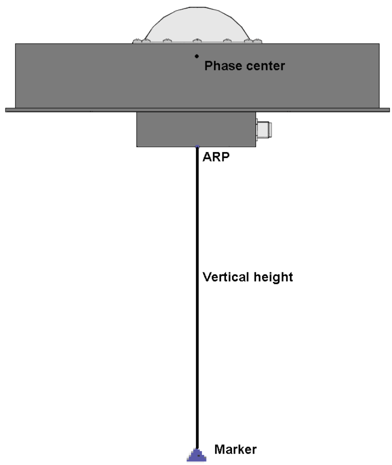


Figure 2-51. GPS Antenna with ARP[1]

- Options – converter options. Use appropriate keys and switches (see Appendix II for details);
- Run by – name of person/agency converting the *.tps file to the RINEX format;
- Observer – name of observer;
- Agency – name of agency;

4. Press **OK**. The information will be stored in the RINEX header file.

The *Download Center* displays the process of uploading RINEX files to the FTP server (Figure 2-52):

Original File	Current File	FTP Server	FTP Directory	State
D:\Raw Data\IPM0\04-02-26\IPM0-0226m.tps	IPM0-0226m.03O	ftp.yourftp.you	Raw Data/RINEX/ 04-02-26	Uploaded 165807 bytes
D:\Raw Data\DON0\04-02-26\DON0-0226l.tps		ftp.yourftp.you	Raw Data/04-02-26/DON0	
D:\Raw Data\DON5\04-02-24\DON5-0224v.tps		ftp.yourftp.you	Raw Data/04-02-24/DON5	
D:\Raw Data\DON2\04-02-20\DON2-0220t.tps		ftp.yourftp.you	Raw Data/04-02-20/DON2	
D:\Raw Data\DON0\04-02-26\DON0-0226m.tps		ftp.yourftp.you	Raw Data/04-02-26/DON0	

Figure 2-52. Example – Uploading RINEX File

1. ARP = Antenna Reference Point

NOTICE

*The RINEX files that result from the *.tps files are removed from local disk after uploading them to the FTP server.*

Tasks for Storing Files

Using the Download Center one can convert *.tps files, compress them, upload to an FTP server, or store locally.

To select a task click **Setup ▸ Tasks** (Figure 2-53)**.**

Figure 2-53. Setup Menu, Upload Selection

The **Tasks** dialog box displays a list of created tasks. Each task contains a set of parameters of uploading and converting the files.

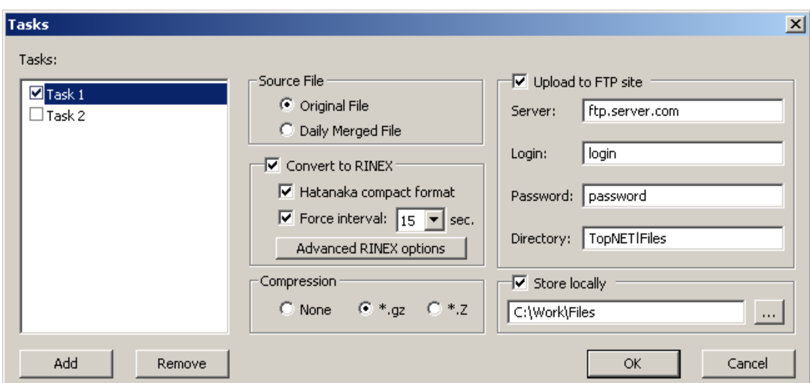


Figure 2-54. Tasks

At a first opening of the **Tasks** dialog box it is empty and all the settings are disabled.

To create a task:

1. Press **Add** and name a task. After a task is named, all the settings become enabled.

The following three steps describe actions that can be performed on the receiver file that is downloaded to its temporary location as specified in Download Setup dialog. (See “Downloading Raw Data to a PC” on page 33.)

2. Select the *Source file*: Original File as it is stored on the receiver, or Daily Merged File that consists of all files received for one day combined into a single merged file.
3. Check the *Convert to RINEX* check box if it is desired to convert a file to the RINEX format, and select the compacting option(s):
   - Hatanaka compact format - the size of the RINEX observation file is reduced with the help of Hatanaka algorithm;
   - Force interval - the size of the RINEX file is reduced, because only the epochs created on the defined time marks (multiple to the selected from the drop-down list) are stored.
   - If additional settings are required, press the **Advanced RINEX options** button and place checkmarks to customize the configuration (Figure 2-55):

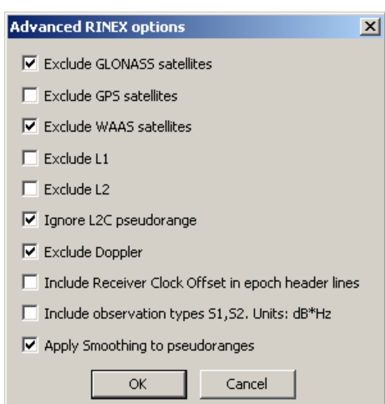


Figure 2-55. Advanced RINEX options

Note that these options correspond to those from TPS2RIN (see Appendix II).

– Exclude GLONASS satellites;

– Exclude GPS satellites;

– Exclude WAAS satellites;

– Exclude L1;

– Exclude L2;

– Ignore L2C pseudorange;

– Exclude Doppler;

– Include Receiver Clock Offset in epoch header lines;

– Include observation types S1, S2. Units: dB*Hz;

– Apply Smoothing to pseudoranges.

4. Select the compression type if necessary:
   - None,
   - *.gz - GZIP compress format
   - *.Z - UNIX-style compress format. This compression algorithm is not as efficient as gzip, but it is selected as a standard format for NGS CORS, for example. Note that *.tps files compressed by this algorithm can turn out larger than their originals. In this case the original *.tps file is stored (uploaded to FTP, or stored locally) and the compressed *.Z file is deleted.

Next two steps describe the actions that can be performed after the processing of the files is over.

5. Check if necessary the *Upload to FTP Site* check box. Insert address of the FTP server, user name, password for access to the FTP server and a target directory on the site. Use macros to enter automatically the date of the file and the name of the receiver into the directory name (see page 2-35 for details).
6. Check if necessary the *Store locally* field and type in or select a path for the files to be stored on your PC. Usual rules of macro substitution work here. Example of a path with macros:

`c:\DATA\%RCV%\%DATE%`

This is considered a permanent storage. Files from this storage are never deleted by the program, so it is operator's responsibility to make sure that there is enough disk space on the permanent storage drive(s).

7. To disable a task during a session, check it off from the list of tasks.
8. Press OK to save changes and close the screen.

To create several tasks just press **Add** and enter a name for the new task (tasks). The **OK** button saves the settings for all the tasks being changed simultaneously.

CAUTION **CAUTION**

For the Tasks operate properly, a receiver must be configured to record files only on "file/a". The activation of both "file/a" and "file/b" is strongly prohibited.

In case of Daily Merged Files storing, the AFRM period (Figure 2-61 on page 2-54) should be constant during the day. If the period is changed, the *Daily Merged File* feature will be available only next day (starting from 00.00.00 UTC).

After activating *Service* by clicking the **Start** button on the Main screen (Figure 2-30 on page 2-29) and establishing a connection to the FTP server, the Download Center will search for new files in the selected receivers. If new files are found, they are downloaded to the local disk according to the active Task in the **Tasks** dialog box.

NOTICE *NOTICE*

The settings of the upload to FTP server can be applied to the files only before the download process starts. These settings will not affect the files that are already downloaded.

The selected files will queue for uploading to the FTP server and the *FTP Status Area* will reflect the uploading process (Figure 2-56).

Original File	Current File	FTP Server	FTP Directory	State
D:\Raw Data\IPM0\04-02-26\IPM0-0226m.tps	IPM0-0226m.tps	ftp.yourftp.you	Raw Data/04-02-26/IPM0	Uploaded 536314 bytes
D:\Raw Data\DON0\04-02-26\DON0-0226l.tps		ftp.yourftp.you	Raw Data/04-02-26/DON0	
D:\Raw Data\DON5\04-02-24\DON5-0224v.tps		ftp.yourftp.you	Raw Data/04-02-24/DON5	
D:\Raw Data\DON2\04-02-20\DON2-0220t.tps		ftp.yourftp.you	Raw Data/04-02-20/DON2	
D:\Raw Data\DON0\04-02-26\DON0-0226m.tps		ftp.yourftp.you	Raw Data/04-02-26/DON0	

Figure 2-56. Example – Uploading File from Queue

If the FTP server settings are invalid, the files waiting to be uploaded will remain in the queue and will be marked as seen in Figure 2-57. Every three minutes, the Download Center will repeat trying to connect to the FTP server.

Original File	Current File	FTP Server	FTP Directory	State
D:\Raw Data\CNII\04-02-23\CNII-0223i.tps		ftp.rrrrr.you		The server name or address could not be resolved...

Figure 2-57. Example – Incorrect FTP Server

That does not affect the uploading process to the correctly defined FTP servers.

NOTICE

NOTICE

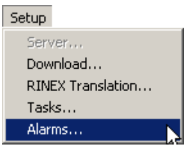
*When uploaded to an FTP server, the *.tps files remain unchanged on the local disk in the Download Folder according to the File Download Setup (see Figure 2-39 on page 2-34).*

If for some reason, a Task was not performed (FTP settings are invalid, the uploading process was interrupted, etc.), and its source file is deleted from temporary storage (manually, or automatically by Download Center), the Task will be removed from the queue and will not be completed. If the file is present in the temporary storage, the Task will be performed after the application restarts.

Alarms

An alarm is used for notification about some event that may request reaction from the customer. The notification is sent by e-mail.

Click **Setup ▸ Alarms** to display the **Alarm Mail Setup** dialog box (Figure 2-58).



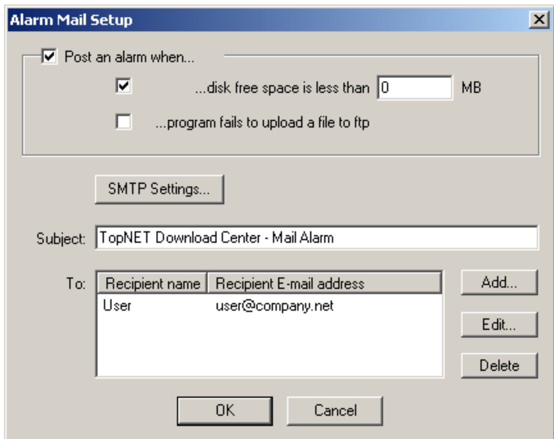


Figure 2-58 Alarms Mail Setup

The **Alarm Mail Setup** screen consists of two parts.

The first part of the screen (Figure 2-58) helps to select the contents of the alarm mail. To make application send notification e-mails, select the **Post an alarm when...** check box.

This tool can be used for notification about:

- disk free space fall below minimum amount;
- application fails to upload a file.
  This alarm is posted only once for each Task that could not upload its files to FTP. When more files are added to the same Task, new alarms are not posted, and e-mail is not sent. If there are several Tasks, Alarms are posted for them independently.
  If Download Center service is stopped and restarted (by pressing Stop/Start, or by restarting the program), the program attempts to perform each pending Task again and Alarms are posted for each failed Task even if it has already been posted before the restart.

The second part of the **Alarm Mail Setup** screen includes common parameters of the sent e-mails similar to those of TopNET-R application (see "Setup Alarm Mail" on page 2-20).

Notes:

File Parameters

File Parameters is a TopNET-R client that manages observational data to be recorded to the TPS receivers' files connected to the TopNET-S server. When connected, the **File Parameters** screen displays the current receiver status and settings for observation data recording to the receiver's files.

When connected to the TopNET-S server, the **File Parameters** screen has the following functions:

- turning on/off AFRM (Automatic File Rotation Mode) and AFRM parameters;
- programming of the receiver's behavior should a power failure occur while recording data;
- setting parameters for data recording.

Starting File Parameters

With TopNET-R running and this client available, select the **Tools ▸ File Parameters** menu or click the **File Parameters** button on the TopNET-R toolbar (Figure 2-59).

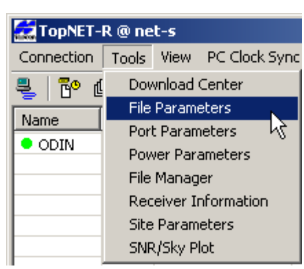


Figure 2-59. Open File Parameters

The **File Parameters** main screen has the following components (Figure 2-60):

- Information pages – sets the parameters or displays information about the ports on the receivers
- Control buttons for the page – applies changes made on individual pages
- Control buttons for all pages – applies changes made on one page to all pages
- Status bar – displays messages for file activities

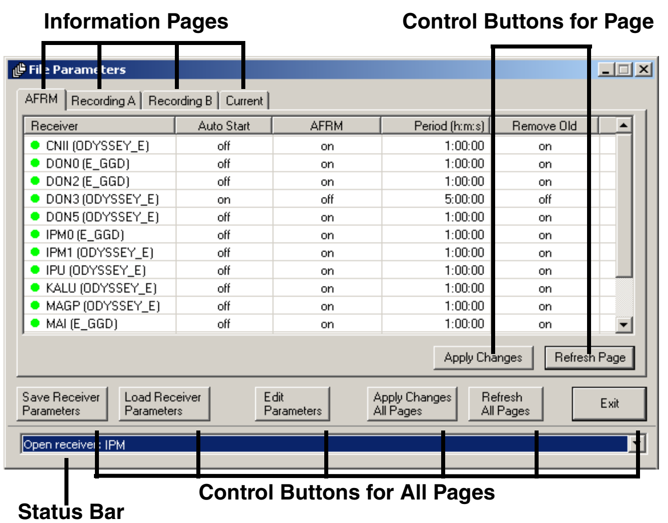


Figure 2-60. Main Screen Components

File Parameters polls the receivers connected to the TopNET-S server and retrieves necessary information. An icon next to the receiver's name indicates the status of information from the receiver:

- – initial stage of polling the receiver
- – data have been successfully retrieved (green icon)
- – the receiver information cannot be retrieved (red icon)

After polling each receiver File Parameters remains connected to the server.

Control Buttons

The control bar contains six buttons to control receiver parameters. Table 2-7 describes the control buttons.

Table 2-7. File Parameters Control Buttons

Button	Description
Save Receiver Parameters	Saves the modified file parameters in every page for all selected receivers, but does not pass them to a server to apply to the receivers.
Load Receiver Parameters	Inserts the saved file parameters to every page for all selected receivers.
Edit Parameters	Opens a dialog box to edit file parameters for all selected receivers.
Apply Changes All Pages	Applies file parameters modified in all pages to every receiver when passed through the server. If an **Apply Changes** button is pressed instead, only parameters modified on the current page will be applied; changes on all other pages will be lost.
Refresh All Pages	Returns all pages to last state, applied to the receiver.
Exit	Closes the File Parameters program without saving any changes.

Status Bar

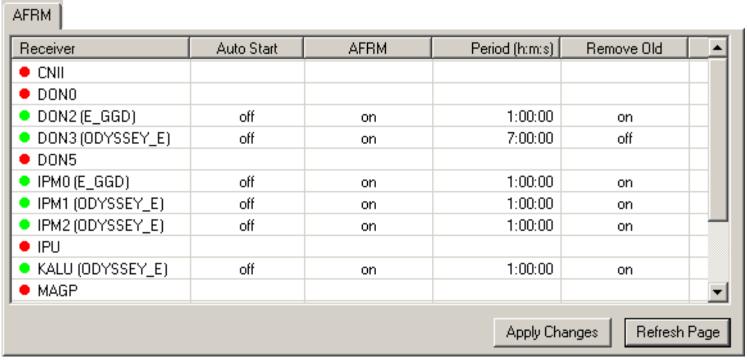
The status bar provides the information about the performed actions and saves the history. The history is available in the drop down list.

AFRM Page

In the Automatic File Rotation Mode (AFRM), the receiver periodically closes/opens files at scheduled intervals.

The *AFRM* page displays the following information (Figure 2-61 on page 2-54):

- Auto Start – whether AutoStart is ON or OFF (for more detailed information, refer to page 2-56)
- AFRM – whether AFRM is enabled (ON) or disabled (OFF)
- Period h:m:s – the time duration in hours, minutes and seconds of each of the multiple files created in AFRM
- Remove old – set the receiver to erase automatically the oldest files if there is insufficient space in the receiver memory for data recording (ON), or not (OFF) (for more detailed information, refer to page 2-56)



Receiver	Auto Start	AFRM	Period (h:m:s)	Remove Old
CNII				
DON0				
DON2 (E_GGD)	off	on	1:00:00	on
DON3 (ODYSSEY_E)	off	on	7:00:00	off
DON5				
IPM0 (E_GGD)	off	on	1:00:00	on
IPM1 (ODYSSEY_E)	off	on	1:00:00	on
IPM2 (ODYSSEY_E)	off	on	1:00:00	on
IPU				
KALU (ODYSSEY_E)	off	on	1:00:00	on
MAGP				

Figure 2-61. AFRM Page

Press **Apply Changes** to apply the parameters edited in the *AFRM* page to the selected receivers.

To retrieve last applied information in the AFRM page for the selected receivers, press **Refresh Page**.

Recording A (B) Page

TPS receivers can create two files (A and B) simultaneously. Each file has its own parameters.

The *Recording A (B)* page displays the following information (Figure 2-62 on page 2-55):

- Activated – whether observation data recording mode is active (Y) or not (N)
- Interval – the recording interval
- Elevation Mask – the minimum elevation angle to consider the satellites' data to be valid.
- File Name Prefix – the prefix that will be added to the names of created receiver files.

Press **Apply Changes** to save the changed parameters to the selected receivers.

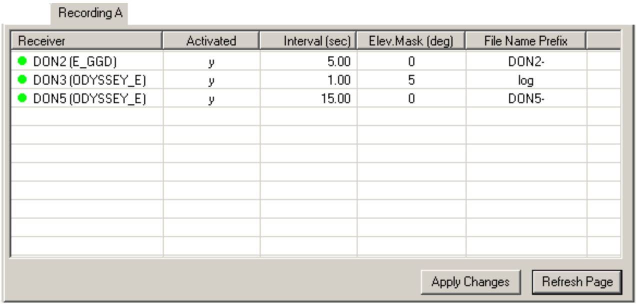


Figure 2-62. Recording A (B) Page

Current Page

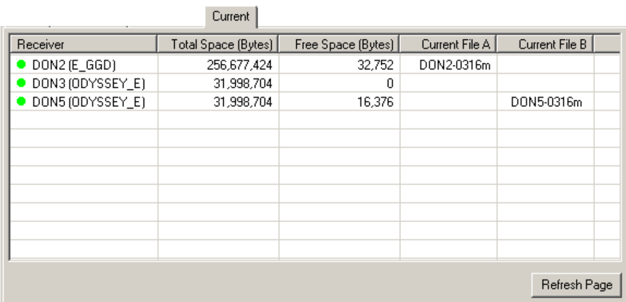
Information about the files A and B currently being created, and total and available capacity of the receiver's internal memory is displayed in the third page of the File Parameters client.

The *Current* page displays the following information (Figure 2-63):

- Total Space – total memory capacity in the receiver, in bytes

- Free Space – memory capacity currently available for observation data recording, in bytes
- Current File A and B – the name of the current file

To retrieve last applied information in the *Current* page for the selected receivers, press **Refresh Page**.



Receiver	Total Space (Bytes)	Free Space (Bytes)	Current File A	Current File B
DON2 (E_GGD)	256,677,424	32,752	DON2-0316m	
DON3 (ODYSSEY_E)	31,998,704	0		
DON5 (ODYSSEY_E)	31,998,704	16,376		DON5-0316m

Figure 2-63. Current Page

Edit Parameters

To edit parameters of the selected receivers, press **Edit Parameters** on the control bar in the main screen.

The **Edit Parameters** dialog box displays the following parameters (Figure 2-64 on page 2-58):

- AFRM field:
  - – enables/disables AFRM. If AFRM is disabled, the observation data file is recorded continuously until space in the receiver's memory exists; if AFRM is enabled, files are recorded according to the settings specified as described below;
  - – specify the interval at which the receiver will close/open observation data files when operating in AFRM mode;
  - – enable/disable automatic removal of the earliest created files if there is insufficient memory in the receiver to continue recording of observational data files.

- Data recording auto start field – programs the receiver behavior in case of a power failure. Table 2-8 describes the settings. Note, if *Always* selected, the receiver will automatically start recording data (to a newly created or an existing file) after resetting the receiver.

Table 2-8. Data Recording and Power Failure

Scenario	"Off"	"On"	"Always"
Before a power failure occurred, receiver data were recorded to a default-named file.	Data recording will not resume automatically after the power is restored.	A new file will be opened after the power is restored and the receiver will start recording data in this file.	A new file will be opened after the power is restored and the receiver will start recording data in this file.
Before a power failure occurred, receiver data recording was stopped.	Data recording will not resume when power is restored.	Data recording will not resume when power is restored.	After the power is restored, a new (default-named) file will be opened and data recording will be started.

- Recording parameters field
  – enable/disable observation data recording into files A and B.
  – set the recording interval.
  – set the minimum elevation angle for the satellites whose observation data will be recorded to the files.
  – set a prefix to the name of the created receiver files. The prefix can be up to 20 characters long. The default value for the Filename Prefix is 'log'. Log file names have the following structure:

  <prefix><month><day><sequential alphabet letter>

To avoid confusion between files created on the same day, the file name has both the creation time (month and day) and additional letter suffixes.

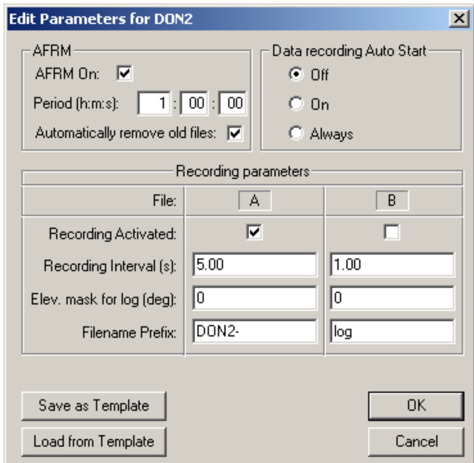


Figure 2-64. Edit Parameters

To save the modified parameters for the selected receivers, press **OK** on the *Edit Parameters* page. All the modified parameters will appear with an asterisk at their values in corresponding pages until the changes are applied (Figure 2-65).

Receiver	Auto Start	AFRM	Period (h:m:s)	Remove Old
DON3 (ODYSSEY_E)	on *	on *	7:00:00 *	on *

Figure 2-65. Example of Saved but not Applied Parameters

Press either **Apply Changes** on the page or **Apply Changes All Pages** on the control bar to apply the modified parameters to the receiver(s).

Figure 2-66 graphically describes the cycle of opening / closing (‘rotating’) files. The horizontal axis represents time duration simulated with beginning from 0h00m00s and divided into intervals equal to the period specified in the **Edit Parameters** dialog box.

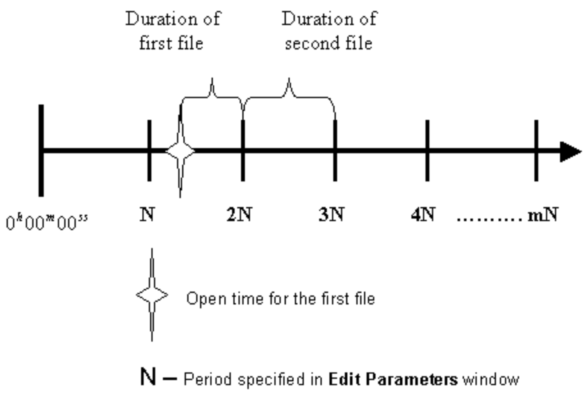


Figure 2-66. Generation of Multiple Files

The first of the TPS files will be opened immediately after clicking **Apply Changes** or **Edit Parameters**.

- Once the following period starts (in our example when t=2N), the current file will be closed, and another file will be opened for data recording.
- Time duration of the next and each following file is equal to the specified file period.
- When AFRM is turned off, the current file will be closed automatically.

The **Edit Parameters** dialog box stores a set of modified parameters as a template.

- To save the parameters as a template, press **Save as Template**.
- To load a set of parameters, press **Load from Template**.

Notes:

Port Parameters

Port Parameters is a TopNET-R client that specifies all necessary settings for serial ports A, B, C and D, and the USB port of TPS receivers connected to a TopNET-S server. When connected to the server, Port Parameters provides the following functions:

- displays the current status of the receiver serial ports A, B, C, D, USB port and TCP port (A,B,C,D,E);
- sets parameters for all receiver ports.

Starting Port Parameters

With TopNET-R running, click **Tools** ▸ **Port Parameters** or click **Port Parameters** on the toolbar (Figure 2-67).

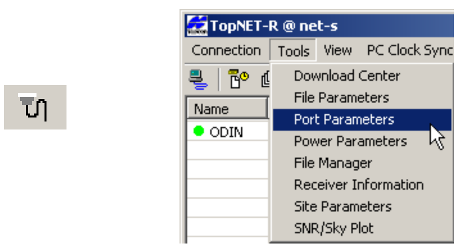


Figure 2-67. Start Port Parameters

The **Port Parameters** main screen has the following components (Figure 2-68 on page 2-62):

- Information pages – set parameters or display information about each port on the connected receiver(s)
- Control buttons for the page – applies changes to current page
- Control buttons for all pages – applies changes to all pages
- Status bar – displays messages about port activities

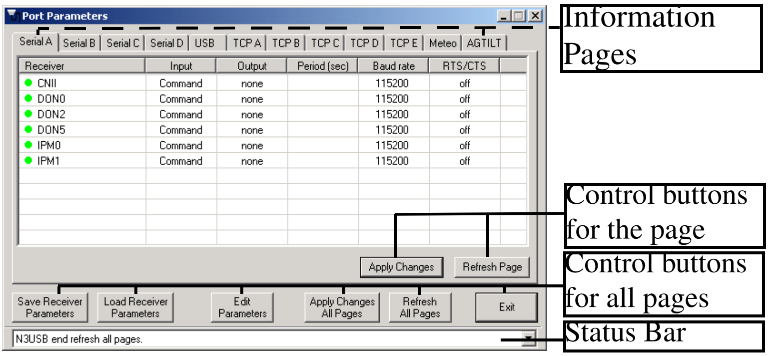


Figure 2-68. Main Screen Components

After starting, Port Parameters polls the receivers connected to the TopNET-S server to retrieve the necessary information. An icon next to the receiver's name indicates the status of information from the receiver:

- – initial stage of polling the receiver
- – data have been successfully retrieved (green icon)
- – the receiver information cannot be retrieved (red icon)

After polling each receiver, Port Parameters remains connected to the TopNET-S server.

TPS receivers can be connected to the server via a serial port (A,B,C,D), a USB port, or via Ethernet. The port connecting the receiver to the TopNET-S server is the current port. When activating the corresponding tab, the port's status will be displayed in the Receiver field (Figure 2-69 on page 2-62).

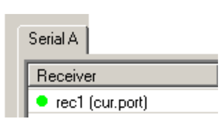


Figure 2-69. Serial A is the Current Port for an Example Receiver

The current port parameters cannot be changed.

Control Buttons

The control bar contains six buttons to control the port parameters of network receivers. Table 2-9 describes the control buttons.

Table 2-9. Port Parameters Control Buttons

Button	Description
Save Receiver Parameters	Saves modified port parameters in every page for all selected receivers to a local file. These parameters are not sent to receivers.
Load Receiver Parameters	Reads the saved port parameters and displays them on all pages for all selected receivers
Edit Parameters	Opens the Edit Parameters screen.
Apply Changes All Pages	Applies modified port parameters in all pages to all selected receivers. If an **Apply Changes** button is pressed instead, only parameters modified on the current page will be applied; changes on all other pages will be lost.
Refresh All Pages	Returns all pages to last state, applied to the receiver.
Exit	Closes the “Port Parameters” program without saving changes
Apply Changes	Applies Changes only on the current page.
Refresh Page	Reloads parameters for the current page

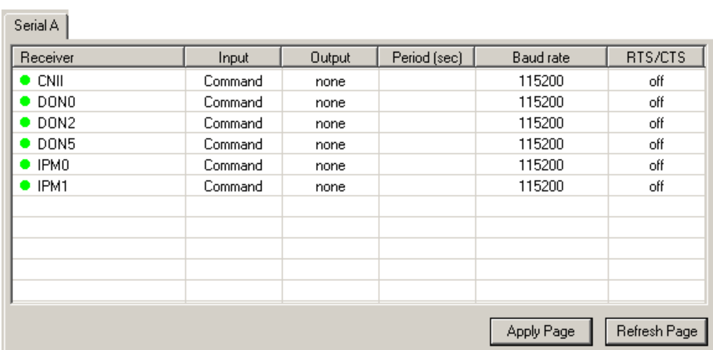
Status Bar

The status bar provides the information about the performed actions and saves the history. The history is available in the drop down list.

Serial A(B,C,D) Page

TPS receivers have up to four RS-232 serial ports. The *Serial A* (or B,C,D) page displays settings for each port on each connected receiver (Figure 2-70 on page 2-64).

- Input – displays what type of data to input on the selected port:
  - – None: the port ignores any incoming data.
  - – Command: the port recognizes user commands.
  - – Echo: the receiver port redirects all incoming data to an "output stream". Note that the "output stream" may be either a port or the current log file.
  - – RTCM: the port is in RTCM input mode.
  - – CMR: the port is in CMR input mode.
  - – TPS: the port is in TPS message input mode.
  - – OmniStar: incoming data to the port will be used by OmniStar VBS (Virtual Base Station) to produce DGPS Solution type.



Receiver	Input	Output	Period (sec)	Baud rate	RTS/CTS
CNII	Command	none		115200	off
DON0	Command	none		115200	off
DON2	Command	none		115200	off
DON5	Command	none		115200	off
IPM0	Command	none		115200	off
IPM1	Command	none		115200	off

Figure 2-70. Serial A (B,C,D) Page

- Output – displays the type of data to output on the selected port. The numbers in braces designate the message types:
  - – None: the port outputs nothing.
  - – DGPS RTCM {1,31,3}
  - – DGPS RTCM {9,34,3}

– RTK RTCM {18,19,22,3}

– RTK RTCM {20,21,22,3}

– RTK RTCM {18,19,23,24}

– RTK RTCM {20,21,23,24}

– RTK RTCM3 GD min - outputs the following RTCM3 messages: 1003 by a user defined period, 1011 by a user defined period, 1006 by period 10.00 s, and 1008 by period 10.00 s.

– RTK RTCM3 GD full - outputs the following RTCM3 messages: 1004 by a user defined period, 1012 by a user defined period, 1006 by period 10.00 s, 1008 by period 10.00 s.

– RTK RTCM3 GGD min - outputs the following RTCM3 messages: 1003 by a user defined period, 1006 by period 10.00 s, 1008 by period 10.00 s.

– RTK RTCM3 GGD full - outputs the following RTCM3 messages: 1004 by a user defined period, 1006 by period 10.00 s, 1008 by period 10.00 s.

– RTK CMR {10,0,1}

– RTK CMR + {10, 0, 9}

– TPS RTK, minimum correction set - outputs the following TPS messages: RT, SI, rc, cp, 2r, 2p, BI, and ET all by a user defined period.

– TPS RTK, maximum correction set - outputs the following messages: RT, SI, rc, cp, DC, EC, 2r, 2p, D2, E2, BI, and ET all by a user defined period.

– Meteo (see description in Appendix V);

– AGTILT (see description in Appendix VI);

– User Defined: the port outputs user-specified data (the user defined some arbitrary message set that will be outputted through the port).

- Period (sec) – displays the message output interval (in seconds).

- Baud rate – displays the baud rate for the indicated receiver port.
- RTS/CTS – displays hardware handshaking status for the port.

USB Page

In addition to serial ports, the TPS receiver may also have a USB port. USB is used for high-speed data transfer and communication between the receiver and an external device.

The *USB* page displays the settings for USB port (Figure 2-71).

The types of incoming and outgoing data for USB port are the same as for corresponding data types for Serial ports of TPS receiver (see “Serial A(B,C,D) Page” on page 2-64 for details).

- Input – displays what type of data to input on the selected port.
- Output – displays what type of data to output via the selected port.
- Period (sec) – displays the message output interval (in seconds).

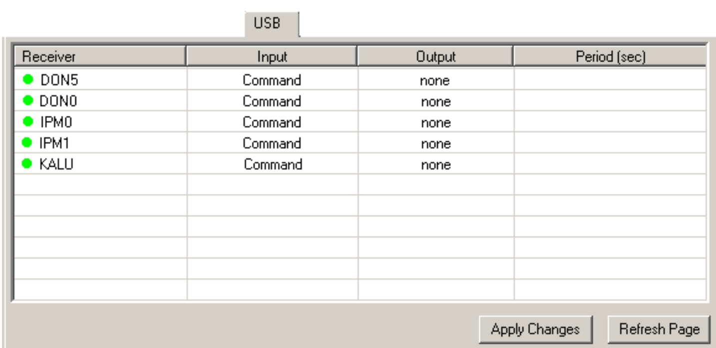


Figure 2-71. USB Page

TCP A (B,C,D,E) Page

The *TCP A* (or B,C,D,E) page displays the type of data to input/output over the corresponding TCP/IP stream (Figure 2-72). These settings are the same as those in the USB page. See “USB Page” on page 2-66 for a description of the fields.

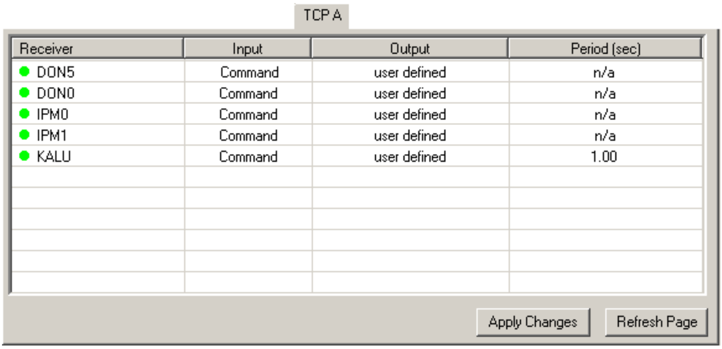


Figure 2-72. TCP Page

Meteo Page

The receiver can be connected to a meteorological device[1] providing the information about temperature, pressure and humidity.

Check, if your meteorological device is connected correctly.

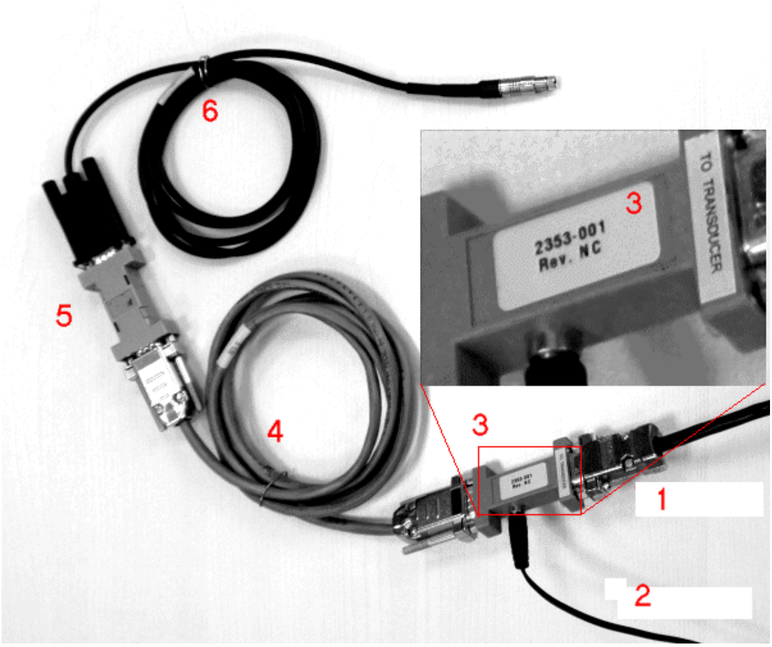


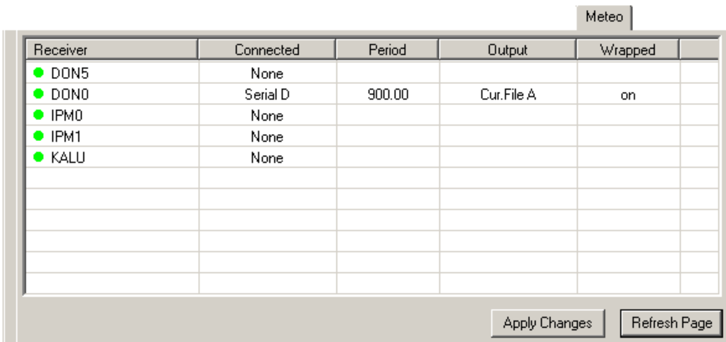
Figure 2-73. Connection of the Meteorological Device to an Odyssey Receiver

The example displays the connection of a MET3A device. Here:

9. RS232 cable to MET3A (comes with MET3A);
10. Cable to 9V DC power adapter (comes with MET3A);
11. RS232+9V Mixer (comes with MET3A);
12. RS232 Male-female extension cable;
13. Male-male null adapter;
14. DB9-Fisher RS232 cable (comes with Topcon receiver).

1. Current version of the application supports communication to Paroscientific MET3 and MET3A meteorological devices only.

The *Meteo* page displays the current settings of the data transfer: the connected port (the **Connected** column), the period of the data transfer (the **Period** column), the output port or file (the **Output** column) and the status of the **Wrapped** flag.



Receiver	Connected	Period	Output	Wrapped
DON5	None			
DON0	Serial D	900.00	Cur.File A	on
IPM0	None			
IPM1	None			
KALU	None			

Figure 2-74. Meteo Page

To make the meteorological device provide the necessary data, the application instructs receiver to send an appropriate command to the port set in the **Connected** column. (The meteorological device is supposed to be connected to that port of the receiver.)

The response received from the meteorological device is an NMEA 0183 sentence, starting with $WIXDR[1] and containing the values of temperature, pressure and relative humidity. It is redirected by receiver to the port or file set in the **Output** column.

The command to the meteorological device and the response from it are sent and received once in the **Period.**

The **Wrapped Echo** mode should be enabled if the redirected information comes to a file along with GPS measurements and the file is supposed to be converted to RINEX.

To change current settings, press the **Edit Parameters** button (Figure 2-68).

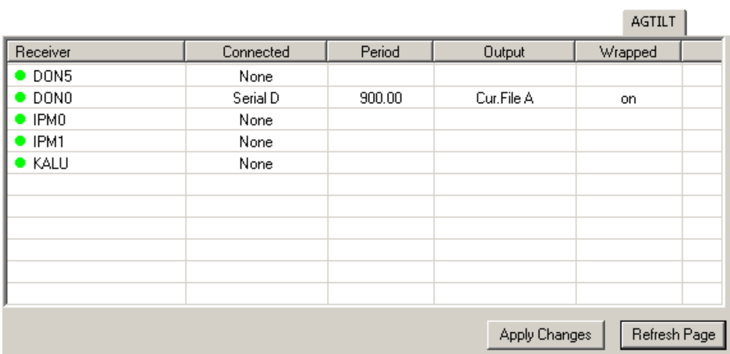
1. WIXDR - Weather Instruments Tranducer Measurement

AGTILT Page

The receiver can be connected to a tiltmeter device[1] providing the information about tilt angles.

In a typical application, the tiltmeter attached to the antenna tower automatically tracks for changes in antenna tower inclination. The measured tilt angles can be sent to the GPS receiver in an NMEA message and used to compensate for the tower inclination in computation of position at the base of the tower.

The *AGTILT* page displays the current settings of the data transfer: the connected port (the **Connected** column), the period of the data transfer (the **Period** column), the output port or file (the **Output** column) and the status of the **Wrapped** flag.



Receiver	Connected	Period	Output	Wrapped
DON5	None			
DON0	Serial D	900.00	Cur.File A	on
IPM0	None			
IPM1	None			
KALU	None			

Figure 2-75. AGTILT Page

To make the tiltmeter device provide the necessary data, the application instructs receiver to send an appropriate command to the port set in the **Connected** column. (The tiltmeter device is supposed to be connected to that port of the receiver.)

The response received from the clinometer device is an NMEA 0183 sentence, starting with $YXXDR and containing the values of two tilt angles. It is redirected by receiver to the port or file set in the **Output** column.

1. The software has been tested using Applied Geomechanics clinometer “IRIS” MD900-T.

The command to the AGTILT device and the response from it are sent and received once in the **Period.**

The **Wrapped Echo** mode should be enabled if the redirected information comes to a file along with GPS measurements.

To change current settings, press the **Edit Parameters** button (Figure 2-68).

Edit Parameters

To edit parameters of the selected receivers, press **Edit Parameters** on the control bar in the main screen. The **Edit Parameters** dialog box displays the parameters given in Figure 2-76. Edit these parameters as needed for each port.

- Input – specifies the input mode of the selected port (Figure 2-72). It is recommended to set Echo input mode if a Meteo or AGTILT device is attached to the corresponding port.
- Output – specifies the message set to output via the selected port (Figure 2-72). If a Meteo or AGTILT device is attached to the port, the corresponding Output field displays Meteo or AGTILT. If the message set transmitted through the port does not match any of the predefined templates listed in the drop-down list, the "user defined" entry is selected automatically.

NOTICE *NOTICE*

The Meteo *and* AGTILT *entries can not be selected manually here - only via corresponding fields below.*

- Period(s) – specifies the message output interval for serial, USB, and TCP (in seconds). For Meteo and AGTILT devices this parameter is displayed automatically after you specify the value in the corresponding field at the bottom of the screen.
- Baud rate – specifies the baud rate for the corresponding receiver serial port
- RTS/CTS – enables/disables hardware handshaking for the serial port.

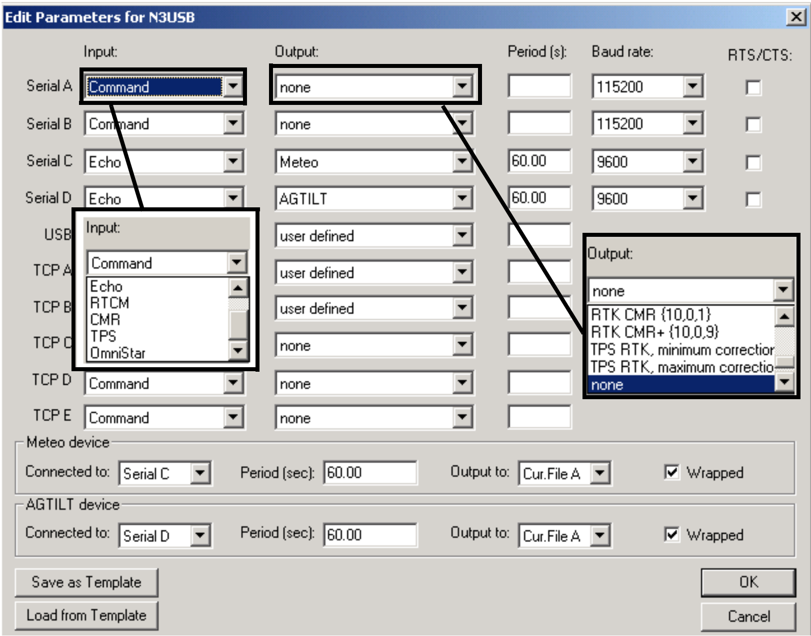


Figure 2-76. Edit Parameters

- Meteo device - specifies the parameters of the meteorological device connected: the connection port, period between sent commands, and the output port or file (optional). Place the **Wrapped** check mark to turn on the Wrapped Echo mode.
- AGTILT device - similar to Meteo device, specifies the parameters of the tiltmeter device connected: the connection port, period between sent commands, and the output port or file (optional). Place the **Wrapped** check mark to turn on the Wrapped Echo mode.

To save the modified parameters for the selected receivers, press **OK** on the *Edit Parameters* page. All the modified parameters will appear with an asterisk at their values in corresponding pages until the changes are applied (Figure 2-77).

Serial A

Receiver	Input	Output	Period (sec)	Baud rate	RTS/CTS
DON0	Command	DGPS full *	10 *	115200	off

Figure 2-77. Example of Saved but not Applied Parameters

Press either **Apply Changes** on the page or **Apply Changes All Pages** on the control bar to apply the modified parameters to the receiver(s). The asterisk will be removed from the parameter values.

The **Edit Parameters** dialog box stores a set of modified parameters in the Port Parameters program as a template.

- To save the parameters as a template, press **Save as Template**.
- To load a set of parameters, press **Load from Template**.

Notes:

Power Parameters

Power Parameters is a TopNET-R client that manages the power of receivers connected to a TopNET-S server. When connected to the server, the **Power Parameters** screen displays the current receiver status and settings for the receiver's power.

The **Power Parameters** application has the following functions:

- control the power usage of each of receivers;
- charge the batteries of the receiver if necessary;
- control power on slots and ports of the receiver;
- control board temperature.

Starting Power Parameters

If the **Power Parameters** client is available, in TopNET-R main screen select the **Tools ▸ Power Parameters** menu or press the **Power Parameters** button on the TopNET-R toolbar (Figure 2-78).

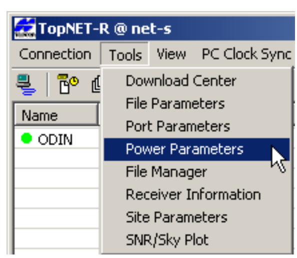


Figure 2-78. Open File Parameters

The **Power Parameters** main screen has the following components (Figure 2-79):

- Information pages – sets the parameters or displays information about the ports on the receivers;

- Control buttons for the page – applies changes made on individual pages;
- Control buttons for all pages – applies changes made on one page to all pages;
- Status bar – displays messages for file activity.

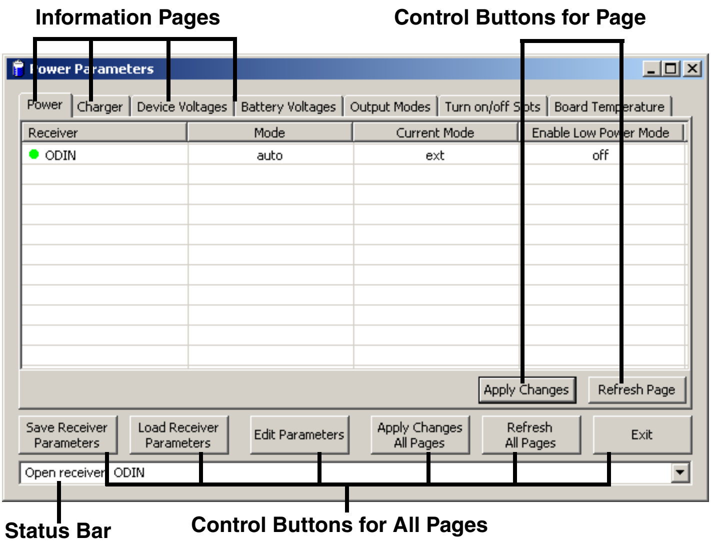


Figure 2-79. Main Screen Components

Power Parameters polls the receivers connected to the TopNET-S server and retrieves necessary information. An icon next to the receiver's name indicates the status of information from the receiver:

- – initial stage of polling the receiver
- – data have been successfully retrieved (green icon)
- – the receiver information cannot be retrieved (red icon)

After polling each receiver File Parameters remains connected to the server.

Control Buttons

The control bar contains six buttons to control receiver parameters. Table 2-10 describes the control buttons.

Table 2-10. File Parameters Control Buttons

Button	Description
Save Receiver Parameters	Saves the modified receiver parameters in every page for all selected receivers, but does not pass them to the TopNET-S server to apply to the receivers.
Load Receiver Parameters	Inserts the saved receiver parameters to every page for all selected receivers.
Edit Parameters	Opens a dialog box to edit receiver parameters for all selected receivers.
Apply Changes All Pages	Applies receiver parameters modified in all pages to every receiver when passed through the server. If an **Apply Changes** button is pressed instead, only parameters modified on the current page will be applied; changes on all other pages will be lost.
Refresh All Pages	Returns all pages to last state, applied to the receiver.
Exit	Closes the Power Parameters program without saving any changes.

Status Bar

The status bar provides the information about the performed actions and saves the history. The history is available in the drop down list.

Power Page

The *Power* page displays the following information (Figure 2-80):

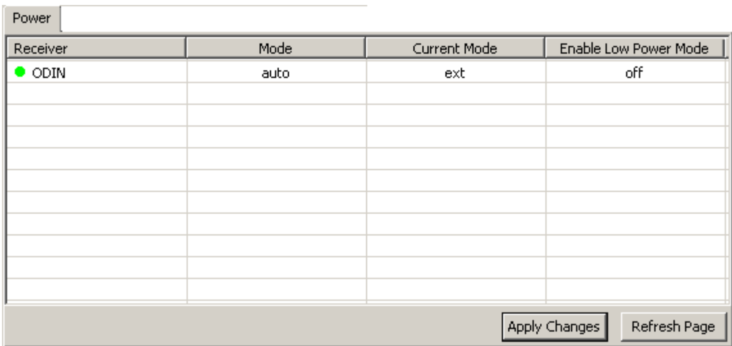


Figure 2-80. Power Page

- Receiver- receiver name;
- Mode - the desired power source for the connected receiver.
- Current Mode - current power source.
- Enable Low Power Mode - shows the enabled low power consumption mode for the receiver's processor.

By default, Mode and Current Mode positions are identical. These and other values are changeable with the help of in the Edit Parameters screen (see "Edit Parameters" on page 2-85). If the Mode is changed, the different Mode and Current Mode values are shown until these changes are successfully applied to the receiver.

Press **Apply Page** to apply the parameters to the selected receivers.

To retrieve last applied information in the *Power* page for the selected receivers, press **Refresh Page**.

Charger Page

The *Charger* page displays the following information (Figure 2-81):

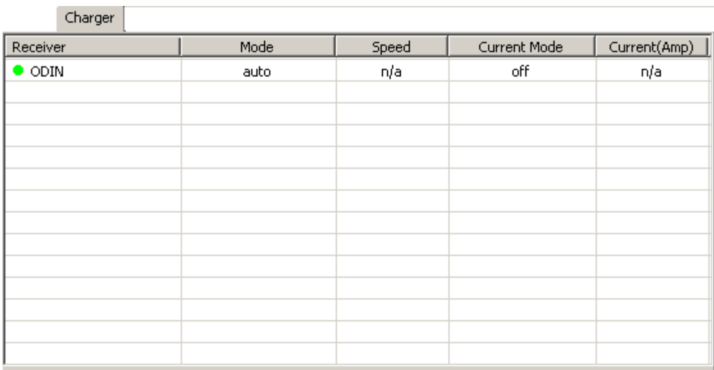


Figure 2-81. Charger Page

- Receiver- receiver name;
- Mode - the battery charging mode;
- Speed - the speed for a battery charge cycle;
- Current Mode - displays the battery currently being charged;
- Current (Amp) - displays the actual battery charging current in Amperes.

To change any values, select the receiver in the list and press the **Edit Parameters** button (see “Edit Parameters” on page 2-85 for details).

Press **Apply Page** to apply the parameters edited in the *Charger* page to the selected receivers.

To retrieve last applied information in the *Charger* page for the selected receivers, press **Refresh Page**.

Device Voltages Page

The *Device Voltages* page displays the following information (Figure 2-82):

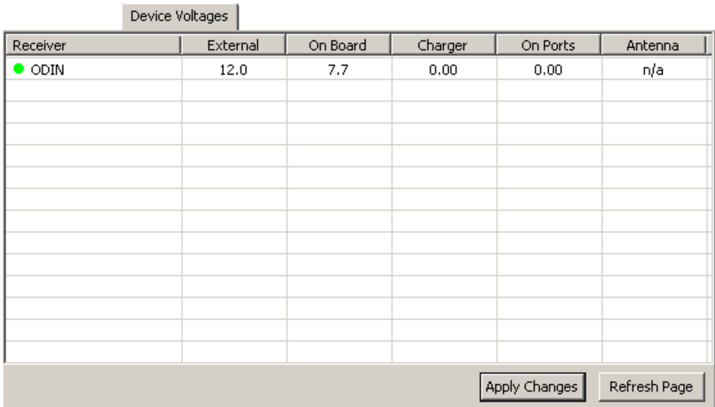


Figure 2-82. Device Voltages Page

- Receiver name;
- External power;
- Power on board;
- Power on Charger;
- Power on Ports;
- Antenna power.

NOTICE

These values are informational and cannot be changed from the Power Parameters client.

To retrieve last applied information in the *Device Voltages* page for the selected receivers, press **Refresh Page**.

Battery Voltages Page

The *Battery Voltage* Page displays the following information:

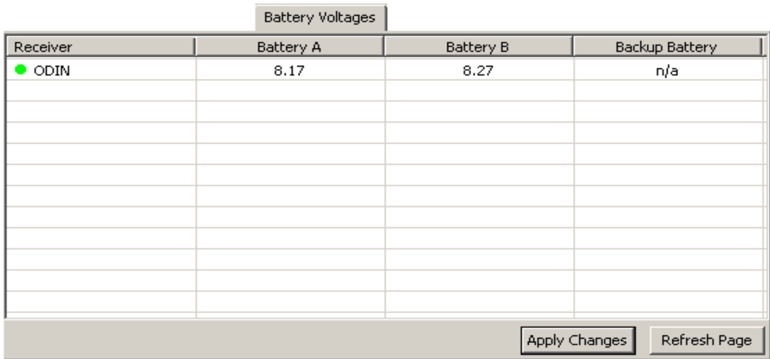


Figure 2-83. Battery Voltages Page

- Receiver name;
- Power on Battery A;
- Power on Battery B;
- Power on Backup Battery.

NOTICE

NOTICE

These values are informational and cannot be changed from the Power Parameters client.

If a receiver uses battery power, check its voltage is not lower than 6.2 - 6.4V. If the charge falls below this value, the receiver switches off.

To retrieve last applied information in the *Battery Voltages* page for the selected receivers, press **Refresh Page**.

Output Modes Page

Output Modes govern power output to ports and slots. This option is only applicable to Odyssey, Odyssey-E, and HiPer family receivers. The corresponding page displays the following information (Figure 2-84):

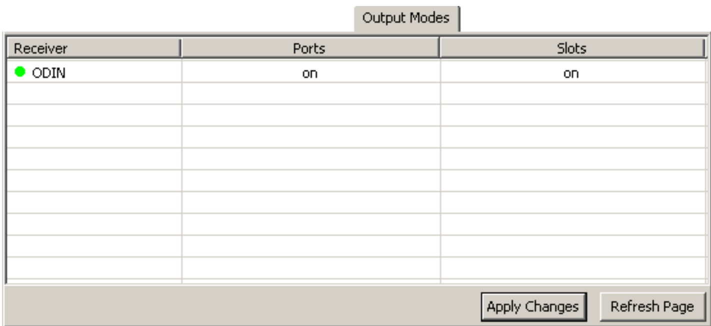


Figure 2-84. Output Modes Page

- Receiver- receiver name;
- Ports - the mode of the power on serial ports;
- Slots- the mode of the power on internal slots.

To change mode of a port or slot, select the receiver in the list and press the **Edit Parameters** button (see "Edit Parameters" on page 2-85 for details).

Press **Apply Page** to apply the parameters edited in the page to the selected receivers.

To retrieve last applied information in the *Output Modes* page for the selected receivers, press **Refresh Page**.

Turn On/Off Slots Page

The *Turn On/Off Slots* page displays the following information (Figure 2-85):

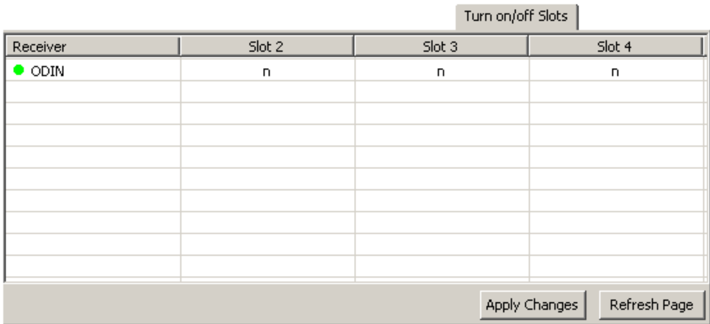


Figure 2-85. Turn On/Off Slots Page

- Receiver- receiver name;
- Slot 2 (3, 4) - reflects, if the corresponding internal slot is disabled or enabled. To change the status of the slot, select the receiver in the list and press the **Edit Parameters** button (see “Edit Parameters” on page 2-85 for details).

Press **Apply Page** to apply the parameters edited in the page to the selected receivers.

To retrieve last applied information in the *Turn On/Off Slots* page for the selected receivers, press **Refresh Page**.

Board Temperature Page

The *Board Temperature* page displays the current temperature on the receiver board:

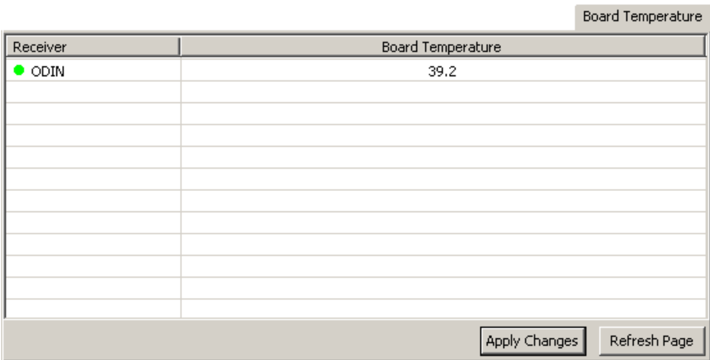


Figure 2-86. Board Temperature Page

NOTICE

These value is informational and cannot be changed from the Power Parameters client.

To retrieve last applied information in the *Board Temperature* page for the selected receivers, press **Refresh Page**.

Edit Parameters

To edit parameters of the selected receivers, press **Edit Parameters** on the control bar in the main screen.

The **Edit Parameters** dialog box displays the following parameters:.

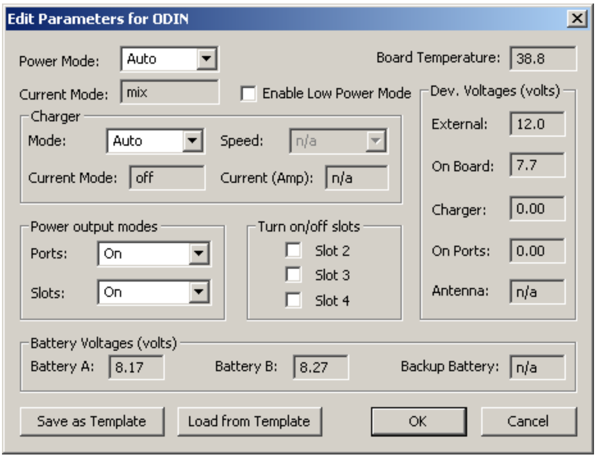


Figure 2-87. Edit Parameters

- Power Mode: sets the desired power source for the connected receiver.
  - – Auto – receiver will automatically select the power source.
  - – Mix – receiver will automatically consume power from the source with the highest voltage.
  - – Battery A – receiver will consume power only from battery A.
  - – Battery B – receiver will consume power only from battery B.
  - – External – receiver will use an external power supply.
- Current Mode: shows the current power source.
- Enable Low Power Mode – enables the receiver's processor for low power consumption.

- Charger – applies charging settings for the battery.
  - Mode: the battery charging mode.
    - – Off – receiver will not charge the batteries.
    - – Charge A – receiver will charge only battery A.
    - – Charge B – receiver will charge only battery B.
    - – Auto – receiver will automatically detect and charge the detected batteries.
  - Speed: sets the speed for a battery charge cycle to either “Normal” of “Fast”. This option is not available for several receiver models.
  - Current mode: displays the battery currently being charged.
  - Current (Amp): displays the actual battery charging current in Amperes
- Power output modes – only applicable to Odyssey, Odyssey-E, and HiPer family receivers, governs power output to ports and slots.
  - Ports: governs power output on the serial ports.
    - – On – when the receiver is turn on, the power board power all available serial ports. If the receiver is turn off, no power will be sent to the ports.
    - – Off – power will not be powered, even if the receiver is turned on.
    - – Always – all serial ports will be powered even if the receiver is turned off.
  - Slots: governs power output to the receiver’s internal slots.
    - – On – when the receiver is on, all internal slots will receive power. If the receiver is turned off, no power will be sent to the slots.
    - – Off – internal slots will not be powered, even if the receiver is turned on.
    - – Always – all internal slots will be powered even if the receiver is turned off.

- Dev. (Device) and Battery Voltages – the current power voltage state of the external power supply (only if connected), receiver board (actual voltage), batteries, charger (internal charger), and ports (first pin of each port).
- Turn on/off Slots – enables the corresponding internal slots. This option is applicable to Odyssey-E and HiPer family receivers only.

To save the modified parameters for the selected receivers, press **OK** on the *Edit Parameters* page. All the modified parameters will appear with an asterisk at their values in corresponding pages until the changes are applied (Figure 2-88).

Receiver	Auto Start	AFRM	Period (h:m:s)	Remove Old
DON3 (ODYSSEY_E)	on *	on *	7:00:00 *	on *

Figure 2-88. Example of Saved but not Applied Parameters

Press either **Apply Changes** on the page or **Apply Changes All Pages** on the control bar to apply the modified parameters to the receiver(s).

Notes:

TopNET-R

File Manager

File Manager is a TopNET-R client that provides the manual control of the receiver file system. It performs the following functions:

- displays a list of the files contained in the internal memory of the network receivers
- displays the current files A / B[1] that are currently being logged
- downloads files from the network receiver to a computer
- removes files from the internal memory of the network receiver
- starts/stops recording of A/B files to the internal memory of the network receiver

Starting File Manager

With TopNET-R running and this client available, click **Tools ▸ File Manager** or click **File Manager** on the toolbar (Figure 2-89).

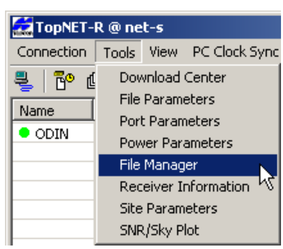


Figure 2-89. Open File Manager

The **Receiver File Manager** main screen has the following components (Figure 2-90):

1. TPS receivers are able to create two files simultaneously. These files are referred to as files A and B. Observation data recording into the files can be managed differently. Each file can have its own parameters.

- Title bar – displays the program name and receiver name
- Menu bar – contains three menus for File Manager functions
- File list – contains information about raw data files
- Toolbar – contains shortcut buttons to frequently used options
- Status bar – provides information about the number of files and the space available in the receiver memory

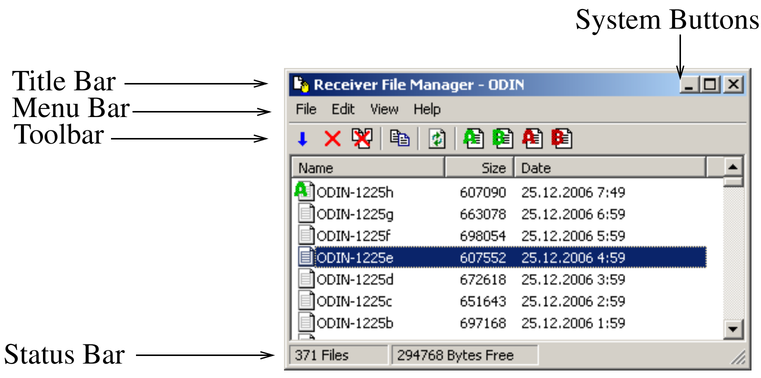


Figure 2-90. Main Screen Components

When started Receiver File Manager polls the receivers connected to the TopNET-S server, retrieves information about the files, and displays the file properties on the main screen of the program.

After polling each receiver, Receiver File Manager disconnects from the server.

Menu Bar

The menu bar provides access to most options available using Receiver File Manager in three drop-down menus:

File Menu

The File menu (Figure 2-91) includes the following menu items:

- Download – downloads a selected file(s) from the internal memory of the receiver to the PC
- Delete – removes the highlighted file(s), except the current file, from the internal receiver memory
- Initialize File System - erases all of the files inside the receiver.
- Start file A/B – creates a new (A or B) file for observation data logging,
- Stop file A/B – stops data recording to the (A or B) file
- Exit – closes File Manager

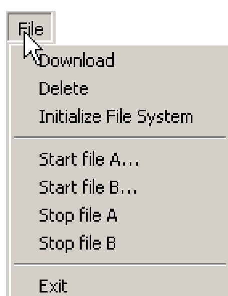


Figure 2-91. File Menu

Edit Menu

The Edit menu (Figure 2-92) includes the following menu items:

- Copy – copies the selected files to the clipboard. Use the Paste command of Windows Explorer to put the data in a new location.
- Select All – selects all files of the receiver

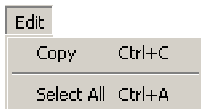


Figure 2-92. Edit Menu

Help menu

The Help menu (Figure 2-93) provides information about the File Manager version and build date.

Figure 2-93. Help Menu

Toolbar

The toolbar contains the buttons to conduct the File Manager main functions (Figure 2-94).

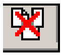
Figure 2-94. Toolbar

Table 2-11 describes the toolbar buttons.

Table 2-11. Toolbar Buttons

Button	Description
	Downloads selected files to the PC.
	Deletes selected files from the internal receiver memory.

Table 2-11. Toolbar Buttons (Continued)

Button	Description
	Initializes File System: erases all files from the receiver.
	Initiates polling the receivers connected to the TopNET-S server and updates the retrieved information in the receivers table.
	Copies the selected files to the clipboard.
	Creates a new A file for observational data recording.
	Creates a new B file for observational data recording.
	Stops observational data logging into the file **A**.
	Stops observational data logging into the file **B**.
	Displays information about the File Manager version and build date.

File list

The file list contains the following columns to display information about raw files:

- Name – the names of the files stored in the internal memory of the receiver. The icons indicate the following information about each file:
  -  : file is closed
  -  : file A is being recorded
  -  : file B is being recorded

- Size – the size of the file in bytes
- Date – the time of the last record in the file

File Operations

Receiver File Manager performs the following operations on the files of the receivers connected to the TopNET-S server:

- downloads a TPS file(s) from internal receiver memory to PC
- starts new files
- deletes TPS files from the receiver's internal memory

These operations can be done either through the File and Edit menu commands or with the help of corresponding toolbar buttons.

Downloading TPS Files to a PC

The files downloaded into the PC will have the same names and extensions as the original files in the internal memory of the receiver.

Receiver File Manager provides three methods for downloading TPS files from the internal receiver memory to the PC.

Download Method A

1. Highlight the desired files in the file list (Figure 2-95).
2. Press the **Download** button (Figure 2-95) (or select File ▸ Download).

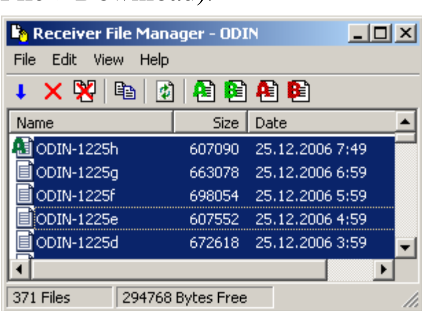


Figure 2-95. Selected Files for Download

3. In the **Browse For Folder** dialog box, select a folder on the local computer to store the downloaded files (Figure 2-96).

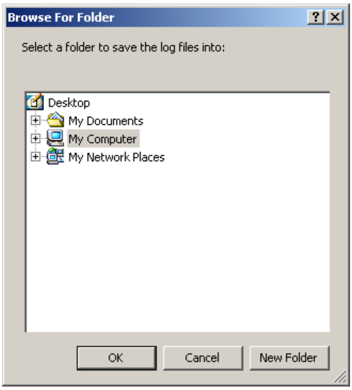


Figure 2-96. Browse For Folder

4. Press **OK** to start downloading TPS files from the receiver(s) to the specified directory (Figure 2-97).

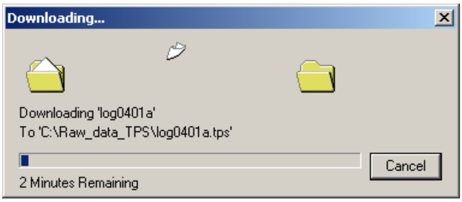


Figure 2-97. Downloading Progress

Download Method B

1. Highlight the desired files in the file list in the main screen (Figure 2-98).
2. Click **Copy** icon (or **Edit ▸ Copy** menu) (Figure 2-98).

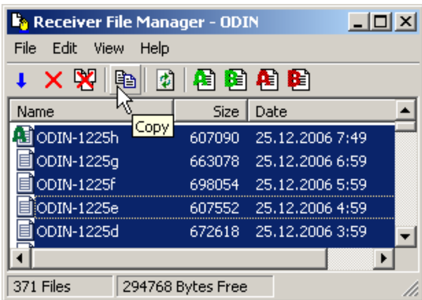


Figure 2-98. Copy Selected Files

3. Run Windows Explorer on the local computer. Select a folder to store the downloaded files and click **Edit ▸ Paste** (Figure 2-99).

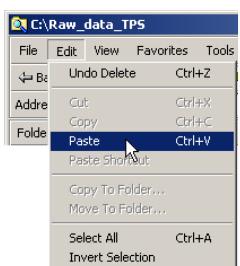


Figure 2-99. Paste Files

Download Method C

1. Run Windows Explorer on the local computer.
2. Highlight the desired files in the file list in the main screen.
3. Drag-and-drop the files to the Windows Explorer (Figure 2-100 on page 2-97).

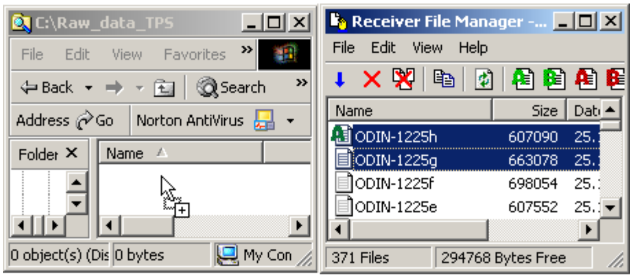


Figure 2-100. Drag-and-Drop the Selected Files

NOTICE

While downloading the current file to the local computer, the TPS receiver continues to log raw data into the current file.

Deleting TPS Files from Internal Memory

1. Highlight the desired files in the file list (Figure 2-101).
2. Press **Delete** (or File ▸ Delete) (Figure 2-101).

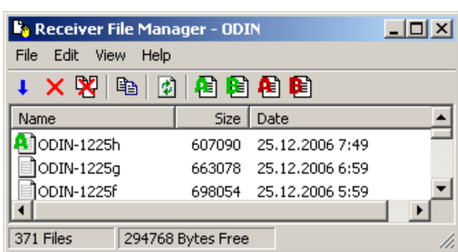


Figure 2-101. Selected Files in File Manager

3. Press **Yes** at the confirmation dialog box (Figure 2-102). The selected files are deleted.

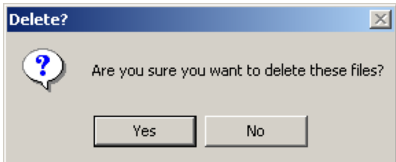


Figure 2-102. Confirmation Dialog Box

If you are removing current files from the receiver memory (with the “A” or “B” file icons), a supplementary delete confirmation message is displayed after the delete confirmation dialog box (Figure 2-103). Press **Yes** to stop data recording and to delete the file.

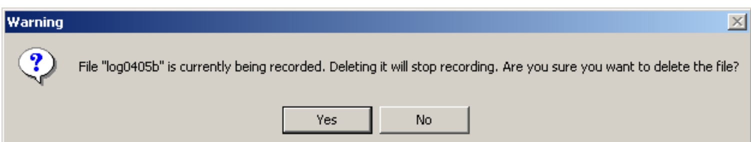


Figure 2-103. Confirm Deletion of Current File

Start New Files

The Receiver File Manager program creates two independent files in the selected receiver using different settings.

To create a new A/B file press the **Create New File** button (Figure 2-104) (or **File ▸ Start file A/Start file B**).

Figure 2-104. Create a New A or B File

Stop Current File

1. To stop data recording into the current A/B file click the stop file recording button (Figure 2-105) (or **File ▸ Stop file A/Stop file B**).

Figure 2-105. Stop Recording File A or B

2. Press **Yes** at the confirmation dialog box (Figure 2-106).

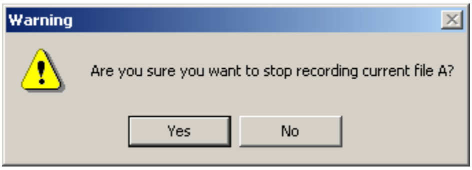


Figure 2-106. Confirm Stop Recording

The data recording will be stopped, and the file will be marked with the “file closed” icon.

Notes:

Receiver Information

Receiver Information is a TopNET-R client that provides information about the TPS receivers connected to the TopNET-S server. Receiver Information displays the following information:

- serial number of receiver, receiver model, type of GPS board, firmware version installed on the TPS receiver's GPS board
- receiver options

Receiver Information performs the following operations on receivers:

- clears the NVRAM[1]
- resets the receiver
- uploads firmware
- uploads an Option Authorization File (OAF)

Starting Receiver Information

With TopNET-R running, click **Tools ▸ Receiver Information** or click **Receiver Information** on the toolbar (Figure 2-107).

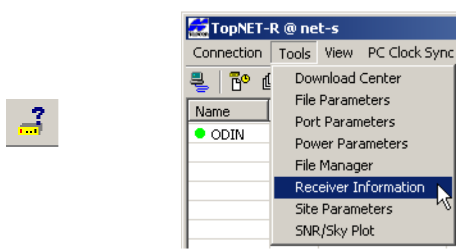


Figure 2-107. Start Receiver Information

1. NVRAM - Non-Volatile Random Access Memory

The **Receiver Information** main screen (Figure 2-108) has the following components:

- Title bar – displays the program name and the Server address
- Menu bar – contains three menus for Receiver Information functions
- Toolbar– contains three shortcut buttons to execute the Receiver Information functions
- Receiver Table – displays information on each connected receiver
- Status bar – provides basic information about the operational state of the program and selected toolbar button

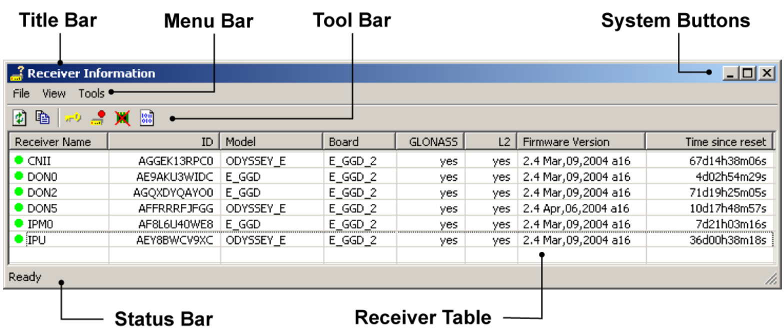


Figure 2-108. Main Screen Components

After starting, Receiver Information polls the receivers connected to the TopNET-S server to retrieve the necessary information. An icon next to the receiver's name indicates the status of information from the receiver:

- – initial stage of polling the receiver
- – data have been successfully retrieved
- – the receiver information cannot be retrieved

After polling each receiver, File Parameters disconnects from the TopNET-S server and the status bar displays a “Ready” message.

Menu Bar

The menu bar provides access to most options available using Receiver Information in three drop-down menus (Figure 2-109).

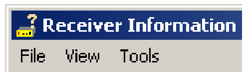


Figure 2-109. Menu Bar

File Menu

The File menu (Figure 2-110) consists of the **Exit** option that closes Receiver Information.

Figure 2-110. File Menu

View Menu

The View menu (Figure 2-111) includes two items:

- Refresh – initiates polling the receivers and updates information in the receiver table according to the polling carried out. Once receiver polling and data retrieving is completed, Receiver Information disconnects itself from the TopNET-S server.
- Copy – copies selected lines to Windows Clipboard.

Figure 2-111. View Menu

Tools Menu

The Tools menu (Figure 2-112 on page 2-104) includes four items:

- Receiver Options – displays the receiver’s options
- Reset Receiver – executes a receiver reset
- Clear NVRAM – clears the receiver's non-volatile RAM

- Upload Firmware – loads firmware to the receiver

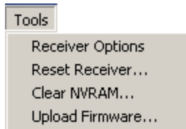


Figure 2-112. Tools Menu

Toolbar

The toolbar contains five buttons for choosing the Receiver Information main functions (Figure 2-113).

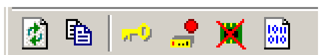


Figure 2-113. Toolbar

Table 2-12 describes the toolbar buttons.

Table 2-12. Toolbar Buttons

Button	Description
	Refresh – Initiates the polling of receivers connected to the TopNET-S server program and updates the retrieved information in the receivers table.
	Copy – Copies the information located in the selected area to the Windows Clipboard
	Receiver Options – Displays receiver options
	Reset receiver – Resets the selected receiver
	Clears NVRAM – Clears the NVRAM of the selected receiver
	Upload Firmware – Downloads firmware to the receiver

Receiver Table

The Receiver table contains the following fields to display information on every receiver:

- Receiver name – the name of receiver
- ID – internal serial number of receiver
- Model – receiver model
- Board – type of GPS board
- GLONASS – whether the receiver tracks GLONASS signal (Yes), or not (No)
- L2 – whether the receiver tracks L2 signal (Yes), or not (No)
- Firmware version – firmware version installed on the TPS receiver’s GPS board
- Time since reset – time span since the last receiver reset in days (d), hours (h), minutes (m), and seconds (s)

Receiver Operations

Receiver Information performs the following operations on a TPS receiver connected to the TopNET-S server:

- displays receiver options
- resets a receiver
- clears a receiver's NVRAM
- uploads firmware to a receiver board

These operations can be done either with the Tools menu commands or with corresponding toolbar buttons.

Receiver Options

To view options authorized for a receiver, select the receiver and click **Tools ▸ Receiver Options** (Figure 2-114) or press the **Receiver Options** button on the toolbar.

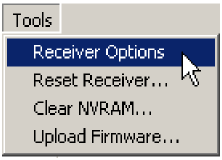
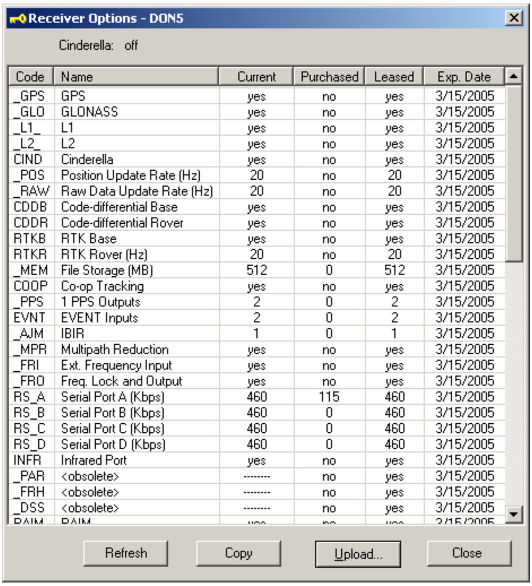


Figure 2-114. Open Receiver Options

The **Receiver Options** dialog box displays a list of all options available for TPS receivers and the status of each option for the receiver (Figure 2-115 on page 2-106).

- To update information, press **Refresh**.
- To copy information to Windows Clipboard, press **Copy**.
- To upload a new OAF from the computer to the receiver, press **Load**. See the procedure on the following page for details.



Code	Name	Current	Purchased	Leased	Exp. Date
_GPS	GPS	yes	no	yes	3/15/2005
_GLO	GLONASS	yes	no	yes	3/15/2005
_L1_	L1	yes	no	yes	3/15/2005
_L2_	L2	yes	no	yes	3/15/2005
CIND	Cinderella	yes	no	yes	3/15/2005
_POS	Position Update Rate (Hz)	20	no	20	3/15/2005
_RAW	Raw Data Update Rate (Hz)	20	no	20	3/15/2005
CDDB	Code-differential Base	yes	no	yes	3/15/2005
CDDR	Code-differential Rover	yes	no	yes	3/15/2005
RTKB	RTK Base	yes	no	yes	3/15/2005
RTKR	RTK Rover (Hz)	20	no	20	3/15/2005
_MEM	File Storage (MB)	512	0	512	3/15/2005
COOP	Co-op Tracking	yes	no	yes	3/15/2005
_PPS	1 PPS Outputs	2	0	2	3/15/2005
EVNT	EVENT Inputs	2	0	2	3/15/2005
_AJM	IBIR	1	0	1	3/15/2005
_MPR	Multipath Reduction	yes	no	yes	3/15/2005
_FRI	Ext. Frequency Input	yes	no	yes	3/15/2005
_FRO	Freq. Lock and Output	yes	no	yes	3/15/2005
RS_A	Serial Port A (Kbps)	460	115	460	3/15/2005
RS_B	Serial Port B (Kbps)	460	0	460	3/15/2005
RS_C	Serial Port C (Kbps)	460	0	460	3/15/2005
RS_D	Serial Port D (Kbps)	460	0	460	3/15/2005
INFR	Infrared Port	yes	no	yes	3/15/2005
_PAR	<obsolete>	--------	no	yes	3/15/2005
_FRH	<obsolete>	--------	no	yes	3/15/2005
_DSS	<obsolete>	--------	no	yes	3/15/2005

Figure 2-115. Receiver Options

If downloading an OAF:

1. Press **Upload** on the **Receiver Options** dialog box.
2. On the **Open** dialog box, navigate to and select the desired OAF (Figure 2-116).

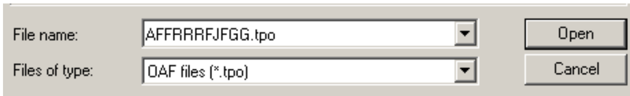


Figure 2-116. Select OAF

3. Press **Open** to begin the upload process (Figure 2-117).

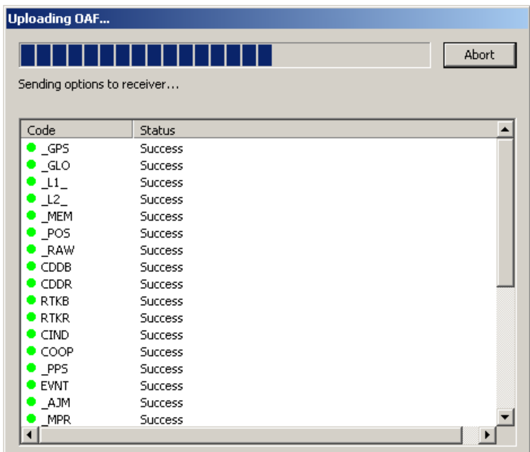


Figure 2-117. Uploading OAF

4. When the download completes, press **Close** (Figure 2-118).

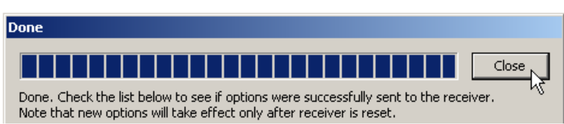


Figure 2-118. OAF Upload Complete

5. Reset the receiver to have the new options take affect.

Receiver Reset

1. To reset a receiver, select the receiver and click **Tools ▸ Reset** (Figure 2-119) or press the **Reset Receiver** button on the toolbar.

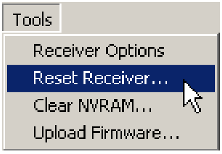


Figure 2-119. Tools Menu, Reset Receiver Selection

2. Press **OK** at the confirmation message to begin the reset process (Figure 2-120). Receiver reset is similar to powering off and on the receiver. The NVRAM is not cleared with this command.

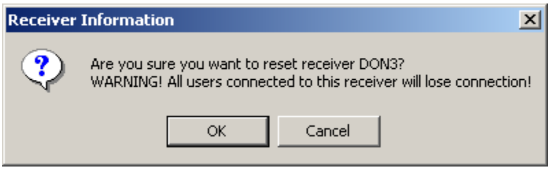


Figure 2-120. Warning Before a Reset

The **Reset** dialog box is displayed (Figure 2-121). If needed, press **Cancel** to abort the reset.

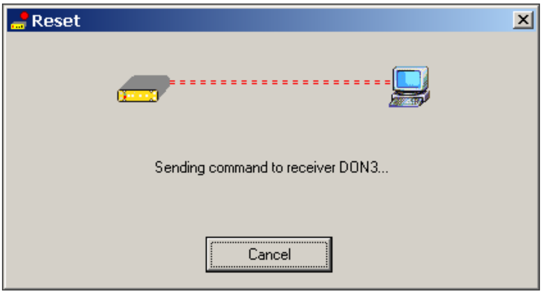


Figure 2-121. Reset in progress

Once the receiver reset is completed, the **Reset** dialog box displays a message that prompts to refresh information in the receiver table (Figure 2-122).

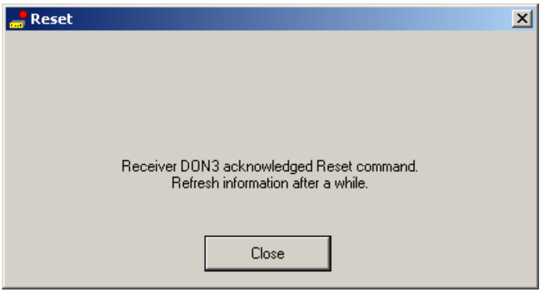


Figure 2-122. Reset Complete Notification

3. Press **Close**. Receiver Information disconnects from the TopNET-S server.

Clearing the NVRAM

1. To clear a receiver's non-volatile RAM, select the receiver and click **Tools ▸ Clear NVRAM** (Figure 2-123) or press **Clear NVRAM** on the toolbar.

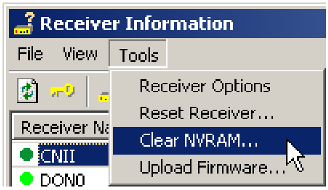


Figure 2-123. Clear NVRAM

2. At the confirmation message, press one of the following:
   - **OK** to begin clearing the NVRAM.
   - **Cancel** to abort the operation.

NOTICE

NOTICE

Clearing the NVRAM sets the receiver's settings to default values and erases the almanac. After clearing the NVRAM, *your receiver will require some time to collect the new ephemerides and complete almanacs (about 15 minutes).*

The **Clear NVRAM** dialog box displays (Figure 2-124). If needed, click **Cancel** to abort the operation.

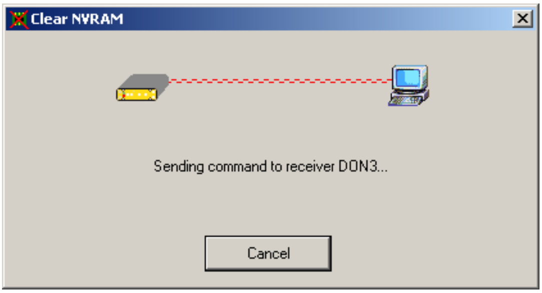


Figure 2-124. Clear NVRAM in Progress

Once the clear NVRAM operation completes, the **Clear NVRAM** dialog box displays a message that prompts to refresh information in the receiver table and reconfigure the receiver (Figure 2-125).

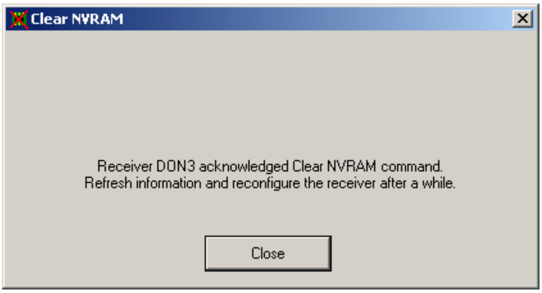


Figure 2-125. Clear NVRAM Complete

3. Click Close. **Receiver Information** disconnects from the TopNET-S server.

NOTICE *NOTICE*

After clearing the NVRAM the receiver requires about 15 minutes to collect new ephemerides and complete almanacs, and to determine its position.

Uploading Firmware

Receiver Information also uploads firmware to a network receiver. The procedures for uploading firmware are different for the receiver connected to the TopNET-S server and receiver connected to PC directly, using USB or COM.

When uploading firmware through the TopNET-S server, the firmware file is sent first to the PC, where TopNET-S is installed, next loaded to Compact Flash Card of the receiver, and then to the receiver's internal memory.

Before uploading the firmware, Receiver Information checks the receiver's free space. If the receiver does not have sufficient space for the new firmware file, the firmware will not be uploaded (Figure 2-126).

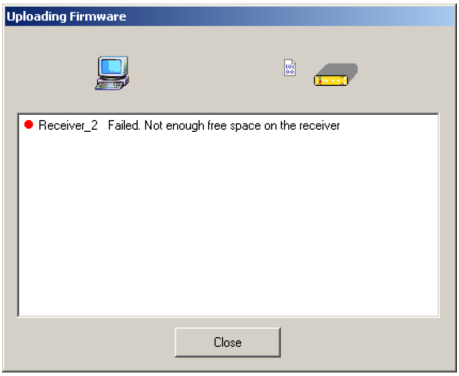


Figure 2-126. Not Enough Free Space on the Receiver to Upload Firmware

Firmware is released as a ZIP archive. It can be downloaded from TPS web site and must be decompressed. This archive contains the following three files:

- ramimage.ldr – the Receiver board RAM file;
- main.ldp – the Receiver board Flash file;
- powbrd.ldr – the Power board RAM file.

NOTICE

You need to load the main.ldp file only when changing firmware in a network receiver.

1. To upload firmware to a network receiver, select the receiver(s) in the Receiver Table and click **Tools ▸ Upload Firmware** (Figure 2-127) or click **Upload Firmware** on the toolbar.

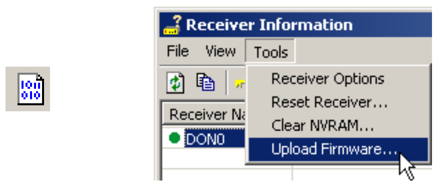


Figure 2-127. Start Uploading Firmware

2. Click **Yes** at the confirmation message to continue (Figure 2-128). Click **No** to exit the firmware upload function.

Figure 2-128. “Uploading Firmware” Confirmation

3. On the **Open** dialog box, navigate to and select the desired firmware file and click **Open** (Figure 2-129).

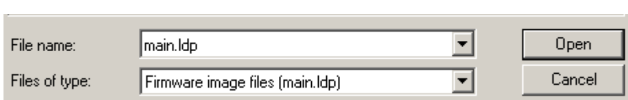


Figure 2-129. Selecting a Firmware File

In the **Uploading Firmware** dialog box (Figure 2-130) click **Close** to stop the firmware upload process, if necessary.

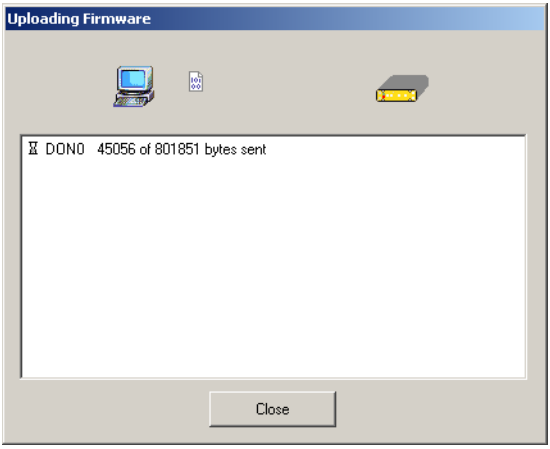


Figure 2-130. Firmware Uploading in progress

4. When the upload of main.ldp to the receiver's Compact Flash Card completes, click **Close** (Figure 2-131).

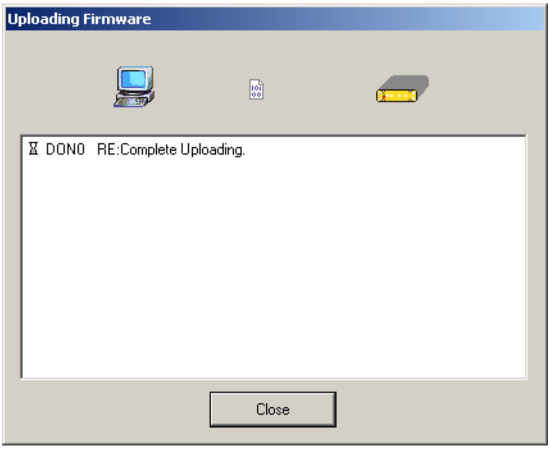


Figure 2-131. A Firmware File is Successfully Downloaded

After the firmware is successfully uploaded to the receiver's Compact Flash Card, the firmware will be automatically installed in the receiver's internal memory and the installation completes (Figure 2-132 on page 2-114).

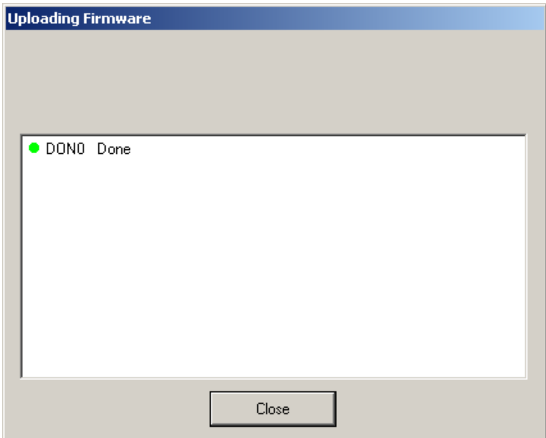


Figure 2-132. Firmware Uploading Completed

5. Click **Close** to disconnect Receiver Information from the TopNET-S server.
6. Use the File Manager client to remove main.ldp from the receiver's compact flash card. See Chapter 2 for details.

NOTICE

To ensure the receiver's memory has sufficient space, remove the main.ldp file from the receiver's compact flash card when done.

Site Parameters

Site Parameters is a TopNET-R client that controls site parameters for TPS receivers connected to a TopNET-S server. When connected to the server, Site Parameters provides the following functions:

- displays/sets a reference station's position in the WGS-84 coordinate system
- displays/set antenna parameters
- displays/sets CMR settings

Starting Site Parameters

With TopNET-R running, click **Tools ▸ Site Parameters** or click the **Site Parameters** button on the toolbar (Figure 2-133).

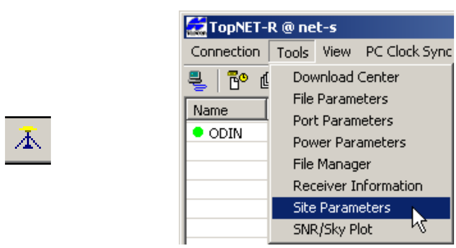


Figure 2-133. Start Site Parameters

The **Site Parameters** main screen has the following components (Figure 2-134 on page 2-116):

- Information pages – sets parameters or displays information about the ports on the receivers
- Control buttons for the page – applies changes made on the current page

- Control buttons for all pages – applies changes made on one page to all pages
- Status bar – displays messages for file activities.

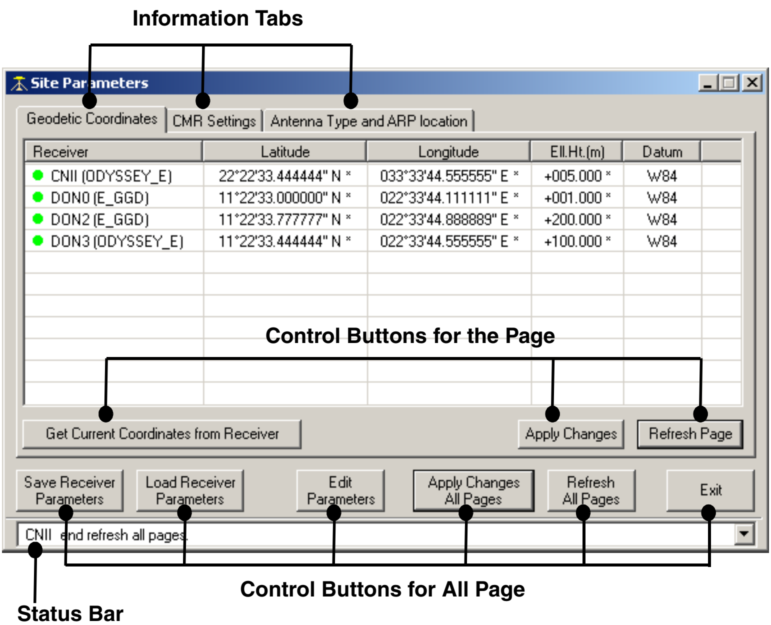


Figure 2-134. Main Screen Components

After starting, Site Parameters polls the receivers connected to the TopNET-S server to retrieve the necessary information. An icon next to the receiver's name indicates the status of information from the receiver:

- – initial stage of polling the receiver
- – data have been successfully retrieved (green icon)
- – the receiver information cannot be retrieved (red icon)

After polling each receiver, Site Parameters remains connected to the server.

Control Buttons

The control bar contains six buttons to control receiver parameters. Table 2-13 describes the control buttons.

Table 2-13. Site Parameters Control Buttons

Button	Description
Save Receiver Parameters	Saves modified site parameters from every page for all selected receivers to the PC, but does not send them to receivers.
Load Receiver Parameters	Restores from the PC the saved site parameters to every page for all selected receivers.
Edit Parameters	Opens a dialog box to edit site parameters for all selected receivers.
Apply Changes All Pages	Applies site parameters modified in all pages to selected receivers. If an **Apply Changes** button is pressed instead, only parameters modified on the current page will be applied; changes on all other pages will be lost.
Refresh All Pages	Returns all pages to last state, applied to the receiver.
Exit	Closes the site parameters program without saving any changes.

Status Bar

The status bar provides the information about the performed actions and saves the history. The history is available in the drop down list.

Geodetic Coordinates Page

The *Geodetic Coordinates* page displays reference positions in WGS-84 for TPS receivers connected to the TopNET-S server (Figure 2-135).

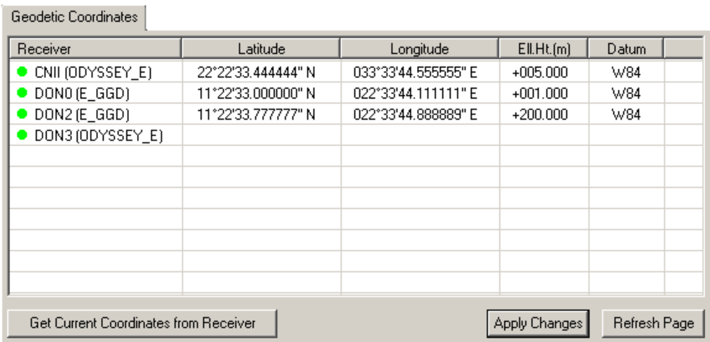


Figure 2-135. Geodetic Coordinates Page

The Site Parameters tool retrieves the coordinates through the TopNET-S server from the receivers.

In the *Geodetic Coordinates* page, appropriate columns will be empty if reference coordinates are not set in the receiver (the receiver DON3 in this example). Site Parameters provides the following two options to enter reference coordinates:

- To use the current position, press 

. If available, the current geodetic coordinates of the receiver's antenna - as estimated by the receiver itself - will be retrieved from the highlighted receiver(s) and displayed in the corresponding columns. If the current coordinates are not available yet, instead of coordinates the message "Not available" will be displayed. As soon as the coordinates become available, they are retrieved and displayed automatically.

- Press Edit Parameters and enter the desired reference position for the receiver, then click **OK** (Figure 2-136).

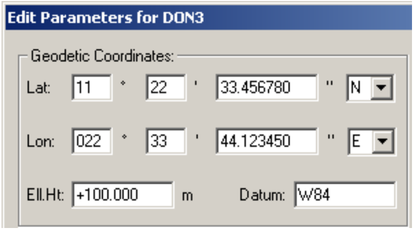


Figure 2-136. Edit Geodetic Coordinates for the Receiver(s)

The *Datum* field contains the code of the datum as it is used by the receiver. For example, W84 stands for the WGS-84 system.

If you want to use any other datum, contact your dealer for the appropriate code. For the entire list of supported datums and their identifiers see GRIL Manual.

In either case, the coordinates reflected on the *Geodetic Coordinate* page will have asterisks at their values (Figure 2-137) until the changes have been transmitted to the receiver and saved as the reference station position in the receiver.

Receiver	Latitude	Longitude	Ell.Ht.(m)	Datum
DON3 (ODYSSEY_E)	11°22'33.456780" N *	022°33'44.123450" E *	+100.000 *	W84 *

Figure 2-137. Example of Saved Parameters Not Applied to Receiver

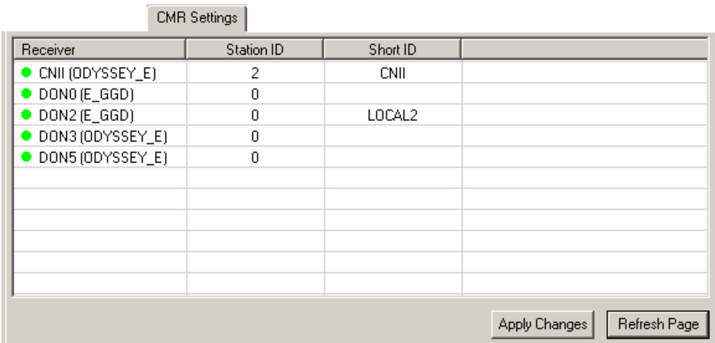
Click either Apply Changes on the page or Apply Changes All Pages on the control bar to apply the modified parameters to the receiver(s). The asterisk will be removed from the parameter values.

CMR Settings Page

The *CMR Settings* page displays the following information about CMR messages set in the receivers (Figure 2-138):

- Station ID – the identifier for the receiver is an integer from 0 to 31, unique for every receiver to distinguish the receivers working in a common area.

- Short ID – an alphanumerical code (up to 8 characters) used to describe the object.



CMR Settings

Receiver	Station ID	Short ID	
CNII (ODYSSEY_E)	2	CNII	
DON0 (E_GGD)	0		
DON2 (E_GGD)	0	LOCAL2	
DON3 (ODYSSEY_E)	0		
DON5 (ODYSSEY_E)	0		

Figure 2-138. CMR Settings Page

1. To edit/enter ID values in the *CMR Settings* page, press

   Edit Parameters .

2. In the *CMR* section of the **Edit Parameters** dialog box, enter applicable information and click **OK** (Figure 2-139).

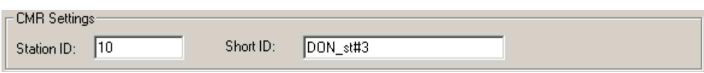
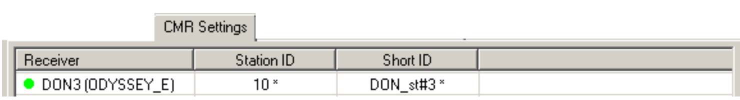


Figure 2-139. CMR Settings Area of Edit Parameters Window

The modified settings on the *CMR Settings* page will have an asterisk next to the value until the changes are applied (Figure 2-140).



CMR Settings

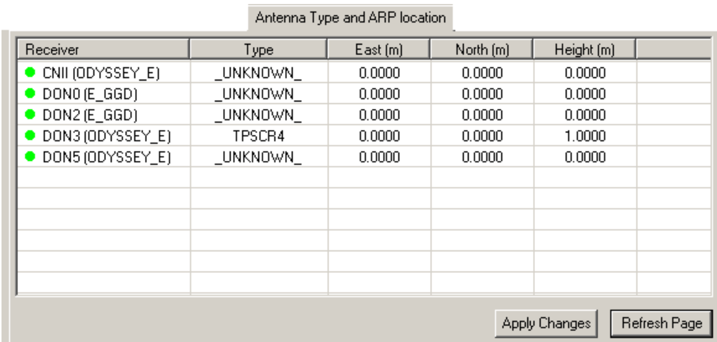
Receiver	Station ID	Short ID	
DON3 (ODYSSEY_E)	10 *	DON_st#3 *	

Figure 2-140. Example of Saved but not Applied Parameters

Click either Apply Changes on the page or Apply Changes All Pages on the control bar to apply the modified parameters to the receiver(s). The asterisk will be removed from the parameter values.

Antenna Type and ARP Location Page

The *Antenna Type and ARP location* page displays information about the type of GPS antenna of the receiver and ARP offsets from the marker (Figure 2-141).



Receiver	Type	East (m)	North (m)	Height (m)
CNII (ODYSSEY_E)	_UNKNOWN_	0.0000	0.0000	0.0000
DON0 (E_GGD)	_UNKNOWN_	0.0000	0.0000	0.0000
DON2 (E_GGD)	_UNKNOWN_	0.0000	0.0000	0.0000
DON3 (ODYSSEY_E)	TPSCR4	0.0000	0.0000	1.0000
DON5 (ODYSSEY_E)	_UNKNOWN_	0.0000	0.0000	0.0000

Figure 2-141. Antenna Type and ARP Location Page

1. To edit/enter ID values in the *Antenna Type* page, press

.

2. In the *Antenna Parameters* section of the **Edit Parameters** dialog box, enter applicable information:

   Type - the standard antenna name as mentioned in the NGS[1] list.

   ARP offsets from marker - offset of the ARP on the antenna from the Marker: east and north distances of the sub-antenna point and the vertical height of the ARP.

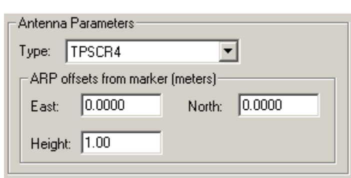


Figure 2-142. Antenna Parameters area of Edit Parameters screen

1. See, for example, Complete Antenna Calibration Results at http://www.ngs.noaa.gov/ANTCAL.

3. Click **OK** (Figure 2-142).

The modified settings will have an asterisk next to the value on the *Antenna Type and ARP location* page until the changes are applied (Figure 2-143).

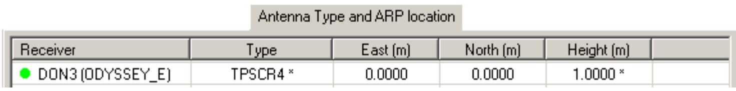


Figure 2-143. Example of Saved but not Applied Parameters

Click either 

on the page or 

on the control bar to apply the modified parameters to the receiver(s). The asterisk will be removed from the parameter values.

Using the Map View

The Map view displays the network Map and shows the relative position of the reference stations. A change in icon color indicates a change in status.

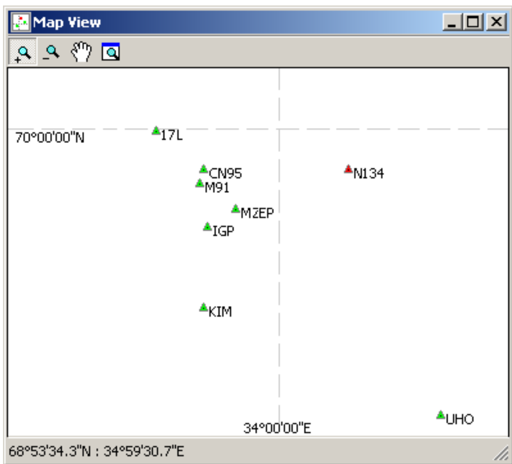


Figure 2-144. Map

Table 2-14 lists the conventions used in the map.

Table 2-14. Map View Conventions

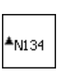
Icon	Description
	A connected reference station displays as a green triangle with a point inside and a name to the right.
	An offline reference station (e.g., disconnected by the operator or out of order) is displayed as a black triangle with a name to the right.
	A reference station disabled by the operator, or a station with a bad or corrupted signal is displayed as a red triangle with a point inside and a name to the right.

Table 2-14. Map View Conventions (Continued)

Icon	Description
	The numerical coordinate values are located near each parallel and meridian on the left and bottom side of the map to form a coordinate grid.

Table 2-15 describes the toolbar buttons used for controlling the Map view.

Table 2-15. Map View Toolbar Buttons

Button	Description
	Zoom In Press the button and click the map to enlarge the view Hold the left mouse button and click-and-drag a rectangle to enlarge a selected part of the screen. Also you can use mouse wheel if you have one.
	Zoom Out Press the button and click the map to decrease the view or rotate mouse wheel.
	Pan Press the button to activate the Pan control that allows one to move the observed area using left mouse button being hold.
	Zoom All Press the button to display all the objects.

SNR/Sky Plot

SNR/Sky Plot is a TopNET-R client that displays the following information on the satellites being tracked for every network receiver:

- the number of satellites tracked (GPS and GLONASS)
- Signal-to-noise ratio (SNR) in the C/A,P1 and P2 channels [dB*Hz]
- satellites' position on the sky
- time of the last message from the receiver
- elevation mask

Starting SNR/Sky Plot

With TopNET-R running, click **Tools ▸ SNR/Sky Plot** or click **SNR/Sky Plot** on the toolbar (Figure 2-145).

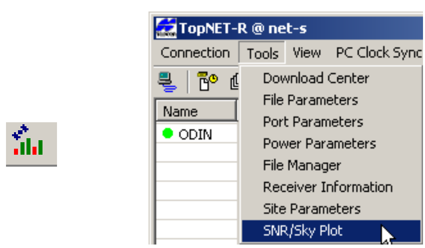


Figure 2-145. Start SNR/Sky Plot

The **SNR/Sky Plot** main screen (Figure 2-146 on page 2-126) has the following components:

- Title bar – displays the receiver's name, the program name, and the Server address.
- Receiver status – indicates the status of communication between TopNET-R (SNR/Sky Plot) and the receiver:

– Connecting... : initial stage of communication with the receiver

– Connected : the receiver is “on-line“; data are coming from this receiver during a defined time period. (This time period is defined in Set Periods and Timers screen. See Figure 2-25 on page 2-23)

– Disconnected : the receiver is “off-line”; no data from receiver came during a defined time period. (This time period is defined in Set Periods and Timers screen. See Figure 2-25 on page 2-23)

- Receiver Time – the time of the last message from the receiver.
- Show SNR - switches the sky plot for each SNR type. See “Sky Plot” on page 2-127 for details.
- The number of GPS and GLONASS satellites tracked.
- Elevation Mask – the minimum elevation for the satellites whose data are received by TopNET-R.

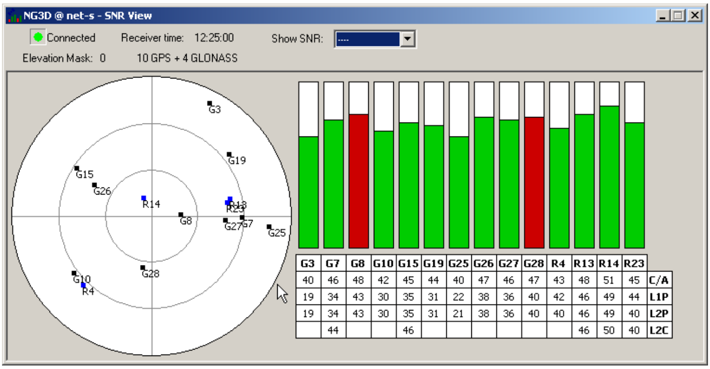


Figure 2-146. SNR View Main Screen

To view or set delays for the connection statuses, use the **Setup ▸ Timers** function in TopNET-R.

Note that GPS and GLONASS satellites are marked with G# and R#, respectively, where # is satellite number (PRN for GPS and slot number for GLONASS). If the GLONASS satellite slot number is unknown, the table shows lowercase 'r' and frequency number of the received signal (e.g. “r12”).

Sky Plot

The Sky Plot displays satellite constellation on a plot of visible sky. Concentric circles represent elevation angles at intervals of 30 degrees. This plot can display the following:

- 

– GPS satellites
- 

– GLONASS satellites
-  – Satellite trajectory

Depending upon the selection in the **Show SNR** field, Sky Plot shows colored traces of the satellites.

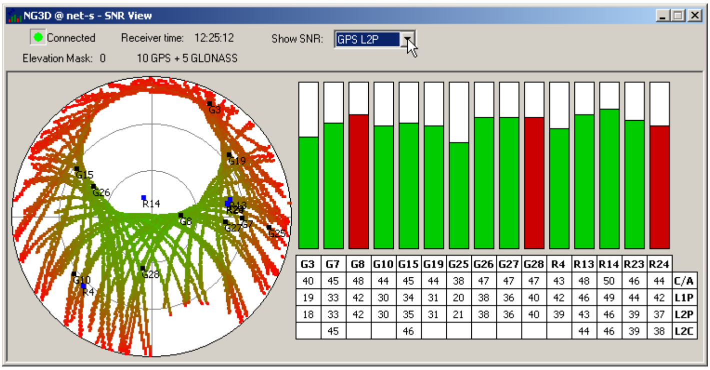


Figure 2-147. An Example SNR View

The traces are stored in separate files for each receiver. Recording of a trace is performed only when the SNR/Sky application is running. The storage files are cumulative, and when you open the application next time, the new data will be added to the existing.

Trace color corresponds to the average strength of the signal, from 10 dB*Hz (bright red) to 60 dB*hz (bright green) at each elevation azimuth. Each point in the trace obtains its color proportionally to the mean SNR value of all the satellites ever coming through this point. To clean this information, delete all the *.snr files in the TopNET directory on your PC.

SNR Bar Chart

The SNR Bar Chart (Figure 2-146 on page 2-126) represents the signal-to-noise ratio in the C/A channel. Each column corresponds to one satellite and each satellite's number is shown in the header of SNR Table. The satellite SNR values are compared with known thresholds in order to estimate the validity of the satellite signal. The following graph displays the signal-to-noise thresholds in the SNR/ Sky Plot program for all possible values of satellite elevations. (Figure 2-148 on page 2-128).

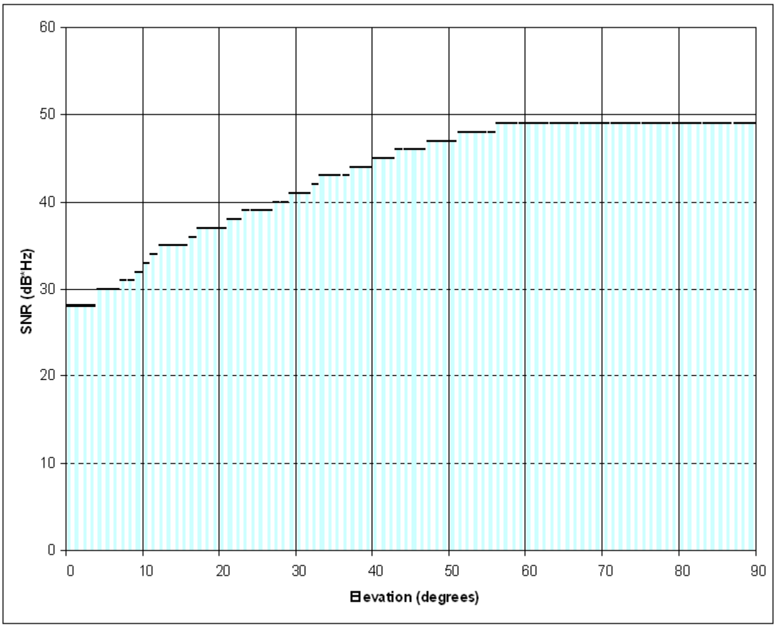


Figure 2-148. Thresholds of SNRs via Elevations

If the SNR value for a satellite is less than the threshold for satellite's current elevation angle, the corresponding column in the Bar Chart is colored red. This is a visual indication alerting that the signal strength of the satellite fell below the expected value (Figure 2-149).

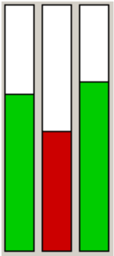


Figure 2-149. SNR for GPS Satellite 20 Under Normal Value for Receiving Signal

To view basic tracking information about a satellite, set the cursor over the desired satellite's bar or place the mouse over the satellite in the Sky Plot. A pop-up screen displays the applicable information (Figure 2-150).

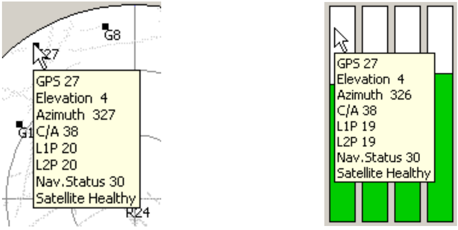


Figure 2-150. Satellite Information

The information on the popup screen also contains the explanation of the satellite validity: Navigation Status and Satellite Health. If a satellite's health is bad, its number in the SNR table (see "SNR Table" on page 2-130) is colored red and the pop-up screen entry Satellite

Healthy changes to Satellite Unhealthy (N), where N is the satellite health code. (See Appendix IV for the satellites health codes.)

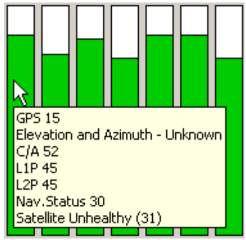


Figure 2-151. Pop-up Information About Unhealthy Satellite

Navigation Status entry stands for the Satellite Navigation Status Code that represents the type of the data that can use the signal of this satellite. See Appendix III for details.

SNR Table

The SNR table displays value of the signal-to-noise ratio in the C/A, L1P, L2P and L2C channels for each tracked satellite in dB*Hz. (Figure 2-152)

G6	G7	G10	G13	G15	G16	G21	G24	G25	G31	R6	R7	R21	R22	R23	
49	49	46	35	46	47	49	37	38	40	52	49	33	49	42	C/A
43	45	34		35	37	43		15	17	53	49	33	48	42	L1P
43	46	34		36	37	43		15	18		43	39	50	37	L2P
									36	46	42			37	L2C

Figure 2-152. SNR Table

Here L2C stands for both GPS L2C and GLONASS L2 C/A. These values are shown only for the satellites that transmit signals on these channels.

These signals are available only for G3 receivers (Net-G3, GR-3) after L2C tracking is activated in these receivers with the following commands:

%%set,/par/lock/gps/l2c,on

%%set,/par/lock/glo/l2c,on

TOPCON

POSITIONING SYSTEMS

Chapter 3. TopNET-N

User's Manual

List of Contents

Introduction

Welcome to TopNET-N, a client program offered as an extension to Topcon's unique and powerful Reference Station Software Suite. As a part of the TopNET RTK package, the TopNET-N application is intended to control a network of reference stations and provide corrections to any rover, located within the coverage area of the network, and is used in conjunction with the TopNET-S server and TopNET-R client programs.

TopNET-N can be configured to support DGPS and RTK modes of operation in any combinations; the user simply decides which data stream is compatible with the mobile equipment in use and requests this through the appropriate communication link.

Surveying with a Single Reference Station

Single reference station surveys are performed with the help of RTK or DGPS algorithms.

The RTK algorithm uses the concept of Real-Time Kinematics (RTK). Using the standard format correction data supplied from the nearest reference station in the network (the pseudorange and phase measurements), a rover receiver derives a standard RTK solution. In the RTK mode TopNET determines the nearest reference station based on the standard NMEA[1]-0183 GGA message sent to the central computer by each mobile unit when it makes the communication connection. Local RTK is usually limited in range. By creating a network of reference stations over a larger geographical area, mobile

1. For this and further abbreviations see "Abbreviations"

users can connect to any one of the reference stations in the network, extending the operation range of mobile units.

The Differential GPS (DGPS) mode operates similar to the RTK mode. It provides sub-meter accuracy (depending on the user equipment). The difference is in the correction data supplied from the nearest reference station in the network. For the DGPS mode that is the pseudorange correction calculated in the TopNET-N application from raw measurements provided by the reference station nearest to the rover. This mode provides less precise data than RTK, however, DGPS is an L1 only solution (code) that can be used by a wider variety of less expensive equipment to support applications such as mapping, GIS, navigation, positioning, agriculture, etc. where centimeter level accuracy (and the associated high end user equipment) is not required. DGPS also works at a much greater range from the reference stations than RTK.

During the RTK or DGPS survey the rover receives correction data from the closest reference station. If another reference station appears to be closer, the TopNET-N switches the rover automatically.

If the rover moves along a middle line between two reference stations, the switching looks as follows:

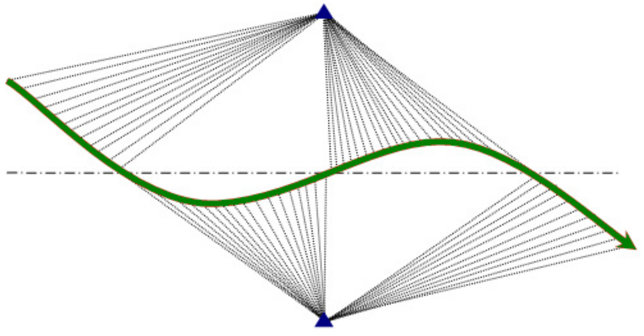


Figure 3-1. Zones of Reference Station Tracking. Equidistant Tracking without "Hysteresis" Effect

However, one should note that each time the rover is switched, it loses Fixed solution and has to restart RTK engine.

In order to avoid multiple switching between reference stations when the rover is near the middle line, the TopNET-N application uses the "hysteresis" effect. The reference station is changed only when the distance difference exceeds a threshold value of 100 m.

In this case the zones of reference station tracking look as follows:

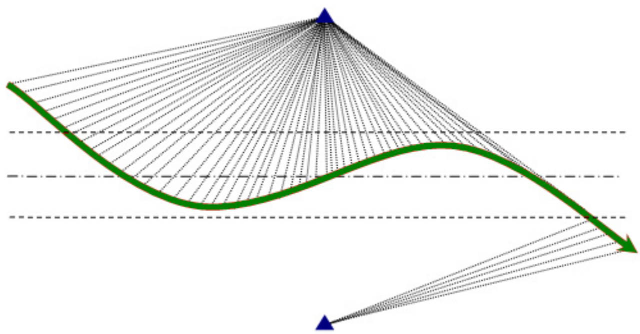


Figure 3-2. Zones of Reference Station Tracking. Equidistant Tracking

Implementation

The Rover connects with TopNET-N using the TCP/IP protocol, through different ports for DGPS and RTK, or via Ntrip caster.

In the case of connection through the ports, the TopNET-N operator sets up these ports once, when starting work for the first time, then provides the settings to all the TopNET-N users. By default, the ports are 8204 for DGPS mode and 8205 for RTK mode.

Operating in Ntrip caster mode, the services are provided through the mountpoints with corresponding names.

The selection of the port (and service provided) is performed when setting up the Rover Receiver (for example, with TopSURV).

Figure 3-3 illustrates the connections between hardware and software with TopNET-N as a central data processor.

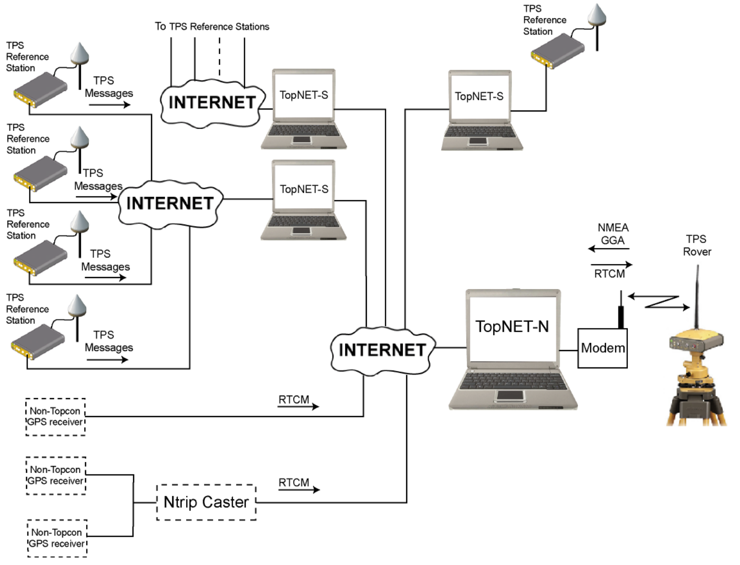


Figure 3-3. TopNET RTK Configuration

The TPS reference stations connect to the TopNET-S server using the Internet, and each TopNET-S server can obtain data from one or more reference stations. The TopNET-N application collects TPS data from TopNET-S servers and RTCM 2.3/3.0 data from Ntrip casters or directly from GPS receivers. These data are processed and transmitted to a Rover using the data communication link.

Getting Started

Main Screen

After startup of the program, the main screen opens. The main screen of the TopNET-N application reflects the current state of the network.

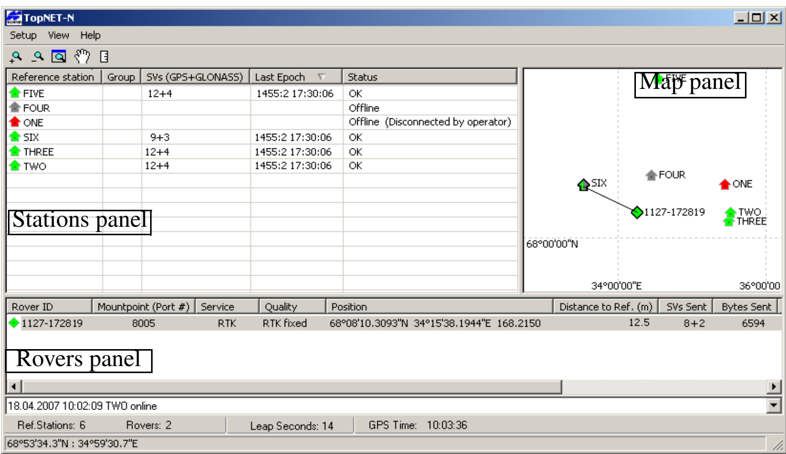


Figure 3-4. Main Screen

When opened for the first time, the screen is empty. To start a network, add at least one reference station (see “Launching Network” on page 3-16 for this procedure). Once added, the reference station is displayed in the Stations panel of the screen. This panel contains the following columns:

- The *Reference station* column lists each station and a status icon that reflects the current status of the corresponding reference station (its interpretation is listed in the last column, *Status*).

 Green icon: the station is available and working properly.

Red icon: the station is available, but it’s coordinates are not valid (the station is offline, position is incorrect, low number of satellites tracked, etc.). If the station was disabled by the operator, the corresponding comment will be displayed.

Dark Grey icon: the station is offline.

Triangle icons stand for non-Topcon reference stations: Ntrip or TCP/IP. Their colors’ meaning is similar to those of the Topcon “house” icons.

- The *SVs* (satellite vehicles) column lists information about the number of satellites received (both GPS and GLONASS).
- The *Last Epoch* column shows the parameters of the last received epoch; the GPS week number, GPS week day (Sunday = 0) and GPS time.
- The *Status* column provides further information on the status of the station.

The right mouse button click on any Reference Station opens the context menu (Figure 3-5):

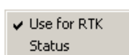


Figure 3-5. Reference Station Context Menu

- *Use for RTK* - if the station operates properly, a checkmark is located near the entry, and the data coming from the reference station is included to the solution. The icon of the reference station is green. To disable the station take off the checkmark, and the icon on the main screen turns red.

- *Status* - shows the **Reference Station Status** screen. It displays the parameters of the reference station:

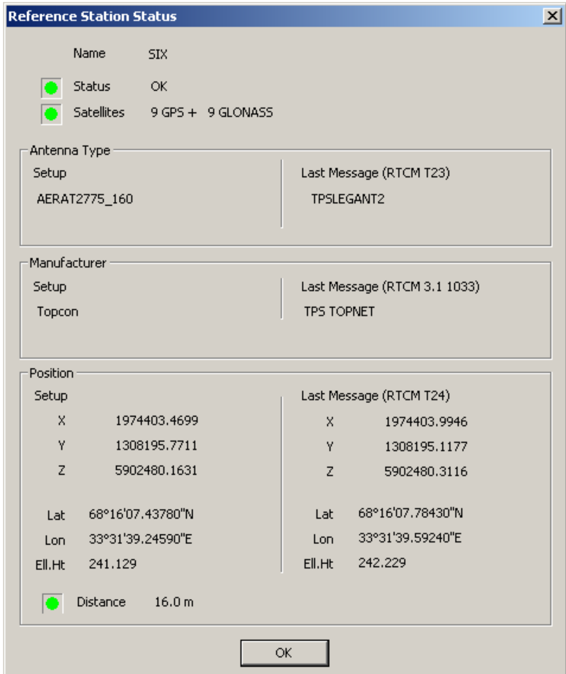


Figure 3-6. Reference Station Status

These are:

- name,
- status of the Reference Station,
- number of received satellites,
- Antenna Type - shows the stored antenna type of the receiver, and that from the last message, if antenna type is transmitted in RTCM.
- Manufacturer - shows the stored Manufacturer's name of the receiver, and that from the last message, if Manufacturer value is transmitted in RTCM.
- Position:
  – the Setup position - the inputted coordinates of the reference station,

– the Last Message position - the coordinates calculated according to the last message received. If there an RTCM 3.0 Message 1005 or 1006 is received, the Last Message field shows the Message Type:

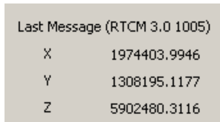


Figure 3-7. Last Message Position with RTCM 3.0 Message

- the Distance between the setup and received positions. Also it shows the icon that displays the connection status:
  – Green color means “OK”.
  – Red - “Offline,” or “Offline (Disconnected by operator)”.

Also this screen shows similar icons for satellites and distance (they turn red if the corresponding conditions are not satisfied).

NOTICE *NOTICE*

The status icon in the main screen turns green only if all the three icons in the Reference Station Status screen are green.

After a rover becomes online, its ID and information appear in the Rovers panel of the Main screen. This panel contains the following columns:

- The *Rover ID* column lists the unique identification number for each rover and a solution status icon (its interpretation is listed in the *Quality* column):

Red icon: the position quality is “Standalone”.

Blue icon: the position quality is “RTK Float”.

Green icon: the position quality is “RTK Fixed”.

Yellow icon: the position quality is “DGPS”.

- The *Mountpoint* (*Port #)* column shows the number of the TCP/IP port the rover is connected to, or an Ntrip Mountpoint name if the rover uses Ntrip protocol for connection.
- The *Service* column displays the type of the service provided to the rover.
- The *Quality* column lists the solution quality i.e. type of rover position as defined in GGA messages coming from the rover.
- The *Position* column shows the geodetic coordinates of the rover.
- The *Distance to Ref.* column displays a distance from a rover to its Reference station.
- The *SVs Sent* column displays the number of satellites in the last epoch sent to the rover. The “0 + 0” value appears if no data is sent.
- The *Bytes Sent* column shows the total number of bytes sent to the rover.
- The *IP Address* column displays the IP Address of the rover.
- The *Ntrip User* column displays the name of the corresponding user if the rover is using Ntrip for connection to TopNET-N.
- The *Duration* column shows the time of the rover connection to TopNET-N.

To sort a list of rovers, press the name of the desired column.

Center the map to the selected rover without changing scale by double-clicking the left mouse button.

To restrict access, a filter can be set on rovers’ IP addresses using the “Allowed IP addresses” tool. For this procedure, see “Preferences” on page 3-47.

The Toolbar contains three menus: **Setup**, **View** and **Help**.

The *Setup* menu contains the following items (Figure 3-8):

- To create or edit a reference station, click **Reference Stations**.
- To set ports and formats, click **Services**.

- To set Ntrip connection parameters, select **Ntrip Access.**

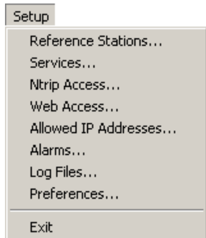


Figure 3-8. Setup Menu

- To set Web Access parameters, select **Web Access...**
- To edit the list of allowed IP addresses, click **Allowed IP Addresses**.
- To set the automatic error notifications by e-mail, click **Alarms**.
- To select location for log files created by the application, click **Log Files**.
- To edit preferences for TopNET-N, click **Preferences**.
- To exit the program, click **Exit**.

The *View* menu allows one to export the reference stations to the Google Earth™ format (see “Export to Google Earth” on page 3-13) and launch Rover Accounting module (see Appendix VII “Rover Accounting”):

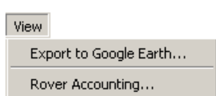


Figure 3-9. View Menu

The **Map** panel shows the relative location of the reference stations and the status of the reference stations. For details see “Using the Map” on page 3-54.

When selecting a rover in the list, the map highlights the icon of the rover with a black border and the Map draws a connector line to the nearest reference station

The *Help* menu (Figure 3-10) provides information about the TopNET-N version and build date, and the Activated Options of the Hardware Key (see “Hardware Key Usage” on page 3-57):

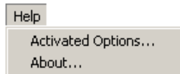


Figure 3-10. Help Menu

The bottom of the main screen displays the Event Log and the Information bar.

The Event Log contains information about connections/reconnections with the receivers (reference stations and rovers), error messages, the selection of a nearest reference station and displays the moment of enabling/disabling a reference station for RTK, by an operator.

The Information Bar displays information on Leap Seconds[1] and GPS time.

Export to Google Earth

Using TopNET-N, you can save coordinates of the reference stations to a Google Earth file for future use and, if the Google Earth application is installed on your PC, immediately obtain the image of your reference stations’ location on the Earth.

To export the coordinates to Google Earth, select **View ▸ Export to Google Earth** menu item.

1. Leap Seconds - the difference between UTC and GPS time.

In the standard Windows **Save As** dialog select a name and location for the file:

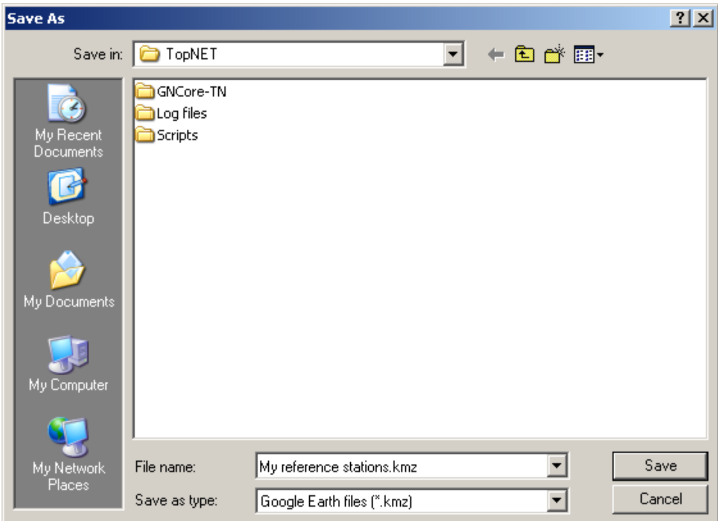


Figure 3-11. Export to Google Earth

The *.kmz file contains the complete information about your network, including icon images.

If **Google Earth** has been installed on your PC, this application is launched after you press the **Save** button. The TopNET-N folder containing the list of your reference stations and rovers, appears in Temporary Places:.

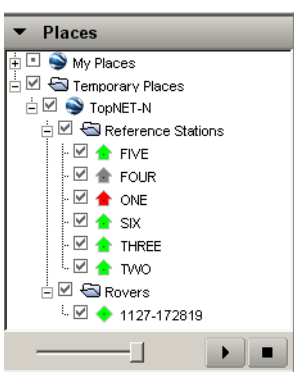


Figure 3-12. Google™ Places

The Google Earth map shows the location of the stations with icons corresponding to those in TopNET-N

Figure 3-13. Google Earth Map

For details on how to work with Google Earth see Google Earth web page.

Launching Network

Click **Setup ▸ Reference Stations** to view the list of available reference stations and parameters: name, coordinates, and antenna type (Figure 3-14 on page 3-16).

- Select a radio button to switch display between *Cartesian (XYZ)* and *Geodetic (BLH)* coordinates. Depending on the selected coordinates, the table displays either X/Y/Z or Latitude/ Longitude/Ellipsoid Height columns.

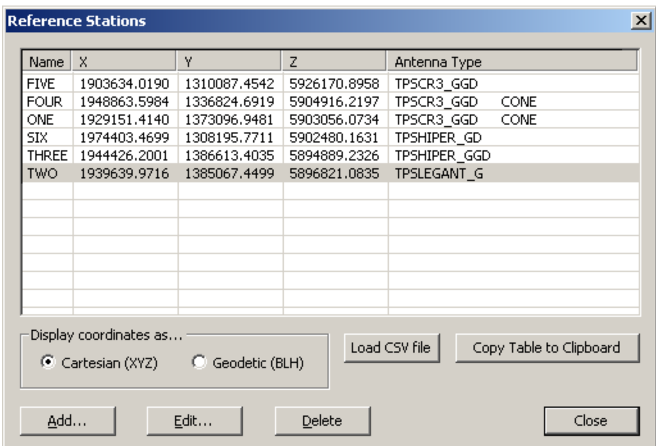


Figure 3-14. Reference Stations

- If you have a CSV file with coordinates of the existing reference stations, you can load it using the **Load CSV file** button. For details and description of the CSV format see "Loading a CSV file" section.
- To copy the list of stations and their coordinates to the clipboard, press the **Copy Table to Clipboard** button.
- To add a reference station, click the **Add** button. Follow the instructions in the Add Reference Station wizard.

NOTICE

A least one of the added reference stations must be a Topcon receiver. Otherwise the RTK service will not work[1].

- To edit a reference station, select the station and click **Edit**. Follow the instructions in the **Edit Reference Station** wizard.
- To delete a reference station, select the station and click **Delete**.
- To return to the Main screen, click **Close**.

Adding/Editing a Reference Station

To add or edit a reference station, click the corresponding button in the **Reference Stations** dialog box. To return to the saved settings, click **Revert to Saved**. To close the Wizard without making any changes, click **Cancel**.

In case of adding a station, select a type of receiver connection: via TopNET-S (only for Topcon receivers), an Ntrip caster, or any receiver streaming data to the Internet:

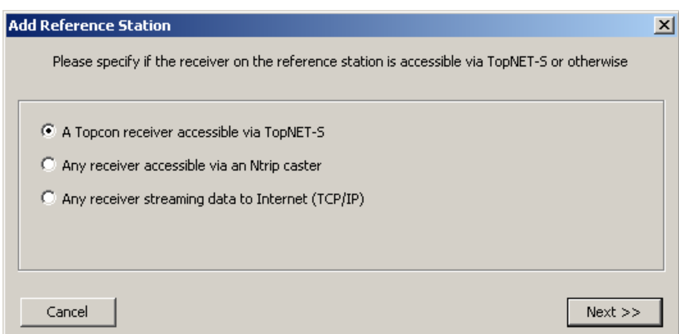


Figure 3-15. Add Reference Station

Click **Next** to select a server. See the following sections for the explanation on how to add three different Reference Stations types.

1. That is due to the fact that time in RTCM messages is transferred modulo one hour, so the application takes absolute time from the Topcon receiver for appropriate RTK functionality.

In case of editing a station, the first Wizard screen corresponds to the selected type of station.

In Wizard, current header of the dialog box always shows the type and display name of the selected station - each starting from the screen where it is selected. For example, *Add Topcon Reference Station FIVE*.

To close the Wizard without making changes, click **Cancel**.

For TopNet-S Reference Stations

The **Add/Edit Reference Station** dialog box contains the selection of the server and receiver for the chosen reference station. When opened in *Add Reference Station* mode, the list of receivers is empty until the **Get Receivers** button is pressed, and the **Revert to Saved** button is displayed as **Revert to Default.**

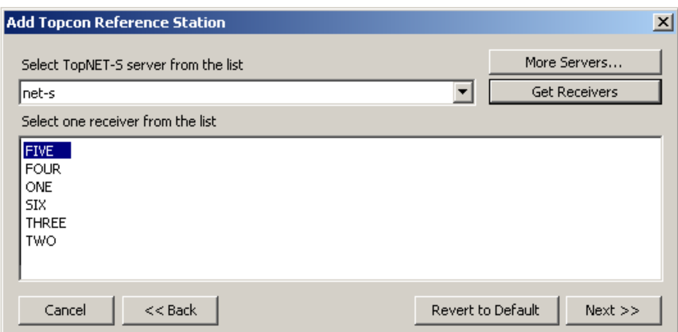


Figure 3-16. Add Topcon Reference Station

When opened in *Edit Reference Station* mode, the list of receivers shows only one selected receiver and the **Get Receivers** button is disabled.

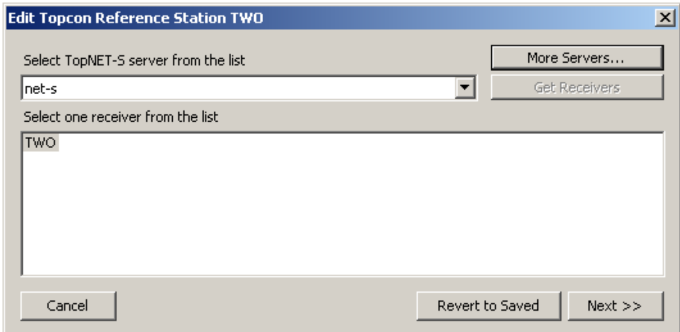


Figure 3-17. Edit Topcon Reference Station

1. In order to set or change server for the station, in the *Select TopNET-S server from the list* list select the host name or IP address of the server. If there is no desired server in the list or no servers at all, click **More Servers** and setup a new server. See “To set up a new server:” on page 3-20 for details.

   In case of adding a receiver:

   – Click **Get Receivers** to display the list of available receivers connected to the selected server.

   – Select the receiver intended to be the reference station.

2. Click **Next** and open a screen where you can change a display name for the station.

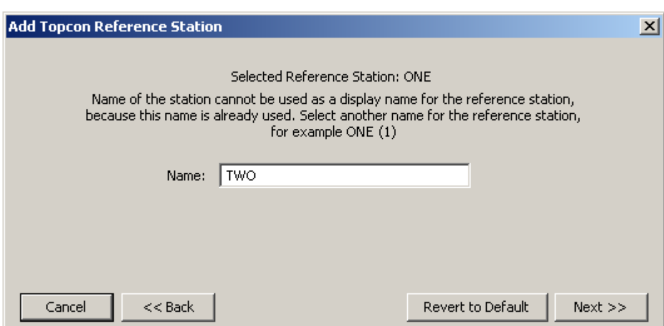


Figure 3-18. Add/Edit Topcon Reference Station - Display Name

If a station with the same display name already exists, you need to select another display name to add the station.

NOTICE

Name of the station should not contain more than 31 symbols.

3. Click **Next** and continue with “Setting the Coordinates of the Reference Station” on page 3-25 or click **Back** to make changes in previous screen.

To set up a new server:

In order to add a server, in **Add Topcon Reference Station** click **More Servers.**

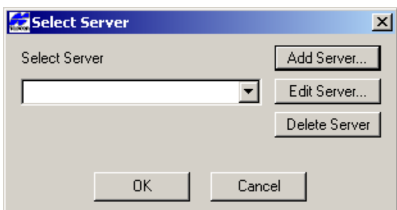


Figure 3-19. Select Server - New

When creating a reference station for the first time, the list of servers is empty.

1. Click **Add Server** to add server properties.
2. Enter the properties of the server: host name or IP address, user name (login), and password.

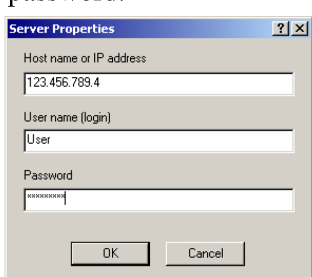


Figure 3-20. Server Properties

3. To set a port for connection, define its number in the address field using colon, i.e. 123.456.456.321:8010. By default, the port number 8000 is used for connection.
4. Click **OK**. The server appears in the **Select Server** list. The latest added or edited server becomes the default one.

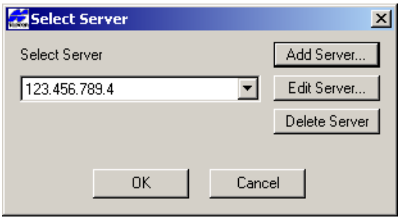


Figure 3-21. Select Server

To edit server's properties, click **Edit Server** and make necessary changes in the **Server Properties** dialog box (Figure 3-20).

To delete the selected server from the list, click **Delete Server**.

For Ntrip Reference Station

The **Add/Edit Ntrip Reference Station** dialog box contains the server's address and port, mountpoint name, and login and password fields.

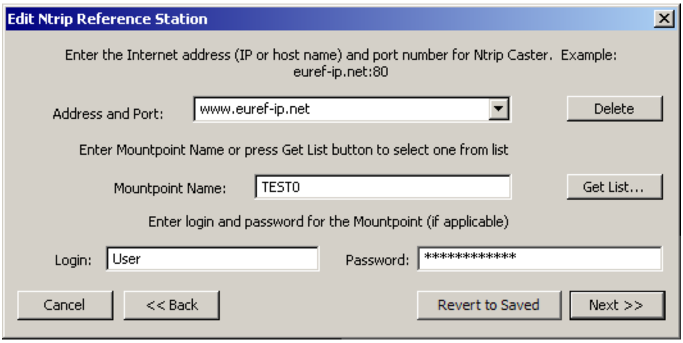


Figure 3-22. Edit Ntrip Reference Station

To add a new Ntrip reference station:

1. Type in the address of an Ntrip Caster (for example, http://www.euref-ip.net), as shown in Figure 3-22, and, if necessary, enter the port number separated by colon.
2. If the *Mountpoint* name is known, enter it in the corresponding field.

   Otherwise, press the **Get List...** button to obtain a list of available casters, networks, and mountpoints from the caster:

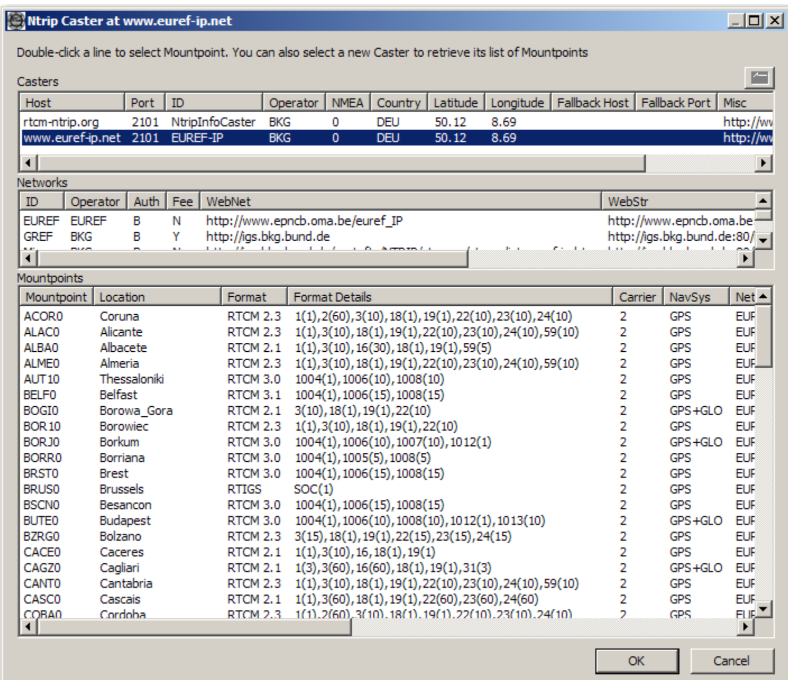


Figure 3-23. Ntrip Caster Example

Select a caster, network and mountpoint (your selection will be highlighted) and press **OK**.

3. Click **Next** in the **Add/Edit Ntrip Reference Station** dialog box.If a new ntrip reference station is created, the following screen is shown:

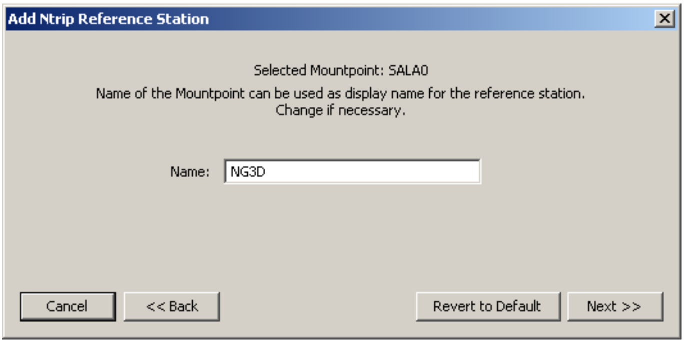


Figure 3-24. Add/Edit Ntrip Reference Station - Display Name

Here you can change a display name for the station, if desired. If a station with the same display name already exists, you need to select another display name to add the station.

If a reference station is being edited, this step is skipped.

4. Click **Next** in the **Add/Edit Ntrip Reference Station** dialog box

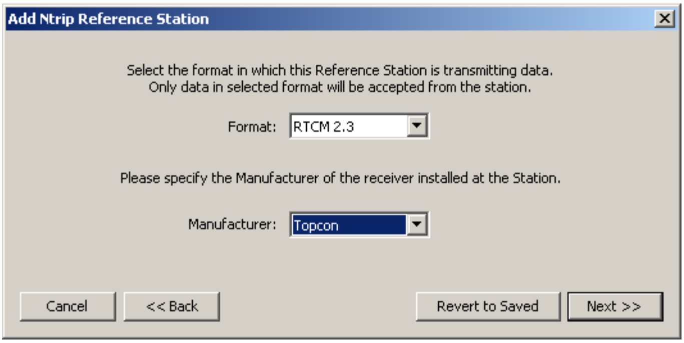


Figure 3-25. Add/Edit Ntrip Reference Station - Data Format

5. Select a format used by your reference station for data transmission. It can be either RTCM 2.3 or RTCM 3.0

   To be used by the TopNET-N application, the RTCM 2.3 message received from an Ntrip Caster or TCP/IP reference station must contain the Types 18 and 19 Messages (carrier phase and pseudorange values). And the RTCM 3.0 message must contain the Type 1003 or 1004 Message (GPS measurements) and Type 1011 or 1012 Message (GLONASS measurements); presence of Type 1005 or 1006 Message (ARP coordinates) and 1007 or 1008 (Antenna Descriptor) is optional.

6. From the drop-down list, select a Manufacturer of the receiver.
7. Click **Next** and continue with "Setting the Coordinates of the Reference Station" on page 3-25.

For Reference Stations Streaming Data via TCP/IP

The **Add/Edit TCP Reference Station** dialog box contains the address/port of the server.

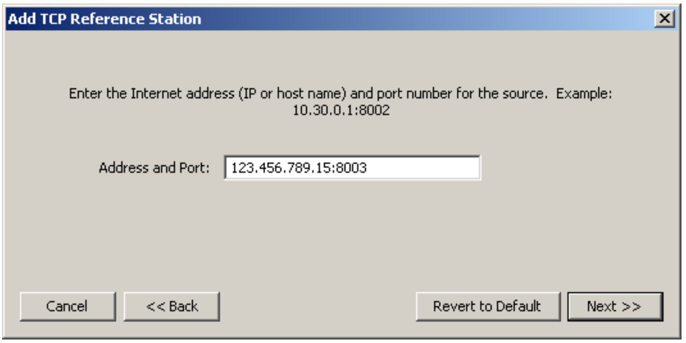


Figure 3-26. Edit TCP Reference Station – Address and Port

1. Click **Next** and enter/change a name for the reference station.

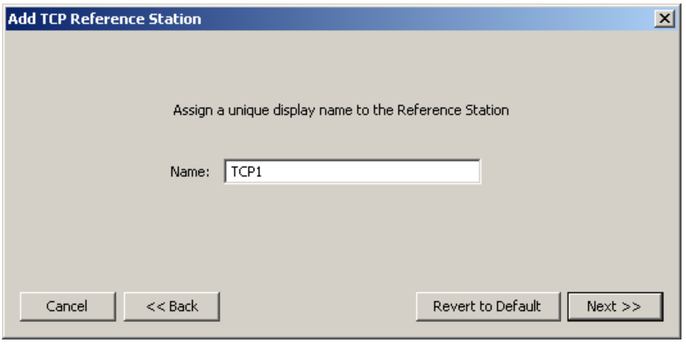


Figure 3-27. Add/Edit TCP Reference Station - Display Name

If a reference station is being edited, this step is skipped.

2. Click **Next** and select a format used by your reference station for data transmission and a receiver's Manufacturer, as described in step 3 on page 3-23.

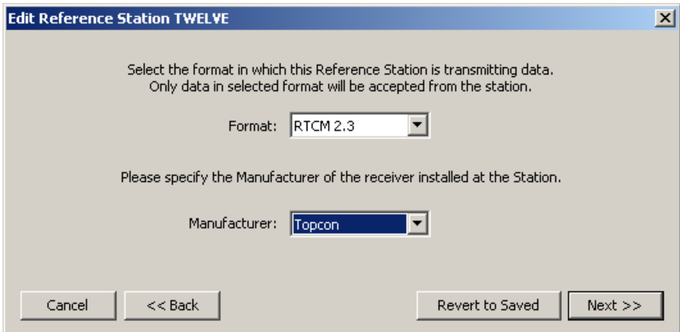


Figure 3-28. Edit TCP Reference Station – Format and Manufacturer

3. Click **Next** and continue with "Setting the Coordinates of the Reference Station" on page 3-25.

Setting the Coordinates of the Reference Station

The **Reference Station Coordinates** screen represents the coordinates of the Reference Station to be added or edited. To undo changes and return to the saved coordinates, click **Revert to Saved**.

NOTICE *NOTICE*

Coordinates of the Reference Stations should be surveyed in with sufficient accuracy and precision. Unreliable coordinates will limit the performance of the network.

1. Select the type of coordinates to be entered: *Cartesian* or *Geodetic*. When editing the coordinates of the existing reference station, enter changes in any form.
   - For *Cartesian*, insert (or change) the X,Y, Z coordinates.

- For *Geodetic*, insert (or change) the Latitude, Longitude and Ellipsoid Height.

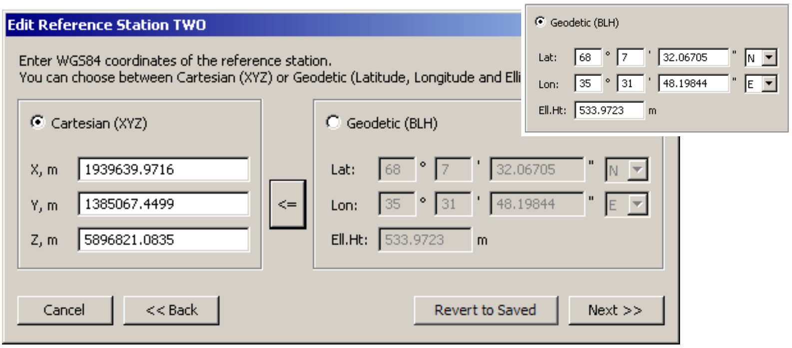


Figure 3-29. Edit Reference Station Coordinates

NOTICE

Note that the coordinates of the Reference Stations should be surveyed in with sufficient accuracy and precision. Unreliable coordinates will limit the performance of the network.

2. Click **Next** and continue with "Setting the Antenna Parameters" on page 3-26.

Setting the Antenna Parameters

After setting up the reference station and its coordinates, specify the antenna used at the reference station.

1. Select **Antenna Type** from the list of NGS[1] standard antenna names and insert **Antenna Serial Number**, if available.

---

1. See, for example, Complete Antenna Calibration Results at http://www.ngs.noaa.gov/ANTCAL.

For each Antenna Type the phase center offsets are available for your reference.

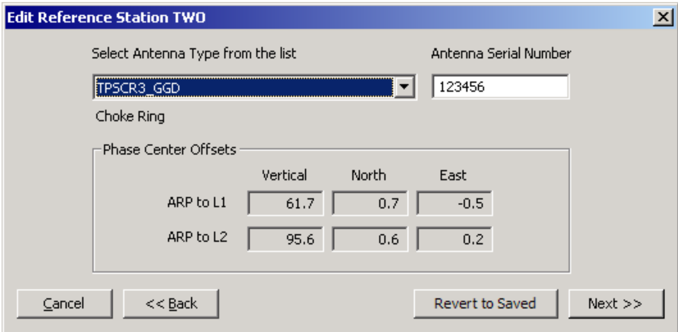


Figure 3-30. Edit Reference Station – Antenna Type

2. Click **Next** and continue with “Setting Group” on page 3-27.

Setting Group

After setting up the Antenna Type, select a group if desired. The group name is shown as a separate column in the reference station list:

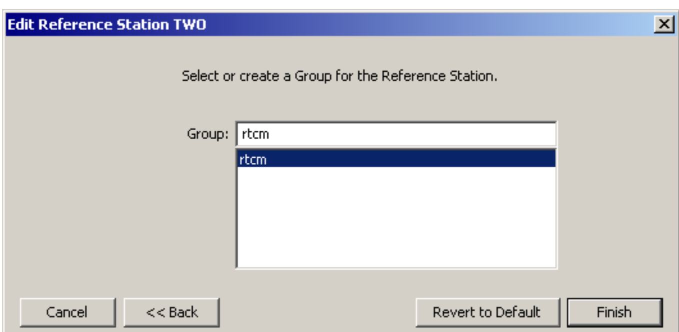


Figure 3-31. Edit Reference Station – Groups

1. Select a group name from the list, or create a new group name, or leave the field empty.
2. Click **Finish** to save the changes and return to the **Reference Stations** screen.

TopNET-N

A newly created reference station appears on the Map and Stations panels and, by default, is activated for RTK service.

Loading a CSV file

CSV file stands for a text file with comma-separated values and *.csv extension. It is used, for example, for applying changes to the parameters of one or several reference stations.

To be correctly interpreted by the application, the CSV file must have a header line which specifies what fields are in the file. The following fields are valid:

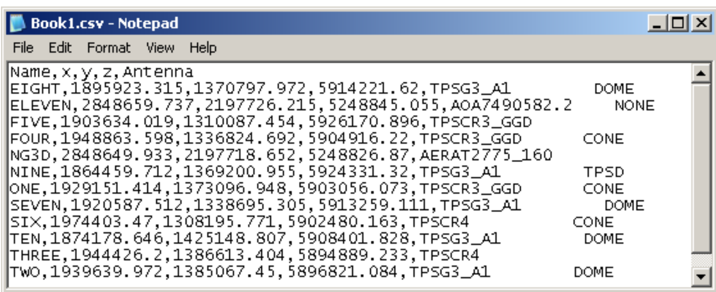
- "Name" - required field;
- "X"
- "Y"
- "Z"
- "LatD"- the name stands for the latitude represented in the decimal degree form. If latitude of the EIGHT station is 68^{o}32'53.9399, then the LatD value is 68.5483166389 degrees.
- "LatDMS0" - the name stands for the latitude represented in the DDDMMSS.ssss form. So, if latitude of the EIGHT station is 68^{o}32'53.9399, then the LatDMS0 value is 683253.9399.
- "LatDMS1" - the name stands for the latitude represented in the DDD.MMSSssss form. So, if latitude of the EIGHT station is 68^{o}32'53.9399, then the LatDMS0 value is 68.32539399.
- "LonD" - the name stands for the longitude represented in the decimal degree form, similar to LatD value.
- "LonDMS0" - the name stands for the longitude represented in the DDDMMSS.ssss form, similar to LatDMS0 value.
- "LonDMS1" - the name stands for the longitude represented in the DDD.MMSSssss form, similar to LatDMS1value.
- "Height"
- "Antenna"

- “AntSerNum”
- “Manufacturer”
- “Group”

NOTICE

All the coordinates present in the file must have one and the same format.
Space is not interpreted as a delimiter.

Once created, the file can be edited in Notepad. An example of CSV file is represented below:



Book1.csv - Notepad

File Edit Format View Help

```
Name,x,y,z,Antenna
EIGHT,1895923.315,1370797.972,5914221.62,TPSG3_A1        DOME
ELEVEN,2848659.737,2197726.215,5248845.055,AOA7490582.2    NONE
FIVE,1903634.019,1310087.454,5926170.896,TPSCR3_GGD
FOUR,1948863.598,1336824.692,5904916.22,TPSCR3_GGD      CONE
NG3D,2848649.933,2197718.652,5248826.87,AERAT2775_160
NINE,1864459.712,1369200.955,5924331.32,TPSG3_A1        TPSD
ONE,1929151.414,1373096.948,5903056.073,TPSCR3_GGD      CONE
SEVEN,1920587.512,1338695.305,5913259.111,TPSG3_A1        DOME
SIX,1974403.47,1308195.771,5902480.163,TPSCR4          CONE
TEN,1874178.646,1425148.807,5908401.828,TPSG3_A1        DOME
THREE,1944426.2,1386613.404,5894889.233,TPSCR4
TWO,1939639.972,1385067.45,5896821.084,TPSG3_A1        DOME
```

Figure 3-32. Example of a CSV file

To load a CSV file, select **Setup ▸ Reference Stations** and press the **Load CSV file** button in the **Reference Stations** dialog box ().

Select a file from your PC and press OK.

If the list of stations in the file corresponds to the list of existing stations in the application, a success message appears.

If the list of stations in the file does not correspond to the list of existing stations in the application, or contents of the file does not agree with the header line fields, an error log is created in the location set by the **Select folders for log files** screen (Figure 3-51 on page 3-52), and displayed.

TopNET-N

Notes:

TopNET-N

Settings

Services

TopNET-N provides RTK correction data to the clients through the internet using the TCP/IP protocol. Each service type is provided through one of the TCP/IP ports.

By default, the services are provided through the following ports (the default formats for each service are italicized):

Table 3-1. Default ports for the provided services

Service	Port	Formats
DGPS	8204	*RTCM 2.3 DGPS*
RTK	8205	CMR CMR+ *RTCM 2.3 RTK* RTCM 3.0

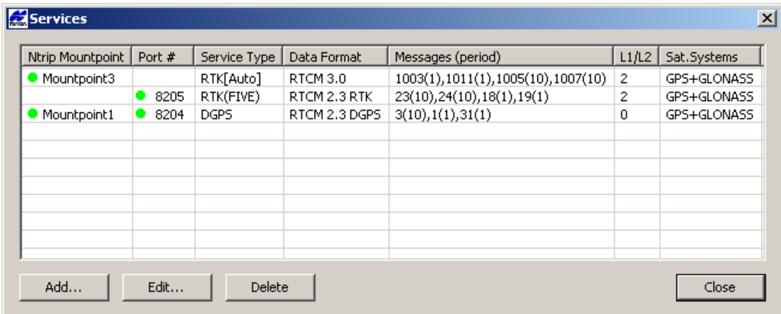
The Message Types for the formats depend upon the settings in the **Service Details** screen (Figure 3-36 on page 3-34).

The TopNET-N application can also operate as an Ntrip caster and broadcast the information using the Ntrip protocol.

To change the default services settings, select **Setup ▸ Services** menu.

Managing Services

1. Click **Setup** ▸ **Services** menu (Figure 3-33).



Ntrip Mountpoint	Port #	Service Type	Data Format	Messages (period)	L1/L2	Sat.Systems
Mountpoint3		RTK[Auto]	RTCM 3.0	1003(1),1011(1),1005(10),1007(10)	2	GPS+GLONASS
	8205	RTK(FIVE)	RTCM 2.3 RTK	23(10),24(10),18(1),19(1)	2	GPS+GLONASS
Mountpoint1	8204	DGPS	RTCM 2.3 DGPS	3(10),1(1),31(1)	0	GPS+GLONASS

Figure 3-33. Services

The Services screen displays a table of available services and their features:

- Ntrip Mountpoint (if available);
- Port # (if available);
- Service Type - shows the Service Type with a note nearby. An [Auto] note appears if no specific reference station is assigned to the service, otherwise a name of the station in parentheses is shown, for example, *RTK(FIVE)*;
- Data Format;
- Messages and their periods (in seconds);
- L1/L2 usage parameter:
  "0" corresponds to a *Code Only* case,
  "1" - to *L1 Code&Phase*,
  "2" - to *L1&L2 Code&Phase;*
- Satellite System.

2. To sort the table by any column, press its header.
3. To delete any service highlight it or select several of them holding the Shift key, and press the **Delete** button.

4. To create a new Service press **Add**, to edit an existing service, press **Edit** or double-click the entry in the table.

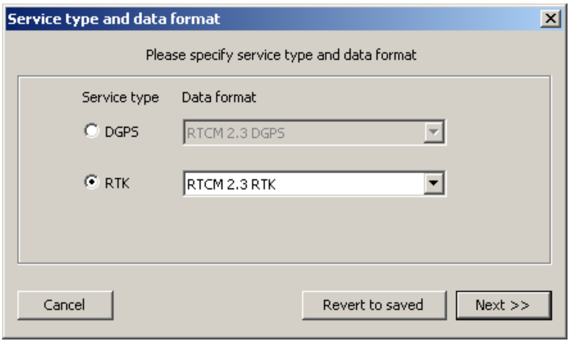


Figure 3-34. Service Type and Data Format

5. On the **Service type and data format** screen select the desired **Service type** and an appropriate **Data format** from the list of available formats. Press **Next**.
6. If an RTK Service Type selected on the previous step, a **Select Reference Station** screen is displayed. Here you can set if the chosen service will be provided through a reference station closest to a rover, or only through the selected one.

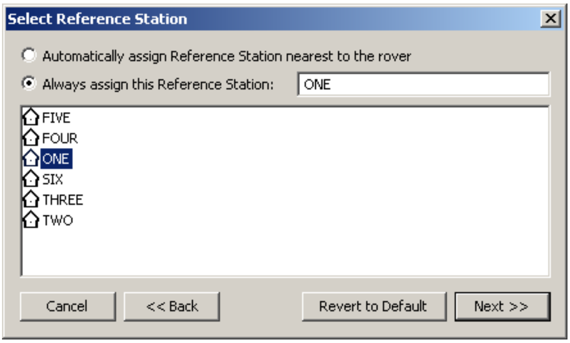


Figure 3-35. Select Reference Station

NOTICE

If a specific reference station is assigned to a Service, and this station is not available (is offline) at the time when rover connects to this service, no data will be sent to the rover.

Press **Next**.

7. Select the desired satellite systems in the **Service Details** screen (Figure 3-36).

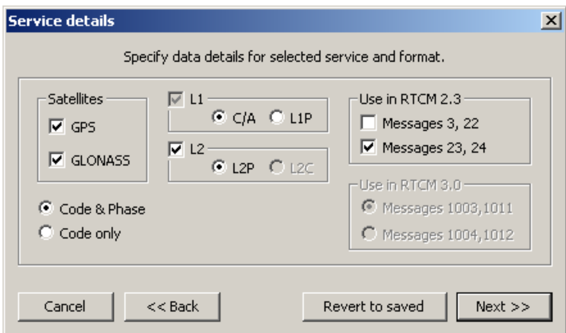


Figure 3-36. Service details

Check the code and carrier phase settings:

- Normally, both Code & Phase are used, but for special applications a code-only data stream may be needed.
- If necessary, change the code type for the L1 frequency.
- By default, the L2 frequency is enabled (only L2P code type is available for now). Remove the checkmark if desired.

Depending upon the selected service and data format, you can select the Message Types to be transmitted:

- The data for DGPS sent in RTCM 2.3 format contain the following Message Types:
  – Type 1 Message: GPS corrections (if GPS enabled);
  – Type 3 Message: Antenna L1 phase center coordinate accurate to the nearest centimeter;

  – Type 31 Message: GLONASS code corrections (if GLONASS enabled).
- The data sent for RTK in RTCM 2.3 format contains one of the following combination of Message Types:
  – 18 (phase), 19 (pseudorange), 23 (antenna type) and 24 (antenna ARP position);
  – 18, 19, 3 (L1 phase center, same as for DGPS) and 22 (correction to position sent in Type 3 Message);
  – 18, 19, 3, 22, 23, 24.
- The data sent in RTCM 3.0 format contains a combination of the following Message Types:
  – Type 1001 Message: GPS L1 code and phase;
  – Type 1003 Message: GPS L1&L2 code and phase;
  – Type 1004 Message: GPS L1&L2 code, phase and L1 pseudorange ambiguity;
  – Type 1005 Message: antenna ARP position;
  – Type 1007 Message: antenna type;
  – Type 1009 Message: GLONASS L1 code and phase;
  – Type 1011 Message: GLONASS L1&L2 code and phase;
  – Type 1012 Message: GLONASS L1&L2 code, phase and L1 pseudorange ambiguity.

NOTICE *NOTICE*

The difference between 1003 and 1004 and between 1011 and 1012 is in “Integer L1 Pseudorange Modulus Ambiguity” which is not required for RTK operations but may be useful if a client wants to restore full code and phase measurements from RTCM 3.0 data.

If the L2 carrier frequency is disabled, the transferred Message Types are predefined and cannot be changed. If L2 is activated, the additional selection for Message Types is available. (For details see Table 3-2).

By enabling and disabling the corresponding fields, you can select between the following Message Types combinations:

Table 3-2. RTCM Message Types

Selected options	RTCM 3.0 Message Types
GPS is enabled, *GLONASS* is disabled, *L2* is disabled *Use for RTCM 3.0* box is disabled	1005, 1007, 1001
GPS is disabled, *GLONASS* is enabled, *L2* is disabled	1005, 1007, 1009
GPS is enabled, *GLONASS* is enabled, *L2* is disabled	1005, 1007, 1001, 1009
GPS is enabled, *GLONASS* is disabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1003
GPS is disabled, *GLONASS* is enabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1011
GPS is enabled, *GLONASS* is enabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1003, 1011
GPS is enabled, *GLONASS* is disabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1004
GPS is disabled, *GLONASS* is enabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1012
GPS is enabled, *GLONASS* is enabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1004, 1012

To discard all the changes, press the **Revert to Saved** button.

8. Press **Next**. In the **TCP/IP Port and Mountpoint** screen place the check box in the corresponding field to activate the desired access method.

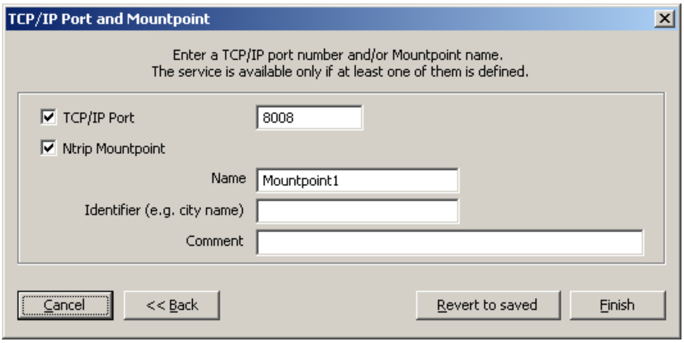


Figure 3-37. TCP/IP Port and Mountpoint

9. Set the number for TCP/IP Port and the parameters of the mountpoint:
   - after you set the unique name of the mountpoint in the **Name** field, it will be available on the network;
   - you can also set a short description of the mountpoint in the **Identifier** field;
   - fill the **Comment** string - extended description of the mountpoint, if desired.

All the other parameters of the connection set through the Ntrip server are available on the **Services** screen. See “Setting the Ntrip Access parameters” on page 3-38.

10. To discard all the changes, press the **Revert to Saved** button.
11. Click **Finish** to save settings.
    The added service will be displayed in the **Services** screen (Figure 3-33 on page 3-32). If the service has an activated TCP/IP port and/or mountpoint, the green dot will appear in the corresponding column near the name of the mountpoint and/or number of the TCP/IP port. If the parameters of the TCP/IP port or mountpoint are inserted but the access method is not activated by the checkmark, the dot will be red.

CAUTION

Notify all TopNET-N field users about TCP/IP changes. They must update communication settings in order to ensure continued network access.

Setting the Ntrip Access parameters

1. To activate the function of broadcasting via Ntrip, click **Setup ▸ Ntrip Access...** menu, check the *Accept Ntrip Request* checkbox in the **Ntrip Settings** screen (Figure 3-38) and set a *TCP Port Number.*

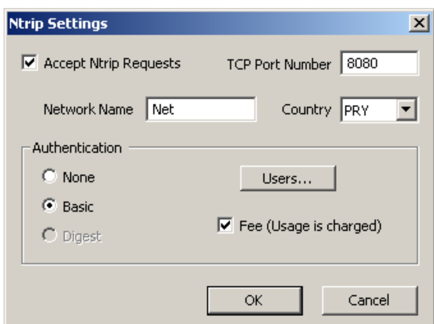


Figure 3-38. Ntrip Parameters

2. Type in the name of the **Network** and select the **Country** from the drop-down list.
3. In the **Authentication** field select the mode of authentication:
   - **None** - no password required;
   - **Basic** - login and password required. In this mode one can select the users that have access to the caster (Figure 3-39 on page 3-39).
   - **Digest** mode is not implemented yet;
4. Place the **Fe**e checkmark if access to the caster is chargeable. This checkbox toggles “Fee” field of STR record in Ntrip Sourcetable between “N” and “Y”. This field is used only to deliver this information to a rover operator.

Managing users of Ntrip caster

To control user access to the Ntrip caster, press the **Users...** button in the **Ntrip Settings** screen (Figure 3-38 on page 3-38).

The **Ntrip Users** screen displays the list of the users that are allowed to access the services grouped according to the access rights (Figure 3-39).

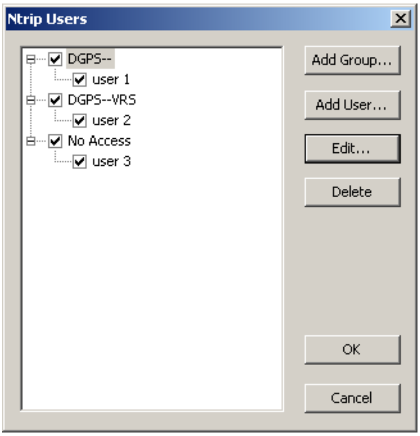


Figure 3-39. Ntrip Users

If you open the Ntrip Users screen for the first time and it is not empty, it means that you had a previous version of TopNET-N being installed, and the application has detected existing users when the first run.

To convert old user entries to the new format, the TopNET-N application grouped them according to their access rights for the mountpoints' service types. The groups created by default, are named after the services permitted to the users in the group, and there can be up to four of them:

1. No access
2. DGPS-
3. DGPS-RTK
4. -RTK

All the users in one group have same access rights, so if you want to change the rights for one particular user, it is necessary to create a group with these rights and assign it to this user.

To create a new group:

1. Press **Add Group...** button in the **Ntrip Users** screen (Figure 3-39 on page 3-39).

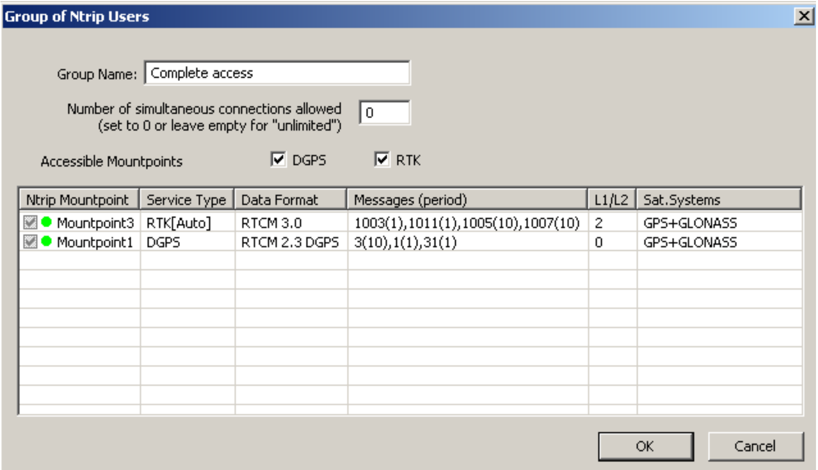


Figure 3-40. Group of Ntrip Users

2. In the **Group of Ntrip Users** screen (Figure 3-40) set the *Group Name* and the *Number of simultaneous connections* allowed for each member of this group. If no restrictions exist for the number of connections, set the corresponding field to zero or leave it blank.

NOTICE

All the users in one Group have the same allowed number of connections. If a set of users with different assigned numbers appeared to get into one group (during conversion from the older version of TopNET-N), this information will be lost.

For this Group, you can check a combination of “master” checkboxes labeled DGPS/RTK to enable/disable access to all existing and future mountpoints of corresponding Service Type. The list of mountpoints shows the mountpoints’ parameters and a color icon that displays if this mountpoint is available for the users assigned to the Group. Three sequentially toggled states of a checkbox at the line start are intended to manage a mountpoint’s individual availability:

 The mountpoint is always available.

 The mountpoint is always not available.

 The mountpoint’s availability corresponds to the state of the “master” checkbox at the top of the table. For example, a mountpoint operating with RTK service will be available only if the “master” RTK checkbox is enabled.

3. Press OK. A new Group appears in the **Ntrip Users** screen.

To add a user to Users list:

1. In the **Ntrip Users** screen (Figure 3-39) press the **Add** button.

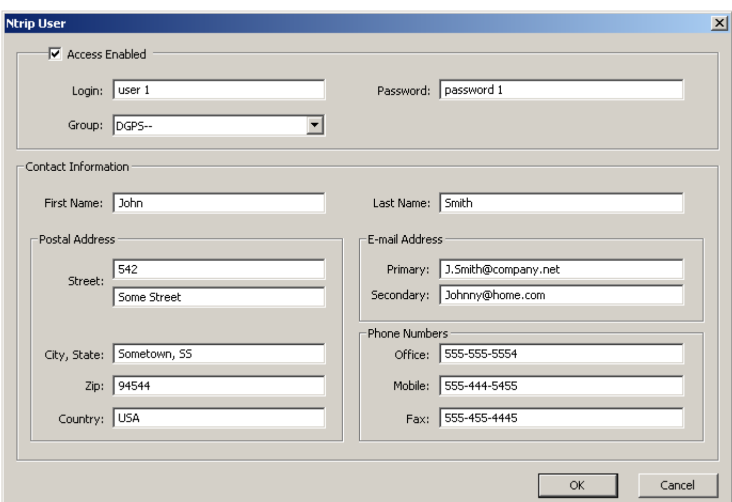


Figure 3-41. Ntrip User

2. Insert Login and Password for the user.
3. Select a group of access rights.
4. Input personal information of the user.
5. Press **OK**. The user will appear in the list (Figure 3-39). A checkbox near user's name allows the administrator to deny access to this user. This checkmark is duplicated on the **Ntrip User** screen as an *Access Enabled* checkmark.
   To deny access to a whole group of users, remove the corresponding checkmark near the *Group Name* in the **Ntrip Users** screen (Figure 3-39).
6. Press **OK** in the **Ntrip Users** screen (Figure 3-39 on page 3-39) to save the changes.

To edit User's Properties:

1. In the **Ntrip Users** screen (Figure 3-39) press the **Edit** button.
2. In the **Ntrip User** screen (Figure 3-41) insert all the necessary changes.
3. Press **OK**. The changes will reflect in the list (Figure 3-39).
4. Press **OK** in the **Ntrip Users** screen (Figure 3-39 on page 3-39) to save the changes.

To delete a user from the list:

1. In the **Ntrip Users** screen (Figure 3-39) select the user to be deleted.
2. Press the **Delete** button. The user will disappear from the list.
3. Press **OK** to save the changes, or **Cancel** to discard.

Setting Web connection parameters

If a local machine running a TopNET-N application has an Apache Tomcat server and TopNET web-application installed[1], it is possible to connect to the TopNET-N with the help of any web-browser and observe the state of the network.

1. How to install an Apache Tomcat server and setup web access for users, is described in Appendix VIII "TopNET Web Interface Setup" .

The option of *Web Access* states the possibility of anonymous access through the web and/or a list of users authenticated for connection.

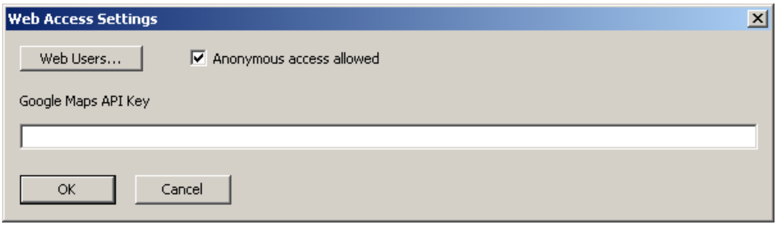


Figure 3-42. Web Access Settings

To allow Anonymous access, select **Setup ▸ Web Access...** menu and place a checkmark in the corresponding field of the **Web Access Settings** screen. The anonymous user will be able to observe the state of the network, including reference stations and ordinary rovers, excluding Ntrip rovers.

To set a list of authorized users, press the **Web Users** button.

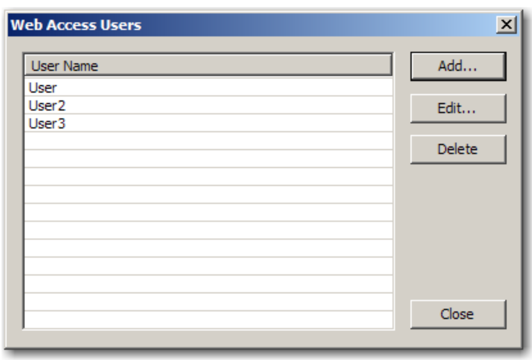


Figure 3-43. Web Access Users

The **Web Access Users** screen shows the list of users allowed to observe the TopNET-N application main screen through the internet. Web interface of TopNET-N provides users with a view on reference stations, ordinary users and some of Ntrip users. The list of Ntrip users available for observation can be set through the **Web Access User Data** screen for each internet user individually upon creation, or later.

To add an internet user for Web Access:

1. Press the **Add** button.

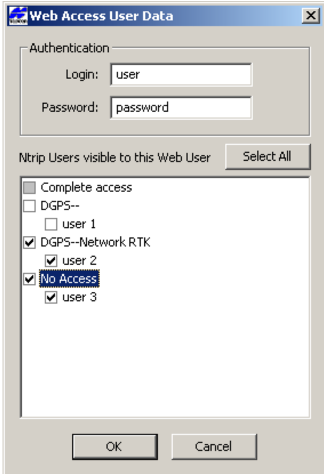


Figure 3-44. Web Access User Data

2. Type in *User name* and *Password*.
3. Place check marks near those Ntrip users whose rovers are visible to the internet user (the list of Ntrip users is similar to that created in “Managing users of Ntrip caster” on page 3-39). To check all entries, press the **Select All** button. If all the entries are selected, the button changes its name to **Unselect All**. Press it to clear all check marks.

TIP

If you need to remove all checkmarks, but not all entries are selected and the button looks like Select All, press it twice.

4. Press **OK**.

To edit parameters of an existing user, select it in the **Web Access Users** screen list and press **Edit**. Make all the necessary changes in the **Web Access User Data** screen.

The *Google Map API* field stores the Google Map API key that is necessary for the web server operation. For the detailed description of web server installation, see Appendix VIII.

For a user to observe the state of the network, provide him or her with the address of the server. Usually, it looks like: http://<server name>:<port number>/TopNETweb, i.e. `http://net-s:1111/TopNETweb`.

The web page looks as follows:

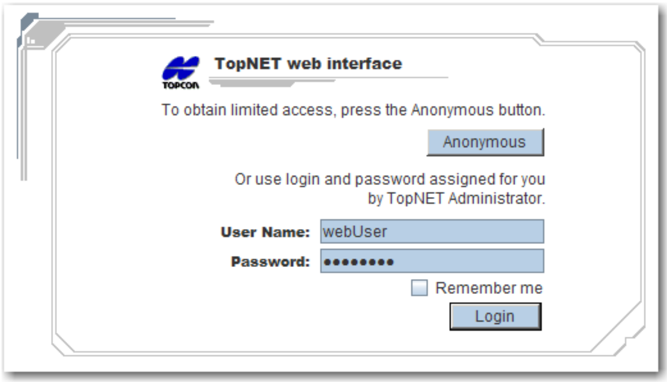


Figure 3-45. Web Interface Welcome Screen

To start working with Topcon web interface one should either press the **Anonymous** button and, if the anonymous access is allowed, get to page, or use login and password assigned by system administrator and press **Login**. These are the login and password created with the help of the dialog box shown on Figure 3-44 on page 3-44.

To make system remember your Login and Password, check the corresponding check box.

All the controls are similar to those on the main screen of the TopNET-N application.

Allowed IP Addresses

To obtain access to the services provided by TopNET-N, the IP address of the rover should be inserted in the list of allowed IP addresses.

1. Click **Setup ▸ Allowed IP Addresses**.

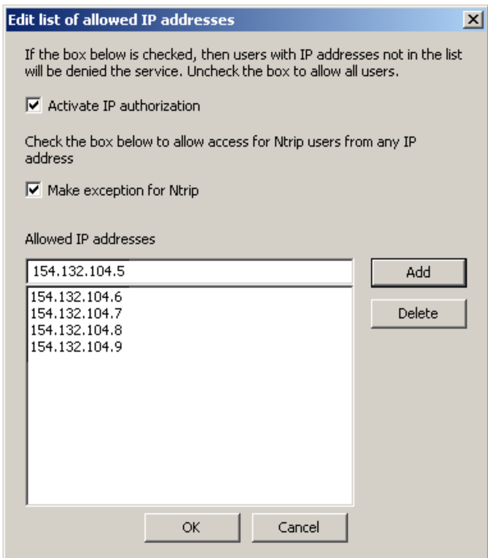


Figure 3-46. Edit List of Allowed IP Addresses

2. To activate the IP authorization, enable the **Activate IP Authorization** check box. In this case the services are provided to those users only whose IP Addresses are listed in the dialog box. If the check box is empty, no restrictions are applied to the users.
3. If necessary, place check mark in the **Make exception for Ntrip** check box. In this case no restrictions are applied to the users of Ntrip caster.
4. Type the IP address in the upper *Allowed IP addresses* field and click **Add**. The IP address appears in the lower field.
5. Repeat step 3 until all addresses have been entered.
6. Click **OK** to save the settings.

7. To remove the address from the list, select it (the chosen address will appear in the upper field) and click **Delete**.

Preferences

To set the preferred settings for the restrictions on reference stations usage, for time periods and timeouts, and for troubleshooting log files, use the **Preferences** screen (Figure 3-47).

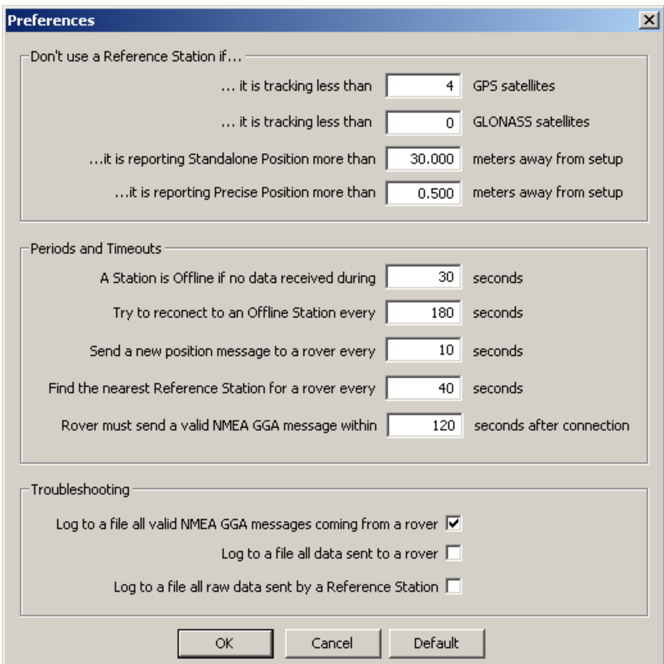


Figure 3-47. Preferences

1. Click **Setup ▸ Preferences**.
2. Insert the conditions on the reference stations usage:
   - the minimum number of GPS satellites being tracked;
   - the minimum number of GLONASS satellites being tracked;
   - the maximum distance of the Standalone Position from the user defined position of the reference station. To switch off

this validity check, enter a large value, e.g. 100 000 m.
The Standalone Position can be transmitted by:

– a Topcon reference station as a [PO] message,

– any receiver connected via Ntrip caster as an RTCM 2.3 or RTCM 3.0 Message,

– by a TCP/IP receiver as an RTCM 2.3 or RTCM 3.0 Message;

- the maximum distance of the Precise Position from the user defined position of the reference station.
  The Precise Position is transferred sometimes by Ntrip and TCP/IP reference stations in the RTCM 2.3 Type 24 Message, or in the RTCM 3.0 Type 1005 or 1006 Message (X, Y, Z - the precise coordinates of the ARP). If the obtained RTCM message does not contain mentioned Message Types, this field is not taken into consideration.

3. Insert the values for periods and timeouts:
   - the maximum time period of not receiving data from the reference station for considering the station to be Offline;
   - the reconnection period for an Offline station;
   - the period of sending new reference station position messages to the rover;
   - the period of searching for the nearest reference station for a rover;
   - the maximum time period of waiting for a valid NMEA GGA message from a Rover that has just connected.
4. Check the check boxes for troubleshooting:
   - to log all valid NMEA GGA messages received from a rover;
   - to log all data sent to a rover;
   - to log all data sent by a reference station.
5. Click **Default** to return to default values.
6. Click **OK** to save the changes.

Alarms

An alarm is used for notification about some event that may request reaction from the customer. The notification is sent by e-mail.

Click **Setup** ▸ **Alarms** to display the **Alarm Mail Setup** dialog box:

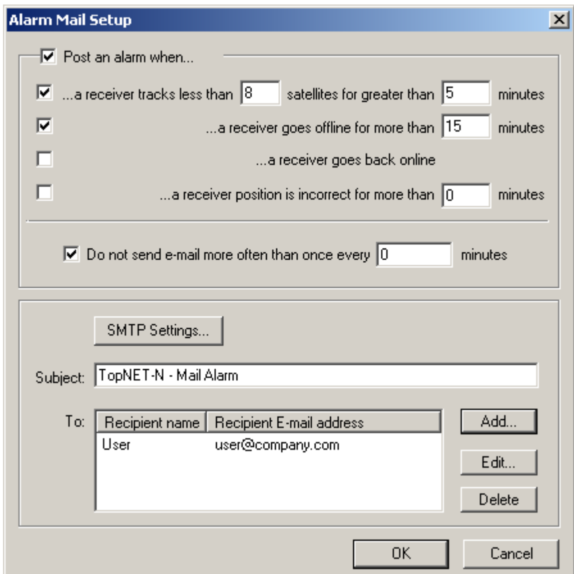


Figure 3-48. Alarm Mail Setup

The **Alarm Mail Setup** screen consists of two parts.

The upper part of the screen (Figure 3-48) helps to select the contents of the alarm mail. To make application send notification e-mails, select the **Post an alarm when...** check box.

This tool can be used for notification about:

- lack of the satellites being tracked, for a considerable period of time. Place a checkmark in the corresponding field and set the minimum number of satellites suitable for a solution, and a maximum admissible tracking period of a less number of satellites.

- a receiver is offline for an extended period of time. Place a checkmark in the corresponding field and set the idle period for a receiver being offline.
- a receiver goes back online. If the previous check box is selected, this field becomes active. It allows you to get notifications about receivers coming back from the offline state.
- a receiver position is incorrect for considerable period of time. Place a checkmark in the corresponding field and set maximum value for the period. The position is considered to be incorrect if the distance between the Precise Position of the reference station and its position defined by the user, exceeds the value set in the **Preferences** screen (Figure 3-47 on page 3-47).

To make the application generate error reports once at a time, i.e. 45 minutes, place a checkmark in the **Do not send e-mail more often than once every ... minutes** field and set the desired period. If there is no checkmark, the notification alarms are sent as soon as event occurs.

The second part of the **Alarm Mail Setup** (Figure 3-48) screen includes common parameters of the sent e-mails:

- an **SMTP Settings** button - opens a **SMTP Settings** screen, where the parameters of the SMTP server and sender's details can be configured:

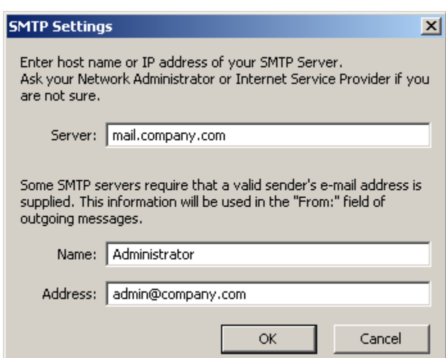


Figure 3-49. SMTP Settings

This module allows a user to send alarm messages, even if no mail client is installed on the PC.
Press **OK** to save the changes and return to the **Alarm Mail Setup** screen.

- *Subject* - the automatic subject of all the e-mails being sent. You can change it, if desired;
- *To* - the mailing list. You can add an address to the list with the help of the **Add** button, or the **Address Book** button. In the latter case, an Address Book of your mailer is shown and you can select address from it. Using the **Add** button, you need to type in user name and address manually, as shown in Figure 3-50.

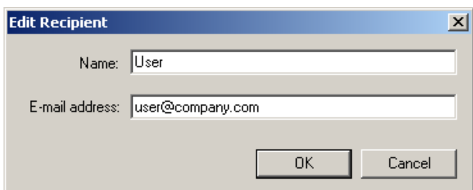


Figure 3-50. Edit Recipient

To make changes in the address entry, select an existing address and press **Edit** button (Figure 3-48) and call the **Edit Recipient** screen (Figure 3-50) again. You can change only one address at a time.

To delete one or several addresses, select the address (if you need to delete several addresses, select them using Shift), and press **Delete** (Figure 3-48). Confirm your decision.

All the generated error reports are displayed in Event log of the main screen and saved in the Log file.

Locating Log Files

While operation, the TopNET-N application produces a set of log files containing information about performed actions, received responses and occurring situations. A new set of log files is created every midnight. To set a specific location for the log files being created, click **Setup ▸ Log Files** menu.

In the **Select folders for log files** screen, set the default location for the log files. Press  to select a folder on your PC (Figure 3-51).

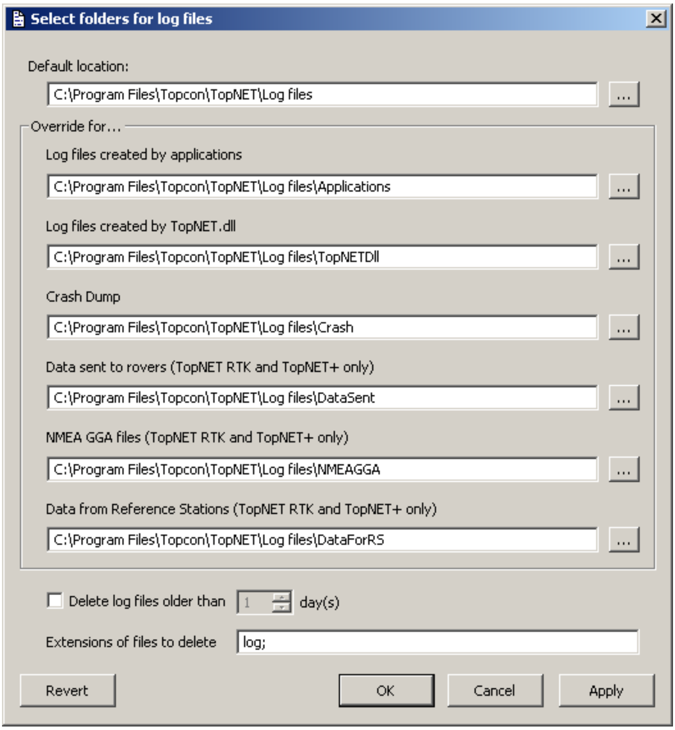


Figure 3-51. Select Folders for Log Files

To select specific folders for different types of log files, enter the overriding locations for the desired log file type:

- Log files created by applications;
- Log files created by TopNET.dll;

- Crash Dump;
- Data sent to rovers;
- NMEA GGA files;
- Data from Reference Stations.

Any field of the **Select folders for log files** screen may display the path to existing folder, to non-existing folder that will be created at the moment of the first corresponding log file being written.

In order not to store large amount of log files, check the **Delete log files older than** field and select the number of days to store the files.

Specify the extensions of the files to be deleted by typing them in the **Extensions of files to delete** field, divided by semicolons.

To save the data, press **Apply**.

To save the data and exit, press **OK**. As soon as a new portion of data will be stored to a log file, the new settings are used.

To revert to the last saved data, press **Revert**.

To exit without saving data, press **Cancel**.

Using the Map

The Map panel displays the network Map and shows the relative position of the reference stations and rovers. A change in icon color indicates a change in status.

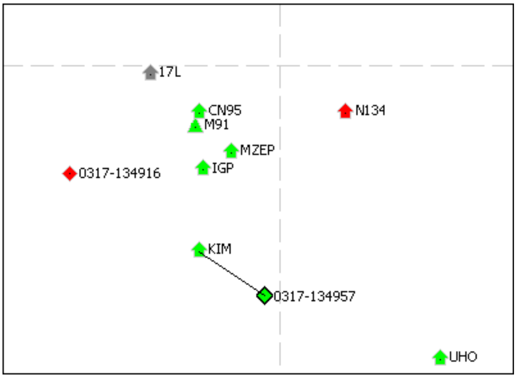


Figure 3-52. Map

When selecting a rover in the list of rovers, the map icon of the selected rover is highlighted with a black border. The map also shows the service provided to the selected rover.

- For the RTK and DGPS services, the map draws a connector line to the nearest reference station and places a black border around the icon.

The map icon of a reference station is highlighted with a black border, if:

- this reference station is selected in the list of stations,
- this station is the nearest reference station for the highlighted rover.

Table 3-3 lists the conventions used in the map.

Table 3-3. Map View Conventions

Icon	Description
	A connected reference station displays as a green icon with a point inside and a name to the right.

Table 3-3. Map View Conventions (Continued)

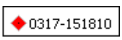
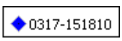
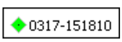
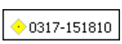
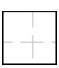
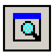
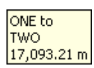
Icon	Description
	An offline reference station (e.g., disconnected by the operator or out of order) is displayed as a black icon with a name to the right.
	A reference station disabled by the operator, or a station with a bad or corrupted signal is displayed as a red icon with a point inside and a name to the right.
	A rover with a “Standalone” position quality is displayed as a red square with a point inside. The rover selected in the list of rovers has a black border around the square.
	A rover with an “RTK Float” position quality is displayed as a blue square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	A rover with an “RTK Fixed” position quality is displayed as a green square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	A rover with a “DGPS” position quality is displayed as a yellow square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	Triangle icons stand for non-Topcon reference stations: Ntrip or TCP/IP. Their colors’ meaning is similar to those of the Topcon “house” icons.
	The numerical coordinate values are located near each parallel and meridian on the left and bottom side of the map to form a coordinate grid.

Table 3-4 describes the toolbar buttons used for controlling the Map view.

Table 3-4. Map View Toolbar Buttons

Button	Description
	Zoom In Click the button and click the map to enlarge the view Hold the left mouse button and click-and-drag a rectangle to enlarge a selected part of the screen up to the whole screen. Also you can use a mouse wheel for this purpose.

Table 3-4. Map View Toolbar Buttons

Button	Description
	Zoom Out Click this button and click the map to decrease the view. Hold the left mouse button and click-and-drag a rectangle to reduce the whole screen to the size of the selected rectangle. Also you can use a mouse wheel for this purpose.
	Zoom All Press the button to display all the objects.
	Pan Press the button to activate the Pan control that allows one to move the observed area using left mouse button being hold.
	Distance Click the button to activate the tool. A Ruler appears near the cursor. Click at the start point and move the cursor in any direction. The floating information box shows the coordinates of the start point and current point and distance between them. If one of the points is located near a reference station or rover, instead of the coordinates, the information box shows the name of this station (rover). For example:  To switch off the connector, press Esc. To switch off the tool, click the button one more time.

Hardware Key Usage

Any computer running TopNET requires a Hardware Protection Key. A Hardware Protection Key enabled for the program must be inserted into the appropriate USB port to run the TopNET RTK or TopNET CORS modules on the computer.

To view the Hardware Protection Key options, select the **Help ▸ Activated Options** menu.

The **Authorization Options** screen shows the options enabled for the selected Hardware Protection Key:

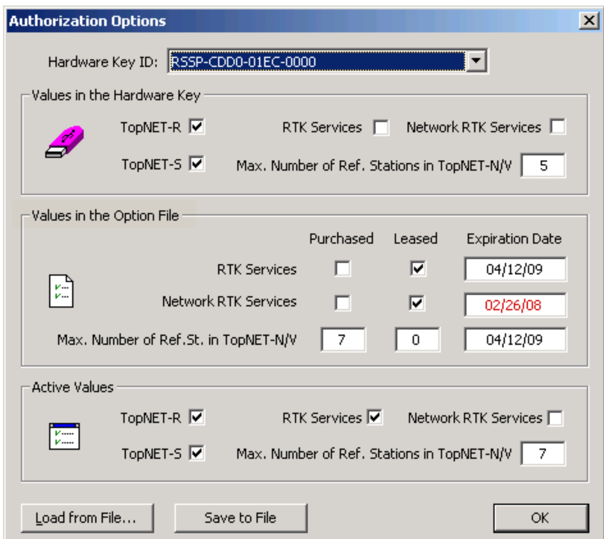


Figure 3-53. Authorization Options

TIP

Authorization Options is an information screen.
All the services and options can be changed only with the help of the Option File (see below).

- *Hardware Key ID* - the unique number of a Hardware Protection Key applied to the PC.

- *Values in the Hardware Key* - services and maximum number of reference stations permitted by the Hardware Key, are displayed;
- *Values in the Option File* - services and maximum number of reference stations permitted by the Option File, are displayed. Each option can be *Purchased* and/or *Leased*. The check mark designates the status of the option. The Expiration Date field shows the Lease expiration date for each option. If there is no Option file on the PC, the *Values in the Option File* field is empty.
- *Active Values* - the current values of the options. These include the maximum of the values displayed in the fields above, including Leased options that have not expired. (If there is no Option File on the PC, the Active Values field duplicates the Values in the Hardware Key field.)
  So, in the example shown above, the following values are set (assuming today is March, 1, 2008):
  - – TopNET-R option is available, as defined by the Hardware Key;
  - – TopNET-S option is available, as defined by the Hardware Key;
  - – RTK Services are available, as defined by the Option File leased option;
  - – Network RTK Services are NOT available, because the lease of corresponding option has expired;
  - – Max. Number of Ref.Stations in TopNET-N/V is 7, as defined by the Option File purchased option.

NOTICE *NOTICE*

RTK Services option includes Ntrip Caster functionality. If RTK Services are disabled, by options, rovers will not be able to connect to Ntrip TCP/IP port.

If no Hardware Protection Key is applied to the PC, RTK services will be available for an hour after application startup. Then the services will be disabled and the new rovers will not be able to connect. To enable services again, apply the Hardware Protection Key and restart the application. Network RTK services will not be available.

To purchase more services:

- press the **Save to File** button in the **Authorization Options** screen. The option file will be saved. An announcement screen will display the location of the option file on your PC. The name of the file is the same as the number of the Hardware Protection Key with *.opt extension;
- open the location and send the corresponding Option File to your dealer along with your order;
- receive the new Option File from the dealer;
- save it to the same location;
- in the **Authorization Options** screen select the necessary Hardware Protection Key;
- press the **Load from file** button and select the location of the Option File;
- press **OK**, the Option File will be activated;
- press **OK** in the **Authorization Options** screen to save the changes and close the screen;
- restart the application.

If more than one Hardware Protection Key are applied to the PC, the **Hardware Key ID** field (Figure 3-53 on page 3-57) shows a drop-down list of available keys.

Each key has its own Option File, so the **Values in Hardware Key** and **Values in Option File** fields (Figure 3-53) correspond to the currently selected key. The **Active Values** field in this case stays the same for all the keys. It shows the maximum of all the values displayed in the fields above for all the available keys.

NOTICE *NOTICE*

Not more than four Hardware Protection Keys should be applied to one PC.

Notes:

TOPCON
POSITIONING SYSTEMS

Chapter 4. TopNET-V

User's Manual

List of Contents

Introduction

Welcome to TopNET-V, a client program offered as an extension to Topcon's unique and powerful Reference Station Software Suite. As a part of the TopNET+ package, the TopNET-V application is intended to control a network of reference stations and provide corrections to any rover, located within the coverage area of the network, and is used in conjunction with the TopNET-S server and TopNET-R client programs.

TopNET-V can be configured to support RTK, DGPS and Network RTK modes of operation in any combinations; the user simply decides which data stream is compatible with the mobile equipment in use and requests this through the appropriate communication link.

Survey with a single reference station

Single reference station surveys are performed with the help of RTK or DGPS algorithms.

The RTK algorithm uses the concept of Real-Time Kinematics (RTK). Using the standard format correction data supplied from the nearest reference station in the network (the pseudorange and phase measurements), a rover receiver derives a standard RTK solution. Operating in RTK mode TopNET-V determines the nearest reference station based on the standard NMEA[1]-0183 GGA message sent to the central computer by each mobile unit when it makes the communication connection. Local RTK is usually limited in range. By creating a network of reference stations over a larger geographical

1. For this and further abbreviations see "Abbreviations"

area, mobile users can connect to any one of the reference stations in the network, extending the operation range of mobile units.

The Differential GPS (DGPS) algorithm operates similar to RTK. It provides sub-meter accuracy (depending on the user equipment). The difference is in the correction data supplied from the nearest reference station in the network. For the DGPS mode that is the pseudorange correction calculated in the TopNET-V application from raw measurements provided by the reference station nearest to the rover. This mode provides less precise data than RTK, however, DGPS is an L1 only solution (code) that can be used by a wider variety of less expensive equipment to support applications such as mapping, GIS, navigation, positioning, agriculture, etc. where centimeter-level accuracy (and the associated high end user equipment) is not required. DGPS also works at a much greater range from the reference stations than RTK.

During the RTK or DGPS survey the rover receives correction data from the closest reference station. If another reference station appears to be closer, the TopNET-V switches the rover automatically.

If the rover moves along a middle line between two reference stations, the switching looks as follows:

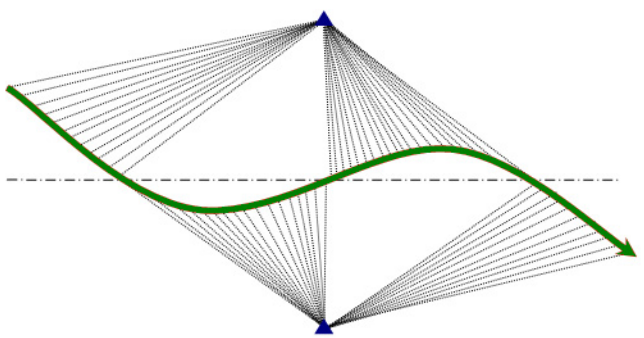


Figure 4-1. The Zones of Reference Station Tracking. Equidistant Tracking without "Hysteresis" Effect

However, one should note that each time the rover is switched, it loses Fixed solution and has to restart RTK engine.

In order to avoid multiple switching between reference stations when the rover is near the middle line, the TopNET-V application uses the “hysteresis” effect. The reference station is changed only when the distance difference exceeds a threshold value of 100 m.

In this case the zones of reference station tracking looks as follows:

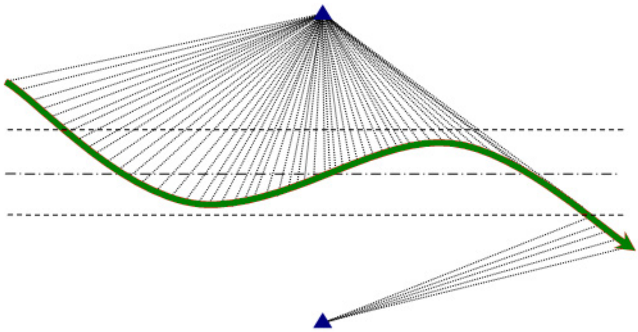


Figure 4-2. The Zones of Reference Station Tracking. Equidistant Tracking

Survey with multiple reference stations

Multiple reference stations surveys are performed with the help of the Network RTK algorithm. In the Network RTK mode the data from all stations come to the server and form the correction field, where precise values can be calculated at any point inside the field.

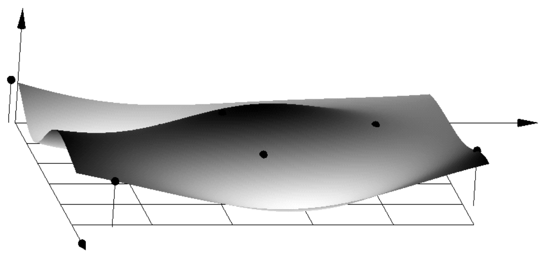


Figure 4-3. Network RTK Mode - An Example Correction Field Form

The TopNET-V application collects data from all the available Reference Stations with the help of the TopNET-S server or directly, and interpolates Ionospheric (dispersive) and Tropospheric (non-dispersive) corrections. From these corrections and user's coordinates sent by the Rover as a standard NMEA-0183 GGA message, the application reconstructs the measurements of pseudorange and carrier phase as if they were measured by a Reference Station receiver located close to the Rover. The generated data are sent to the Rover as RTCM or CMR messages and used for obtaining precise solutions.

Implementation

The Rover connects with TopNET-V using the TCP/IP protocol either through different ports for different services, or via Ntrip caster.

In the case of connection through the ports, the TopNET-V operator sets up services (see "Services" on page 4-47), then provides the settings to all the TopNET-V users.

Operating in Ntrip caster mode, the services are provided through the mountpoints with corresponding names.

The selection of the port (and service provided) is performed when setting up the Rover Receiver (for example, with TopSURV).

Figure 4-4 illustrates the connections between hardware and software with TopNET-V as a central data processor.

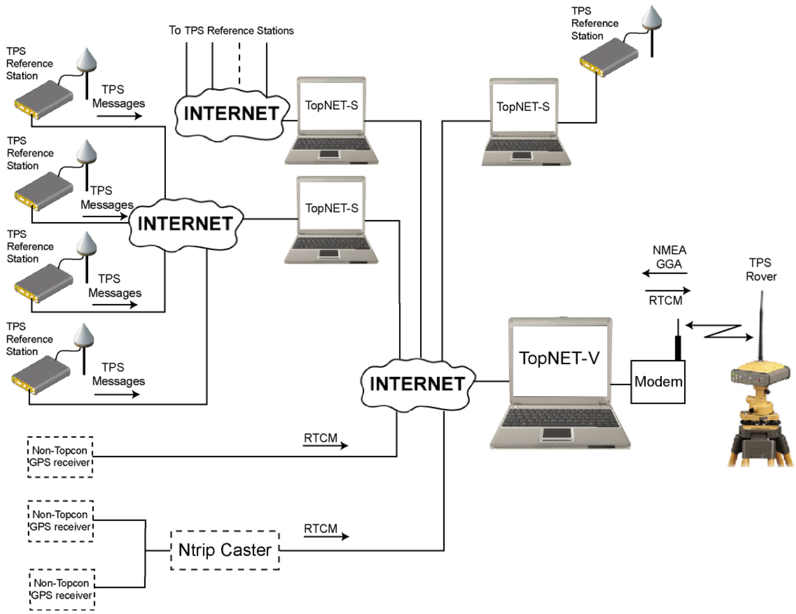


Figure 4-4. TopNET+ Configuration

The TPS reference stations connect to the TopNET-S server using the Internet, and each TopNET-S server can obtain data from one or more reference stations. The TopNET-V application collects TPS data from TopNET-S servers and RTCM 2.3/3.0 data from Ntrip casters or directly from GPS receivers. These data are processed and transmitted to a Rover using the data communication link.

Notes:

TopNET-V

Getting Started

Main Screen

The main screen of the TopNET-V application reflects the current state of the network.

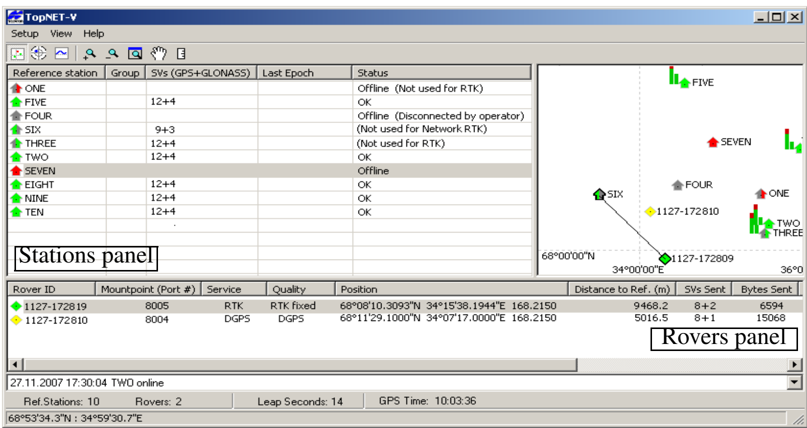


Figure 4-5. Main Screen

When opened for the first time, the screen is empty. To start a network, add at least one reference station (see “Launching Network” on page 4-24 for this procedure). Once added, the reference station is displayed in the Stations panel of the screen. This panel contains the following columns:

- The *Reference station* column lists each station and an icon that reflects the current status of the corresponding reference station (its interpretation is listed in the last column, *Status*).

Green icon: the station is available for RTK and Network RTK, and working properly.

Red icon: the station is available for RTK and Network RTK, but it’s coordinates are not valid, or the satellites number does not satisfy the conditions set in Preferences, or it was disabled or disconnected by an operator. The corresponding comment will be displayed.

Left half of the icon is green: the station is available for RTK and is working properly.

Right half of the icon is green: the station is available for Network RTK and is working properly.

Left half of the icon is red: the station can be used for RTK and but the data are corrupted or unavailable for some reason.

Right half of the icon is red: the station can be used for Network RTK and but the data are corrupted or unavailable by some reason.

Dark Grey icon: the station is not available for both RTK and Network RTK, so considered to be offline.

Triangle icons stand for non-Topcon reference stations: Ntrip or TCP/IP. Their colors’ meaning is similar to those of the Topcon “house” icons.

- The *SVs* (satellite vehicles) column lists information about the number of satellites received (both GPS and GLONASS).
- The *Last Epoch* column shows the parameters of the last epoch received by the station; the GPS week number, GPS week day (Sunday = 0) and GPS time.
- The *Status* column provides further information on the status of the station.

The left mouse button double-click centers the map to the selected station without changing the scale.

The right mouse button click on the Reference Station opens the context menu of the following items (Figure 4-6):

Figure 4-6. Reference Station Context Menu

- *Use for RTK*, *Use for Network RTK* - place a checkmark near entry to activate the corresponding service for the reference station. If the station operates properly and a checkmark is located near the corresponding entry, the data coming from the reference station is included to the solution. The corresponding part of the reference station icon becomes green. To disable the station, take off both checkmarks, and the icon on the main screen turns dark grey.
- *Status* - shows the **Reference Station Status** screen. It displays the parameters of the reference station:

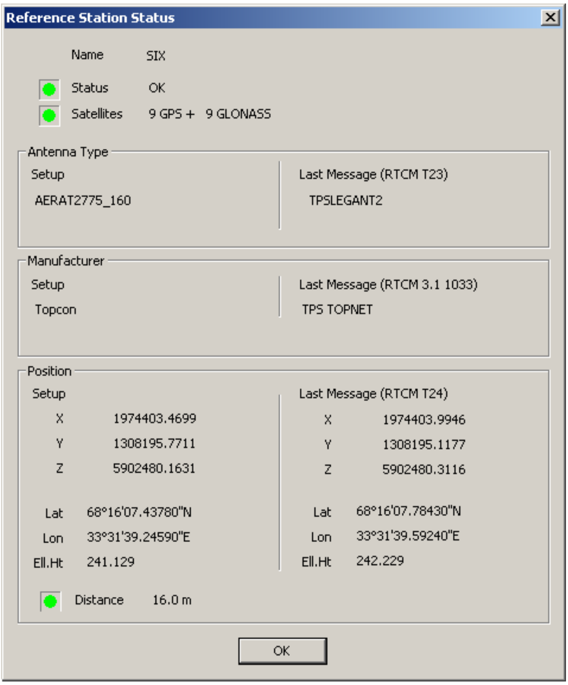


Figure 4-7. Reference Station Status

These are:

- name,
- status of the Reference Station,
- number of received satellites,
- Antenna Type - shows antenna type of the receiver, and that from the last message of the corresponding type, if antenna type is transmitted in RTCM.
- Manufacturer - shows the stored Manufacturer's name of the receiver, and that from the last message, if Manufacturer value is transmitted in RTCM.
- Position:

– the Setup position - the supplied coordinates of the reference station,

– the Last Message position - the coordinates calculated according to the last message of the corresponding type being received. If there an RTCM 3.0 Message 1005 or 1006 is received, the Last Message field shows the Message Type:

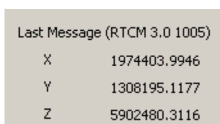


Figure 4-8. Last Message Position with RTCM 3.0 Message

- the Distance between the setup and received positions.Also it shows the icon that displays the connection status:
  – Green color means "OK", or in the case of only Network RTK service enabled, "(Not used for RTK)", or the case of only RTK service enabled "(Not used for Network RTK).
  – Red - "Offline," or "Offline (Disconnected by operator)".

Also this screen shows similar icons for satellites and distance (they turn red if the corresponding conditions are not satisfied).

NOTICE

The status icon in the main screen turns green only if all the three icons in the Reference Station Status screen are green.

After a rover becomes online, its ID and information appear in the Rovers panel of the Main screen. This panel contains the following columns:

- The *Rover ID* column lists the unique identification number for each rover and a solution type icon (its interpretation is listed in the *Quality* column):

  - Red icon: the position quality is “Standalone”.
  - Blue icon: the position quality is “RTK Float”.
  - Green icon: the position quality is “RTK Fixed”.
  - Yellow icon: the position quality is “DGPS”.

- The *Mountpoint* (*Port #)* column shows the number of the TCP/IP port the rover is connected to, or an Ntrip Mountpoint name if the rover uses Ntrip protocol for connection.
- The *Service* column displays the type of the service provided to the rover.
- The *Quality* column lists the solution quality, i.e. type of rover position as defined in GGA messages coming from the rover.
- The *Position* column shows the geodetic coordinates of the rover.
- The *Distance to Ref.* column displays a distance from a rover to its Reference station.
- The *SVs Sent* column displays the number of satellites in the last epoch sent to the rover. The “0 + 0” value appears if no data is sent.

- The *Bytes Sent* column shows the total number of bytes sent to the rover.
- The *IP Address* column displays the IP Address of the rover.
- The *Ntrip User* column displays the name of the corresponding user if the rover is using Ntrip for connection to TopNET-V.
- The *Duration* column shows the time of the rover connection to TopNET-V.

To sort a list of rovers, press the name of the desired column.

Center the map to the selected rover without changing scale by double-clicking the left mouse button.

To restrict access, a filter can be set on rovers' IP addresses using the "Allowed IP addresses" tool. For this procedure, see "Services" on page 4-47.

The bottom of the main screen displays the Event Log and the Information bar.

The Event Log contains information about connections/reconnections with the receivers (reference stations and rovers), error messages, the selection of a nearest reference station and displays the moment of enabling/disabling a reference station for RTK and/or Network RTK, by an operator.

The Information Bar displays information on Leap Seconds[1] and GPS time.

The Toolbar contains three menus: **Setup**, **View** and **Help**.

1. Leap Seconds - the difference between UTC and GPS time.

Setup menu

The *Setup* menu contains the following items:

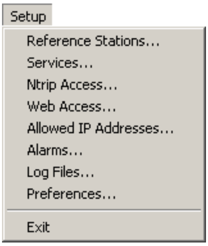


Figure 4-9. Setup Menu

- To create or edit a reference station, click **Reference Stations**.
- To set the ports and formats, click **Services**.
- To set Ntrip connection parameters, select **Ntrip Access.**
- To set Web Access parameters, select **Web Access...**
- To edit the list of allowed IP addresses, click **Allowed IP Addresses**.
- To set the automatic error notifications by e-mail, click **Alarms**.
- To select location for log files created by the application, click **Log Files**.
- To edit preferences for TopNET-N, click **Preferences**.
- To exit the program, click **Exit**.

View menu

The *View* menu allows one to:

- toggle between three different views (see “Views” on page 4-39),
- observe Engine Status (see “Engine Status” on page 4-18),
- export the reference stations to the Google Earth™ format (see “Export to Google Earth” on page 4-22), and
- launch Rover Accounting module (see Appendix VII “Rover Accounting”):

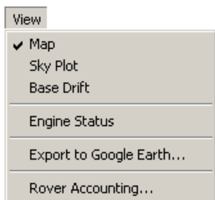


Figure 4-10. View Menu

Toggling between the views can be also performed using icons on the Toolbar:

Icon	View
	Map view
	Sky Plot view
	Base Drift view

Map view

Map view shows the relative location of the reference stations, the status of the reference stations and the chart of the visible satellites solution quality, for GPS and GLONASS systems, correspondingly. For details see “Map View” on page 4-39.

If you place the cursor on the satellite area, a floating information box appears:

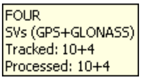


Figure 4-11. Information Box

It displays:

- name of the reference station;
- the number of satellites being tracked;
- the number of satellites being processed by engine.

If the **Map view** is enabled, when selecting a rover in the list, the map highlights the icon of the rover with a black border. For RTK and DGPS services the Map draws a connector line to the nearest reference station. For Network RTK service, the map draws a simulated reference station as an empty triangle and a connector line to it.

The left mouse button double-click on any Reference Station or Rover entry in the list centers the map to the selected object without changing the scale.

The bar charts reflect the quality of the Network RTK solution on the reference station. If the bars are red, the data from this station will not be sent to rover. If all the bars are red, the rover does not receive data and you can see “0 + 0” value in the *SVs Sent* column on the Rovers panel.

Sky Plot view

Sky Plot view shows the constellation of visible satellites and charts reflecting the visibility of these satellites to the reference stations. For details see “Sky Plot View” on page 4-42.

Base Drift view

Base Drift view displays the drift of the reference station position (estimated by the Network RTK engine) with respect to the user defined coordinates. If none of the stations is selected, the Base Drift view displays the plot for all the reference stations. For details see “Base Drift View” on page 4-45 section.

Help

The Help menu (Figure 4-12) provides information about the TopNET-V version and build date, and the Activated Options of the Hardware Key (see “Hardware Key Usage” on page 4-71):

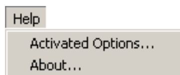


Figure 4-12. Help Menu

Engine Status

The **Engine Status** window can be opened by selecting *Engine Status* from the *View* menu in TopNET-V.

It displays the current status of the preprocessor, network RTK engine, and network adjustment components of the engine. **Engine Status** is an informational tabbed window with tabs each corresponding to one engine component.

The preprocessor tab is shown in Figure 4-13.

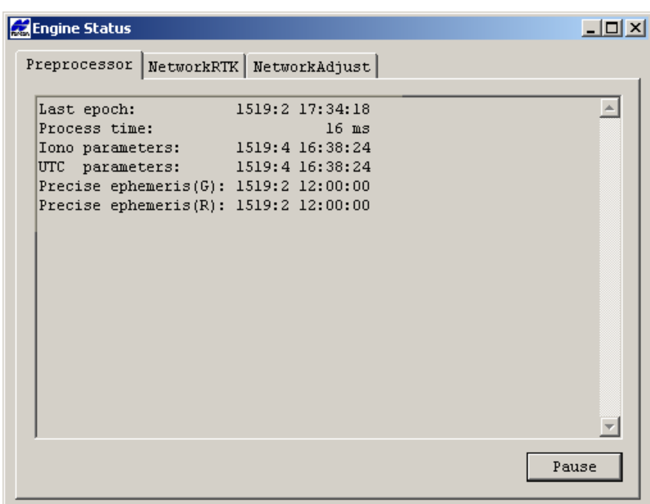


Figure 4-13. Engine Status. Preprocessor tab.

Here:

Last epoch: the time of the last epoch processed by the preprocessor. The times are formatted as the GPS week and day of week, followed by the hours, minutes, and seconds of day.

Process time: the amount of time in milliseconds used by the preprocessor to process the last epoch.

Iono parameters: time taken from the first GPS subframe associated with the ionospheric correction parameters.

UTC parameters: reference time of the Universal Coordinated Time parameters.

Precise ephemeris (G): time of the GPS ultra?rapid precise ephemeris file.

Precise ephemeris (R): time of the GLONASS ultra?rapid precise ephemeris file.

The NetworkRTK tab are shown in Figure 4-14.

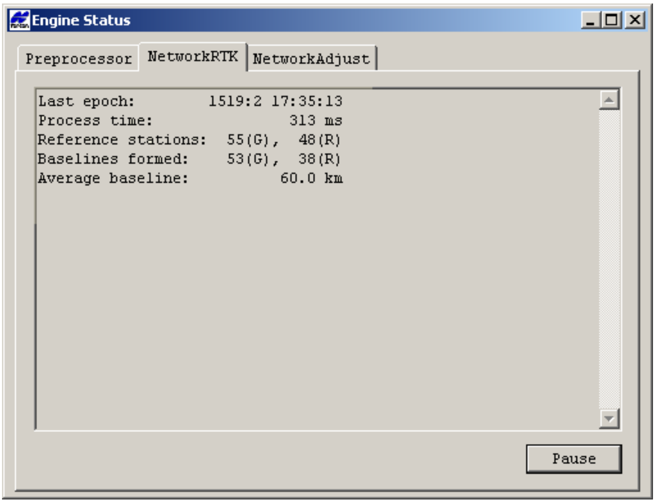


Figure 4-14. Engine Status. NetworkRTK tab.

Here:

Last epoch: the time of the last epoch processed by the network RTK engine. The times are formatted as the GPS week and day of week, followed by the hours, minutes, and seconds of day.

Process time: is the amount of time in milliseconds used by the network RTK engine to process the last epoch.

Reference stations: the number of reference stations with GPS and GLONASS observations used in the last epoch.

Baselines formed: the number of baselines with GPS and GLONASS observations formed in the last epoch.

Average baseline: the average baseline length (in km) of all baselines formed in the last epoch.

The NetworkAdjust tab is shown in Figure 4-15. The network adjustment engine is used to estimate reference stations coordinates in real time for the purpose of comparing to user supplied coordinates. The adjustment used all fixed baselines from every epoch, but is only updated every 15 seconds.

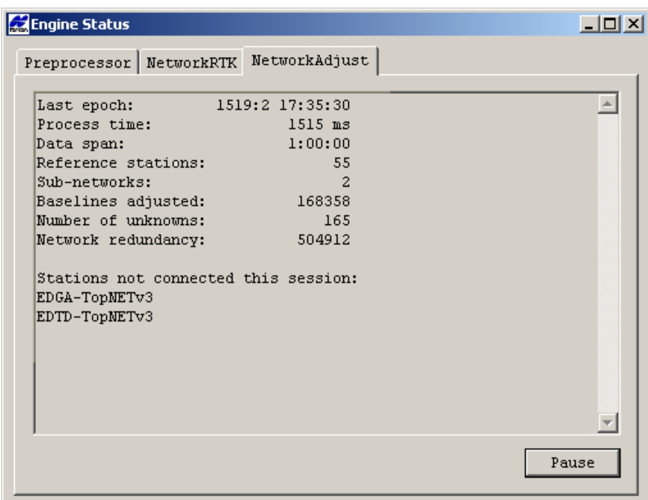


Figure 4-15. Engine Status. NetworkAdjust tab.

Here:

Last epoch: the time of the last epoch processed by the network adjustment engine. The times are formatted as the GPS week and day of week, followed by the hours, minutes, and seconds of day.

Process time: is the amount of time in milliseconds used by the network adjustment engine to process the last epoch.

Data span: the period of data span in hours, minutes, and seconds. The period is currently capped at one hour.

Reference stations: the number of reference stations included in the network adjustment.

Sub-networks: the number of subnetworks included in the network adjustment. Sub-networks are due to the network engine not forming/resolving baselines between station clusters. This is most often due to excessive spacing between clusters.

Baselines adjusted: the number of baselines adjusted in the data span.

Number of unknowns: the number of estimated stations coordinates.

Network redundancy: the redundancy of the network adjustment.

Stations not connected this session: stations whose coordinates have been estimated during the data span but that were not a part of any adjusted baselines in the last session. Note that a session is comprised of all baselines fixed over a 15 second period.

Export to Google Earth

Using TopNET-V, you can save coordinates of the reference stations to a Google Earth file for future use and, if the Google Earth application is installed on your PC, immediately obtain the image of your reference stations' location on the Earth.

To export the coordinates to Google Earth, select **View ▸ Export to Google Earth** menu item.

In the standard Windows **Save As** dialog select a name and location for the file:

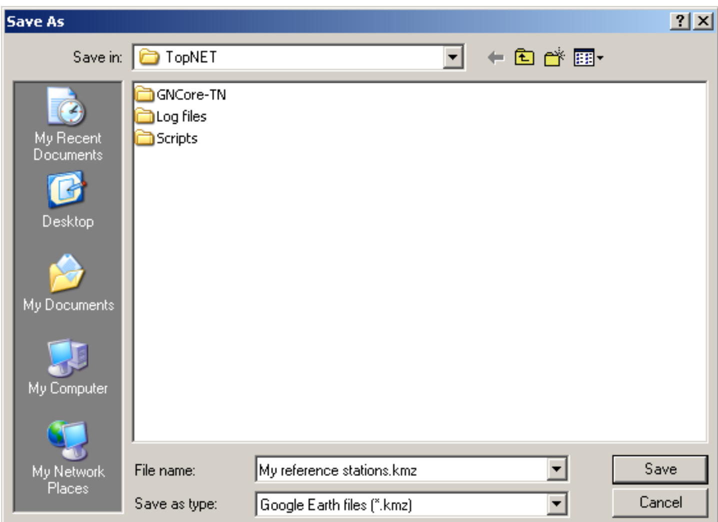


Figure 4-16. Export to Google Earth

The *.kmz file contains the complete information about your network, including icon images.

If **Google Earth** has been installed on your PC, this application is launched after you press the **Save** button. The TopNET-V folder containing the list of your reference stations and rovers, appears in Temporary Places:.

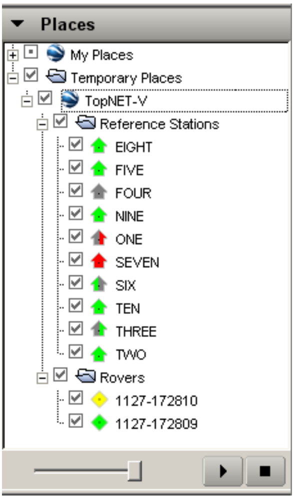


Figure 4-17. Google™ Places

The Google Earth map shows the location of the stations with icons corresponding to those in TopNET-V

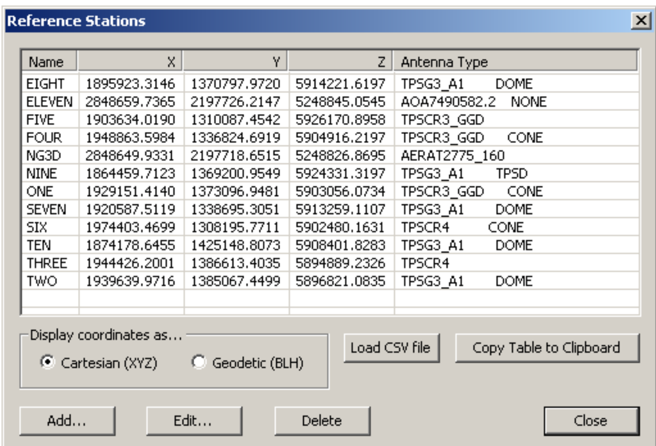
Figure 4-18. Google Earth Map

For details on how to work with Google Earth see Google Earth web page.

Launching Network

Click **Setup ▸ Reference Stations** to view the list of available reference stations and their parameters: name, coordinates, and antenna type (Figure 4-19 on page 4-24).

- Select a radio button to switch shown coordinates between *Cartesian (XYZ)* and *Geodetic (BLH)* types. Depending on the selection, the table displays either *X/Y/Z*, or *Latitude/Longitude/ Ellipsoid Height* columns.



Name	X	Y	Z	Antenna Type
EIGHT	1895923.3146	1370797.9720	5914221.6197	TPSG3_A1 DOME
ELEVEN	2848659.7365	2197726.2147	5248845.0545	AOA7490582.2 NONE
FIVE	1903634.0190	1310087.4542	5926170.8958	TPSCR3_GGD
FOUR	1948863.5984	1336824.6919	5904916.2197	TPSCR3_GGD CONE
NG3D	2848649.9331	2197718.6515	5248826.8695	AERAT2775_160
NINE	1864459.7123	1369200.9549	5924331.3197	TPSG3_A1 TPSD
ONE	1929151.4140	1373096.9481	5903056.0734	TPSCR3_GGD CONE
SEVEN	1920587.5119	1338695.3051	5913259.1107	TPSG3_A1 DOME
SIX	1974403.4699	1308195.7711	5902480.1631	TPSCR4 CONE
TEN	1874178.6455	1425148.8073	5908401.8283	TPSG3_A1 DOME
THREE	1944426.2001	1386613.4035	5894889.2326	TPSCR4
TWO	1939639.9716	1385067.4499	5896821.0835	TPSG3_A1 DOME

Figure 4-19. Reference Stations

- If you have a CSV file with coordinates of any existing reference stations, you can load it using the **Load CSV file** button. For details and description of the CSV format see “Loading a CSV file” section.
- To copy the list of stations and their coordinates to the clipboard, press the **Copy Table to Clipboard** button.
- To edit a reference station, select the station and click **Edit**. Follow the instructions in the **Edit Reference Station** wizard (see “Adding/Editing a Reference Station” on page 4-25).
- To add a new reference station, click the **Add** button. Follow the instructions in the **Add Reference Station** wizard, that is similar to the **Edit Reference Station** wizard, mentioned above.

- To delete one or several reference station, highlight the station(s) and click **Delete**.
- To return to the Main screen, click **Close**.

Adding/Editing a Reference Station

To add or edit a reference station, click the corresponding button in the **Reference Stations** dialog box (Figure 4-19). A Wizard opens and guides you through the process of adding or editing.

In case of adding a station, select a type of receiver connection: via TopNET-S (only for Topcon receivers), an Ntrip caster, or any receiver streaming data to the Internet:

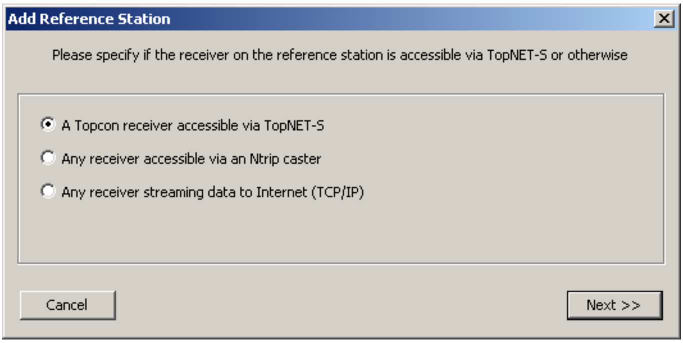


Figure 4-20. Add Reference Station

NOTICE

NOTICE

In case of the Network RTK mode, at least one of the added receivers should be Topcon receiver. Otherwise the Network RTK mode will not work.

Click **Next** to select a server. See the following sections for the explanation on how to add three different Reference Stations types.

In case of editing a station, the first Wizard screen corresponds to the selected type of station.

In Wizard, current header of the dialog box always shows the type and display name of the selected station - each starting from the screen where it is selected. For example, *Add Topcon Reference Station FIVE*.

To close the Wizard without making changes, click **Cancel**.

For TopNet-S Reference Stations

The **Add/Edit Reference Station** dialog box contains the selection of the server and receiver for the chosen reference station. When opened in *Add Reference Station* mode, the list of receivers is empty until the **Get Receivers** button is pressed, and the **Revert to Saved** button is displayed as **Revert to Default**.

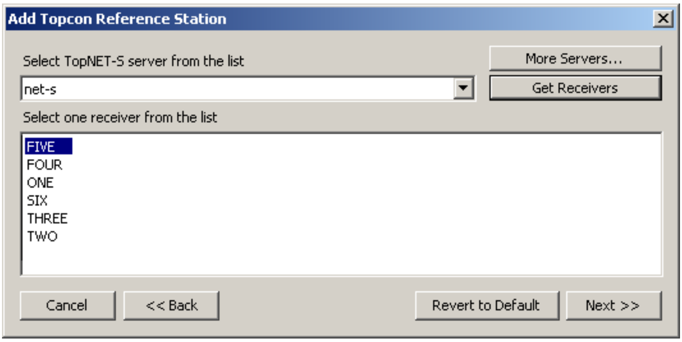


Figure 4-21. Add Topcon Reference Station

When opened in *Edit Reference Station* mode, the list of receivers shows only one selected receiver and the **Get Receivers** button is disabled.

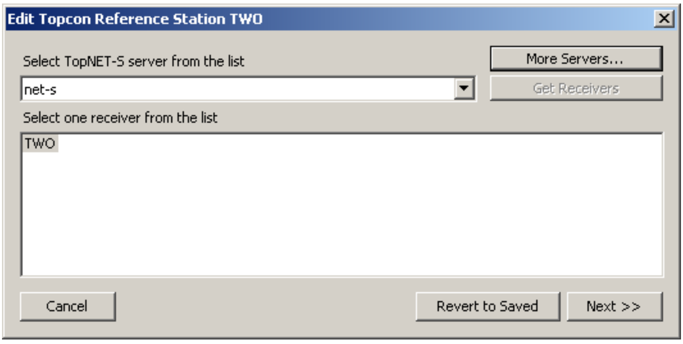


Figure 4-22. Edit Topcon Reference Station

TopNET-V

1. In order to set or change server for the station, in the *Select TopNET-S server from the list* list select the host name or IP address of the server. If there is no desired server in the list or no servers at all, click **More Servers** and setup a new server. See “To set up a new server:” on page 4-28 for details.

   In case of adding a receiver:

   – Click **Get Receivers** to display the list of available receivers connected to the selected server.

   – Select the receiver intended to be the reference station.

2. Click **Next** and open a screen where you can change a display name for the station.

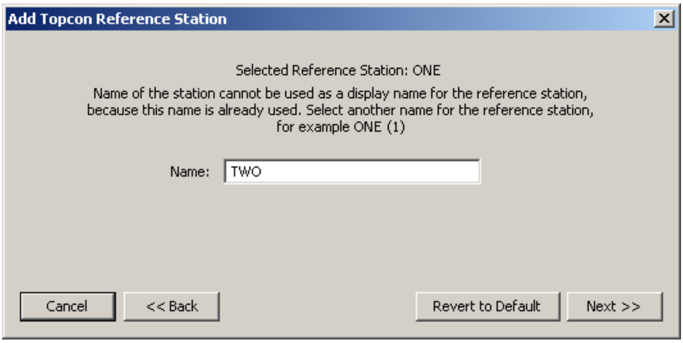


Figure 4-23. Add/Edit Topcon Reference Station - Display Name

If a station with the same display name already exists, you need to select another display name to add the station.

NOTICE

Name of the station should not contain more than 31 symbols.

3. Click **Next** and continue with “Setting Coordinates of Reference Station” on page 4-33 or click **Back** to make changes in previous screen.

To set up a new server:

In order to add a server, in **Add Topcon Reference Station** click **More Servers.**

When creating a reference station for the first time, the list of servers is empty.

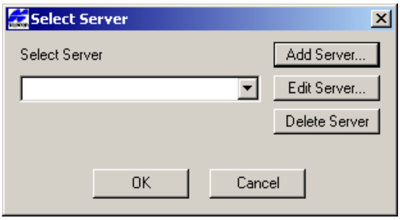


Figure 4-24. Select Server - New

1. Click **Add Server** to add server properties.
2. Enter the properties of the server: host name or IP address, user name (login), and password.

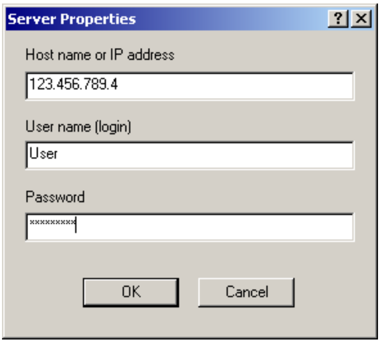


Figure 4-25. Server Properties

3. To set a port for connection, define its number in the address field using colon, i.e. 123.456.456.321:8010. By default, the port number 8000 is used for connection.

4. Click **OK**. The server appears in the **Select Server** list. The latest added or edited server becomes the default one.

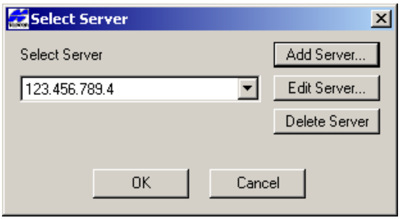


Figure 4-26. Select Server

To edit server's properties, click **Edit Server** and make necessary changes in the **Server Properties** dialog box (Figure 4-25).

To delete the selected server from the list, click **Delete Server**.

For Ntrip Reference Station

The **Add/Edit Ntrip Reference Station** dialog box contains the server's address and port, mountpoint name, and login and password fields.

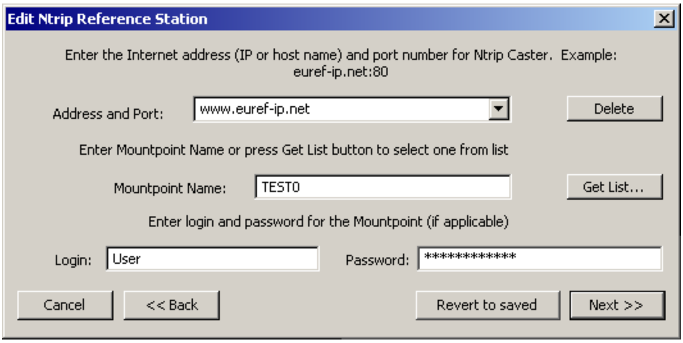


Figure 4-27. Add/Edit Ntrip Reference Station - Select Mountpoint

To add a new Ntrip reference station:

1. Type in the address of an Ntrip Caster (for example, http://www.euref-ip.net) as shown in Figure 4-27, and, if necessary, enter the port number separated by colon.
2. If the *Mountpoint* name is known, enter it in the corresponding field.

Otherwise, press the **Get List** button to obtain a list of available casters, networks, and mountpoints from the caster.

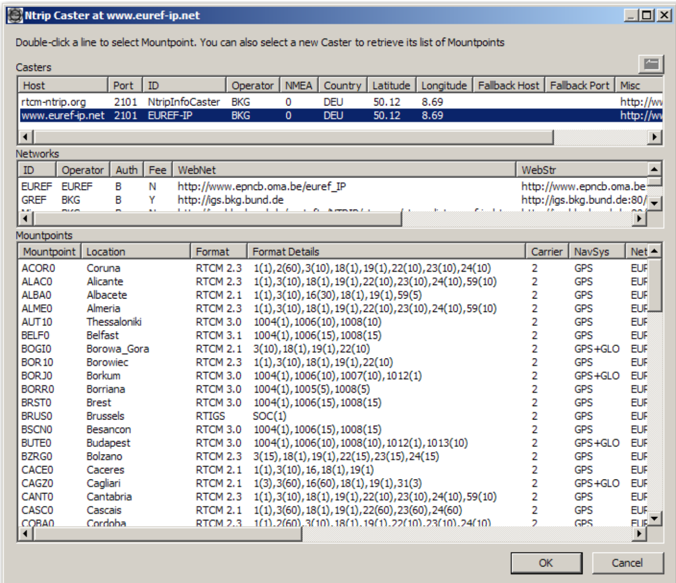


Figure 4-28. Ntrip Caster Example

Select a caster, network and mountpoint (your selection will be highlighted) and press **OK**.

3. Click **Next** in the **Add/Edit Ntrip Reference Station** dialog box. If a new ntrip reference station is created, the following screen is shown:

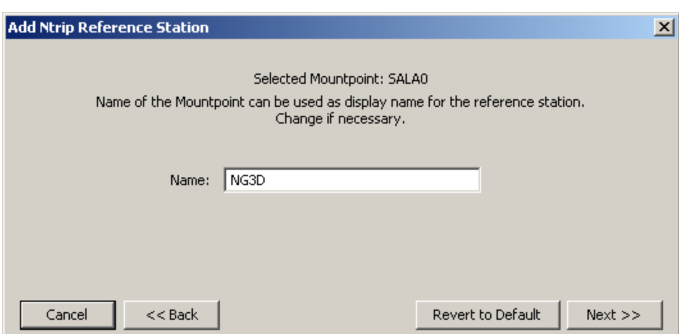


Figure 4-29. Add/Edit Ntrip Reference Station - Display Name

TopNET-V

Here you can change a display name for the station, if desired. If a station with the same display name already exists, you need to select another display name to add the station.

If a reference station is being edited, this step is skipped.

4. Click **Next** in the **Add/Edit Ntrip Reference Station** dialog box

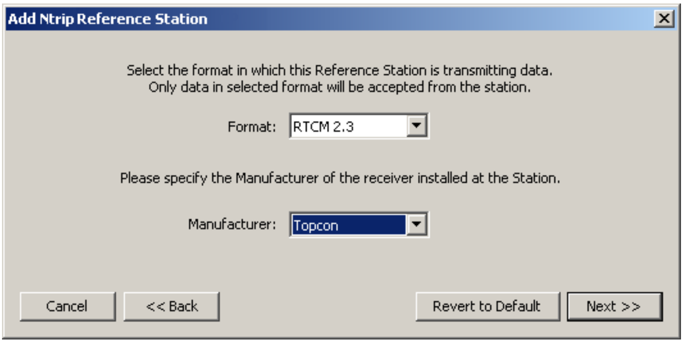


Figure 4-30. Add/Edit Ntrip Reference Station - Data Format

Select a format used by your reference station for data transmission. It can be either RTCM 2.3 or RTCM 3.0.

To be used by the TopNET-V application, the RTCM 2.3 message received from an Ntrip Caster or TCP/IP reference station must contain the Types 18 and 19 Messages (carrier phase and pseudorange values). And the RTCM 3.0 message must contain the Type 1003 or 1004 Message (GPS measurements) and Type 1011 or 1012 Message (GLONASS measurements); presence of Type 1005 or 1006 Message (ARP coordinates) and 1007 or 1008 (Antenna Descriptor) is optional.

5. From the drop-down list, select a Manufacturer of the receiver.
6. Click **Next** and continue with "Setting Coordinates of Reference Station" on page 4-33.

For Reference Stations Streaming Data via TCP/IP

The **Add/Edit TCP Reference Station** dialog box should contain address/port of the server as shown in Figure 4-31

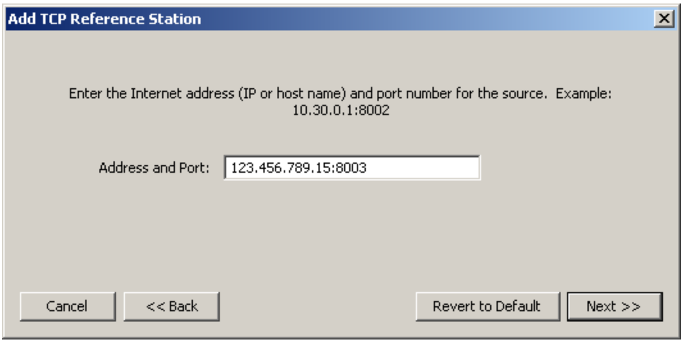


Figure 4-31. Add/Edit TCP Reference Station - Address and Port

1. Click **Next** and enter/change a name for the reference station.

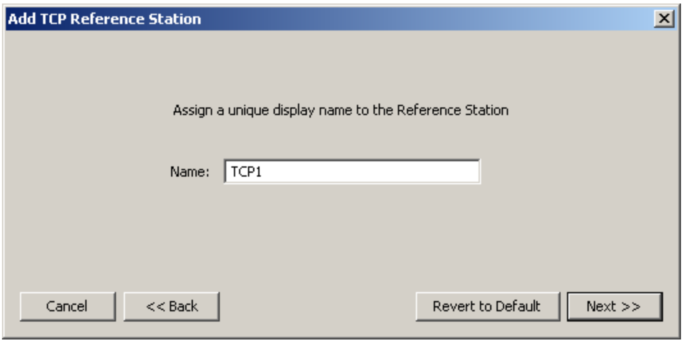


Figure 4-32. Add/Edit TCP Reference Station - Display Name

If a reference station is being edited, this step is skipped.

2. Click **Next** and select a format used by your reference station for data transmission and a receiver's Manufacturer, as described in step 3 on page 4-30.

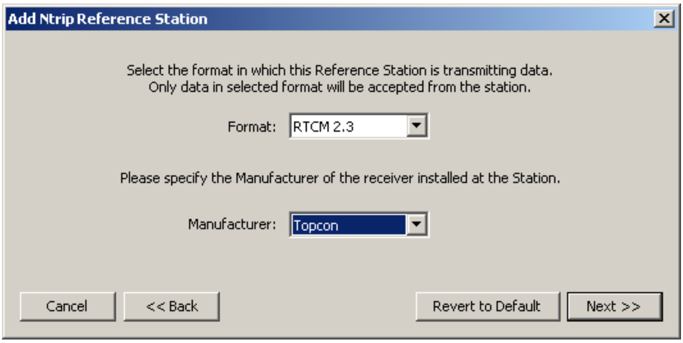


Figure 4-33. Add TCP Reference Station – Format and Manufacturer

3. Click **Next** and continue with "Setting Coordinates of Reference Station" .

Setting Coordinates of Reference Station

The **Reference Station Coordinates** screen represents the coordinates of the Reference Station to be added or edited. To undo changes and return to the saved coordinates, click **Revert to Saved**.

NOTICE

Coordinates of the Reference Stations should be surveyed in with sufficient accuracy and precision. Unreliable coordinates will limit the performance of the network.

1. Select the type of coordinates to be entered, *Cartesian* or *Geodetic*. When editing the coordinates of the existing reference station, enter changes in any form.
   - For *Cartesian*, insert (or change) the X,Y, Z coordinates.
   - For *Geodetic*, insert (or change) the Latitude, Longitude and Ellipsoid Height.

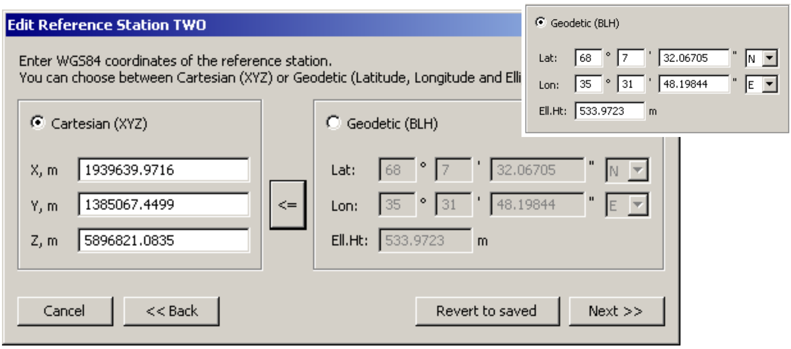


Figure 4-34. Add/Edit Reference Station - Coordinates

Two different reference stations enabled for Network RTK should not be located too close to each other, because that can cause a failure in the system operation.

2. Click **Next** and continue with "Setting Antenna Parameters" below.

Setting Antenna Parameters

After setting up the reference station and its coordinates, specify the antenna used at the reference station.

1. Select **Antenna Type** from the list of NGS[1] standard antenna names and insert **Antenna Serial Number**, if available.

---

1. See, for example, Complete Antenna Calibration Results at http://www.ngs.noaa.gov/ANTCAL.

For each Antenna Type the phase center offsets are available for your reference.

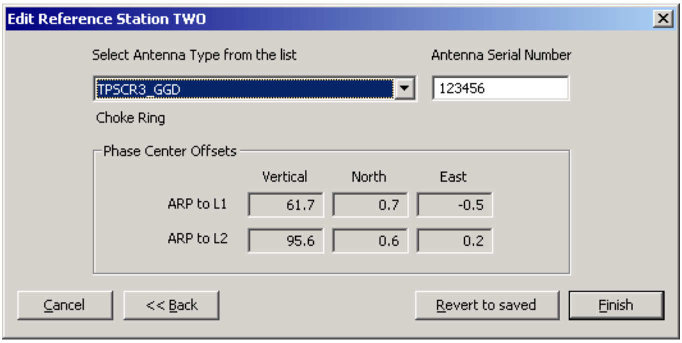


Figure 4-35. Add/Edit Reference Station – Step 3

2. Click **Next** and continue with "Setting Group" on page 4-35.

Setting Group

After setting up the Antenna Type, select a group if desired. The group name is shown as a separate column in the reference station list:

1. Select a group name from the list, or create a new group name, or leave the field empty.

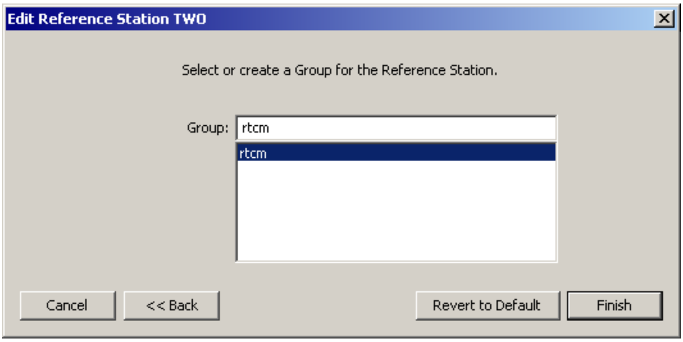


Figure 4-36. Add/Edit Reference Station – Groups

2. Click **Finish** to save the changes and return to the **Reference Stations** screen.

A newly created reference station appears on the Map View and Stations panel and, by default, is activated for both RTK and Network RTK services. Also the information about this reference station becomes displayed on the Base Drift and Sky Plot views.

Loading a CSV file

CSV file stands for a text file with comma-separated values and *.csv extension. It is used, for example, for applying changes to the parameters of one or several reference stations.

To be correctly interpreted by the application, the CSV file must have a header line which specifies what fields are in the file. The following fields are valid:

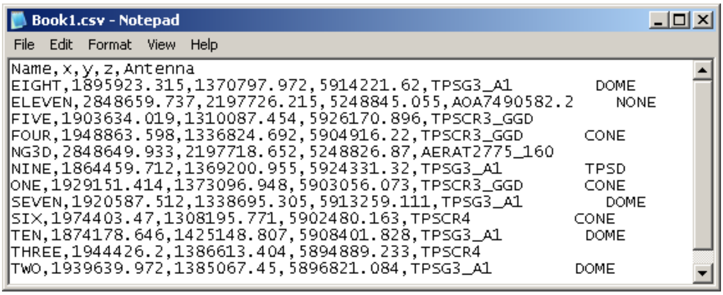
- “Name” - required field;
- “X”
- “Y”
- “Z”
- “LatD”- the name stands for the latitude represented in the decimal degree form. If latitude of the EIGHT station is 68^{o}32’53.9399, then the LatD value is 68.5483166389 degrees.
- “LatDMS0” - the name stands for the latitude represented in the DDDMMSS.ssss form. So, if latitude of the EIGHT station is 68^{o}32’53.9399, then the LatDMS0 value is 683253.9399.
- “LatDMS1” - the name stands for the latitude represented in the DDD.MMSSssss form. So, if latitude of the EIGHT station is 68^{o}32’53.9399, then the LatDMS0 value is 68.32539399.
- “LonD” - the name stands for the longitude represented in the decimal degree form, similar to LatD value.
- “LonDMS0” - the name stands for the longitude represented in the DDDMMSS.ssss form, similar to LatDMS0 value.
- “LonDMS1” - the name stands for the longitude represented in the DDD.MMSSssss form, similar to LatDMS1value.
- “Height”

- "Antenna"
- "AntSerNum"
- "Manufacturer"
- "Group"

NOTICE *NOTICE*

All the coordinates present in the file must have one and the same format.
Space is not interpreted as a delimiter.

Once created, the file can be edited in Notepad. An example of CSV file is represented below:



```
Name,x,y,z,Antenna
EIGHT,1895923.315,1370797.972,5914221.62,TPSG3_A1        DOME
ELEVEN,2848659.737,2197726.215,5248845.055,AOA7490582.2    NONE
FIVE,1903634.019,1310087.454,5926170.896,TPSCR3_GGD
FOUR,1948863.598,1336824.692,5904916.22,TPSCR3_GGD      CONE
NG3D,2848649.933,2197718.652,5248826.87,AERAT2775_160
NINE,1864459.712,1369200.955,5924331.32,TPSG3_A1        TPSD
ONE,1929151.414,1373096.948,5903056.073,TPSCR3_GGD      CONE
SEVEN,1920587.512,1338695.305,5913259.111,TPSG3_A1        DOME
SIX,1974403.47,1308195.771,5902480.163,TPSCR4           CONE
TEN,1874178.646,1425148.807,5908401.828,TPSG3_A1        DOME
THREE,1944426.2,1386613.404,5894889.233,TPSCR4
TWO,1939639.972,1385067.45,5896821.084,TPSG3_A1         DOME
```

Figure 4-37. Example of a CSV file

To load a CSV file, select **Setup ▸ Reference Stations** and press the **Load CSV file** button in the **Reference Stations** dialog box (Figure 4-19 on page 4-24).

Select a file from your PC and press OK.

If the list of stations in the file corresponds to the list of existing stations in the application, a success message appears.

If the list of stations in the file does not correspond to the list of existing stations in the application, or contents of the file does not agree with the header line fields, an error log is created in the location set by the **Select folders for log files** screen (Figure 4-65 on page 4-69), and displayed.

TopNET-V

Notes:

Views

Map View

The Map view displays the network Map and shows the relative position of the reference stations and rovers. A change in icon color indicates a change in status.

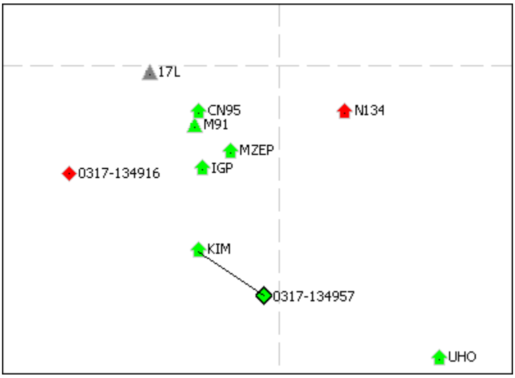


Figure 4-38. Map

When selecting a rover in the list of rovers, the map icon of the selected rover is highlighted with a black border. The map also shows the service provided to the selected rover.

- For the RTK and DGPS services, the map draws a connector line to the nearest reference station and places a black border around the icon. (Figure 4-38).
- For Network RTK service, the map draws a simulated reference station as an empty triangle and a connector line to it. If a rover is located closely to the simulated station, then the application places the simulated reference station to the exactly same location where the rover is.

The map icon of a reference station is highlighted with a black border, if:

- this reference station is selected in the list of stations,
- this station is the nearest reference station for the highlighted rover.

Table 4-1 lists the conventions used in the map.

Table 4-1. Map View Conventions

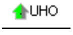
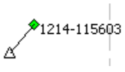
Icon	Description
	A connected reference station used for both RTK and Network RTK is displayed as a green icon with a point inside and a name to the right.
	A reference station not used for RTK and Network RTK is considered to be disconnected and is displayed as a gray icon with a point inside and a name to the right.
	A reference station used for both RTK and Network RTK with a bad or corrupted signal is displayed as a red icon with a point inside and a name to the right.
	A reference station used for RTK is displayed as a half-green and half-grey icon if it operates properly, and half-red and half-grey icon if its signal is corrupted.
	A reference station used for Network RTK is displayed as a half-grey and half-green icon if it operates properly, and half-grey and half-red icon if its signal is corrupted.
	A simulated reference station is displayed as an empty triangle.
	Bar charts reflect the quality of the Network RTK solution. Left bar is proportional to the number of GPS satellites used by Network RTK engine, right bar - to the number of corresponding GLONASS satellites. Green part of the bar stands for the number of integer ambiguities resolved by Network RTK engine, red part - for the number of those not resolved.
	Colored triangle icons stand for non-Topcon reference stations: Ntrip or TCP/IP. Their colors' meaning is similar to those of the Topcon "house" icons.

Table 4-1. Map View Conventions (Continued)

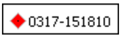
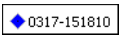
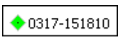
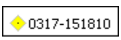
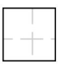
Icon	Description
	A rover with a “Standalone” position quality is displayed as a red square with a point inside. The rover selected in the list of rovers has a black border around the square.
	A rover with an “RTK Float” position quality is displayed as a blue square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	A rover with an “RTK Fixed” position quality is displayed as a green square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	A rover with a “DGPS” position quality is displayed as a yellow square with a point inside. The rover selected in the list of rovers also has a black border around the square.
	The numerical coordinate values are located near each parallel and meridian on the left and bottom side of the map to form a coordinate grid.

Table 4-2 describes the toolbar buttons used for controlling the Map view.

Table 4-2. Map View Toolbar Buttons

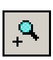
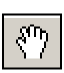
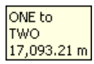
Button	Description
	Zoom In Click the button and click the map to enlarge the view Hold the left mouse button and click-and-drag a rectangle to enlarge a selected part of the screen up to the whole screen. Also you can use a mouse wheel for this purpose.
	Zoom Out Click this button and click the map to decrease the view. Hold the left mouse button and click-and-drag a rectangle to reduce the whole screen to the size of the selected rectangle. Also you can use a mouse wheel for this purpose.
	Pan Press the button to activate the Pan control that allows one to move the observed area using left mouse button being hold.
	Zoom All Click the button to display all the objects.

Table 4-2. Map View Toolbar Buttons

Button	Description
	Distance Click the button to activate the tool. A Ruler appears near the cursor. Click at the start point and move the cursor in any direction. The floating information box shows the coordinates of the start point and current point and distance between them. If one of the points is located near a reference station or rover, instead of the coordinates, the information box shows the name of this station (rover). For example:  To switch off the connector, press Esc. To switch off the tool, click the button one more time.

Sky Plot View

Sky Plot View is an information screen that displays the constellation of visible satellites and charts reflecting the visibility of these satellites to the reference stations.

To open the Sky Plot view, select **View ▸ Sky Plot** menu:

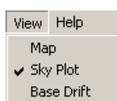


Figure 4-39. View Menu – Sky Plot Selection

The right part of the main view shows the plot of the satellites positions:

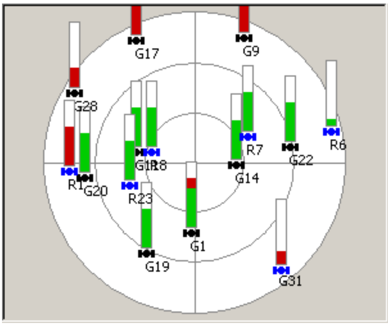


Figure 4-40. Sky Plot View for All Reference Stations

If none of the stations is selected in the Stations panel, the Sky Plot view displays all the satellites tracked by all the reference stations.

When selecting a reference station in the list of stations, the sky plot changes.

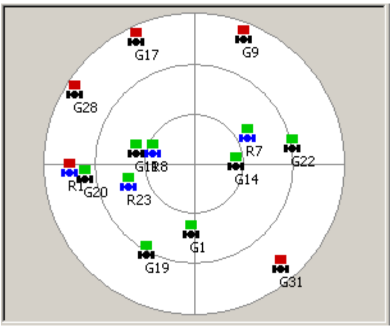
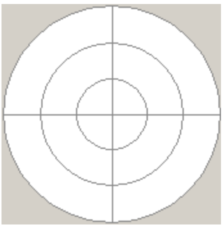
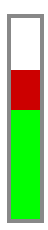


Figure 4-41. Sky Plot View for One Reference Station

This screen shows only satellites visible at selected reference station. Green rectangle above a satellite icon indicates that Network RTK engine processes data from this satellite at this reference station. If Network RTK engine rejected the data (for any reason), the corresponding rectangle above the satellite icon changes color to red.

Table 4-3 shows the description of view elements:

Table 4-3. Sky Plot Conventions

Icon	Description
	The sky plot is oriented to the north and marked with concentric circles illustrating elevation. The edge is 0 degrees, the center is 90 degrees above the horizon.
	On the general Sky Plot view, a bar near each satellite corresponds to the total number of the reference stations. Here: the green part of the bar displays the number of reference stations where the satellite data are processed; the red part shows the number of stations where the satellite is visible, but rejected by Network RTK engine, and the white part stands for the stations where the satellite is not visible, and disconnected stations.
	A black or blue satellite icon represents the position of the satellite on the sky plot. Blue icons correspond to GLONASS satellites, black ones - to GPS satellites.
R8 G20	The letters near each satellite show the satellite system which the satellite belongs to and its number. “R8” stands for the satellite #8 in GLONASS system, “G20” - for the satellite #20 in GPS system.

On the general Sky Plot view, if you place the cursor on the satellite area, a floating information box appears:

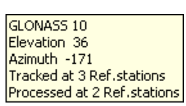


Figure 4-42. Information Box

It displays:

- the satellite system and PRN number of the satellite[1];

- the elevation value;
- the azimuth value;
- the number of stations for which this satellite signal is processed.

Base Drift View

Base Drift View is an information screen that displays the plot of the the reference stations coordinates instability.

To open the Base Drift view, select **View ▸ Base Drift** menu.

Figure 4-43. View Menu – Base Drift Selection

The Base Drift view shows the coordinates offsets from the user defined position along each direction correspondingly. It displays the plot of data stored since application started, but not more than for a day.

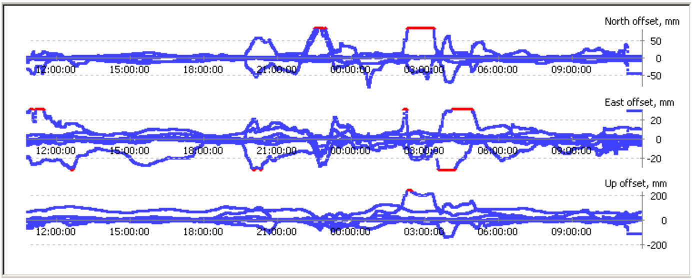


Figure 4-44. Base Drift View for All Reference Stations

The vertical axis represents the North, East or Vertical offset as estimated by engine. The horizontal axis represents the GPS time.

If none of the stations is selected in the Stations panel, the Base Drift view displays the offset variations for all the reference stations. If you

1. PRN is Pseudo-Random Noise that stands for the satellite identifier in the GPS system.

have a wheel mouse, place a cursor in the graph area and change the vertical scale by rotating the mouse wheel.

If a scale is too small and some of the peak values do not fit in, the corresponding section is flat and colored with red. Reduce the scale to see the graph entirely.

When selecting a reference station in the list of stations, the base drift plot changes. In this case it shows the coordinates drift of the selected reference station:

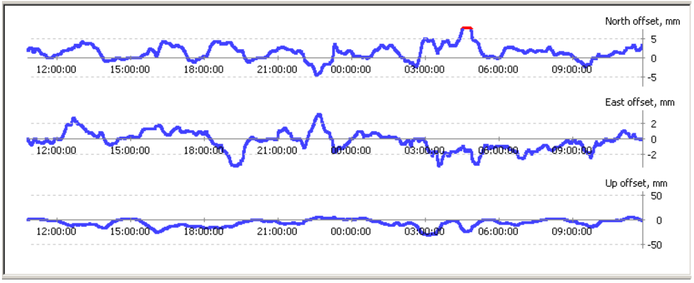


Figure 4-45. Base Drift View for One Reference station

On this screen the drift for one reference station is displayed. The vertical scale stands for the maximum offset of the selected reference station in the corresponding direction.

This view can be used for continuous monitoring of the difference between user defined position of the reference station and its estimation made by the TopNET engine. This difference can be produced either by measurement errors, or physical displacement of the station.

Settings

Services

TopNET-V provides RTK correction data to the clients through the internet using the TCP/IP protocol.

Each service type is provided through one of the TCP/IP ports. By default, the services are provided through the following ports (the default formats for each service are marked with asterisk):

Table 4-4. Default ports for the provided services

Service	Port	Available Formats
DGPS	8204	*RTCM 2.3 DGPS
RTK	8205	CMR CMR+ *RTCM 2.3 RTK RTCM 3.0
Network RTK	8207	CMR CMR+ *RTCM 2.3 RTK RTCM 3.0

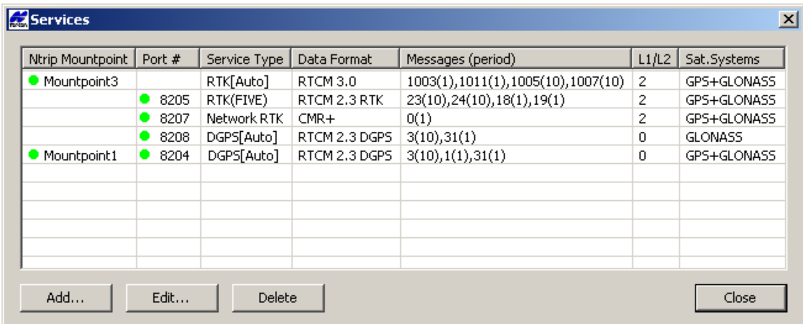
The Message Types for the formats depend upon the settings in the **Service Details** screen (Figure 4-49 on page 4-50).

The TopNET-V application can also operate as an Ntrip caster and broadcast the information using the Ntrip protocol.

To change the default services settings, select **Setup ▸ Services** menu.

Managing Services

1. Click **Setup ▸ Services** menu (Figure 4-46).



Ntrip Mountpoint	Port #	Service Type	Data Format	Messages (period)	L1/L2	Sat.Systems
Mountpoint3		RTK[Auto]	RTCM 3.0	1003(1),1011(1),1005(10),1007(10)	2	GPS+GLONASS
	8205	RTK(FIVE)	RTCM 2.3 RTK	23(10),24(10),18(1),19(1)	2	GPS+GLONASS
	8207	Network RTK	CMR+	0(1)	2	GPS+GLONASS
	8208	DGPS[Auto]	RTCM 2.3 DGPS	3(10),31(1)	0	GLONASS
Mountpoint1	8204	DGPS[Auto]	RTCM 2.3 DGPS	3(10),1(1),31(1)	0	GPS+GLONASS

Figure 4-46. Services

The Services screen displays a table of available services and their features:

- Ntrip Mountpoint (if available);
- Port # (if available);
- Service Type - shows the Service Type with a note nearby. An [Auto] note appears if no specific reference station is assigned to the service, otherwise a name of the station in parentheses is shown, for example, *RTK(FIVE)*;
- Data Format;
- Messages and their periods (in seconds);
- L1/L2 usage parameter:
  "0" corresponds to a *Code Only* case,
  "1" - to *L1 Code&Phase*,
  "2" - to *L1&L2 Code&Phase;*
- Satellite System.

2. To sort the table by any column, press its header.
3. To delete any service, highlight it or select several of them holding the Shift key, and press the **Delete** button.

4. To create a new service, press **Add**, to edit an existing service, press **Edit**. **Service type and data format** screen is opened (Figure 4-47).

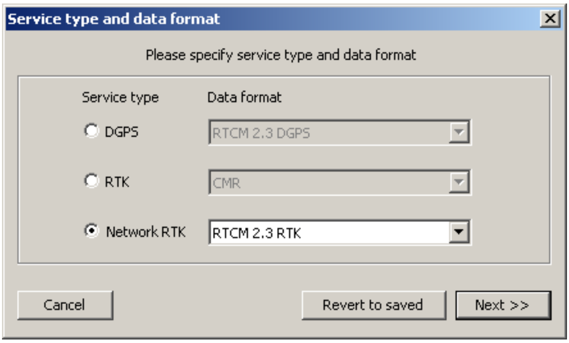


Figure 4-47. Service Type and Data Format

5. Select the desired **Service type** and an appropriate **Data format** from the list of available formats. Press **Next**.
6. If an RTK Service Type selected on the previous step, a **Select Reference Station** screen is displayed. Here you can set if the chosen service will be provided through a reference station closest to a rover, or only through the selected one.

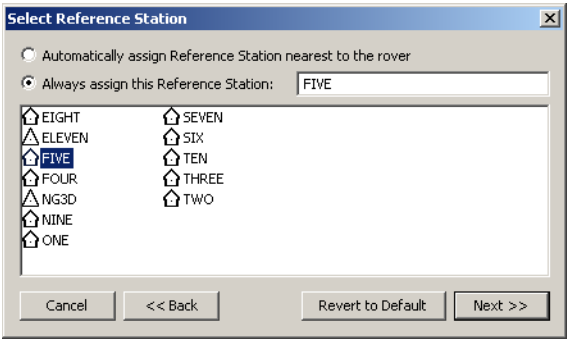


Figure 4-48. Select Reference Station

Press **Next**.

7. Select the desired satellite systems in the **Service Details** screen (Figure 4-49).

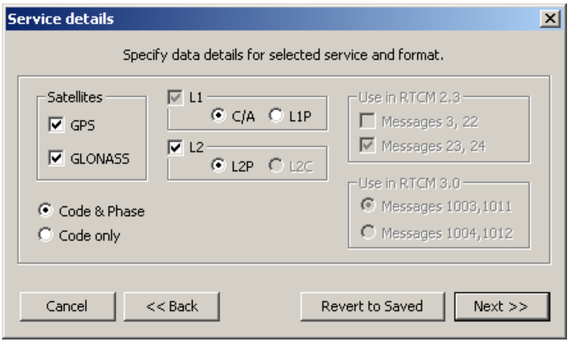


Figure 4-49. Service Details

Check the code and carrier phase settings:

- Normally, both Code & Phase are used, but for special applications a code-only data stream may be needed.
- If necessary, change the code type for the L1 frequency.
- By default, the L2 frequency is enabled (only L2P code type is available for now). Remove the checkmark if desired.

Depending upon the selected service and data format, you can select the Message Types to be transmitted:

- The data for DGPS sent in RTCM 2.3 format contain the following Message Types:
  - Message Type 1: GPS corrections (if GPS enabled);
  - Message Type 3: Antenna L1 phase center coordinate accurate to the nearest centimeter;
  - Message Type 31: GLONASS code corrections (if GLONASS enabled).
- The data sent for RTK in RTCM 2.3 format contains one of the following combination of Message Types:
  - 18 (phase), 19 (pseudorange), 23 (antenna type) and 24 (antenna ARP position);

– 18, 19, 3 (L1 phase center) and 22 (correction to position sent in Message Type 3);

– 18, 19, 3, 22, 23, 24.

- The data sent in RTCM 3.0 format contains a combination of the following Message Types:
  - – Type 1001 Message: GPS L1 code and phase;
  - – Type 1003 Message: GPS L1&L2 code and phase;
  - – Type 1004 Message: GPS L1&L2 code, phase and L1 pseudorange ambiguity;
  - – Type 1005 Message: antenna ARP position;
  - – Type 1007 Message: antenna type;
  - – Type 1009 Message: GLONASS L1 code and phase;
  - – Type 1011 Message: GLONASS L1&L2 code and phase;
  - – Type 1012 Message: GLONASS L1&L2 code, phase and L1 pseudorange ambiguity.

NOTICE *NOTICE*

The difference between 1003 and 1004 and between 1011 and 1012 is in "Integer L1 Pseudorange Modulus Ambiguity" which is not required for RTK operations but may be useful if a client wants to restore full code and phase measurements from RTCM 3.0 data.

If the L2 carrier frequency is disabled, the transferred Message Types are predefined and cannot be changed. If L2 is activated, the additional selection for Message Types is available. (For details see Table 4-5).

By enabling and disabling the corresponding fields, you can select between the following Message Types combinations:

Table 4-5. RTCM Message Types

Selected options	RTCM 3.0 Message Types
GPS is enabled, *GLONASS* is disabled, *L2* is disabled *Use for RTCM 3.0* box is disabled	1005, 1007, 1001
GPS is disabled, *GLONASS* is enabled, *L2* is disabled	1005, 1007, 1009
GPS is enabled, *GLONASS* is enabled, *L2* is disabled	1005, 1007, 1001, 1009
GPS is enabled, *GLONASS* is disabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1003
GPS is disabled, *GLONASS* is enabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1011
GPS is enabled, *GLONASS* is enabled, *L2* is enabled, Messages 1003, 1011 are selected	1005, 1007, 1003, 1011
GPS is enabled, *GLONASS* is disabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1004
GPS is disabled, *GLONASS* is enabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1012
GPS is enabled, *GLONASS* is enabled, *L2* is enabled, Messages 1004, 1012 are selected	1005, 1007, 1004, 1012

An alternative way to make rover estimate ionosphere is to configure the rover by setting to zero the threshold distance to reference station. Use the following command:

/par/pos/pd/ionr,0

To discard all the changes, press the **Revert to Saved** button.

8. Press **Next**. In the **TCP/IP Port and Mountpoint** screen place the check box in the corresponding field to activate the desired access method.

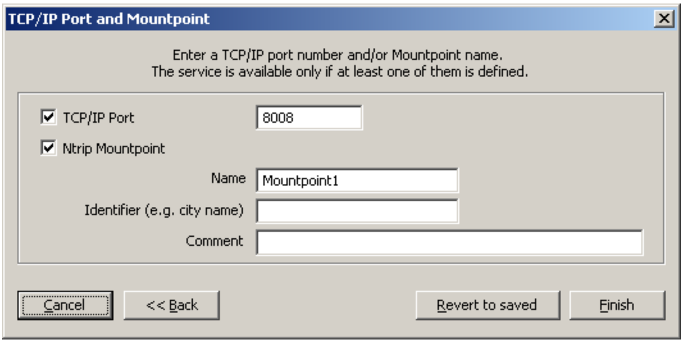


Figure 4-50. TCP/IP Port and Mountpoint

9. Set the number for TCP/IP Port and/or the parameters of the Mountpoint:
   - after you set the unique name of the mountpoint in the **Name** field, it will be available on the network;
   - you can also set a short description of the mountpoint in the **Identifier** field;
   - fill the **Comment** string - extended description of the mountpoint, if desired.

   All the other parameters of the connection set through the Ntrip server are available on the **Services** screen. See “Setting the Ntrip Access parameters” on page 4-54.
10. To discard all the changes, press the **Revert to Saved** button.
11. Click **Finish** to save settings.
    The added service will be displayed in the **Services** screen

(Figure 4-46 on page 4-48). If the service has an activated TCP/IP port and/or mountpoint, the green dot will appear in the corresponding column near the name of the mountpoint and/or number of the TCP/IP port. If the parameters of the TCP/IP port or mountpoint are inserted but the access method is not activated by the checkmark, the dot will be red.

CAUTION

Notify all TopNET-V field users about TCP/IP changes. They must update communication settings in order to ensure continued network access.

Setting the Ntrip Access parameters

1. To activate the function of broadcasting via Ntrip, click **Setup ▸ Ntrip Access...** menu, check the *Accept Ntrip Request* checkbox in the **Ntrip Settings** screen (Figure 4-51) and set a *TCP Port Number.*

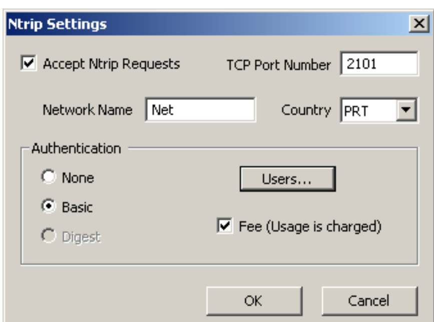


Figure 4-51. Ntrip Settings

2. Type in a **Network Name** and select the **Country** from the drop-down list.
3. In the **Authentication** field select the mode of authentication:
   - **None** - no password required;
   - **Basic** - login and password required. In this mode one can select the users that have access to the caster (Figure 4-52 on page 4-55).

- **Digest** mode is not implemented yet;

Place the **Fee** check mark if access to the caster is chargeable. This checkbox toggles "Fee" field of STR record in Ntrip Sourcetable between "N" and "Y". This field is used only to deliver this information to a rover operator.

Managing users of Ntrip caster

To control user access to the Ntrip caster, press the **Users...** button in the **Ntrip Settings** screen (Figure 4-51 on page 4-54).

The **Ntrip Users** screen displays the list of the users that are allowed to access the services grouped according to the access rights (Figure 4-52).

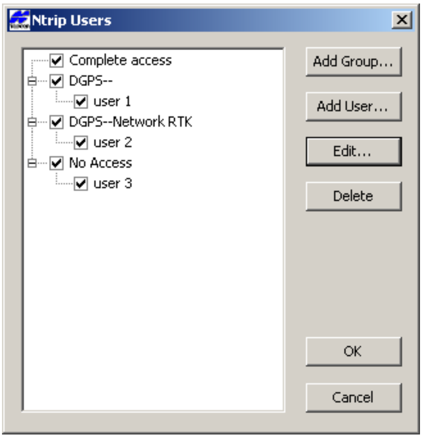


Figure 4-52. Ntrip Users

If you open the Ntrip Users screen for the first time and it is not empty, it means that you had a previous version of TopNET-V being installed, and the application has detected existing users when the first run.

To convert old user entries to the new format, the TopNET-V application grouped them according to their access rights for the mountpoints' service types. The groups created by default, are named after the services permitted to the users in the group, and there can be up to eight of them:

1. No access

2. DGPS--
3. DGPS-RTK-
4. DGPS--Network RTK
5. DGPS-RTK-Network RTK
6. -RTK-
7. -RTK-Network RTK
8. --Network RTK

All the users in one group have same access rights, so if you want to change the rights for one particular user, it is necessary to create a group with these rights and assign it to this user.

To create a new group:

1. Press **Add Group...** button in the **Ntrip Users** screen (Figure 4-52 on page 4-55).

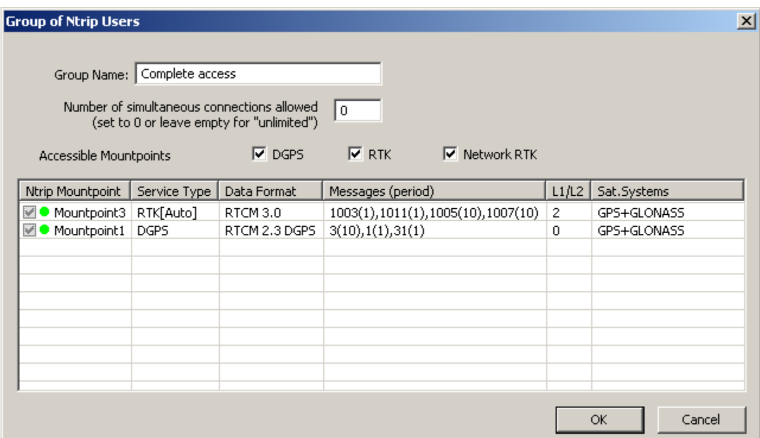


Figure 4-53. Group of Ntrip Users

2. In the **Group of Ntrip Users** screen (Figure 4-53) set the *Group Name* and the *Number of simultaneous connections* allowed for each member of this group. If no restrictions exist for the number of connections, set the corresponding field to zero or leave it blank.

NOTICE

All the users in one Group have the same allowed number of connections. If a set of users with different assigned numbers appeared to get into one group (during conversion from the older version of TopNET-V), this information will be lost.

For this Group, you can check a combination of "master" checkboxes labeled DGPS/RTK/Network RTK to enable/disable access to all existing and future mountpoints of corresponding Service Type.
The list of mountpoints shows the mountpoints' parameters and a color icon that displays if this mountpoint is available for the users assigned to the Group. Three sequentially toggled states of a checkbox at the line start are intended to manage a mountpoint's individual availability:

- ☑ The mountpoint is always available.
- ☐ The mountpoint is always not available.
- ☑ The mountpoint's availability corresponds to the state of the "master" checkbox at the top of the table. For example, a mountpoint operating with RTK service will be available only if the "master" RTK checkbox is enabled.

3. Press **OK**. A new Group appears in the **Ntrip Users** screen.

To add a user to Users list:

1. In the **Ntrip Users** screen (Figure 4-52) press the **Add User...** button. An **Ntrip User** screen appears

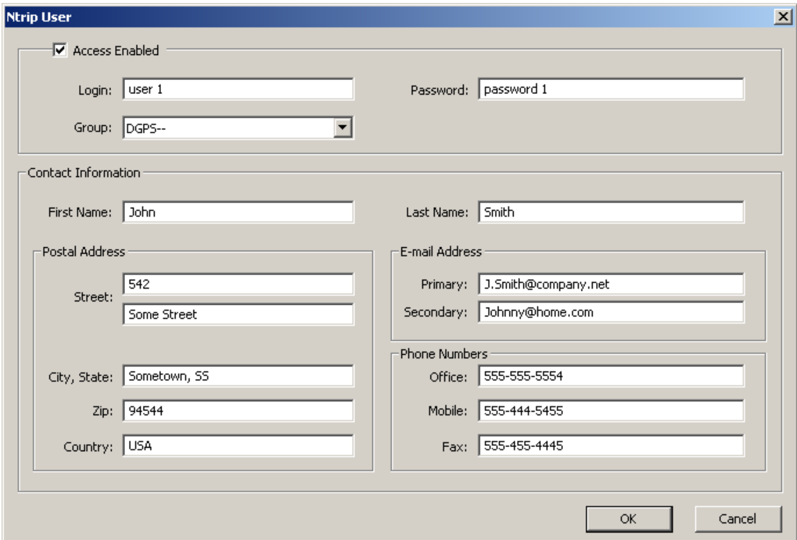


Figure 4-54. Ntrip User

2. Insert Login and Password for the user.
3. Select a group of access rights.
4. Input personal information of the user.
5. Press **OK**. The user will appear in the list (Figure 4-52). A checkbox near user's name allows the administrator to deny access to this user. This checkmark is duplicated on the **Ntrip User** screen as an *Access Enabled* checkmark.
   To deny access to a whole group of users, remove the corresponding checkmark near the *Group Name* in the **Ntrip Users** screen (Figure 4-52).
6. Press **OK** in the **Ntrip Users** screen (Figure 4-52 on page 4-55) to save the changes.

To edit User's Properties:

1. In the **Ntrip Users** screen (Figure 4-52) press the **Edit** button.

2. In the **Ntrip User** screen (Figure 4-54) insert all the necessary changes.
3. Press **OK**. The changes will reflect in the list (Figure 4-52).
4. Press **OK** in the **Ntrip Users** screen (Figure 4-52 on page 4-55) to save the changes.

To delete a user from the list:

1. In the **Ntrip Users** screen (Figure 4-52) select the user to be deleted.
2. Press the **Delete** button. The user will disappear from the list.
3. Press **OK** to save the changes, or **Cancel** to discard.

Setting Web connection parameters

If a local machine running a TopNET-V application has an Apache Tomcat server and TopNET web-application installed[1], it is possible to connect to the TopNET-V with the help of any web-browser and observe the state of the network.

The option of *Web Access* states the possibility of anonymous access through the web and/or a list of users authenticated for connection.

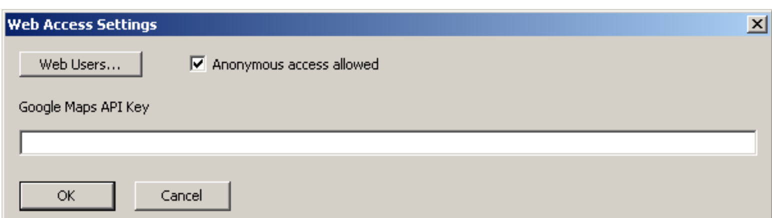


Figure 4-55. Web Access Settings

To allow Anonymous access, select **Setup ▸ Web Access...** menu and place a checkmark in the corresponding field of the **Web Access Settings** screen. The anonymous user will be able to observe the state of the network, including reference stations and ordinary rovers, excluding Ntrip rovers.

1. How to install an Apache Tomcat server and setup web access for users, is described in Appendix VIII “TopNET Web Interface Setup” .

To set a list of authorized users, press the **Web Users** button.

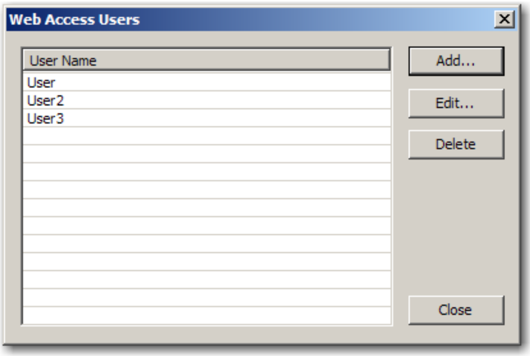


Figure 4-56. Web Access Users

The **Web Access Users** screen shows the list of users allowed to observe the TopNET-V application main screen through the internet. Web interface of TopNET-V provides users with a view on reference stations, ordinary users and some of Ntrip users. The list of Ntrip users available for observation can be set through the **Web Access User Data** screen for each internet user individually upon creation, or later.

To add an internet user for Web Access:

1. Press the **Add...** button.

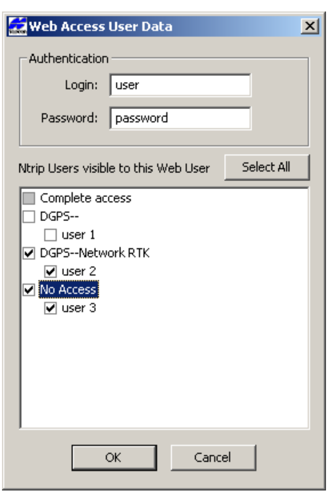


Figure 4-57. Web Access User Data

2. In the **Web Access User Data** dialog box, type in *User name* and *Password*.
3. Place check marks near those Ntrip users whose rovers are visible to the internet user (the list of Ntrip users is similar to that created in “Managing users of Ntrip caster” on page 4-55). To check all entries, press the **Select All** button. If all the entries are selected, the button changes its name to **Unselect All**. Press it to clear all check marks.

TIP

If you need to remove all checkmarks, but not all entries are selected and the button looks like Select All, press it twice.

4. Press **OK**.

To edit parameters of an existing user, select it in the **Web Access Users** screen list and press **Edit**. Make all the necessary changes in the **Web Access User Data** screen.

The *Google Map API* field stores the Google Map API key that is necessary for the web server operation. For the detailed description of web server installation, see Appendix VIII.

For a user to observe the state of the network, provide him or her with the address of the server. Usually, it looks like: http://<server name>:<port number>/TopNETweb, i.e. `http://net-s:1111/TopNETweb`.

The web page looks as follows:

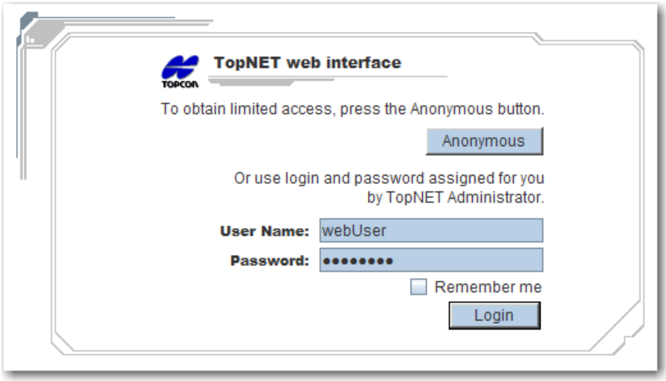


Figure 4-58. Web Interface Welcome Screen

To start working with Topcon web interface one should either press the **Anonymous** button and, if the anonymous access is allowed, get to page, or use login and password assigned by system administrator and press **Login**. These are the login and password created with the help of the dialog box shown on Figure 4-57 on page 4-60.

To make system remember your Login and Password, check the corresponding check box.

The page with Topcon Web Interface looks, for example, as follows:

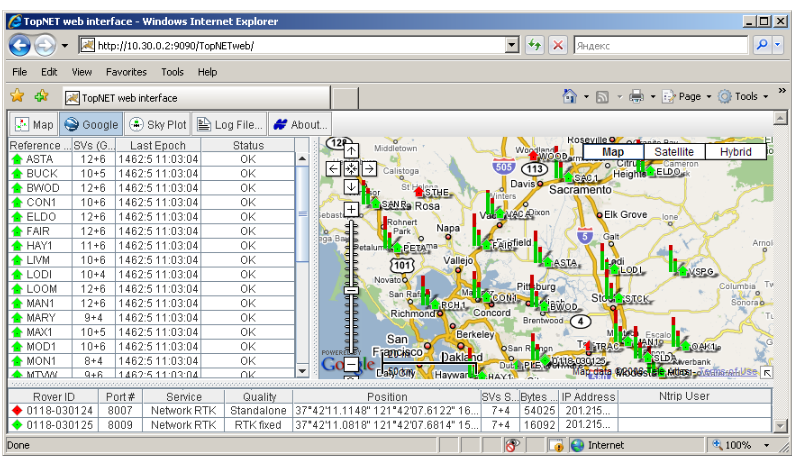


Figure 4-59. TopNET Web Interface

Allowed IP Addresses

To obtain access to the services provided by TopNET-V, the IP address of the rover should be inserted in the list of allowed IP addresses.

1. Click **Setup ▸ Allowed IP Addresses**.

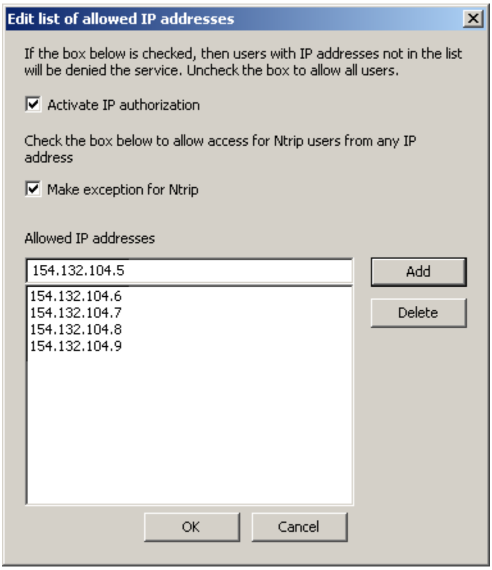


Figure 4-60. Edit List of Allowed IP Addresses

2. To activate the IP authorization, enable the **Activate IP Authorization** check box. In this case the services are provided to those users only whose IP Addresses are listed in the dialog box. If the check box is empty, no restrictions are applied to the users.
3. If necessary, place check mark in the **Make exception for Ntrip** check box. In this case no restrictions are applied to the users of Ntrip caster.
4. Type the IP address in the upper *Allowed IP addresses* field and click **Add**. The IP address appears in the lower field.
5. Repeat step 4 until all addresses have been entered.
6. Click **OK** to save the settings.

7. To remove the address from the list, select it (the chosen address will appear in the upper field) and click **Delete**.

Preferences

To set the preferred settings for the restrictions on reference stations usage, for time periods and timeouts, and for troubleshooting log files, use the **Preferences** screen.

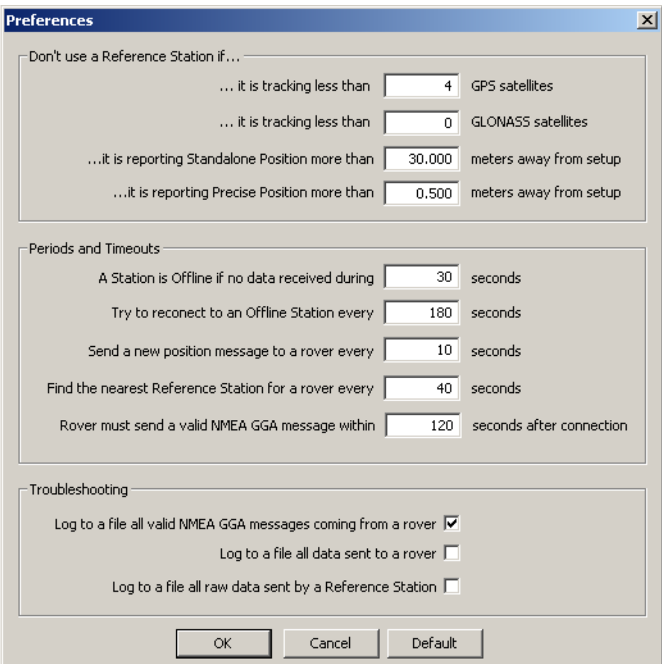


Figure 4-61. Preferences

1. Click **Setup ▸ Preferences**.
2. Insert the conditions on the reference stations usage:
   - the minimum number of GPS satellites being tracked;
   - the minimum number of GLONASS satellites being tracked;
   - the maximum distance of the Standalone Position from the user defined position of the reference station. To switch off

this validity check, enter a large value, i.e. 100 000 m. The Standalone Position can be transmitted by:

– a Topcon reference station as a [PO] message,

– any receiver connected via Ntrip caster as RTCM 2.3 or RTCM 3.0 Message, or

– by a TCP/IP receiver as RTCM 2.3 or RTCM 3.0 Message;

- the maximum distance of the Precise Position from the user defined position of the reference station. The Precise Position is transferred sometimes by Ntrip and TCP/IP reference stations in the RTCM 2.3 Type 24 Message, or in the RTCM 3.0 Type 1005 or 1006 Message (X, Y, Z - the precise coordinates of the ARP). If the obtained RTCM message does not contain mentioned Message Types, this field is not taken into consideration.

3. Insert the values for periods and timeouts:
   - the maximum time period of not-receiving data from the reference station for considering the station to be Offline;
   - the reconnection period for an Offline station;
   - the period of sending of new reference station position messages to the rover;
   - the period of searching for the nearest reference station for a rover;
   - the maximum time period of waiting for a valid NMEA GGA message from a Rover that has just connected.
4. Check the check boxes for troubleshooting:
   - to log all valid NMEA GGA messages received from a rover;
   - to log all data sent to a rover;
   - to log all data sent by a reference station.
5. Click **Default** to return to default values.
6. Click **OK** to save the changes.

Alarms

An alarm is used for notification about some events that may request reaction from the customer. The notification is sent by e-mail.

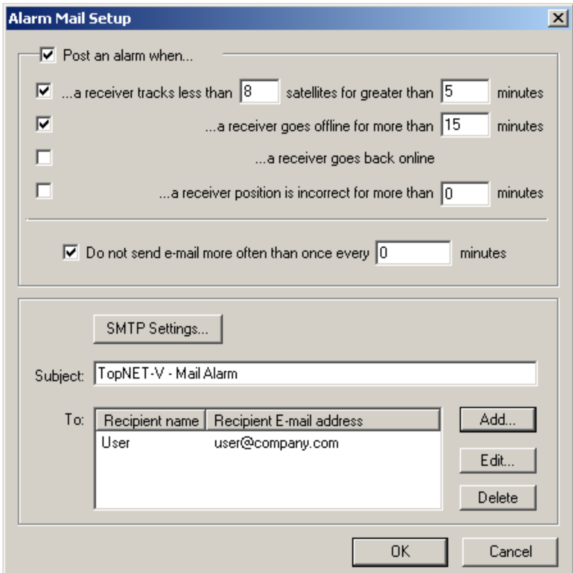


Figure 4-62. Alarm Mail Setup

The **Alarm Mail Setup** screen consists of two parts.

The first part of the screen (Figure 4-62) helps to select the contents of the alarm mail. To make application send notification e-mails, select the **Post an alarm when...** check box.

This tool can be used for notification about:

- lack of the satellites being tracked, for a considerable period of time. Place a checkmark in the corresponding field and set the minimum number of satellites and a maximum admissible tracking period of a less number of satellites.
- a receiver is offline for an extended period of time. Place a checkmark in the corresponding field and set the idle period for a receiver being offline.

- a receiver goes back online. If the previous check box is selected, this field becomes active. It allows you to get notifications about receivers coming back from the offline state.
- a receiver position is incorrect for a noticeable period of time. Place a checkmark in the corresponding field and set maximum value for the period. The position is considered to be incorrect if the distance between the Precise Position of the reference station and its position defined by the user, exceeds the value set in the **Preferences** screen (Figure 4-61 on page 4-64).

To make the application generate error reports once at a time, i.e. 45 minutes, place a checkmark in the **Do not send e-mail more often than once every ... minutes** field and set the desired period. If there is no checkmark, the notification alarms are sent as soon as event occurs.

The second part of the **Alarm Mail Setup** (Figure 4-62) screen includes common parameters of the sent e-mails:

- **SMTP Settings** button - opens a SMTP Settings screen, where the parameters of the SMTP server and sender's details can be configured:

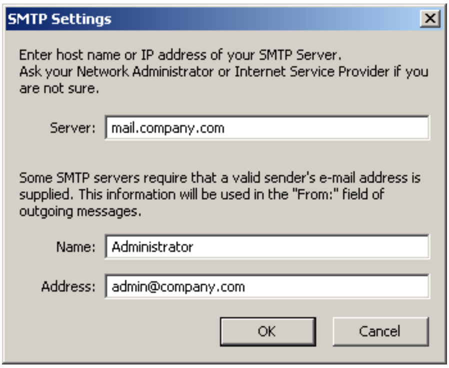


Figure 4-63. SMTP Settings

This module allows a user to send alarm messages, even if no mail client is installed on the PC.
Press **OK** to save the changes and return to the **Alarm Mail Setup** screen.

- Subject - the automatic subject of all the e-mails being sent. You can change it, if desired;
- To - the mailing list. You can add an address to the list with the help of the **Add** button, or the **Address Book** button. In the latter case, an Address Book of your mailer is shown and you can select address from it. Using the **Add** button, you need to type in user name and address manually, as shown in Figure 4-64.

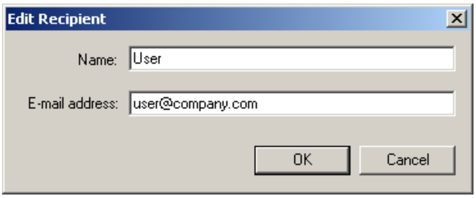


Figure 4-64. Edit Recipient

To make changes in the address entry, select an existing address and press **Edit** button (Figure 4-62) and call the **Edit Recipient** screen (Figure 4-64) again. You can change only one address at a time.

To delete one or several addresses, select the address (if you need to delete several addresses, select them using Shift), and press **Delete** (Figure 4-62). Confirm your decision.

All the generated error reports are displayed in Event log of the main screen and saved in the Log file.

Locating Log Files

While operation, the TopNET-V application produces a set of log files containing information about performed actions, received responses and occurring situations. A new set of log files is created every midnight. To set a specific location for the log files being created, click **Setup ▸ Log Files** menu.

In the **Select folders for log files** screen, set the default location for the log files. Press ... to select a folder on your PC (Figure 4-65).

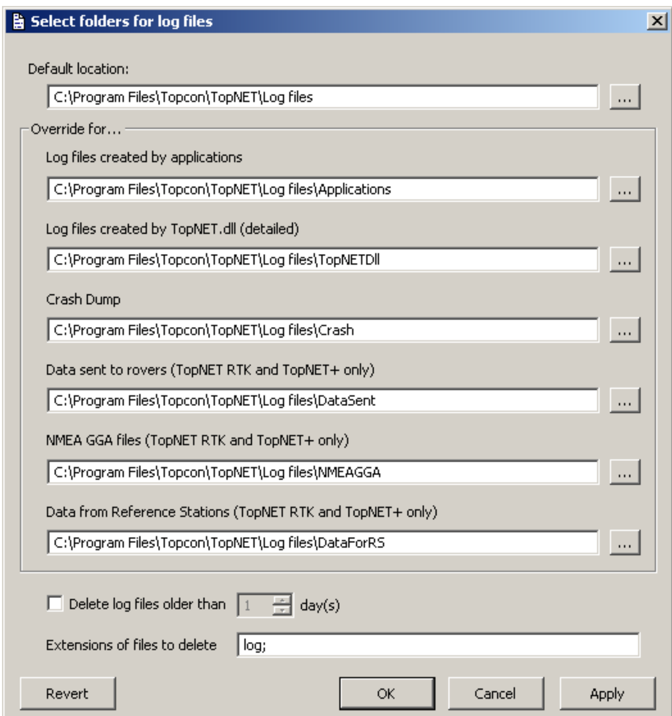


Figure 4-65. Select Folders for Log Files

To select specific folders for different types of log files, enter the overriding locations for the desired log file type:

- Log files created by applications;
- Log files created by TopNET.dll;

- Crash Dump;
- Data sent to rovers;
- NMEA GGA files;
- Data from Reference Stations.

Any field of the **Select folders for log files** screen may display the path to existing folder, to non-existing folder that will be created at the moment of the first corresponding log file being written.

In order not to store large amount of log files, check the **Delete log files older than** field and select the number of days to store the files.

Specify the extensions of the files to be deleted by typing them in the **Extensions of files to delete** field, divided by semicolons.

To save the data, press **Apply**.

To save the data and exit, press **OK**. As soon as a new portion of data will be stored to a log file, the new settings are used.

To revert to the last saved data, press **Revert**.

To exit without saving data, press **Cancel**.

Hardware Key Usage

Any computer running TopNET requires a Hardware Protection Key. A Hardware Protection Key enabled for the program must be inserted into the appropriate USB port to run the TopNET + or TopNET CORS modules on the computer.

To view the Hardware Protection Key options, select the **Help ▸ Activated Options...** menu.

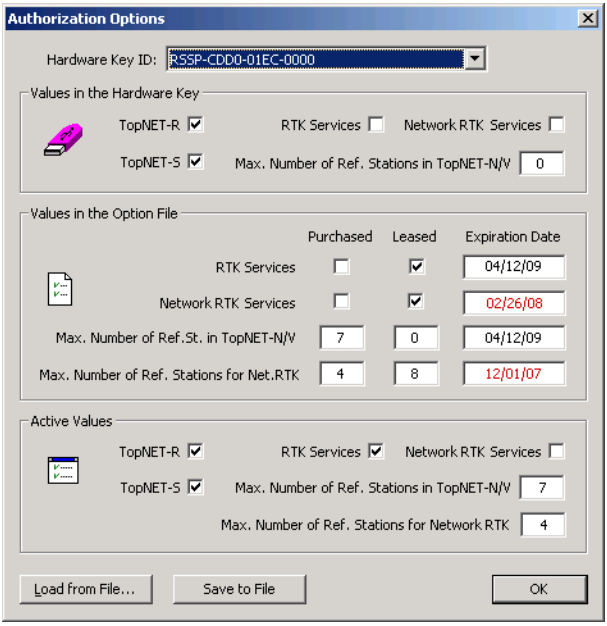


Figure 4-66. Authorization Options

TIP

Authorization Options is an information screen.
All the services and options can be changed only with the help of the Option File (see below).

- *Hardware Key ID* - the unique number of a Hardware Protection Key applied to the PC.
- *Values in the Hardware Key* - services and maximum number of reference stations permitted by the Hardware Key, are displayed;
- *Values in the Option File* - services and maximum number of reference stations permitted by the Option File, are displayed. Each option can be *Purchased* and/or *Leased*. The check mark designates the status of the option. The Expiration Date field shows the Lease expiration date for each option. If there is no Option file on the PC, the *Values in the Option File* field is empty.
- *Max. number of Ref. Stations for Network RTK* - defines the maximum number of reference stations that can be used for Network RTK operation.
  If the number of Reference Stations enabled for Network RTK exceeds the specified option value, the program will automatically disable some of the stations. In this case a message with warning pops up.
  This option is controlled by Topcon. Please contact Topcon to get an option file approving the number of purchased or leased stations that you are going to use for Network RTK needs. This file has a name of the following type: RSSP-CDD0-XXXX-NRSV.opt where XXXX is a unique identifier of the hardware key for which the option file is generated. This file must be placed in TopNET-V executable files folder (typically "C:\Program Files\Topcon\TopNET").

NOTICE *NOTICE*

Do not use the Load from File *button in the* Activated Options *dialog box to operate with this additional option file. Application will attach it automatically.*

- *Active Values* - the current values of the options. These include the maximum of the values displayed in the fields above, including Leased options that have not expired. (If there is no Option File on the PC, the Active Values field duplicates the Values in the Hardware Key field.)

So, in the example shown above, the following values are set (assuming today is March, 1, 2008):

– TopNET-R option is available, as defined by the Hardware Key;

– TopNET-S option is available, as defined by the Hardware Key;

– RTK Services are available, as defined by the Option File leased option;

– Network RTK Services are NOT available, because the lease of corresponding option has expired;

– Max. Number of Ref.Stations in TopNET-N/V is 7, as defined by the Option File purchased option.

– Max. Number of Ref.Stations for Network RTK is 4, as defined by the Option file purchased option. The possibility to use 8 reference stations for the Network RTK has expired.

NOTICE *NOTICE*

RTK Services option includes Ntrip Caster functionality. If RTK Services are disabled, by options, rovers will not be able to connect to Ntrip TCP/IP port.

If no Hardware Protection Key is applied to the PC, RTK services will be available for an hour after application startup. Then the services will be disabled and the new rovers will not be able to connect. To enable services again, apply the Hardware Protection Key and restart the application. Network RTK services will not be available.

To purchase more services:

- press the **Save to File** button in the **Authorization Options** screen. The option file will be saved. An announcement screen will display the location of the option file on your PC. The name of the file is the same as the number of the Hardware Protection Key with *.opt extension;

- open the location and send the corresponding Option File to your dealer along with your order;
- receive the new Option File from the dealer;
- save it to the same location;
- in the **Authorization Options** screen select the necessary Hardware Protection Key;
- press the **Load from file** button and select the location of the Option File;
- press **OK**, the Option File will be activated;
- press **OK** in the **Authorization Options** screen to save the changes and close the screen;
- restart the application.

If more than one Hardware Protection Key are applied to the PC, the **Hardware Key ID** field (Figure 4-66 on page 4-71) shows a drop-down list of available keys.

Each key has its own Option File, so the **Values in Hardware Key** and **Values in Option File** fields (Figure 4-66) correspond to the currently selected key. The **Active Values** field in this case stays the same for all the keys. It shows the maximum of all the values displayed in the fields above for all the available keys.

NOTICE *NOTICE*

Not more than four Hardware Protection Keys should be applied to one PC.

Appendices

List of Appendices:

Configuring a TPS Receiver

The following sections describe preliminary configurations for a TPS receiver to be used with the TopNET-S server.

Configuring the Ethernet Port of a TPS Receiver

To enter the settings for the Ethernet port of the TPS receiver use the PC-CDU software (version 2.1.11 or higher).

1. Connect the computer to the TPS receiver through either the serial or USB port. Run PC-CDU.
2. On the **Connections Settings** dialog box, select the appropriate communications settings (Figure I-1) and click **Connect**.

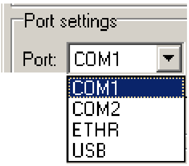


Figure I-1. Select Communication Setting

3. Once PC-CDU connects, click **Configuration ▸ Receiver**.

4. Select the *Ports* tab then the *Ethernet* tab and specify the IP settings for the receiver (Figure I-2).

   If needed, contact your system administrator for the network that includes the TPS receiver to get any unknown settings.

Figure I-2. IP Settings

NOTICE

The receiver must be accessible from the computer running a TopNET-S server. If this computer and the receiver belong to different Local Area Networks (LAN), the receiver's IP should be accessible from outside the receiver's LAN.

Consult your receiver's LAN system administrator about IP addresses. Additional actions may need to be taken, such as configuring Network Address Translation (NAT) service for the LAN, or reserving a direct static IP for the receiver.

5. Configure Telnet settings for the receiver (Figure I-3):
   - TCP Port – 8002 is recommended.
   - Timeout – 600 (in seconds) is recommended.
   - Enter a password for the receiver. Passwords are case sensitive.

Figure I-3. Telnet Settings

6. Configure FTP settings for the receiver (Figure I-4):
   - TCP Port – 21 is recommended
   - Timeout – 600 (in seconds) is recommended

Figure I-4. FTP Settings

7. Click **Apply** to confirm the settings, then **OK** to close the dialog box.
8. Click **Tools** ▸ **Reset receiver** to reset the receiver (Figure I-5).

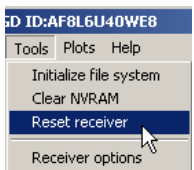


Figure I-5. Reset Receiver

9. Click **OK** at the prompt.

Configuring the Internal GSM Modem

1. Connect the computer to the TPS receiver through either the serial or USB port. Run PC-CDU.
2. On the **Connections Settings** dialog box, select the appropriate communications settings (Figure I-1) and click **Connect**.

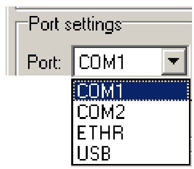


Figure I-6. Select Communication Setting

3. Once PC-CDU connects, click **File ▸ Manual Mode**.
4. On the **Manual Mode** dialog box, enter the commands listed in Table I-1. Press **Enter** or click **Send command** after each command.

Table I-1. Manual Mode Commands

Command	Description
%% set,/par/modem/c/mode,off	sets OFF mode
%% set,/par/modem/c/pin,6492	enters PIN code where 6492 represents the PIN code of your SIM card[a]
%% set,/par/modem/c/rcvtime,5	enters the “wait on data from modem” timeout, in our case 5 seconds
%% set,/par/modem/c/sndtime,2	enters the service word repeat period
%% set,/par/modem/c/mode,slave	sets SLAVE mode for the receiver[b]

a. Assuming a modem is connected to port C of the TPS receiver.
b. In ‘Slave’ mode, the receiver will wait for incoming calls.

When finished, five responses should display on the **Manual Mode** dialog box (Figure I-7 on page I-5).

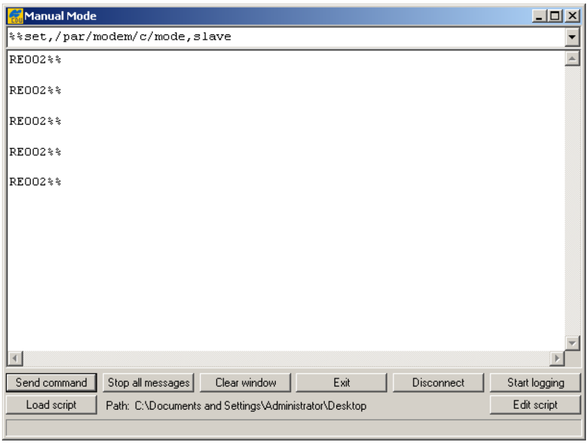


Figure I-7. Manual Mode Responses

5. To check entered parameters, send the commands listed in Table I-2.

Table I-2. Manual Mode Commands

Command	Response	Description
print,/par/modem/c/mode	RE006 slave	mode “slave” is set for the receiver’s port (‘C’ in this case).
print,/par/modem/c/pin	RE006“6492”	PIN code of your SIM card is set.
print,/par/modem/c/state	RE006 detect RE006 ready	detecting the attached modem modem has been initialized and registered in the GSM network. It is ready to receive incoming calls.
print,/par/modem/c:on (continued on next page)	RE04C%%/par/modem/c={mode=slave,port=/dev/ser/c,pin=“6492”,dial=“not applicable”,	The parameter ‘**err**’ describes the last of the errors detected/identified by the modem management. This parameter can have the following values: – “**none**”: No errors have been detected – “**not detected**”: Cannot find a modem on the port

Table I-2. Manual Mode Commands (Continued)

Command	Response	Description
print,/par/modem/c:on	RE02F%%state=ready, err=none,rcvtimeout=5 ,sndtime=2}	– "**init error**": An initialization error has occurred – "**wrong pin**": Wrong PIN code – "**net error**": A network error has occurred – "**no carrier**"': Cannot detect the carrier signal. This "temporary" error can occur if the second modem (at the other end of the radio link) has not been initialized or if there are some problems with the GSM network. The given GSM modem will continue to dial the modem on the other side of the radio link until the carrier signal is detected.

NOTICE

NOTICE

A GSM modem is ready to work ONLY if the response is "RE006 ready".

6. Click **Disconnect** to complete the configuration of the GSM modem built into the TPS receiver.

Appendices

Using MINTER to Check the Modem Status

Use the receiver's MINTER to check modem status.

Press the **FN** button twice to switch to the modem status blink mode:

- If the receiver has not established a connection with the modem, the STAT LED will blink red.
- If the modem is transmitting data, the STAT LED will blink yellow.
- If the modem is receiving data, the STAT LED will blink green.
- If the STAT LED blinks red for 1–2 minutes, turn the receiver off and then turn on (to enable the modem status blink mode, press the button **FN** twice). If the STAT LED continues to blink red, use an appropriate terminal to check the modem status (see step 5 on page I-5).

Notes:

Description of TPS2RIN Options

TPS2RIN is a Win32 console application that converts TPS data files into RINEX format.

In the *Options* field of the **RINEX translation setup** dialog box, additional options can be set for the converter. Table II-1 describes all valid options.

Usage:

```
TPS2RIN -i <input file name> [-o
<destination directory>] [<sw> [<sw>]]
```

With only the input file specified, three files will be produced, assuming the source TPS file has the required information: GPS and GLONASS navigation message files and an observation data file. The profile is a plain text Windows INI file and provides information to enter into RINEX files.

Table II-1. Converter Options

-i	Specify the source file
-h	Display help screen
-o	Specify the destination directory (default: current directory)
-p	Specify profile (.INI) file with additional information
-1	Exclude L1
-2	Exclude L2
-q	Quiet mode (no output to the screen)
-l	Write log file for TPS file (with the same name as the source file)

Table II-1. Converter Options (Continued)

-e	Write epoch statistics into the observation file
-s dd[hh[mm[ss]]]	Start date/time (two digits are required for all numbers): dd - day of the month hh - hour (default is 00) mm - minutes (default is 00) ss - seconds (default is 00) Example: -s 31 - 31-th, 00:00:00 -s 310001 - 31-th, 00:01:00 -s 02220106 - 2-nd 22:01:06
-f dd[hh[mm[ss]]]	End date/time: dd - day of the month hh - hour (default is 23) mm - minutes (default is 59) ss - seconds (default is 59) If only Start time is specified, the following rule is applied to get End time: -s dd End time = dd235959 -s ddhh End time = ddhh5959 -s ddhhmm End time = ddhhmm59
-G	Exclude GPS satellites
-g <nn>	Exclude GPS satellite number nn. To exclude more than one satellite repeat -g switch: -g 1 -g 22
-R	Exclude GLONASS satellites
-r <nn>	Exclude GLONASS satellite number nn. To exclude more than one satellite repeat -r switch: -r 1 -r 22
-M <name>	Force marker name to “name” (up to 20 characters). Overrides marker name contained in the TPS file and in the profile (if any). -M “My marker”

Appendices

Table II-1. Converter Options (Continued)

-m <text>	Force marker number to "text" (up to 20 characters).
-W	Exclude all WAAS satellites
-w <n>	Exclude specific WAAS satellite (with number nn). To exclude more than one satellite use this switch repeatedly (e.g., -w 20 -w 22 will exclude WAAS SVs with code numbers 120 and 122)
-I <ss.ss>	"Alternative" data decimation technique (by using a float parameter in the format SS.SS [in seconds])
-d <n>	n-fold data decimation
-L <n>	If Leap Seconds information is missing, force Leap Seconds to n (default = 15)
-~	Apply Smoothing to pseudoranges
-D	Exclude Doppler
-O	Include Receiver Clock Offset in epoch header lines
-S	Include observation types S1,S2. Units: dB*Hz
-A	Override Antenna type name found in profile or input file. If Type name contains spaces, take the whole name in quote marks (e.g., tps2rin -A "TPSPG_A1 TPSD"). Make sure to use correct number of whitespace symbols to separate antenna name from dome name.
-a	Override Antenna Serial # found in profile or input file.
-C2	Ignore L2C pseudorange (do not put C2 RINEX observable).

Table II-1. Converter Options (Continued)

-=	Ignore User Events. Conversion will ignore any _SIT, _ANT, _ANH and other user events. Also the [JP] message which normally breaks the file will be ignored. This switch will also force the program to accept files without (normally required) [JP] and [MF] messages.

Satellite Navigation Status Codes

Table III-1 lists the codes seen in the information pop-up screens of the SNR/Sky Plot screen.

Table III-1. Satellite Navigation Status Structure

Code	Description
00	L1 C/A data used for absolute position computation
01	L1 P data used for absolute position computation
02	L2 P data used for absolute position computation
03	Ionosphere-free combination used for absolute position computation
04	Measurements unavailable
05	Ephemeris unavailable
06	Unhealthy SV (as follows from operational (=ephemeris) SV health)
07	Time-Frequency parameters from the ephemeris data set may be wrong[a]
08	Initial conditions (position and velocity vectors) from the ephemeris data set may be wrong[a]
09	Almanac SV health indicator unavailable for this satellite[a]
10	Unhealthy SV [as follows from the almanac SV health indicator][a]
11	“Alert” flag (from the word “HOW”) is set[b]
12	URA indicates the absence of accuracy prediction for this SV[b]
13	User excluded this SV from position computation
14	User excluded this SV with this frequency channel number from position computation[a]
15	This SV is excluded from solution since its system number is unknown[a]
16	This SV has an elevation lower than the specified mask angle

Table III-1. Satellite Navigation Status Structure (Continued)

Code	Description
17	Unable to compute position with the given ephemeris data (for example, too many iterations have been produces for solving Kepler's equation).
18	Ephemeris data is too old
19	This SV does not belong to the constellation the user has selected
20	Differential data from Base Station unavailable for given satellite (applicable only when receiver runs in DGPS)
21	Reserved
22	RAIM has detected wrong measurements
23	SNR below specified minimum level
24, 25	Reserved
26	DLL not settled
27	Ionospheric corrections are not received from Base Station
28	Coarse code outlier (hundreds of meters or more) has been detected
29	Ephemeris index unavailable (no ephemeris information; similar to code 05)
30	This SV is not used in RTK processing (similar to code 20 but is used specifically for RTK)
31	This SV is excluded from RTK processing because a wrong combination of measurements has been detected
32	This SV is excluded from OmniStar HP position computation
33-50	Reserved
51	L1 C/A slot is used in RTK processing
52	L1 P slot is used in RTK processing
53	L2 P slot is used in RTK processing
54	L1 P & L2 P measurements are used in RTK processing
55	L1 C/A & L2 P measurements are used in RTK processing
56	This SV is used for OmniStar HP position computation
57-62	Reserved
63	Satellite navigation status is undefined

a. GLONASS only
b. GPS only

Satellite Health Codes

Table IV-1 lists the codes seen in the information pop-up screens of the SNR/Sky Plot screen according to the Interface Control Documents (ICD) of both satellite systems.

Table IV-1. Satellite Navigation Status Structure

Code	GPS	GLONASS
00	All Signals OK[1]	• Cn[2] word indicates malfunction of the satellite.
01	All Signals Weak	• Cn word indicates malfunction of the satellite; • Bn word indicates that satellite is unhealthy.
02	All Signals Dead	• Cn word indicates malfunction of the satellite; • Time-frequency parameters may be wrong.
03	All Signals Have No Data Modulation	• Cn word indicates malfunction of the satellite; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy.
04	L1 P Signal Weak	• Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong
05	L1 P Signal Dead	• Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong; • Bn word indicates that satellite is unhealthy.

Table IV-1. Satellite Navigation Status Structure (Continued)

Code	GPS	GLONASS
06	L1 P Signal Has No Data Modulation	• Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong.
07	L2 P Signal Weak	• Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy.
08	L2 P Signal Dead	All signals OK. (This code does not appear on the SNR/Sky view pop-up screens)
09	L2 P Signal Has No Data Modulation	• Bn word indicates that satellite is unhealthy.
10	L1 C Signal Weak	• Time-frequency parameters may be wrong.
11	L1 C Signal Dead	• Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy.
12	L1 C Signal Has No Data Modulation	• Initial conditions of position and velocity may be wrong.
13	L1 C Signal Has No Data Modulation	• Initial conditions of position and velocity may be wrong; • Bn word indicates that satellite is unhealthy.
14	L2 C Signal Dead	• Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong.

Table IV-1. Satellite Navigation Status Structure (Continued)

Code	GPS	GLONASS
15	L2 C Signal Has No Data Modulation	• Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy
16	L1 & L2 P Signal Weak	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite;
17	L1 & L2 P Signal Dead	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Bn word indicates that satellite is unhealthy
18	L1 & L2 P Signal Has No Data Modulation	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Time-frequency parameters may be wrong.
19	L1 & L2 C Signal Weak	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy.
20	L1 & L2 C Signal Dead	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong.

Table IV-1. Satellite Navigation Status Structure (Continued)

Code	GPS	GLONASS
21	L1 & L2 C Signal Has No Data Modulation	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong. • Bn word indicates that satellite is unhealthy
22	L1 Signal Weak	• SV health status from almanac is available • Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong.
23	L1 Signal Dead	• SV health status from almanac is available; • Cn word indicates malfunction of the satellite; • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy
24	L1 Signal Has No Data Modulation	• SV health status from almanac is available.
25	L2 Signal Weak	• SV health status from almanac is available; • Bn word indicates that satellite is unhealthy
26	L2 Signal Dead	• SV health status from almanac is available • Time-frequency parameters may be wrong

Table IV-1. Satellite Navigation Status Structure (Continued)

Code	GPS	GLONASS
27	L2 Signal Has No Data Modulation	• SV health status from almanac is available; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy
28	SV Is Temporarily Out (Do not use this SV during current pass)**	• SV health status from almanac is available • Initial conditions of position and velocity may be wrong.
29	SV Will Be Temporarily Out (Use with caution)**	• SV health status from almanac is available; • Initial conditions of position and velocity may be wrong; • Bn word indicates that satellite is unhealthy
30	Spare	• SV health status from almanac is available • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong.
31	More Than One Combination Would Be Required To Describe Anomalies (Except those marked by **)	• SV health status from almanac is available • Initial conditions of position and velocity may be wrong; • Time-frequency parameters may be wrong; • Bn word indicates that satellite is unhealthy.

1. For details on GPS ICD, see http://www.navcen.uscg.gov/pubs/gps/icd200/icd200cw1234.pdf.
2. Words Cn and Bn are two different health flags. See GLONASS ICD (http://www.glonass-ianc.rsa.ru/i/glonass/ICD02_e.pdf) for details.

Notes:

Appendices

GPS-MET Integration

Purchase the a Broadband Digiquartz® MET3 or MET3A.

Connect the MET Station to your Topcon Odyssey receiver as shown below:

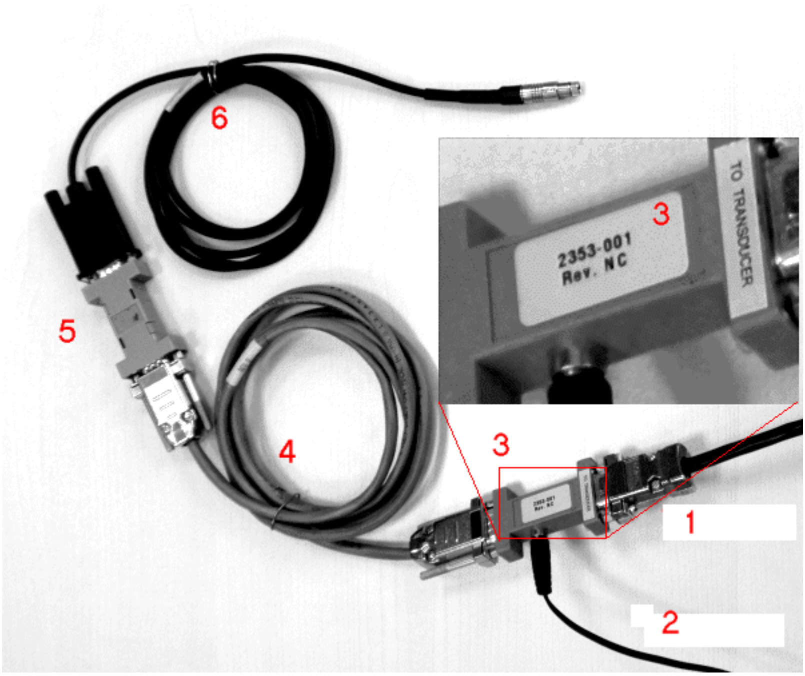


Figure V-1. Connection of the MET to an Odyssey Receiver

Here:

1. RS232 cable to MET (comes with MET);
2. Cable to 9V DC power adapter (comes with MET);
3. RS232+9V Mixer (comes with MET);
4. RS232 Male-female extension cable;
5. Male-male null adapter;

6. DB9-Fisher RS232 cable (comes with a Topcon receiver).
7. If you have a NET-G3 Topcon receiver, you don't need the cables #5 and #6. NET-G3 has three serial ports for direct RS232 connection on its back.

Figure V-2. Net-G3 Receiver Rear Panel

If any cables are missing from your package, please contact Topcon Positioning Systems, Inc. or Paroscientific, Inc.

Setting the GPS receiver

Once the GPS receiver and MET station are connected and powered up, the MET station is ready to respond to a P9 command issued from the GPS receiver. The Topcon GPS receiver must be configured to send a P9 command to the MET station.

1. Set the baud rate[2] to 9600 (by default, the baud rate of any receiver port is 115200) using a command:

```
%%set, dev/ser/c/rate, 9600
```

2. Instruct the receiver to sent a request to the MET device with a specific period (e.g., 60 seconds), if necessary. Period should not be less than 3 seconds, otherwise MET responses will be incomplete. By default, the period is 600 seconds.

```
%%em, dev/ser/c, misc/MET3: 60
```

3. This command sets the port of connection with MET and the period of sending commands.
   In a test mode, you can set the period to 5 sec to make sure that the TX and RX indicators on you MET3 device blink in an appropriate way.

4. Upon activation the `*0100P9<CR><LF>` command[3] will be issued by the GPS receiver to get a response from the MET station. A typical response to a P9 command is as below.

```
$WIXDR,P,<Pressure>,B,<SN>,C,<Temperature>,C,
<SN>,H,<Humidity>,P,<SN><CR><LF>
```

Transducer	Field	Units
Pressure	P	B=Bar
Temperature	C	C=Celsius
Humidity	H	P=Percent

---

2. Here and further the connection port is assumed to be the Serial C port. If you use another port, change the commands correspondingly.
3. The <CR> and <LF> correspond to the carriage return and line feed characters.

<SN> = Transducer Serial Number (Typically - DQ#####)

The MET station returns the pressure, temperature and humidity in standard NMEA[4] format.

5. To redirect all the MET responses to the current file A, use the following command:

```
%%set, dev/ser/c/echo, cur/file/a
```

Also you can redirect all the MET responses to a current file B or another port, if necessary.

6. To switch the Serial C port to echo mode, use the following command:

```
%%set, dev/ser/c/imode, echo
```

In this mode the incoming messages will not be treated as commands.

7. To enable the wrapped echo mode, use the following command:

```
%%set, dev/ser/c/ewrap, on
```

If the redirected information comes to a file along with GPS measurements and a file is supposed to be converted to a RINEX MET file using **tps2rin** application, the responses should be wrapped for correct conversion.

4. The NMEA 0183 (National Marine Electronics Association) Standard for Interfacing Marine Electronics Devices is a voluntary industry standard, first released in March of 1983. The NMEA has become a standard protocol for interfacing navigational devices such as GPS and DGPS receivers. It defines electrical signal requirements, data transmission protocol, timing and specific sentence formats.

All these parameters can be set with the help of Ports Parameters tool of the TopNET-R application.

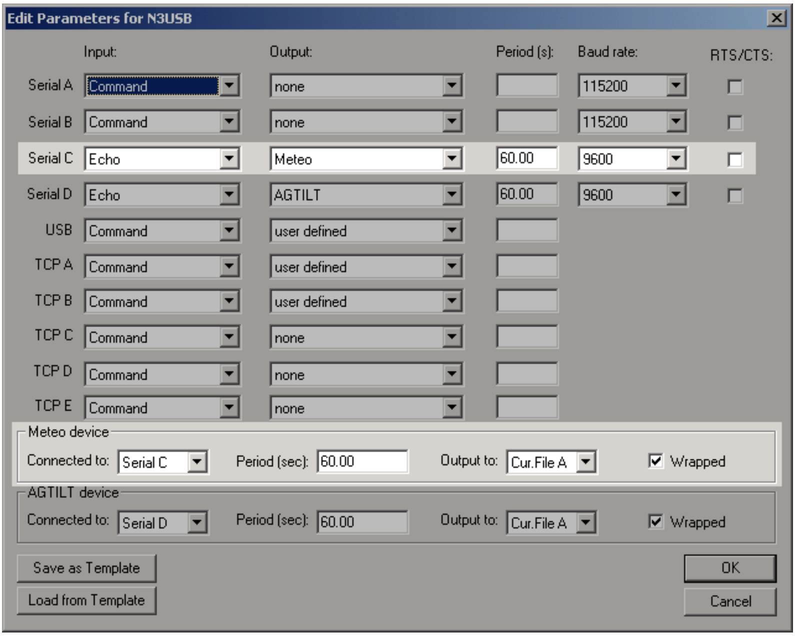


Figure V-3. Edit Parameters

All the parameters for MET (except Baud Rate) are set in the Meteo device field.

Notes:

GPS-AGTILT Integration

Purchase the Applied Geomechanics Digital Clinometer (AGTILT).

Connect AGTILT to any Serial port of Topcon Net-G3 receiver (Figure VI-2) using the Serial Interface of the AGTILT cable (included):

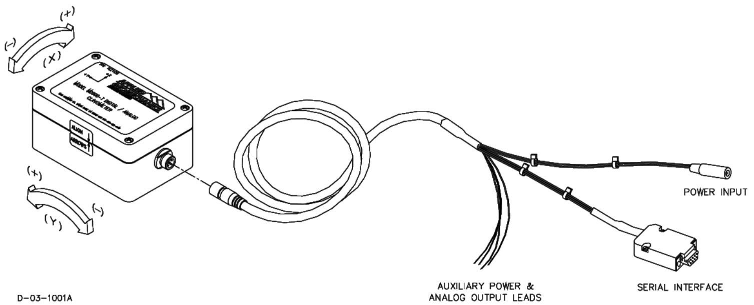


Figure VI-1. Model MD900-Y Digital Clinometer with P/N 84033 Cable Assembly

Figure VI-2. Net-G3 Receiver Rear Panel

Configuration of a GPS receiver to send an XY command to the AGTILT device

Once the GPS receiver and AGTILT device are connected and powered up, AGTILT is ready to respond to an XY command issued from the GPS receiver. The Topcon GPS receiver must be configured to send an XY command to the AGTILT device.

1. Set the baud rate of the port[5] to 9600 (by default, the baud rate of any receiver port is 115200) using the following command:

```
%%set, dev/ser/c/rate, 9600
```

2. Instruct the receiver to send a request to the AGTILT device with a specific period (e.g., 120 seconds), if necessary. By default, the period is 600 seconds.

```
%%em, dev/ser/c, misc/AGTILT: 120
```

   This command sets the port of connection with AGTILT and the period of sending commands.

Appendices

5. Here and further the connection port is assumed to be the Serial C port.

3. Upon activation, the *XY* command will be issued by the GPS receiver to get a response in standard NMEA 0183 format[6]:

$YXXDR,	A,	x.x,	D,	N,	A,	x.x,	D,	E,	C,	x.x,	C,	T,	-sn,	*hh
	Data Type, A=Angular	Y (N) - tilt value	Units, M=microradians, D=degrees	Comment, N for North/South (Y) direction	Data Type, A=Angular	X (E) - tilt value	Units, M=microradians, D=degrees	Comment, E for East/West (E) direction	Data Type, C=Temperature	Temperature of tiltmeter	Units, C=degrees C	Comment, T for Temperature	Serial number	Checksum

The line is terminated by carriage return and line feed characters.

4. To redirect all the AGTILT responses to the current file A, use the following command:

```
%%set, dev/ser/c/echo, cur/file/a
```

Also you can redirect all the AGTILT responses to a current file B or another port, if necessary.

5. To switch the Serial port to echo mode, use the following command:

```
%%set, dev/ser/c/imode, echo
```

In this mode the AGTILT responses to receiver will not be treated as commands.

6. To enable the wrapped echo mode, use the following command:

```
%%set, dev/ser/c/ewrap, on
```


---

6. The NMEA 0183 (National Marine Electronics Association) Standard for Interfacing Marine Electronics Devices is a voluntary industry standard, first released in March of 1983. The NMEA has become a standard protocol for interfacing navigational devices such as GPS and DGPS receivers. It defines electrical signal requirements, data transmission protocol, timing and specific sentence formats.

If the redirected information comes to a file along with GPS measurements and a file is supposed to be converted to a RINEX file using **tps2rin** application, the responses should be wrapped for correct conversion. RINEX does not provide standard format for TILT measurements, so tps2rin will create a “pseudo-RINEX” T-file, containing strings with time tags and NMEA messages of the following type:

```
07  7 30  0 12 30 $YXXDR,A,-00.850,D,N,A,-
00.310,D,E,C,028.051,C,T-N21313*55
```

A time tag comprises: the year (2 digits with leading zero), month (2 digits), day (2 digits), hour (2 digits), minute (2 digits) and second (2 digits) of the message arrival, separated by spaces.

All these parameters can be set with Ports Parameters tool of the TopNET-R application.

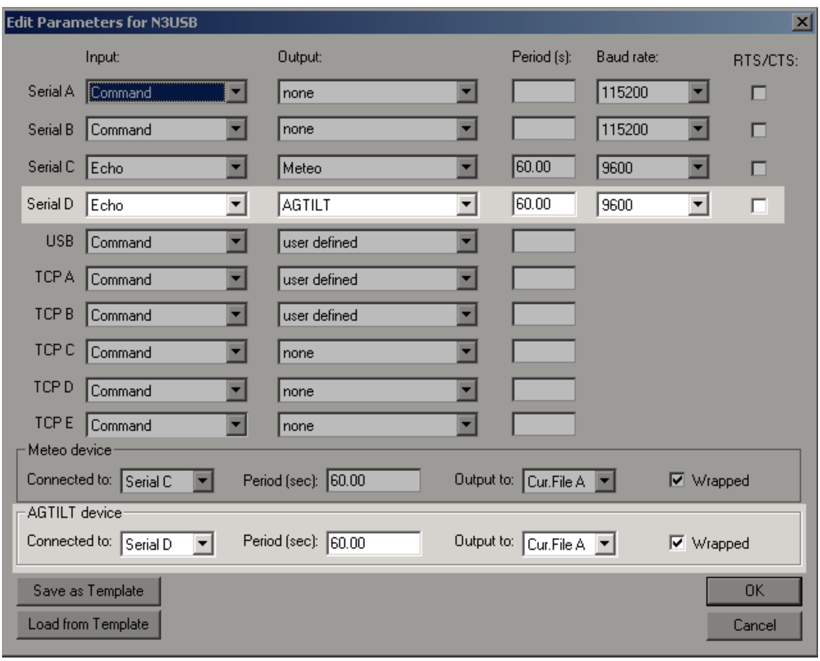


Figure VI-3. Edit Parameters

All the parameters for AGTILT (except Baud Rate) are set in the AGTILT device field. See “AGTILT Page” on page 2-70 for details.

Rover Accounting

With the help of the Rover Accounting application one can create and view reports about TopNET services usage. These reports are based on data stored in the TopNET system while operating, and compile information about time and duration of all the connections with clients' systems, requested service types, and amount of data sent to each client during this connection.

Rover Accounting application provides an opportunity to:

– create reports about all the connections established in the given period of time,

– create summary reports that contain total quantity of connections and entire amount of the received data for the given period, for each client,

– export any of the created reports to the CSV format.

Launching

For now, the Rover Accounting application can be launched through the **Start Up** menu from the **Topcon** group, or using a **View** ▸ **Rover Accounting** menu selection.

Before launching the Rover Accounting module, you need to ensure that your Hardware Authorization Key is attached to the PC and has a corresponding option enabled (leased or purchased). See "Authorization Options" on page VII-6.

If the Rover Accounting option is NOT active, contact your dealer for an Option file update. (See the description of an Option file, for example, in "Hardware Key Usage" section on page 3-57).

Information about the services provided by the TopNET system is stored as a data base. Each day information is saved to a separate file with ADB extension. These files are available for listing through **File ▸ Open DB File(s)**. A standard dialog box opens that suggests you to select a file from PC.

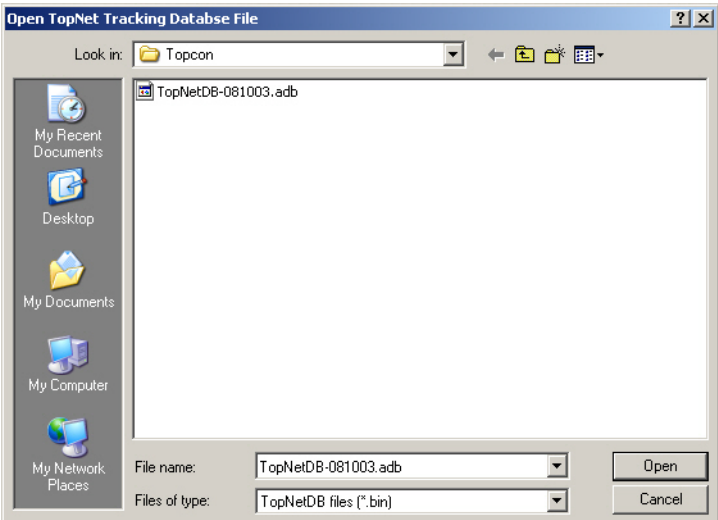


Figure VII-1. Open TopNET Tracking Database File

Select a file and click **Open**. A screen with raw data, corresponding to the chosen date, is shown. It can be saved and sorted as a usual Report screen, as described below.

Operation

At first, main screen of the Rover Accounting application is empty. To create a report about current network operation, select **File ▸ Create Report**:

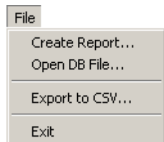


Figure VII-2. File Menu

In the **Create Report** screen set the parameters of the query:

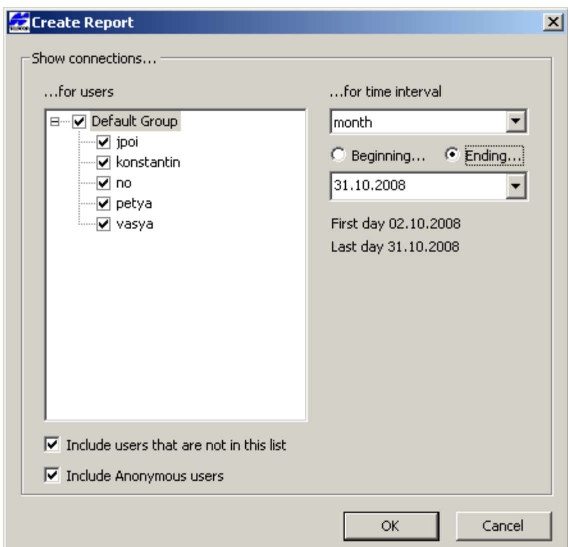


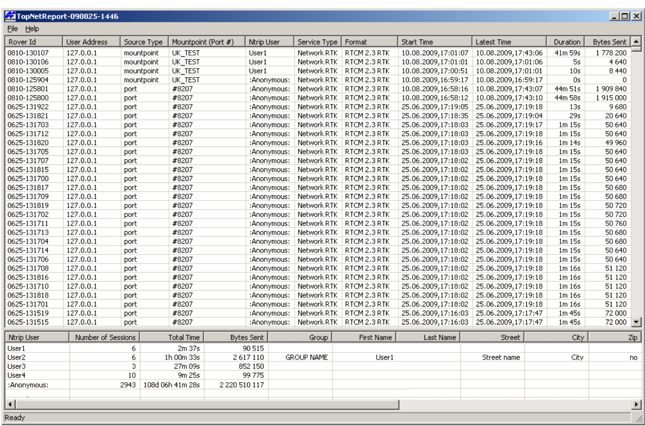
Figure VII-3. Create New Query

- *Show connections for users* - by default, all users are included to the report. To exclude a group or a user from the report, remove the corresponding check mark.
- *Time interval* - a predefined period of time, for which the report is formed. Notice that a month can have different length: 30 or 31 days, so check the dates *First day* and *Last day* displayed below.

- *Beginning/Ending* - select one of these radio buttons depending upon the date set in the calendar: is it the beginning of the time interval, or ending.
- *Include users that are not in the list* - checked by default, includes into Report those users, whose records are absent in the list at the current moment. For example, if a week ago a user was deleted from TopNET-N/-V, their records can be seen in a Report created for the last month.
- *Include Anonymous users* - if checked, the Anonymous users will be included into the Report. These are either Ntrip users that did not send login/password (if Ntrip Authentication is set to None) or non-Ntrip users. All anonymous users are represented as one ":Anonymous:" entry in the *Summary* field.

Press **OK**.

A new report is created according to the parameters of the query:


TopNetReport-090825-1446

File Help

Rover Id	User Address	Source Type	Mountpoint (Port #)	Ntrip User	Service Type	Format	Start Time	Latest Time	Duration	Bytes Sent
0810-130107	127.0.0.1	mountpoint	UK_TEST	User1	Network RTK	RTCM 2.3 RTK	10.08.2009,17:01:07	10.08.2009,17:43:06	41m 59s	1 778 200
0810-130106	127.0.0.1	mountpoint	UK_TEST	User1	Network RTK	RTCM 2.3 RTK	10.08.2009,17:01:01	10.08.2009,17:01:06	5s	4 640
0810-130005	127.0.0.1	mountpoint	UK_TEST	User1	Network RTK	RTCM 2.3 RTK	10.08.2009,17:00:51	10.08.2009,17:01:01	10s	8 440
0810-125904	127.0.0.1	mountpoint	UK_TEST	:Anonymous:	Network RTK	RTCM 2.3 RTK	10.08.2009,16:59:17	10.08.2009,16:59:17	0s	0
0810-125801	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	10.08.2009,16:58:16	10.08.2009,17:43:07	44m 51s	1 909 840
0810-125800	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	10.08.2009,16:58:12	10.08.2009,17:43:10	44m 58s	1 915 000
0625-131922	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:19:05	25.06.2009,17:19:18	13s	9 680
0625-131821	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:35	25.06.2009,17:19:04	29s	20 640
0625-131703	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:03	25.06.2009,17:19:17	1m 15s	50 640
0625-131712	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:03	25.06.2009,17:19:18	1m 15s	50 640
0625-131820	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:03	25.06.2009,17:19:16	1m 14s	49 960
0625-131705	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:03	25.06.2009,17:19:18	1m 15s	50 640
0625-131707	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 640
0625-131815	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 640
0625-131700	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 640
0625-131817	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 680
0625-131709	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 680
0625-131819	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 720
0625-131702	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 720
0625-131711	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 760
0625-131713	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 680
0625-131704	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 680
0625-131714	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 640
0625-131706	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 15s	50 640
0625-131708	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 16s	51 120
0625-131816	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 16s	51 120
0625-131710	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 16s	51 120
0625-131818	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 16s	51 120
0625-131701	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:18:02	25.06.2009,17:19:18	1m 16s	51 120
0625-131519	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:16:03	25.06.2009,17:17:47	1m 45s	72 000
0625-131515	127.0.0.1	port	#8207	:Anonymous:	Network RTK	RTCM 2.3 RTK	25.06.2009,17:16:03	25.06.2009,17:17:47	1m 45s	72 000

Ntrip User	Number of Sessions	Total Time	Bytes Sent	Group	First Name	Last Name	Street	City	Zip
User1	6	2m 37s	90 515						
User2	6	1h 00m 33s	2 617 110	GROUP NAME	User1		Street name	City	no
User3	3	27m 09s	852 150						
User4	10	9m 25s	99 775						
:Anonymous:	2943	108d 06h 41m 28s	2 220 510 117						

Ready

Figure VII-4. Report Screen – Details and Summary

Upper part of the screen represents a detailed list of connections. Each row corresponds to one connection session.

The columns of this table are the same as those of Rover Panel at the Main view in TopNET-N/-V:

- The *Rover ID* column lists the unique identification number for each rover.

- The *User Address* column shows IP number of the rover.
- *Source Type* can be either “*mountpoint*” or “*port*”.
- The *Mountpoint* (*Port #)* column shows the number of the TCP/IP port the rover is connected to, or an Ntrip Mountpoint name if the rover uses Ntrip protocol for connection.
- The *Ntrip User* column displays the name of the corresponding user if the rover is using Ntrip for connection to TopNET-N/-V.
- The *Service Type* column displays the type of the service provided to the rover.
- The *Format* column shows the format of the messages sent to the rover.
- *Start Time* - date and time of the first message sent to the rover.
- *Latest Time* - date and time of the last message sent to the rover.
- The *Bytes Sent* column shows the total number of bytes sent to the rover.
- The *Duration* column shows the time of the rover connection to TopNET-N/-V.

Lower part of the screen shows a summary of the current Report. Each row corresponds to one user, matches data set in the **Ntrip User** screen (i.e., Figure 3-41 on page 3-41), and displays the following parameters of the user connection:

- *Ntrip User* - name of an Ntrip user, or “Anonymous” entry.
- *Number of Sessions*,
- *Total Time*,
- *Bytes Sent*,
- *Group,*
- *First Name,*
- *Last Name,*
- *Street,*
- *City,*

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- *Zip - zip index of a user,*
- *Country,*
- *Primary E-mail,*
- *Secondary E-mail,*
- *Office Phone Number,*
- *Mobile Phone Number,*
- *Fax.*

To update a table using other query parameters, select **File ▸ Create Report...** menu, in the **Create New Query** screen change the parameters and press **OK**.

To arrange a table by any column, press its header.

This table can be exported to CSV format. To export the Report database, select **File ▸ Export to CSV...** menu, enter the name and location of the new file, and press **OK**.

Authorization Options

Information about Rover Accounting option is available through **Help ▸ Activated Options...**

NOTICE *NOTICE*

The Authorization Options screen for Rover Accounting and Authorization Options screen for the main application (see "Hardware Key Usage" on page 3-57 section, for example) use the same Option file, but contain different options.

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The **Authorization Options** screen for Rover Accounting displays the following information:

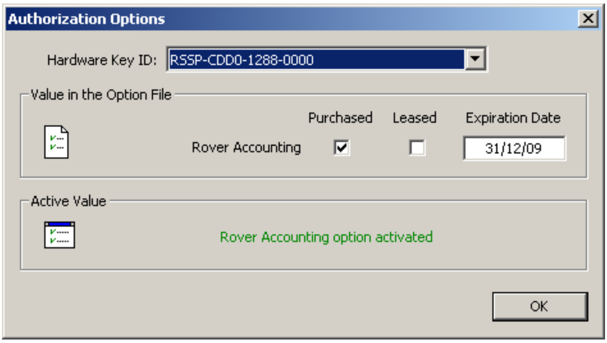

Figure VII-5. Authorization Options

- *Hardware Key ID* - the unique number of a Hardware Protection Key applied to the PC.
- *Values in the Option File* - here the options permitted by the Option File, are displayed. The *Rover Accounting* option can be *Purchased* and/or *Leased*. The check mark designates the status of the option. The Expiration Date field shows the Lease expiration date for each option.
- *Active Values* - the current value of the *Rover Accounting* option shows, whether it is activated or not.

Press **OK** to close the screen.

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TopNET Web Interface Setup

Prerequisitions

- TopNET-N or TopNET-V;
- Java Runtime Environment 6.x or later (see details at http://sun.com/getjava/);
- Web browser: Internet Explorer 7.x or later, Mozilla Firefox 2.x or later, or Opera 9.x or later.

Installation instructions

1. Start TopNET-N or TopNET-V.
2. Go to http://tomcat.apache.org/download-60.cgi.
3. In section *Binary Distributions* click *Windows Service Installer* Save the setup file of the Apache Tomcat server on your PC.

4. Launch the file.

Figure VIII-1. Apache TomCat Server – Page 1

5. Press the **Next** button.

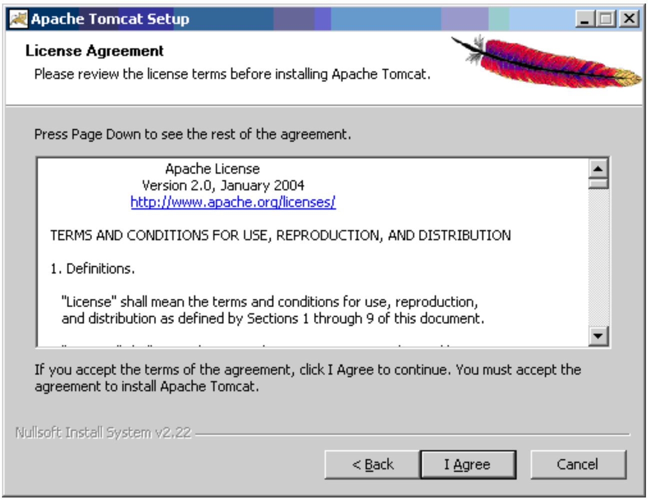

Figure VIII-2. Apache TomCat Server –Page 2

6. Read the license agreement.

7. Press the **I Agree** button.

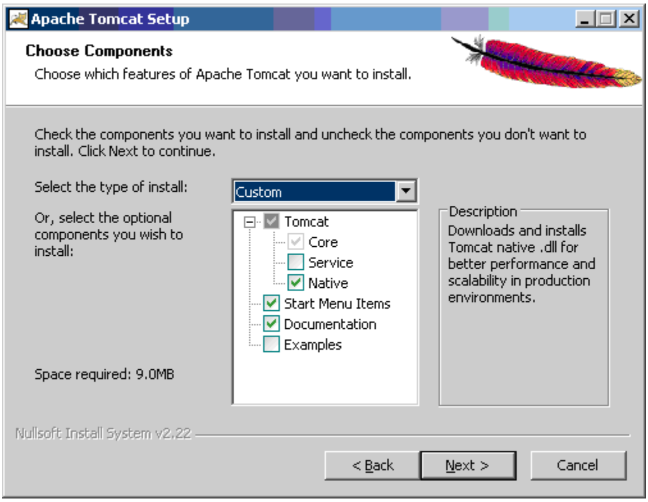

Figure VIII-3. Apache TomCat Server –Page 3

8. Check *Native* for the best productivity of the Apache Tomcat server (an internet connection must be active).

9. Press **Next:**

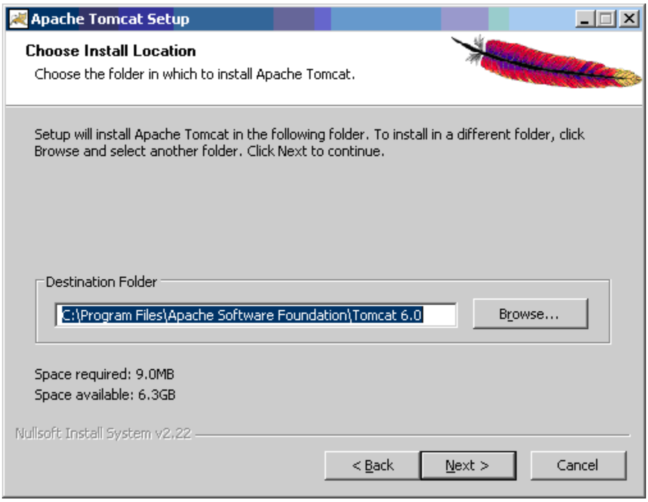

Figure VIII-4. Apache TomCat Server –Page 4

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10. Select a directory where the Apache Tomcat server will be installed or use the default one.
11. Press **Next.**

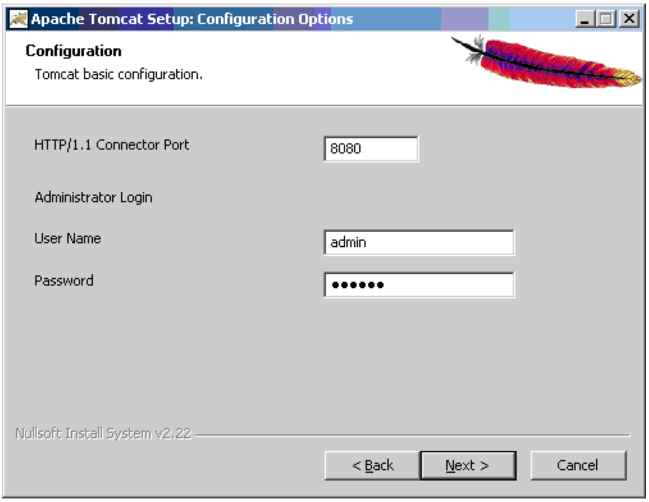

Figure VIII-5. Apache TomCat Server –Page 5

12. Check that the *HTTP/1.1 Connector Port* is 8080 (by default).
13. Set the administrator's name and password, these will be used for login through web-interface.
14. Press the **Next** button.

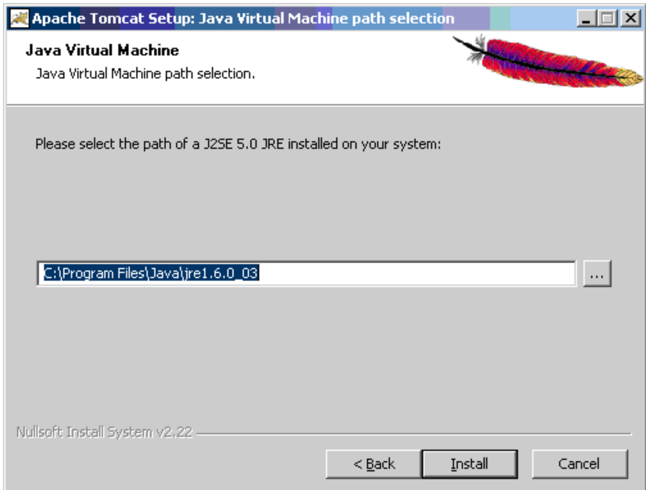

Figure VIII-6. Apache TomCat Server –Page 6

15. Select a path to latest JRE (or use a default one). You will be able to change this later.

WARNING

Even if the Apache Tomcat Server Installation Wizard prompts for J2SE 5.0 JRE, remember that JRE version 6.x or later is required for proper operation of the server.

16. Press the **Install** button.

17. Wait while the setup process is done.

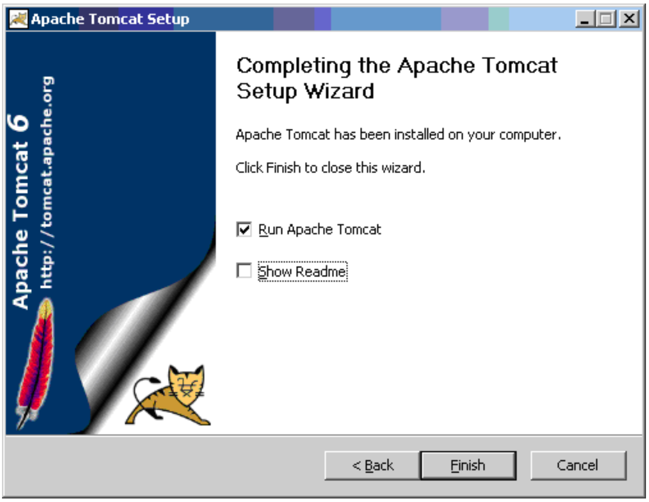

Figure VIII-7. Apache TomCat Server –Page 7

18. Uncheck *Show Readme*.
19. Press *Finish*.

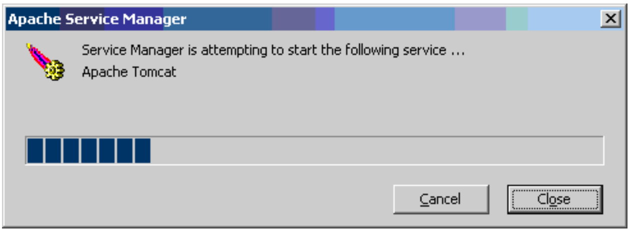

Figure VIII-8. Apache Server Manager

A window indicating the start process of the Apache Tomcat will be shown.

20. Wait for the process is finished and the window is closed.
21. Launch a browser and enter *http://<server_name>:8080/manager/html/* to the address bar.

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To avoid port conflict, check that ports 8080, 8005 and 8009 used by Tomcat server are free, because until recently port 8080 was used by Ntrip in TopNET-N/-V, and 8005 was the default for RTK service.

22. Press *Enter* on your keyboard
23. Enter the name and the password used during the installation of the Apache Tomcat server.

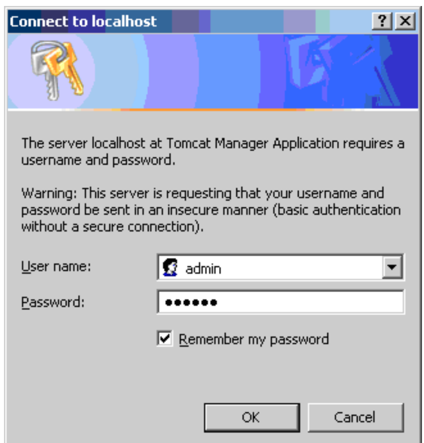

Figure VIII-9. Connect to Localhost

24. Press *Ok*. If the name and the password have been correct there will be a section on the web page:

Figure VIII-10. Select WAR File to Upload

25. Click the **Browse...** button.

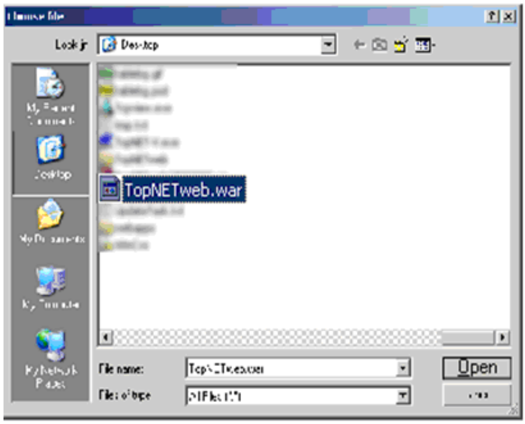

Figure VIII-11. Browsing for the File

26. Select the *TopNETweb.war* file from the installation TopNET directory (usually it is C:\Program Files\Topcon\TopNET). Press **Open** and return to the previous screen:

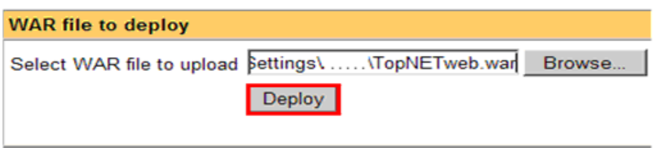

Figure VIII-12. Select WAR File to Upload

27. Press the **Deploy** button. The TopNETweb application will be shown in the list of web-applications:

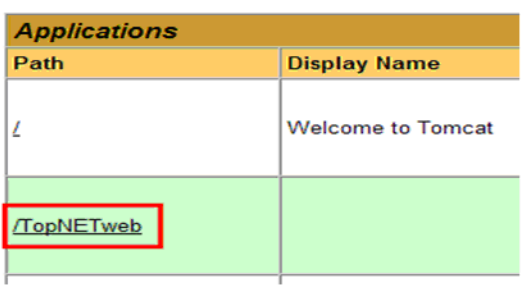

Figure VIII-13. List of Web Applications

28. The web application *TopNET web interface* will be launched by clicking */TopNETweb*. The application shows the same picture similar to TopNET-N/-V in 15-30 seconds.

The application is installed and launched.

Google Maps integration

If it's necessary to use Google Maps, some additional steps should be done:

1. Create the Google account. For that purpose visit https://www.google.com/accounts/Login?continue and login.
2. Go to http://code.google.com/apis/maps/signup.html
3. Read the Google Maps API license agreement and agree by placing a checkmark in the corresponding screen.

Figure VIII-14. Generation of an API Key

NOTICE

NOTICE

Server address you type in "My web site URL" field must include port number used by your Tomcat server (e.g., 123.123.123.123:8080).

The address you type in browser when you go to TopNET Web must match the address you typed in Google, so even if "myserver.com" and "123.123.123.123" is the same computer, web user will get access to Google maps only on one of these (the one you requested the license for).

Note that if your server has two IP addresses (e.g., an external one and one in your corporate network), you can have a license for only one of them, not for both.

4. Enter your server address and press the **Generate API Key** button.

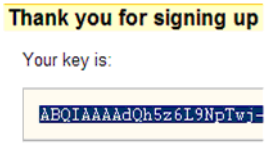

Figure VIII-15. Key Number Sample

5. Copy to the clipboard the acquired API key.
6. Select **Setup ▸ Web Access** in the TopNET-N/-V menu.

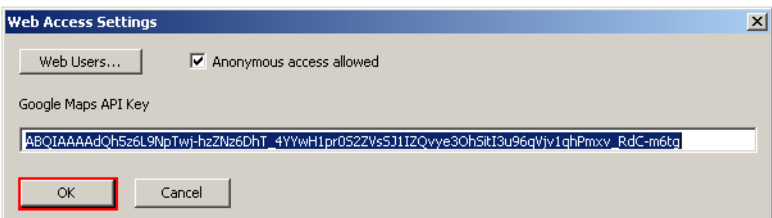

Figure VIII-16. Web Access Settings

7. Paste the Google Maps API key acquired on step 5.
8. Press **OK**.

Now you can use the Google Maps service. To check, enter “http://<server_name>:8080/TopNETweb/” in a browser’s address bar and press Enter.

Abbreviations

ARP - Antenna Reference Point

CMR - Compact Measurement Record format.

CMR+ - improved CMR.

DGPS - Differential GPS method using pseudorange corrections.

GGA - a NMEA message containing Global Positioning System Fix Data.

GNSS - Global Navigation Satellite System. Includes GPS, GLONASS and Galileo.

NMEA - National Marine Electronic Association.

Ntrip - Networked Transport of RTCM via Internet Protocol.

PRN - Pseudo-Random Noise.

RTCM - Radio Technical Commission for Maritime Services.

RTK - Real-Time Kinematic.

TCP/IP -Transmission Control Protocol/ Internet Protocol.

UTC - Coordinated Universal Time.

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FM 68448

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